

# Stochastic Processes

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2026

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# Preface

The following lecture notes are prepared for the course *Stochastic Processes*, which is a 10 ECTS course in probability theory. The prerequisites include a course in measure theory and advanced probability. Some basic knowledge of Markov chains with finite state space is useful as well.

In the course, we study discrete as well as continuous time stochastic processes. The discrete time case is treated first. We will study examples of stochastic processes, their existence (based on Kolmogorov's Consistency Theorem), and derive several interesting properties. We focus particularly on Markov chains and especially on the case when the state space is countable. This part is inspired by many good books in the field. In particular, let me mention Meyn and Tweedie (2009), Durrett (2010), Cappé et al. (2005), Grimmett and Stirzaker (2001), and Robert and Casella (2004).

In the second part of the course, we study continuous time stochastic processes, focusing particularly on Lévy processes such as Brownian motion and the Poisson process. Appropriate references include, e.g., Durrett (2010), Karlin and Taylor (1975), Durrett (2016), Karatzas and Shreve (1991), Schilling and Partzsch (2014), Karlin and Taylor (1981), Revuz and Yor (1999), and Norris (1998).

I thank Henrik Nørgaard Hald for many useful suggestions and for finding many typos in a previous version of the notes. I thank Lars Madsen for extensive help on all kinds of questions related to  $\text{\LaTeX}$ .

Jan Pedersen, December 2019

The notes have been slightly expanded and several exercises have been added.

Jan Pedersen, July 2022

A few corrections have been made. I thank Anders Vendelbo Agernem for providing a detailed list of misprints.

Jan Pedersen, July 2023

I have corrected some typos and added a few examples and exercises.

Jan Pedersen, December 2025

# Preliminaries

Before giving some basic definitions, we need the following.

1. Put

$$\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}, \quad \mathbb{N} = \{1, 2, \dots\}, \quad \text{and} \quad \mathbb{N}_0 = \{0, 1, 2, \dots\}.$$

2. Throughout,  $S$  denotes a non-empty Polish space and  $\mathcal{B}(S)$  is the corresponding Borel  $\sigma$ -field. Since  $S$  is a separable metric space,

$$\mathcal{B}(S^n) = \mathcal{B}(S)^n, \quad \text{for } n \in \mathbb{N}, \tag{1.1}$$

where the left-hand side denotes the Borel- $\sigma$ -field on  $S^n$  and the right-hand side is the product  $\sigma$ -field on  $S^n$ .

3. When  $S$  is countable, we always use the discrete metric, which implies that  $\mathcal{B}(S)$  is the collection of all subsets of  $S$ .
4. Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space. An  $(\mathcal{F}, \mathcal{B}(S))$ -measurable map  $Z : \Omega \rightarrow S$  is called a *random variable* or a *random element* and  $S$  is called the *state space* of  $Z$ . The *distribution* of  $Z$  is the (push-forward) probability measure  $\mathbb{P}_Z$  on  $(S, \mathcal{B}(S))$  defined as  $\mathbb{P}_Z(A) = \mathbb{P}(Z \in A)$  for  $A \in \mathcal{B}(S)$ .
5. Two  $S$ -valued random variables  $Z$  (defined on  $(\Omega, \mathcal{F}, \mathbb{P})$ ) and  $Z'$  (defined on  $(\Omega', \mathcal{F}', \mathbb{P}')$ ) have the same distribution (which we write  $Z \stackrel{d}{=} Z'$ ) if  $\mathbb{P}_Z = \mathbb{P}'_{Z'}$ .
6. Let  $n \in \mathbb{N}$  and consider a map  $Z_j : \Omega \rightarrow S$  for  $j = 1, \dots, n$ . By (1.1),  $Z_j$  is  $(\mathcal{F}, \mathcal{B}(S))$ -measurable for every  $j = 1, \dots, n$  if and only if the vector  $Z$  defined as  $Z = (Z_1, \dots, Z_n)$  is  $(\mathcal{F}, \mathcal{B}(S^n))$ -measurable. That is,  $Z_j$  is an  $S$ -valued random variable for all  $j = 1, \dots, n$  if and only if  $Z$  is an  $S^n$ -valued random variable.

7. When  $Z$  is of the form  $Z = (Z_1, \dots, Z_n)$  for some  $n \in \mathbb{N}$  where  $Z_j$  is a random variable with state space  $S$  for all  $j = 1, \dots, n$ , we also refer to  $Z$  as a *random vector* with state space  $S^n$ .
8. Let  $Z = (Z_1, \dots, Z_n)$  be an  $S^n$ -valued random vector and  $Z' = (Z'_1, \dots, Z'_n) : \Omega' \rightarrow S^n$  (defined on a probability space  $(\Omega', \mathcal{F}', \mathbb{P}')$ ) be another  $S^n$ -valued random vector. By (1.1) and a uniqueness result for measures,  $Z \stackrel{d}{=} Z'$  (i.e.,  $\mathbb{P}(Z \in A) = \mathbb{P}'(Z' \in A)$  for all  $A \in \mathcal{B}(S^n)$ ) if and only if

$$\mathbb{P}(Z_1 \in B_1, \dots, Z_n \in B_n) = \mathbb{P}'(Z'_1 \in B_1, \dots, Z'_n \in B_n)$$

for all  $B_1, \dots, B_n \in \mathcal{B}(S)$ .

## Some definitions and examples

In the following, we define stochastic processes and give some examples.

**Definition 2.1.** Let  $T$  be a non-empty index set. A **stochastic process with state space  $S$  and index set  $T$**  is a collection of  $S$ -valued random variables  $(Y_t)_{t \in T}$  defined on some probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ . When the state space is  $S = \mathbb{R}$  we speak of a **real stochastic process**.

As usual, the probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  will often not be mentioned explicitly when we consider stochastic processes.

**Definition 2.2.** A real stochastic process  $(Y_t)_{t \in T}$  is called **Gaussian** if  $(Y_{t_1}, \dots, Y_{t_k})$  follows a  $k$ -variate normal distribution for all  $k \in \mathbb{N}$  and  $t_1, \dots, t_k \in T$ . (When  $T$  is a subset of  $\mathbb{R}$  it suffices to consider  $t_1 < \dots < t_k$ .)

A stochastic process  $(X_n)_{n \in \mathbb{Z}}$  is **stationary** if

$$(X_0, \dots, X_n) \stackrel{d}{=} (X_k, \dots, X_{k+n}) \quad \text{for every } n \in \mathbb{N}_0 \text{ and } k \in \mathbb{Z}. \quad (2.1)$$

For stochastic processes indexed by  $\mathbb{R}$ ,  $\mathbb{R}_+$ ,  $\mathbb{N}_0$  or  $\mathbb{N}$ , stationarity is defined similarly.

**Example 2.3.** Let us give a few examples.

1. Let  $(X_t)_{t \in \mathbb{R}}$  be Gaussian and define  $Y_t$  as  $Y_t = X_t - 7X_{t-1}$  for  $t \in \mathbb{R}$ . Then  $(Y_t)_{t \in \mathbb{R}}$  is a Gaussian process.
2. Let  $(\epsilon_n)_{n \in \mathbb{N}_0}$  denote an i.i.d. sequence satisfying  $\epsilon_n \sim N(0, 1)$  for  $n \in \mathbb{N}_0$ . Then  $(\epsilon_n)_{n \in \mathbb{N}_0}$  is stationary and Gaussian.

3. The property  $\epsilon_1 \stackrel{d}{=} \epsilon_2 \sim N(0, 1)$  does not imply that  $(\epsilon_1, \epsilon_2)$  is Gaussian. See Exercise 4.
4. Let  $(\epsilon_n)_{n \in \mathbb{Z}}$  denote an i.i.d. sequence such that  $\epsilon_n \sim N(0, 1)$  for  $n \in \mathbb{Z}$ . Let  $(\alpha(j))_{j \in \mathbb{N}_0}$  denote a real sequence satisfying  $\sum_{j=0}^{\infty} \alpha(j)^2 < \infty$ . Define

$$X_n = \sum_{j=0}^{\infty} \alpha(j) \epsilon_{n-j} = \sum_{r=-\infty}^n \alpha(n-r) \epsilon_r \quad \text{for } n \in \mathbb{Z}. \quad (2.2)$$

(To verify the last equality put  $r = n - j$ ). Then

- a)  $X_n$  is well-defined in  $L^2(\mathbb{P})$  for  $n \in \mathbb{Z}$ . (That is, the sums in (2.2) converge in  $L^2(\mathbb{P})$ ).
- b)  $(X_n)_{n \in \mathbb{Z}}$  is a Gaussian process.
- c)  $(X_n)_{n \in \mathbb{Z}}$  is a stationary process.

For a proof see Exercise 14. ■

The process defined by (2.2) is a so-called *Gaussian moving average*. Since moving averages are very important in probability theory and statistics, we will briefly mention some generalizations here.

**Definition 2.4.** A real stochastic process  $(\epsilon_n)_{n \in \mathbb{Z}}$  is called a **white noise** if  $\epsilon_n \in L^2(\mathbb{P})$  for all  $n \in \mathbb{Z}$  and there is a  $\sigma^2 \geq 0$  such that  $E[\epsilon_n] = 0$ ,  $\text{Var}(\epsilon_n) = \sigma^2$ , and  $\text{Cov}(\epsilon_n, \epsilon_m) = 0$  for all  $n, m \in \mathbb{Z}$  with  $n \neq m$ .

A real stochastic process  $(X_n)_{n \in \mathbb{Z}}$  is called **weakly stationary** if  $X_n \in L^2(\mathbb{P})$  for every  $n \in \mathbb{Z}$ ,  $E[X_n] = E[X_0]$  and  $\text{Cov}(X_n, X_m) = \text{Cov}(X_0, X_{m-n})$  for arbitrary  $n, m \in \mathbb{Z}$ .

Whenever  $(\epsilon_n)_{n \in \mathbb{Z}}$  is a white noise and  $(\alpha(m))_{m \in \mathbb{N}_0}$  is a real sequence satisfying  $\sum_{m=0}^{\infty} \alpha(m)^2 < \infty$  the process  $(X_n)_{n \in \mathbb{Z}}$  defined by (2.2) is well-defined and weakly stationary. (See Exercise 9). In fact,

$$\text{Cov}(X_n, X_m) = \sigma^2 \sum_{k=0}^{\infty} \alpha(k) \alpha(k + |n - m|) \quad \text{for } m, n \in \mathbb{Z}. \quad (2.3)$$

Call  $(X_n)_{n \in \mathbb{Z}}$  a moving average driven by  $(\epsilon_n)_{n \in \mathbb{Z}}$ . When  $(\epsilon_n)_{n \in \mathbb{Z}}$  is i.i.d. the moving average  $(X_n)_{n \in \mathbb{Z}}$  is even stationary.

**Remark 2.5.** The *Wold decomposition* cf. Giraitis et al. (2012, Theorem 3.2.1) shows that any weakly stationary sequence  $(X_n)_{n \in \mathbb{Z}}$  satisfying weak regularity conditions is a moving average.

As an example, let  $p, q \in \mathbb{N}$  and consider the polynomials  $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$  and  $\theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$  for  $\theta_1, \dots, \theta_q, \phi_1, \dots, \phi_p \in \mathbb{R}$ . Assume  $(\epsilon_n)_{n \in \mathbb{Z}}$  is a white noise. Recall that if

$$|\phi(z)|, |\theta(z)| > 0 \quad \text{for all } |z| \leq 1 \quad (2.4)$$

and  $\phi$  and  $\theta$  do not have common roots, then there exists a unique stationary ARMA( $p, q$ ) process  $(X_n)_{n \in \mathbb{Z}}$  satisfying

$$\phi(B)X_n = \theta(B)\epsilon_n \quad \text{for } n \in \mathbb{Z}. \quad (2.5)$$

Here  $B$  denotes the shift operator, which, in a non-rigorous sense, is defined as  $BX_n = X_{n-1}$  and more generally  $B^j X_n = X_{n-j}$  for  $j \in \mathbb{N}$ . Thus, the precise way to interpret (2.5) is as

$$X_n - \phi_1 X_{n-1} - \dots - \phi_p X_{n-p} = \epsilon_n + \theta_1 \epsilon_{n-1} + \dots + \theta_q \epsilon_{n-q} \quad \text{for } n \in \mathbb{Z}. \quad (2.6)$$

Further recall that AR is short for *AutoRegressive* while MA stands for *Moving Average*. To represent the ARMA( $p, q$ ) process  $(X_n)_{n \in \mathbb{Z}}$  rewrite (2.5) (in a non-rigorous way) as  $X_n = \frac{\theta(B)}{\phi(B)} \epsilon_n$ . To make this precise, we use that under the assumptions stated above  $z \mapsto \theta(z)/\phi(z)$  has a power series expansion

$$\frac{\theta(z)}{\phi(z)} = \sum_{j=0}^{\infty} \alpha(j) z^j \quad \text{for } |z| \leq r \quad (2.7)$$

for some  $r > 1$  and some sequence  $(\alpha(j))_{j \in \mathbb{N}_0}$ . (See Brockwell and Davis (2006, Theorem 3.1.3), where you will also find more general so-called Laurent expansions). Thus,  $X_n$  must be given by  $\sum_{j=0}^{\infty} \alpha(j) B^j \epsilon_n$  or, in a rigorous sense, by (2.2). By Giraitis et al. (2012, p. 181),  $\alpha(n) \rightarrow 0$  as  $n \rightarrow \infty$  exponentially fast in this case, which in a sense means that ARMA processes have short range dependence.

An interesting and useful generalization of the ARMA framework is the class of ARFIMA( $p, d, q$ ) models (Giraitis et al. (2012, Definition 7.2.2). Here  $d = 0$  corresponds to ARMA( $p, q$ ), and  $(X_n)_{n \in \mathbb{Z}}$  is still a moving average but  $\alpha(n)$  generally converges to zero at a much slower rate (Giraitis et al. (2012, Theorem 7.2.3)) which causes long range dependence.

Brockwell and Davis (2006) is one of the standard references for statistical and probabilistic properties of time series, including ARMA processes. See also Giraitis et al. (2012) for parameter estimation in ARFIMA models and other models with long range dependence. ■

**Example 2.6 (Stationary AR(1) processes).**

Let  $(\epsilon_n)_{n \in \mathbb{Z}}$  be an i.i.d. sequence of real random variables. Assume  $\epsilon_n \in L^2(\mathbb{P})$  and  $\mathbb{E}[\epsilon_n] = 0$  for  $n \in \mathbb{Z}$ . Let  $\theta \in (-1, 1)$ . In the following we show that there is a unique stationary process  $(Y_n)_{n \in \mathbb{Z}}$  satisfying

$$Y_{n+1} = \theta Y_n + \epsilon_{n+1} \quad \text{for } n \in \mathbb{Z}. \quad (2.8)$$

Indeed, we have the following:

1. Let  $Y_n = \sum_{m=0}^{\infty} \theta^m \epsilon_{n-m} = \sum_{k=-\infty}^n \theta^{n-k} \epsilon_k$  for  $n \in \mathbb{Z}$  (where the sums exist in  $L^2(\mathbb{P})$ ). Then  $(Y_n)_{n \in \mathbb{Z}}$  is a stationary square-integrable process satisfying (2.8).
2. Let  $(Y_n)_{n \in \mathbb{Z}}$  be a real stochastic process satisfying (2.8). Then

$$Y_n = \theta^{n-m} Y_m + \sum_{k=m+1}^n \theta^{n-k} \epsilon_k \quad \text{for } n > m. \quad (2.9)$$

3. Let  $(Y_n)_{n \in \mathbb{Z}}$  be a real stationary stochastic process satisfying (2.8). Then  $Y_n = \sum_{k=-\infty}^n \theta^{n-k} \epsilon_k$  for  $n \in \mathbb{Z}$ .

See Exercise 10. ■

**Example 2.7 (GARCH processes).**

Let  $(\epsilon_n)_{n \in \mathbb{Z}}$  be an i.i.d. sequence of real random variables. Assume  $\epsilon_n \in L^2(\mathbb{P})$ ,  $\mathbb{E}[\epsilon_n] = 0$  and  $\text{Var}(\epsilon_n) = 1$  for  $n \in \mathbb{Z}$ . A process  $(X_n)_{n \in \mathbb{Z}}$  is called a GARCH( $p, q$ ) process (with  $p, q \in \mathbb{N}$ ) if it is stationary and

$$X_n = \sigma_n \epsilon_n \quad \text{for } n \in \mathbb{Z}, \quad (2.10)$$

where

$$\sigma_n^2 = \alpha_0 + \sum_{j=1}^p \alpha_j X_{n-j}^2 + \sum_{j=1}^q \beta_j \sigma_{n-j}^2 \quad \text{for } n \in \mathbb{Z}, \quad (2.11)$$

and  $\alpha_0 > 0$ ,  $\alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q \geq 0$ , and  $(\sigma_n)_{n \in \mathbb{Z}}$  is a stationary strictly positive real process.

GARCH processes (including statistical aspects and many interesting generalizations) are well described in Francq and Zakoïan (2010). Let us briefly consider the case  $p = q = 1$ . A sufficient condition for the existence of a GARCH(1, 1) process is that  $\alpha_1 + \beta_1 < 1$ . Assuming this is the case it then follows by backwards iteration that  $\sigma_n^2$  is given by

$$\sigma_n^2 = \alpha_0 + \alpha_0 \sum_{i=1}^{\infty} \prod_{j=1}^i (\alpha_1 \epsilon_{n-j}^2 + \beta_1) \quad \text{for } n \in \mathbb{Z}. \quad (2.12)$$

Moreover,  $(X_n)_{n \in \mathbb{Z}}$  in (2.10) is a white noise. ■

## The canonical path space and Kolmogorov's Consistency Theorem

In the following, we first consider stochastic processes indexed by  $\mathbb{N}_0$ . Such processes will also be denoted  $(Y_n)_{n \in \mathbb{N}_0}$  or, more frequently,  $(Y_n)_{n \geq 0}$ .

We often define our stochastic processes on a particular measurable space  $(S^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$  which can be described as follows: First, let  $S^{\mathbb{N}_0}$  be the set of  $S$ -valued sequences indexed by  $\mathbb{N}_0$ , i.e.,

$$S^{\mathbb{N}_0} = \{(\omega_0, \omega_1, \dots) : \omega_n \in S \text{ for } n \in \mathbb{N}_0\}.$$

The set  $S^{\mathbb{N}_0}$  is referred to as a *canonical path space*. For  $n \in \mathbb{N}_0$ , define  $X_n : S^{\mathbb{N}_0} \rightarrow S$  as  $X_n(\omega_0, \omega_1, \dots) = \omega_n$ . That is,  $X_n$  picks out the  $n$ th coordinate of the sequence  $(\omega_0, \omega_1, \dots)$ . We call  $(X_n)_{n \geq 0}$  the *coordinate process* on  $S^{\mathbb{N}_0}$ . Further, let  $\mathcal{B}(S)^{\mathbb{N}_0} = \sigma(X_n \mid n \in \mathbb{N}_0)$  be the  $\sigma$ -field generated by the coordinate process. Written out in more detail,  $\mathcal{B}(S)^{\mathbb{N}_0}$  is the least  $\sigma$ -field containing all sets of the form

$$\{X_n \in B\} \text{ for some } n \in \mathbb{N}_0 \text{ and some } B \in \mathcal{B}(S). \quad (3.1)$$

Note that  $\mathcal{B}(S)^{\mathbb{N}_0}$  is also the least  $\sigma$ -field containing all sets of the form

$$\{X_0 \in B_0, \dots, X_n \in B_n\} \text{ for some } n \in \mathbb{N}_0 \text{ and } B_0, \dots, B_n \in \mathcal{B}(S). \quad (3.2)$$

The collection of sets given by (3.2) has the advantage over (3.1) that it is stable under intersection.

Given a stochastic process  $(Y_n)_{n \geq 0}$  defined on  $(\Omega, \mathcal{F}, \mathbb{P})$ , define the map  $Y$  from  $\Omega$  to  $S^{\mathbb{N}_0}$  as

$$\begin{aligned} Y : \Omega &\rightarrow S^{\mathbb{N}_0} \\ \omega &\mapsto (Y_0(\omega), Y_1(\omega), \dots). \end{aligned}$$

**Lemma 3.1.** *The map  $Y$  is  $(\mathcal{F}, \mathcal{B}(S)^{\mathbb{N}_0})$ -measurable.*

*Proof.* We must verify that  $\{Y \in A\} \in \mathcal{F}$  for every  $A \in \mathcal{B}(S)^{\mathbb{N}_0}$ , and since this  $\sigma$ -field is generated by sets of the form (3.2), it suffices to consider such a set. (In fact, it would even suffice to consider (3.1)).

Writing things out in detail, we have

$$\begin{aligned} \{X_0 \in B_0, \dots, X_n \in B_n\} \\ &= \{(\omega_0, \omega_1, \dots) \in S^{\mathbb{N}_0} : X_0(\omega_0, \omega_1, \dots) \in B_0, \dots, X_n(\omega_0, \omega_1, \dots) \in B_n\} \\ &= \{(\omega_0, \omega_1, \dots) \in S^{\mathbb{N}_0} : \omega_0 \in B_0, \dots, \omega_n \in B_n\}, \end{aligned}$$

and hence

$$\begin{aligned} \{Y \in \{X_0 \in B_0, \dots, X_n \in B_n\}\} \\ &= \{\omega \in \Omega : (Y_0(\omega), Y_1(\omega), \dots) \in \{X_0 \in B_0, \dots, X_n \in B_n\}\} \\ &= \{\omega \in \Omega : Y_0(\omega) \in B_0, \dots, Y_n(\omega) \in B_n\} = \bigcap_{k=0}^n \{Y_k \in B_k\}. \end{aligned}$$

The set  $\{Y_k \in B_k\}$  is in  $\mathcal{F}$  for  $k = 0, \dots, n$  and  $B_0, \dots, B_n \in \mathcal{B}(S)$ , so it follows that

$$\{Y \in \{X_0 \in B_0, \dots, X_n \in B_n\}\} \in \mathcal{F} \quad \text{for } n \in \mathbb{N}_0, \text{ and } B_0, \dots, B_n \in \mathcal{B}(S).$$

Hence,  $Y$  is measurable. □

By measurability of  $Y$ , this map induces an image measure, to be denoted  $\mathbb{P}_Y$ , on  $(S^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$ . That is,  $\mathbb{P}_Y(A)$  is defined as

$$\mathbb{P}_Y(A) = \mathbb{P}(Y \in A), \quad A \in \mathcal{B}(S)^{\mathbb{N}_0}.$$

The image measure  $\mathbb{P}_Y$  is also referred to as *the distribution* of  $(Y_n)_{n \geq 0}$ . Two stochastic processes  $(Y_n)_{n \geq 0}$  and  $(Y'_n)_{n \geq 0}$ , defined on separate probability spaces  $(\Omega, \mathcal{F}, \mathbb{P})$  and  $(\Omega', \mathcal{F}', \mathbb{P}')$ , are said to be *versions* or to *have the same distribution* if  $\mathbb{P}_Y = \mathbb{P}'_{Y'}$ . We write this as  $Y \stackrel{d}{=} Y'$ . The following shows that this is the case if and only if the so-called *finite-dimensional marginal distributions* are the same:

**Lemma 3.2.** *Two stochastic processes  $(Y_n)_{n \geq 0}$  and  $(Y'_n)_{n \geq 0}$  are versions if and only if  $(Y_0, \dots, Y_n) \stackrel{d}{=} (Y'_0, \dots, Y'_n)$  for every  $n \in \mathbb{N}_0$ .*

*Proof.* By definition, the two processes are versions if and only if  $\mathbb{P}(Y \in A) = \mathbb{P}'(Y' \in A)$  for every  $A \in \mathcal{B}(S)^{\mathbb{N}_0}$ . Since the collection of sets in (3.2) is stable under

intersection and generates  $\mathcal{B}(S)^{\mathbb{N}_0}$ ,  $Y \stackrel{d}{=} Y'$  if and only if  $\mathbb{P}(Y \in A) = \mathbb{P}'(Y' \in A)$  for every  $A$  of the form

$$A = \{X_0 \in B_0, \dots, X_n \in B_n\} \text{ for some } n \in \mathbb{N}_0 \text{ and some } B_0, \dots, B_n \in \mathcal{B}(S). \quad (3.3)$$

The proof of Lemma 3.1 shows that for such an  $A$ ,

$$\{Y \in A\} = \{Y_0 \in B_0, \dots, Y_n \in B_n\} \quad \text{and} \quad \{Y' \in A\} = \{Y'_0 \in B_0, \dots, Y'_n \in B_n\}.$$

Hence, the two processes are versions if and only if

$$\mathbb{P}(Y_0 \in B_0, \dots, Y_n \in B_n) = \mathbb{P}'(Y'_0 \in B_0, \dots, Y'_n \in B_n)$$

for every  $n \in \mathbb{N}_0$  and  $B_0, \dots, B_n \in \mathcal{B}(S)$ , that is if and only if  $(Y_0, \dots, Y_n) \stackrel{d}{=} (Y'_0, \dots, Y'_n)$  for every  $n \in \mathbb{N}_0$ .  $\square$

**Example 3.3.** Let  $(Y_n)_{n \geq 0}$  and  $(Y'_n)_{n \geq 0}$  be real processes. We have the following results:

- (1) If  $(Y_n)_{n \geq 0}$  and  $(Y'_n)_{n \geq 0}$  are versions then  $(Y_n)_{n \geq 0}$  is Gaussian if and only if  $(Y'_n)_{n \geq 0}$  is Gaussian.
- (2) Two Gaussian processes  $(Y_n)_{n \geq 0}$  and  $(Y'_n)_{n \geq 0}$  are versions if and only if  $\mathbb{E}[Y_n] = \mathbb{E}[Y'_n]$  and  $\text{Cov}(Y_n, Y_m) = \text{Cov}(Y'_n, Y'_m)$  for all  $n, m \in \mathbb{N}_0$ .  $\blacksquare$

In the following, we characterize stochastic processes in terms of their finite-dimensional marginal distributions.

**Remark 3.4.** Let  $(Y_n)_{n \geq 0}$  be a stochastic process defined on a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ . For  $n \in \mathbb{N}_0$ , let  $\nu_n$  denote the distribution of  $(Y_0, \dots, Y_n)$ ; that is,

$$\nu_n(A) = \mathbb{P}((Y_0, \dots, Y_n) \in A), \quad A \in \mathcal{B}(S^{n+1}).$$

In particular, if  $A = B_0 \times \dots \times B_n$ , where  $B_0, \dots, B_n \in \mathcal{B}(S)$ , then

$$\nu_n(B_0 \times \dots \times B_n) = \mathbb{P}(Y_0 \in B_0, \dots, Y_n \in B_n).$$

We call  $\{\nu_n : n \in \mathbb{N}_0\}$  the family of finite-dimensional marginal distributions of  $(Y_n)_{n \geq 0}$ .

Note that

$$\begin{aligned} \nu_{n+1}(B_0 \times \dots \times B_n \times S) &= \nu_n(B_0 \times \dots \times B_n) \\ &\text{for every } n \in \mathbb{N}_0 \text{ and } B_0, \dots, B_n \in \mathcal{B}(S). \end{aligned} \quad (3.4)$$

Indeed,

$$\begin{aligned}\nu_{n+1}(B_0 \times \cdots \times B_n \times S) &= \mathbb{P}(Y_0 \in B_0, \dots, Y_n \in B_n, Y_{n+1} \in S) \\ &= \mathbb{P}(Y_0 \in B_0, \dots, Y_n \in B_n) \\ &= \nu_n(B_0 \times \cdots \times B_n)\end{aligned}$$

as was to be proved. ■

Let  $\{\nu_n : n \in \mathbb{N}_0\}$  denote a family of probability measures, where  $\nu_n$  is a probability measure on  $(S^{n+1}, \mathcal{B}(S^{n+1}))$ . This family is called *consistent* if it satisfies (3.4).

By Remark 3.4 the finite-dimensional marginal distributions of a stochastic process are consistent. The next result shows that conversely any consistent family is related to a stochastic process in this way.

**Theorem 3.5 (Kolmogorov's Consistency Theorem).**

*Let  $\{\nu_n : n \in \mathbb{N}_0\}$  be consistent. Then there exists exactly one probability measure  $\mathbb{P}$  on  $(S^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$  satisfying*

$$\mathbb{P}(X_0 \in B_0, \dots, X_n \in B_n) = \nu_n(B_0 \times \cdots \times B_n) \quad \text{for every } n \in \mathbb{N}_0 \text{ and } B_0, \dots, B_n \in \mathcal{B}(S).$$

The uniqueness part is due to the fact that the collection of sets of the form (3.2) is stable under intersection and generates  $\mathcal{B}(S)^{\mathbb{N}_0}$ . The existence part of the theorem is much more difficult and beyond the scope of these notes.

**Remark 3.6.** Remark 3.4 and Kolmogorov's Consistency Theorem show together that there exists a stochastic process  $(Y_n)_{n \geq 0}$  with finite-dimensional marginal distributions  $\{\nu_n : n \in \mathbb{N}_0\}$  if and only if the latter is consistent. We can always let the stochastic process be the coordinate process on  $(S^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$ .

Moreover,  $\{\nu_n : n \in \mathbb{N}_0\}$  determines the distribution of  $(Y_n)_{n \geq 0}$  uniquely by Lemma 3.2. ■

**Example 3.7.** Let  $S = \mathbb{R}^d$  for some  $d \in \mathbb{N}$  and let  $\mu$  be a probability measure on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ . Then there exists an i.i.d. sequence  $(\epsilon_n)_{n \geq 0}$  defined on some probability space such that  $\epsilon_n \sim \mu$  for every  $n \in \mathbb{N}_0$ . Indeed, in this case  $\nu_n = \mu \times \cdots \times \mu$  ( $n+1$  times) for  $n \in \mathbb{N}_0$ . (See Exercise 16). ■

**Proposition 3.8 (Existence of Gaussian processes).**

*Let  $\mu_i \in \mathbb{R}$  and  $\sigma_{ij} \in \mathbb{R}$  for  $i, j \in \mathbb{N}_0$ . There exists a Gaussian process  $(Y_n)_{n \geq 0}$  such that  $\mathbb{E}[Y_i] = \mu_i$  and  $\text{Cov}(Y_i, Y_j) = \sigma_{ij}$  for every  $i, j \in \mathbb{N}_0$  if and only if the  $(n+1) \times (n+1)$  matrix  $\{\sigma_{ij}\}_{i,j=0}^n$  is non-negative definite for every  $n \in \mathbb{N}_0$ .*

*The distribution of  $(Y_n)_{n \geq 0}$  is uniquely determined by  $(\mu_i)_{i \geq 0}$  and  $(\sigma_{ij})_{i,j \geq 0}$ .*

*Proof.* If  $(Y_n)_{n \geq 0}$  is Gaussian such that  $\mathbb{E}[Y_i] = \mu_i$  and  $\text{Cov}(Y_i, Y_j) = \sigma_{ij}$  for every  $i, j \in \mathbb{N}_0$  then, for every  $n \in \mathbb{N}_0$ ,  $(Y_0, \dots, Y_n) \sim N_{n+1}((\mu_i)_{i=0}^n, \{\sigma_{ij}\}_{i,j=0}^n)$ . Since  $\{\sigma_{ij}\}_{i,j=0}^n$  is a covariance matrix it is non-negative definite.

Conversely assume that the  $(n+1) \times (n+1)$  matrix  $\{\sigma_{ij}\}_{i,j=0}^n$  is non-negative definite for every  $n \in \mathbb{N}_0$ . Put  $\nu_n = N_{n+1}((\mu_i)_{i=0}^n, \{\sigma_{ij}\}_{i,j=0}^n)$  and note that  $\{\nu_n : n \in \mathbb{N}_0\}$  is consistent. Indeed, the left-hand side of (3.4) corresponds to removing the last coordinate of an  $N_{n+2}((\mu_i)_{i=0}^{n+1}, \{\sigma_{ij}\}_{i,j=0}^{n+1})$ -distributed random vector, which, by well-known properties of the multivariate normal distribution, gives an  $N_{n+1}((\mu_i)_{i=0}^n, \{\sigma_{ij}\}_{i,j=0}^n)$ -distribution; this corresponds to the right-hand side of (3.4). Thus, by Kolmogorov's Consistency Theorem there exists a stochastic process  $(Y_n)_{n \geq 0}$  satisfying  $(Y_0, \dots, Y_n) \sim \nu_n$  for every  $n$ . This process is Gaussian by Gaussianity of  $\nu_n$ .

The uniqueness part follows from Example 3.3.  $\square$

**Remark 3.9.** Above we have discussed stochastic processes indexed by  $\mathbb{N}_0$ . Most of the results generalize to arbitrary non-empty index sets  $T$  in an obvious way. We collect a few definitions and results below.

- (1) Let  $S^T = \{(\omega_s)_{s \in T} : \omega_s \in S\}$  and for  $t \in T$  define  $X_t : S^T \rightarrow S$  as  $X_t((\omega_s)_{s \in T}) = \omega_t$ . Let  $\mathcal{B}(S)^T$  be the  $\sigma$ -field generated by  $X_t, t \in T$ .
- (2) Let  $(Y_t)_{t \in T}$  be a stochastic process defined on  $(\Omega, \mathcal{F}, \mathbb{P})$ . The map  $Y : \Omega \rightarrow S^T$  defined as  $Y(\omega) = (Y_t(\omega))_{t \in T}$  is  $(\mathcal{F}, \mathcal{B}(S)^T)$ -measurable. The image measure  $\mathbb{P}_Y$  is called *the distribution* or *the law* of  $(Y_t)_{t \in T}$ .
- (3) Two stochastic processes  $(Y_t)_{t \in T}$  and  $(Y'_t)_{t \in T}$  are *versions* or have *the same distribution* if  $Y$  and  $Y'$  induce the same image measure on  $(S^T, \mathcal{B}(S)^T)$ . This is the case if and only if

$$(Y_{t_1}, \dots, Y_{t_n}) \stackrel{d}{=} (Y'_{t_1}, \dots, Y'_{t_n}) \quad \text{for every } n \in \mathbb{N} \text{ and } t_1, \dots, t_n \in T.$$

- (4) We only consider *Kolmogorov's Consistency Theorem* in the case  $T \subseteq \mathbb{R}$  so assume this in the following.

Given a stochastic process  $(Y_t)_{t \in T}$  let  $\nu(t_1, \dots, t_n)$  (for  $n \in \mathbb{N}$  and  $t_1 < \dots < t_n$  where  $t_1, \dots, t_n \in T$ ) denote the distribution of  $(Y_{t_1}, \dots, Y_{t_n})$ , i.e.,

$$\nu(t_1, \dots, t_n)(A) = \mathbb{P}((Y_{t_1}, \dots, Y_{t_n}) \in A), \quad A \in \mathcal{B}(S^n).$$

Then the family of probability measures  $\nu(t_1, \dots, t_n)$  with  $n$  and  $t_1, \dots, t_n$  as before is *consistent* in the sense that

$$\begin{aligned} \nu(t_1, \dots, t_{j-1}, t_j, t_{j+1}, \dots, t_n)(B_1 \times \dots \times B_{j-1} \times S \times B_{j+1} \times \dots \times B_n) \\ = \nu(t_1, \dots, t_{j-1}, t_{j+1}, \dots, t_n)(B_1 \times \dots \times B_{j-1} \times B_{j+1} \times \dots \times B_n) \end{aligned}$$

for every  $n \geq 2$ ,  $t_1 < \dots < t_n$  where  $t_1, \dots, t_n \in T$ ,  $j = 1, \dots, n$  and every sequence  $B_1, \dots, B_{j-1}, B_{j+1}, \dots, B_n \in \mathcal{B}(S)$ .

Conversely, given a consistent family of probability measures  $\nu(t_1, \dots, t_n)$  for  $n \in \mathbb{N}$  and  $t_1 < \dots < t_n$  where  $t_1, \dots, t_n \in T$ , there exists a unique probability measure  $\mathbb{P}$  on  $(S^T, \mathcal{B}(S)^T)$  under which  $(X_{t_1}, \dots, X_{t_n})$  has distribution  $\nu(t_1, \dots, t_n)$  for arbitrary  $n \in \mathbb{N}$  and  $t_1 < \dots < t_n$ .

- (5) There are results corresponding to Example 3.7 and Proposition 3.8 for general index sets  $T$ . ■

Finally let us briefly discuss the existence of a stochastic process  $(Y_t)_{t \in T}$  when the index set  $T$  is arbitrary. Equivalently, we must clarify what is meant by consistency of a family  $\{\nu(t_1, \dots, t_k) : k \in \mathbb{N}, t_1, \dots, t_k \in T\}$  of probability measures, where each  $\nu(t_1, \dots, t_k)$  is a probability measure on  $S^k$ .

Obviously, for arbitrary  $t_1, t_2 \in T$  the distribution of  $(Y_{t_1}, Y_{t_2})$  corresponds to that of  $(Y_{t_2}, Y_{t_1})$  with the coordinates switched. Thus, we must add a condition saying e.g. that  $\nu(t_1, t_2) = \nu(t_2, t_1) \circ f^{-1}$  where  $f : S^2 \rightarrow S^2$  is  $f(x, y) = (y, x)$  and corresponds to shifting coordinates. (We have avoided this condition so far by looking only at increasing time points). The interested reader can find the precise conditions for consistency by googling *Kolmogorov's extension theorem*. (C.f. also Schilling and Partzsch (2014, Section 4.2)). In the Gaussian case we have the following generalization of Proposition 3.8.

**Proposition 3.10 (Existence of Gaussian processes with arbitrary index set).**

*Let  $T$  be an arbitrary non-empty index set. For  $s, t \in T$  let  $\mu_t \in \mathbb{R}$  and  $\sigma_{st} \in \mathbb{R}$ . There exists a Gaussian process  $(Y_t)_{t \in T}$  such that  $\mathbb{E}[Y_t] = \mu_t$  and  $\text{Cov}(Y_s, Y_t) = \sigma_{st}$  for every  $s, t \in T$  if and only if the  $n \times n$  matrix  $\{\sigma_{t_i t_j}\}_{i,j=1}^n$  is non-negative definite for every  $n \in \mathbb{N}$  and  $t_1, \dots, t_n \in T$ . If  $T \subseteq \mathbb{R}$  it suffices to check this condition for  $t_1 \leq t_2 \leq \dots \leq t_n$ .*

## Markov chains: Definitions, examples and properties

In this chapter, we give the definition of a Markov chain and characterize the finite-dimensional marginal distributions. We consider both countable and general state spaces. We further give various extensions of the basic Markov property and consider several examples. The most general result is the important *strong Markov property*, which will be used repeatedly throughout these notes.

### 4.1 General state space

Let us start by giving the basic definitions, assuming  $S$  is an arbitrary (Polish) state space.

**Definition 4.1.** A function  $Q : S \times \mathcal{B}(S) \rightarrow [0, 1]$  is said to be a **transition kernel** (on  $S$ ) if

- (1) For every  $x \in S$ ,  $B \mapsto Q(x, B)$  is a probability measure on  $(S, \mathcal{B}(S))$ ;
- (2) for every  $B \in \mathcal{B}(S)$ ,  $x \mapsto Q(x, B)$  is a measurable function.

Transition kernels will be used to model the evolution of Markov chains “one step ahead” in the sense that, if the current state is  $x \in S$ , then the probability that a chain is in  $B$  after one step is  $Q(x, B)$ .

**Definition 4.2 (Markov chains with a filtration).**

Let  $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n \geq 0}, \mathbb{P})$  be a filtered probability space,  $(Y_n)_{n \geq 0}$  be a stochastic process, and  $Q$  be a transition kernel.

The process  $(Y_n)_{n \geq 0}$  is a **Markov chain with transition kernel  $Q$  with respect to  $(\mathcal{F}_n)_{n \geq 0}$**  if it is adapted and

$$\mathbb{P}(Y_{n+1} \in B \mid \mathcal{F}_n) = Q(Y_n, B) \quad (4.1)$$

for every  $B \in \mathcal{B}(S)$  and  $n \in \mathbb{N}_0$ .

Recall that, when  $(Y_n)_{n \geq 0}$  is a stochastic process, the corresponding natural filtration  $(\mathcal{F}_n^Y)_{n \geq 0}$  is defined as

$$\mathcal{F}_n^Y = \sigma(Y_0, \dots, Y_n) \quad \text{for } n \in \mathbb{N}_0. \quad (4.2)$$

**Definition 4.3 (Markov chains without filtration).**

Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and  $(Y_n)_{n \geq 0}$  be a stochastic process.

The process  $(Y_n)_{n \geq 0}$  is called a **Markov chain with transition kernel  $Q$**  if it is a Markov chain with respect to the natural filtration  $(\mathcal{F}_n^Y)_{n \geq 0}$ , that is, if

$$\mathbb{P}(Y_{n+1} \in B \mid \mathcal{F}_n^Y) = Q(Y_n, B) \quad (4.3)$$

for every  $B \in \mathcal{B}(S)$  and  $n \in \mathbb{N}_0$ . Equivalently, by (4.2),  $(Y_n)_{n \geq 0}$  is a Markov chain with transition kernel  $Q$  if and only if

$$\mathbb{P}(Y_{n+1} \in B \mid Y_0, \dots, Y_n) = Q(Y_n, B) \quad (4.4)$$

for every  $B \in \mathcal{B}(S)$  and  $n \in \mathbb{N}_0$ .

The following shows that Definition 4.2 is more general than Definition 4.3.

**Proposition 4.4.** Let  $(Y_n)_{n \geq 0}$  be a Markov chain with transition kernel  $Q$  with respect to a filtration  $(\mathcal{F}_n)_{n \geq 0}$ .

- (1) Let  $(\mathcal{G}_n)_{n \geq 0}$  be a filtration such that  $(Y_n)_{n \geq 0}$  is adapted to  $(\mathcal{G}_n)_{n \geq 0}$  and  $\mathcal{G}_n \subseteq \mathcal{F}_n$  for all  $n \in \mathbb{N}_0$ . Then  $(Y_n)_{n \geq 0}$  is a Markov chain with transition kernel  $Q$  with respect to  $(\mathcal{G}_n)_{n \geq 0}$ .
- (2) The process  $(Y_n)_{n \geq 0}$  is a Markov chain with transition kernel  $Q$ .

*Proof.* To prove the first result, we have to verify (4.1) with  $\mathcal{F}_n$  replaced by  $\mathcal{G}_n$ . Hence, let  $n \in \mathbb{N}_0$  and  $B \in \mathcal{B}(S)$ . Then

$$\begin{aligned} \mathbb{P}(Y_{n+1} \in B \mid \mathcal{G}_n) &= \mathbb{E}[\mathbf{1}_B(Y_{n+1}) \mid \mathcal{G}_n] = \mathbb{E}[\mathbb{E}[\mathbf{1}_B(Y_{n+1}) \mid \mathcal{F}_n] \mid \mathcal{G}_n] \\ &= \mathbb{E}[Q(Y_n, B) \mid \mathcal{G}_n] = Q(Y_n, B), \end{aligned}$$

where the first equality is by definition of the conditional probability, the second is by the tower principle (Theorem E.10(2)) and the last equality is by Theorem E.10(3).

To prove that  $(Y_n)_{n \geq 0}$  is a Markov chain with transition kernel  $Q$ , note that  $(Y_n)_{n \geq 0}$  is adapted to  $(\mathcal{F}_n)_{n \geq 0}$  if and only if  $\mathcal{F}_n^Y \subseteq \mathcal{F}_n$  for every  $n \in \mathbb{N}_0$ . Hence, we can use the first part with  $\mathcal{G}_n = \mathcal{F}_n^Y$ .  $\square$

**Remark 4.5.** Let us collect a few remarks and further definitions related to Markov chains. Let  $Q$  be a transition kernel.

- (1) For  $f : S \rightarrow \mathbb{R}$  measurable that is either non-negative or bounded (to ensure that the integral below exists), put

$$Qf(x) = \int f(y) Q(x, dy) \quad \text{for } x \in S. \quad (4.5)$$

As a particular case,

$$Q \mathbb{1}_B(x) = \int \mathbb{1}_B(y) Q(x, dy) = Q(x, B) \quad \text{for } x \in S \text{ and } B \in \mathcal{B}(S). \quad (4.6)$$

Hence, we can also state the Markov property (4.1) as

$$\mathbb{E}[\mathbb{1}_B(Y_{n+1}) | \mathcal{F}_n] = Q \mathbb{1}_B(Y_n) \quad (4.7)$$

for every  $B \in \mathcal{B}(S)$  and  $n \in \mathbb{N}_0$ .

An application of the standard proof shows that (4.7) generalizes to

$$\mathbb{E}[f(Y_{n+1}) | \mathcal{F}_n] = Qf(Y_n) \quad \text{for } n \in \mathbb{N}_0 \quad (4.8)$$

and  $f : S \rightarrow \mathbb{R}$  measurable and bounded or non-negative.

- (2) Later, we will use that the natural filtration  $(\mathcal{F}_n^Y)_{n \geq 0}$  is given as

$$\mathcal{F}_n^Y = \left\{ \{(Y_0, \dots, Y_n) \in B\} : B \in \mathcal{B}(S^{n+1}) \right\} \quad \text{for } n \in \mathbb{N}_0. \quad (4.9)$$

- (3) When  $(Y_n)_{n \geq 0}$  is a Markov chain, the distribution of  $Y_0$  is called *the initial distribution*.

- (4) Let  $(Y_n)_{n \geq 0}$  be a Markov chain. Condition (4.3) can be rephrased as

$$\mathbb{P}(Y_{n+1} \in B | Y_n, \dots, Y_0) = Q(Y_n, B)$$

or, using regular conditional probabilities,

$$\mathbb{P}(Y_{n+1} \in B | Y_n = y_n, \dots, Y_0 = y_0) = Q(y_n, B).$$

for  $n \in \mathbb{N}_0$ ,  $B \in \mathcal{B}(S)$ , and  $y_0, \dots, y_n \in S$ . By the standard proof,

$$\mathbb{E}[f(Y_{n+1}) | Y_0 = y_0, \dots, Y_n = y_n] = \int_S f(y) Q(y_n, dy) = Qf(y_n) \quad (4.10)$$

for all  $n \in \mathbb{N}_0$ ,  $f : S \rightarrow \mathbb{R}$  measurable and bounded or non-negative, and  $y_0, \dots, y_n \in S$ .  $\blacksquare$

## 4.2 Countable state space

In this section,  $S$  is countable. Recall that, in this case, we always choose the metric in such a way that  $\mathcal{B}(S)$  is the set of all subsets of  $S$ .

Let  $Q$  be a transition kernel. Put  $p_{ij} = Q(i, \{j\})$  for all  $i, j \in S$  and let  $P = \{p_{ij}\}_{i,j \in S}$ . That is, we can think of  $p_{ij}$  as the probability of a transition from  $i$  to  $j$  in one step. Besides,  $P$  is a *transition matrix*, i.e.,  $p_{ij} \geq 0$  for all  $i, j \in S$ , and  $\sum_{j \in S} p_{ij} = 1$  for all  $i \in S$ .

Conversely, given a transition matrix  $P$  there is a unique transition kernel  $Q$  satisfying  $Q(i, \{j\}) = p_{ij}$  for all  $i, j \in S$ . Indeed, if  $Q$  is such a transition kernel then

$$Q(i, B) = Q(i, \cup_{j \in B} \{j\}) = \sum_{j \in B} Q(i, \{j\}) = \sum_{j \in B} p_{ij}$$

for all  $i \in S$  and  $B \subseteq S$ .

When  $S$  is countable, we will often work with  $P$  instead of  $Q$  and this is reflected in the following.

### Definition 4.6 (Markov chains with countable state space).

Let  $Q$  be a transition kernel and  $P = \{p_{ij}\}_{i,j \in S}$  be the corresponding transition matrix ( $p_{ij} = Q(i, \{j\})$  for all  $i, j \in S$ ). Further, let  $(Y_n)_{n \geq 0}$  be a stochastic process.

- (1) Assume  $(Y_n)_{n \geq 0}$  is adapted to a filtration  $(\mathcal{F}_n)_{n \geq 0}$ . We say that  $(Y_n)_{n \geq 0}$  is a **Markov chain with transition matrix  $P$  with respect to  $(\mathcal{F}_n)_{n \geq 0}$**  if  $(Y_n)_{n \geq 0}$  is a Markov chain with transition kernel  $Q$  with respect to  $(\mathcal{F}_n)_{n \geq 0}$ .
- (2) We say that  $(Y_n)_{n \geq 0}$  is a **Markov chain with transition matrix  $P$**  if  $(Y_n)_{n \geq 0}$  is a Markov chain with transition kernel  $Q$ .

**Proposition 4.7.** Let  $S$  be countable,  $Q$  be a transition kernel and  $P = \{p_{ij}\}_{i,j \in S}$  be the corresponding transition matrix ( $p_{ij} = Q(i, \{j\})$  for all  $i, j \in S$ ). Further, let  $(Y_n)_{n \geq 0}$  be a stochastic process.

- (1) Assume  $(Y_n)_{n \geq 0}$  is adapted to a filtration  $(\mathcal{F}_n)_{n \geq 0}$ . Then  $(Y_n)_{n \geq 0}$  is a Markov chain with transition matrix  $P$  with respect to  $(\mathcal{F}_n)_{n \geq 0}$  if and only if

$$\mathbb{P}(Y_{n+1} = j \mid \mathcal{F}_n) = p_{Y_n j} \quad \text{for all } n \in \mathbb{N}_0 \text{ and } j \in S. \quad (4.11)$$

- (2) The following statements are equivalent.

- (a)  $(Y_n)_{n \geq 0}$  is a Markov chain with transition matrix  $P$ .  
 (b) For all  $n \in \mathbb{N}_0$  and  $j \in S$

$$\mathbb{P}(Y_{n+1} = j \mid Y_0, \dots, Y_n) = p_{Y_n j}. \quad (4.12)$$

- (c) For all  $n \in \mathbb{N}_0$  and  $i_0, \dots, i_n, j \in S$

$$\mathbb{P}(Y_{n+1} = j \mid Y_0 = i_0, \dots, Y_n = i_n) = p_{i_n j}. \quad (4.13)$$

*Proof. Item (1):* Condition (4.1) implies (4.11) by letting  $B = \{j\}$ . Conversely, assuming (4.11) holds for all  $n$  and  $j$  we get

$$\mathbb{P}(Y_{n+1} \in B \mid \mathcal{F}_n) = \sum_{j \in B} \mathbb{P}(Y_{n+1} = j \mid \mathcal{F}_n) = \sum_{j \in B} p_{Y_n j} = Q(Y_n, B)$$

for all  $n \in \mathbb{N}_0$  and  $B \subseteq S$ . Hence we also have (4.1).

*Item (2):* Since  $\mathcal{F}_n^Y = \sigma(Y_0, \dots, Y_n)$ , item (1) shows that (a) and (b) are equivalent.

To prove that (b) and (c) are equivalent, fix  $n \in \mathbb{N}_0$  and  $j \in S$ . From (E.6) with  $Y = (Y_0, \dots, Y_n)$  and  $Z = \mathbb{1}_{\{Y_{n+1}=j\}}$  it follows that

$$\mathbb{P}(Y_{n+1} = j \mid Y_0, \dots, Y_n) = \sum_{i_0, \dots, i_n \in S} \mathbb{1}_{\{Y_0=i_0, \dots, Y_n=i_n\}} \mathbb{P}(Y_{n+1} = j \mid Y_0 = i_0, \dots, Y_n = i_n). \quad (4.14)$$

Moreover, the right-hand side of (4.11) can be decomposed as

$$p_{Y_n j} = \left( \sum_{i_0, \dots, i_n \in S} \mathbb{1}_{\{Y_0=i_0, \dots, Y_n=i_n\}} \right) p_{Y_n j} = \sum_{i_0, \dots, i_n \in S} \mathbb{1}_{\{Y_0=i_0, \dots, Y_n=i_n\}} p_{i_n j}, \quad (4.15)$$

where the first equality follows from the fact that the sum equals 1. Thus, (b) and (c) are equivalent.  $\square$

The above result shows that, in the countable state space case, the definition of a Markov chain given here corresponds to the one often given in a first year course.

**Remark 4.8.** Let  $P = \{p_{ij}\}_{i,j \in S}$  denote the transition matrix corresponding to  $Q$  ( $p_{ij} = Q(i, \{j\})$  for  $i, j \in S$ ). Recall that if  $(Y_n)_{n \geq 0}$  is a Markov chain with transition matrix  $P$  with respect to  $(\mathcal{F}_n)_{n \geq 0}$ , then (4.8) also applies, where  $Qf$  is defined in

(4.5). In Exercise 22 it is shown that for  $i \in S$  and a bounded (or non-negative) function  $f : S \rightarrow \mathbb{R}$ ,

$$Qf(i) = \sum_{j \in S} f(j)p_{ij}, \quad (4.16)$$

Further, by Proposition 4.7(2) and Equation (4.16),

$$\begin{aligned} \mathbb{E}[f(Y_{n+1}) | Y_0 = i_0, \dots, Y_n = i_n] &= \sum_{j \in S} f(j)\mathbb{P}(Y_{n+1} = j | Y_0 = i_0, \dots, Y_n = i_n) \\ &= Qf(i_n), \end{aligned} \quad (4.17)$$

for all  $n \in \mathbb{N}_0$ ,  $f : S \rightarrow \mathbb{R}$  bounded or non-negative, and  $i_0, \dots, i_n \in S$ . ■

### 4.3 Examples

In the following, we collect a set of examples – both with general and countable state spaces. Many of these are probably more or less well known.

We start by giving a useful recipe for constructing Markov chains.

**Example 4.9.** Let  $(\epsilon_n)_{n \geq 1}$  be an i.i.d. sequence with (Polish) state space  $\bar{S}$ , independent of  $Y_0$ , which has state space  $S$ . Let  $F : S \times \bar{S} \rightarrow S$  be measurable.

Define  $(Y_n)_{n \geq 1}$  recursively as

$$Y_{n+1} = F(Y_n, \epsilon_{n+1}), \quad n \in \mathbb{N}_0. \quad (4.18)$$

Then  $(Y_n)_{n \geq 0}$  is a Markov chain with respect to the filtration

$$\mathcal{F}_n = \sigma(Y_0, \epsilon_1, \dots, \epsilon_n), \quad n \in \mathbb{N}_0.$$

The transition kernel is

$$Q(x, B) = \mathbb{P}(F(x, \epsilon_1) \in B) \quad \text{for } x \in S \text{ and } B \in \mathcal{B}(S).$$

It is quite intuitive that  $(Y_n)_{n \geq 0}$  is a Markov chain since by (4.18) the state “one step ahead” is fully determined by “present state” and some “i.i.d. noise”.

To actually prove the Markov property of  $(Y_n)_{n \geq 0}$ , first use recursion to infer that  $(Y_n)_{n \geq 0}$  is adapted to  $(\mathcal{F}_n)_{n \geq 0}$ . Then note that for  $n \in \mathbb{N}_0$  and  $B \in \mathcal{B}(S)$ ,

$$\begin{aligned} \mathbb{P}(Y_{n+1} \in B | \mathcal{F}_n) &= \mathbb{P}(F(Y_n, \epsilon_{n+1}) \in B | \mathcal{F}_n) \\ &= \mathbb{P}(F(x, \epsilon_{n+1}) \in B) \Big|_{x=Y_n} = \mathbb{P}(F(x, \epsilon_1) \in B) \Big|_{x=Y_n} = Q(Y_n, B), \end{aligned}$$

where the second equality is by the useful rule (see Theorem E.7) and the third is due to  $\epsilon_1 \stackrel{d}{=} \epsilon_{n+1}$ . ■

**Example 4.10 (AR(1) processes).**

Let  $(\epsilon_n)_{n \geq 1}$  be an i.i.d. sequence of real random variables independent of  $Y_0$ , which is real as well. Let  $\theta \in \mathbb{R}$ . Define  $(Y_n)_{n \geq 1}$  as

$$Y_{n+1} = \theta Y_n + \epsilon_{n+1}.$$

The process  $(Y_n)_{n \geq 0}$  is called an AR(1) process. It is a Markov chain due to Example 4.9.

When  $\epsilon_n \sim N(0, 1)$ , the transition kernel is  $Q(x, \cdot) = N(\theta x, 1)$  (why?). More generally, if  $\epsilon_n$  has density  $p$  with respect to the Lebesgue measure on  $\mathbb{R}$  then  $Q(x, \cdot)$  is absolutely continuous with respect to the Lebesgue measure with density  $Q(x, dy) = p(y - \theta x)dy$ . ■

**Example 4.11 (Random walk).**

Let  $(\epsilon_n)_{n \geq 1}$  be an i.i.d. sequence of  $\mathbb{R}^d$ -valued random variables independent of  $Y_0$  which is  $\mathbb{R}^d$ -valued as well. Define

$$Y_n = Y_0 + \sum_{i=1}^n \epsilon_i.$$

Then  $(Y_n)_{n \geq 0}$  is called a random walk. It is a Markov chain due to Example 4.9. Further,  $Q(x, B) = \mu(B - x)$ , where  $\mu$  denotes the distribution of  $\epsilon_n$  (why?). ■

**Example 4.12 (A discretely observed Black–Scholes model).**

Consider i.i.d. random variables  $(\epsilon_n)_{n \geq 1}$  such that  $\epsilon_n \sim N(\mu, \sigma^2)$  for all  $n \in \mathbb{N}$ . Let  $S_0 \in (0, \infty)$  be a constant, and define  $(S_n)_{n \geq 0}$  by

$$S_n = S_0 e^{\epsilon_1 + \dots + \epsilon_n}, \quad n \in \mathbb{N}. \quad (4.19)$$

Then  $(S_n)_{n \geq 0}$  is a Markov chain with state space  $S = (0, \infty)$  and transition kernel given by

$$Q(x, B) = \mathbb{P}(xe^{\epsilon_1} \in B), \quad x \in S, B \in \mathcal{B}(S).$$

Indeed, this follows from Example 4.9 with  $F(x, y) = xe^y$ , since  $S_{n+1} = S_n e^{\epsilon_{n+1}}$ .

The model (4.19) can be seen as a discrete-time analogue of the Black–Scholes model. In continuous time, the (geometric) Black–Scholes model for a stock price is

$$S_t = S_0 \exp\left(\left(\tilde{\mu} - \frac{1}{2}\sigma^2\right)t + \sigma B_t\right), \quad (4.20)$$

where  $(B_t)_{t \geq 0}$  is a standard Brownian motion (see Chapter 10). Observing (4.20) at the integer times  $t = 1, 2, \dots$  yields a process of the form (4.19) with i.i.d. Gaussian log-returns  $\epsilon_n \sim N(\mu, \sigma^2)$ , where  $\mu = \tilde{\mu} - \frac{1}{2}\sigma^2$  and  $\epsilon_n = \mu + \sigma(B_n - B_{n-1})$ .

The Black–Scholes model is one of the most important and widely used models for the evolution of stock prices in mathematical finance. One reason for this is that it provides a tractable, arbitrage-free benchmark with explicit formulas for pricing and hedging (in particular via delta hedging and the Greeks), and it is therefore a standard point of reference in both theory and practice; see Björk (2019); Hull (2021). Empirically, however, its key assumptions often fail: log-returns are generally not independent and are poorly described by a normal distribution. Nevertheless, the model remains widely used because market option prices are routinely summarized and communicated through Black–Scholes implied volatilities. Thus, even richer models are often discussed relative to the implied-volatility surface; see Gatheral (2006); Hull (2021). ■

**Example 4.13 (CIR processes).**

A Cox–Ingersoll–Ross (CIR) process  $(Z_t)_{t \in [0, \infty)}$  solves a stochastic differential equation of the form

$$dZ_t = \alpha(\gamma - Z_t)dt + \eta\sqrt{Z_t}dB_t, \quad (4.21)$$

where  $\alpha, \gamma, \eta > 0$  and  $(B_t)_{t \geq 0}$  is a standard Brownian motion (see Chapter 10). Loosely speaking, (4.21) means that, for  $dt$  small and positive,

$$Z_{t+dt} \approx Z_t + \alpha(\gamma - Z_t)dt + \eta\sqrt{Z_t}(B_{t+dt} - B_t),$$

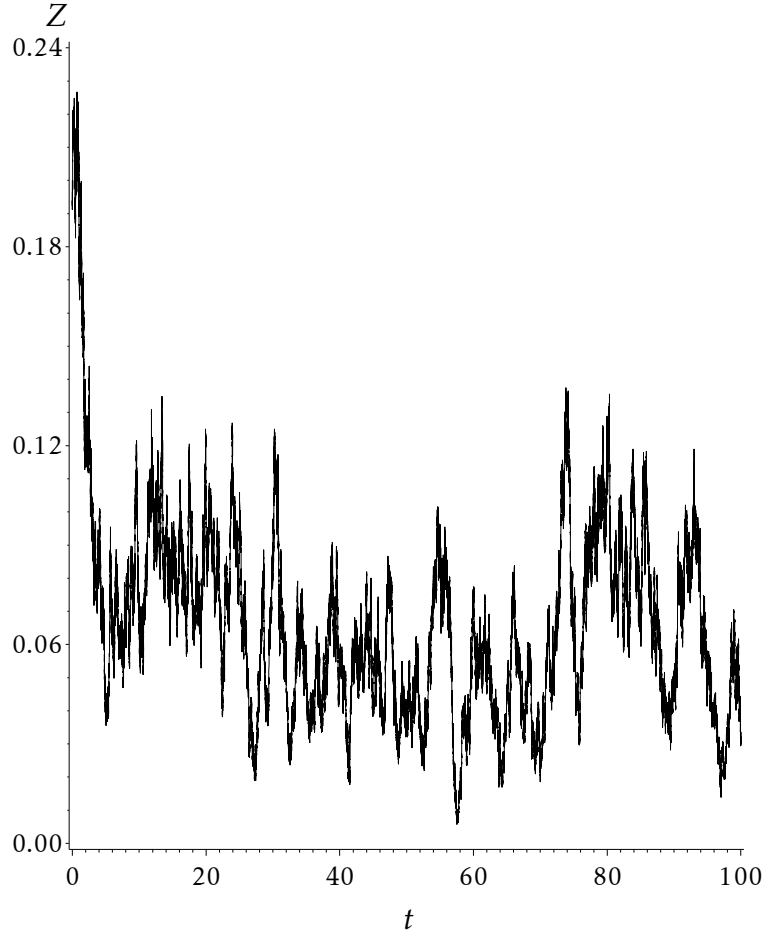
where  $B_{t+dt} - B_t \sim N(0, dt)$  for  $dt > 0$ . This idea can be used to simulate a CIR process using the so-called *Euler scheme* in the following way. Choose a large  $n \in \mathbb{N}$  and fix an initial value  $Z_0 \in (0, \infty)$ . Now simulate  $Z_{\frac{k}{n}}$  recursively for  $k = 0, 1, 2, \dots$  as

$$Z_{\frac{k+1}{n}} = Z_{\frac{k}{n}} + \alpha(\gamma - Z_{\frac{k}{n}}) \cdot \frac{1}{n} + \eta\sqrt{Z_{\frac{k}{n}}} \cdot \frac{U_{k+1}}{\sqrt{n}},$$

where  $U_1, U_2, \dots$  are i.i.d. with  $U_k \sim N(0, 1)$ . It can be shown that the process defined by the Euler scheme converges to the CIR process (in a suitable sense) as  $n \rightarrow \infty$ .

Figure 4.1 shows a simulation with parameters  $\alpha = 0.5$ ,  $\gamma = 0.06$ ,  $\eta = 0.11$  and initial condition  $Z_0 = 0.2$ . It seems that the path  $t \mapsto Z_t$  is continuous but “rough”; in fact, it can be shown that the paths are continuous but not differentiable. Moreover, it seems that the CIR process tends to fluctuate around a long-run level  $\gamma = 0.06$ , and even though the initial value ( $Z_0 = 0.2$ ) is far from  $\gamma$ , the drift towards  $\gamma$  seems strong.

CIR processes are popular models for interest rates in mathematical finance. It is therefore important that it can be shown that a CIR process with  $Z_0 > 0$  is



**Figure 4.1:** A simulation of a CIR process.

strictly positive (a.s.) if the parameters satisfy

$$2\alpha\gamma \geq \eta^2. \quad (4.22)$$

The Euler scheme does not preserve positivity and may produce negative values, even when  $Z_0 > 0$ . In practice one often uses a positivity-preserving modification or an exact simulation method for CIR.

Solutions to stochastic differential equations are Markov processes under mild regularity conditions, but in most cases the solution is not known in any explicit form. The transition densities of CIR processes are, however, known explicitly. More precisely, if a CIR process with parameters satisfying (4.22) is observed at discrete time points  $0, t, 2t, \dots$  for some fixed  $t > 0$ , then the resulting process  $(Z_{nt})_{n \geq 0}$  is a Markov chain with state space  $S = (0, \infty)$ , where the transition kernel  $Q$  is  $Q(x, dy) = p(t, x, y) dy$  and the transition density  $p(t, x, y)$  is given by

$$p(t, x, y) = \frac{e^{\alpha t}}{2c} \left( \frac{ye^{\alpha t}}{x} \right)^{\nu/2} \exp\left(-\frac{x + ye^{\alpha t}}{2c}\right) I_{\nu}\left(\frac{\sqrt{xye^{\alpha t}}}{c}\right), \quad \text{for } x, y > 0, \quad (4.23)$$

where  $c = \frac{\eta^2}{4\alpha}(e^{\alpha t} - 1)$ ,  $\nu = \frac{2\alpha\gamma}{\eta^2} - 1$ , and  $I_\nu$  is the modified Bessel function of the first kind of order  $\nu$ . ■

**Example 4.14 (ARMA and GARCH processes).**

Generally, ARMA and GARCH processes (cf. Remark 2.5 and Example 2.7) are not Markov chains (with a few exceptions – see e.g. Example 4.10). But a Markov chain can be obtained by including, e.g., also some past values. This observation is very useful in analyzing the asymptotic properties of (estimates in) GARCH and ARMA models, see e.g. Boussama et al. (2011) or Francq and Zakoïan (2010), and references therein. ■

**Example 4.15 (Non-Markovian models).**

Often Markov chains are used as starting point to construct non-Markovian processes, and the following is an example.

Let  $(Z_n)_{n \geq 0}$  be a CIR process (cf. Example 4.13) observed discretely at time points  $n \in \mathbb{N}_0$  and define  $Y_n$  as  $Y_n = Z_n \epsilon_n$  for  $n \in \mathbb{N}_0$ , where  $(\epsilon_n)_{n \geq 0}$  is an i.i.d. sequence of  $N(0, 1)$ -distributed random variables independent of  $(Z_n)_{n \geq 0}$ . Then the process  $(Z_n, Y_n)_{n \geq 0}$  is a Markov chain but  $(Y_n)_{n \geq 0}$  is not.

Processes such as  $Y_n$  that appear as a function of a Markov chain (such as  $Z_n$ ) and of some independent noise (such as  $\epsilon_n$ ) are often referred to as *state space models* or *hidden Markov models*. The particular case above originates from mathematical finance, where it is an example of a discretely observed so-called Heston model and of a stochastic volatility model.

There are many interesting problems related to state space models. For example, it is generally quite hard to estimate the parameters in such models cf. e.g., Jensen and Petersen (1999) or Cappé et al. (2005) and the references therein. However, often the asymptotic properties of the underlying Markov chain are inherited to the state space models cf. Genon-Catalot et al. (2000). ■

The state space is countable in the next examples.

**Example 4.16 (Simple random walk on  $\mathbb{Z}$ ).**

Let  $S = \mathbb{Z}$  and specify the transition matrix  $P = \{p_{ij}\}_{i,j \in S}$  as

$$p_{i,i+1} = p, \quad p_{i,i-1} = 1 - p, \quad p_{ij} = 0 \text{ otherwise,}$$

where  $p \in (0, 1)$ . Then we speak of a simple random walk on  $\mathbb{Z}$ . ■

**Example 4.17 (Gambler's ruin).**

In this example let  $S = \{0, \dots, N\}$  for some integer  $N \in \mathbb{N}$ , and specify the transition matrix  $P = \{p_{ij}\}_{i,j \in S}$  as

$$\begin{aligned} p_{i,i+1} &= p, & p_{i,i-1} &= 1 - p \text{ for } i = 1, \dots, N-1, \\ p_{00} &= 1, & p_{NN} &= 1, \text{ and } p_{ij} = 0 \text{ otherwise} \end{aligned}$$

where  $p \in (0, 1)$ .

We can interpret the corresponding Markov chain in the following way. Suppose you play the roulette. Let  $p$  denote the probability of winning a given game and  $1 - p$  be the probability of losing. When losing a game, you hand in DKK 1, and when winning, you receive DKK 1. Stop playing in case of ruin (no money left) or if you reach a predefined maximal capital  $N$ . Assuming that the outcomes of consecutive games are independent, a Markov chain with transition matrix  $P$  models the development of your fortune over time. The initial distribution models the distribution of your initial capital.

For applications of this transition matrix see e.g., Durrett (2016). ■

**Example 4.18 (Random walk on  $\mathbb{N}_0$  with 0 as absorbing state).**

Let  $S = \mathbb{N}_0$  and specify the transition matrix  $P = \{p_{ij}\}_{i,j \in S}$  as

$$\begin{aligned} p_{i,i+1} &= p, & p_{i,i-1} &= 1 - p, & \text{for } i = 1, 2, \dots, \\ p_{00} &= 1, & p_{ij} &= 0 \text{ otherwise,} \end{aligned}$$

where  $p \in (0, 1)$ . We speak of a random walk on  $\mathbb{N}_0$  with 0 as absorbing state. ■

**Example 4.19 (Branching processes).**

Let  $S = \{0, 1, \dots\}$  and specify the transition probabilities as

$$p_{ij} = \mathbb{P}\left(\sum_{m=1}^i \xi_m = j\right)$$

where  $(\xi_m)_{m \geq 1}$  are i.i.d. non-negative integer-valued random variables.

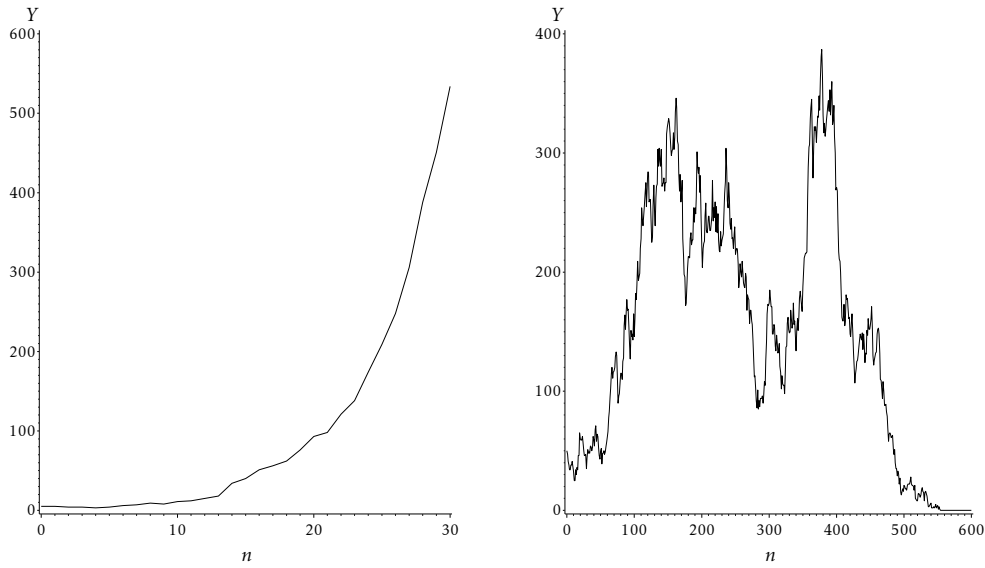
One way to represent the corresponding Markov chain, which is referred to as a *branching process*, is as

$$Y_{n+1} = \begin{cases} \sum_{k=1}^{Y_n} \xi_k^{n+1}, & \text{if } Y_n > 0, \\ 0, & \text{if } Y_n = 0, \end{cases} \quad (4.24)$$

where  $\xi_k^n$ , for  $k, n \in \mathbb{N}$ , are i.i.d. random variables (independent of  $Y_0$ ) having the same distribution as the  $\xi_m$ 's.

We can think of the branching process as a model for the evolution of a population. In this case,  $Y_n$  denotes the size of the population at time  $n$ , and  $\xi_k^{n+1}$  is the number of offspring for individual  $k$  at time  $n$ . An interesting question is whether the population goes extinct ( $Y_n = 0$  for some  $n$ ) or grows to infinity.

Assume, e.g.,  $\xi_m \sim \text{po}(\lambda)$ . If the number of individuals in some generation is  $i \in \mathbb{N}$ , the number of individuals in the next generation is Poisson distributed with mean  $i\lambda$ . Figure 4.2 displays simulations of a branching process when  $\lambda = 1.2$  and  $Y_0 = 5$  (left), and when  $\lambda = 1$  and  $Y_0 = 50$  (right). ■



**Figure 4.2:** Left: A simulation of a branching process when the average number of offspring per individual is greater than 1. Right: A simulation of a branching process when the average number of offspring per individual is 1.

**Example 4.20 (Birth and Death processes – discrete time).**

Let  $S = \mathbb{N}_0$  and

$$p_{ij} = \begin{cases} p_i, & \text{if } i \in \mathbb{N}_0 \text{ and } j = i + 1, \\ q_i, & \text{if } i \in \mathbb{N} \text{ and } j = i - 1, \\ 1 - p_i - q_i, & \text{if } i \in \mathbb{N} \text{ and } j = i, \\ 1 - p_0, & \text{if } i = j = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Here  $p_i \in [0, 1)$  for  $i \in \mathbb{N}_0$ ,  $q_i \in [0, 1)$  for  $i \in \mathbb{N}$  and  $p_i + q_i \leq 1$  for  $i \in \mathbb{N}$ . We speak of a *birth and death process*. ■

## 4.4 Characterization and existence of Markov chains

In the next results we characterize the finite-dimensional marginal distributions of Markov chains and show that Markov chains actually exist when the transition kernel and the initial distribution are given.

**Theorem 4.21 (The finite-dimensional marginal distributions of a Markov chain).**

Let  $(Y_n)_{n \geq 0}$  be a stochastic process, let  $Q$  be a transition kernel on  $S$  and let  $\mu$  be a probability measure on  $(S, \mathcal{B}(S))$ . The following statements are equivalent:

- (1)  $(Y_n)_{n \geq 0}$  is a Markov chain with transition kernel  $Q$  and initial distribution  $\mu$ .
- (2) For every  $n \in \mathbb{N}_0$  and  $B_0, \dots, B_n \in \mathcal{B}(S)$ ,

$$\begin{aligned} \mathbb{P}(Y_0 \in B_0, \dots, Y_n \in B_n) \\ = \int_{B_0} \int_{B_1} \cdots \int_{B_{n-1}} \int_{B_n} 1 Q(y_{n-1}, dy_n) Q(y_{n-2}, dy_{n-1}) \cdots Q(y_0, dy_1) \mu(dy_0). \end{aligned} \quad (4.25)$$

Sometimes, equation (4.25) is written compactly as

$$\mathbb{P}(Y_0 \in dy_0, Y_1 \in dy_1, \dots, Y_n \in dy_n) = \mu(dy_0) Q(y_0, dy_1) \cdots Q(y_{n-1}, dy_n). \quad (4.26)$$

When  $S$  is countable the precise content of this equation is as follows.

**Corollary 4.22 (The countable state space case).**

Let  $S$  be countable,  $P = \{p_{ij}\}_{i,j \in S}$  be a transition matrix,  $\mu$  be a probability measure on  $S$ , and  $Y = (Y_n)_{n \geq 0}$  be a stochastic process. Then  $Y = (Y_n)_{n \geq 0}$  is a Markov chain with transition matrix  $P$  and initial distribution  $\mu$  if and only if for all  $n \in \mathbb{N}_0$  and  $i_0, \dots, i_n \in S$ ,

$$\mathbb{P}(Y_0 = i_0, \dots, Y_n = i_n) = \mu(\{i_0\}) p_{i_0 i_1} \cdots p_{i_{n-1} i_n}. \quad (4.27)$$

**Remark 4.23.** We can rewrite (4.25) as

$$\begin{aligned} \mathbb{E}[\mathbb{1}_{B_0 \times \cdots \times B_n}(Y_0, \dots, Y_n)] \\ = \iint \cdots \int \mathbb{1}_{B_0 \times \cdots \times B_n}(y_0, \dots, y_n) Q(y_{n-1}, dy_n) \cdots Q(y_0, dy_1) \mu(dy_0). \end{aligned}$$

Thus, if (4.25) holds for some  $n$  and all  $B_0, \dots, B_n \in \mathcal{B}(S)$ , an application of the Monotone Class Theorem (Theorem H.1), with  $\mathcal{A}$  denoting the collection of all subsets of the form  $B_0 \times \cdots \times B_n$  for  $B_0, \dots, B_n \in \mathcal{B}(S)$ , shows that

$$\begin{aligned} \mathbb{E}[f(Y_0, \dots, Y_n)] \\ = \iint \cdots \int f(y_0, \dots, y_n) Q(y_{n-1}, dy_n) \cdots Q(y_0, dy_1) \mu(dy_0) \end{aligned}$$

for every measurable  $f : S^{n+1} \rightarrow \mathbb{R}$  which is either non-negative or bounded. ■

**Theorem 4.24 (Existence of Markov chains).**

Let  $Q$  be a transition kernel and  $\mu$  be a probability measure on  $(S, \mathcal{B}(S))$ . Then there exists a unique probability measure  $\mathbb{P}_\mu$  on  $(S^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$  under which the coordinate process  $(X_n)_{n \geq 0}$  is a Markov chain with transition kernel  $Q$  and initial distribution  $\mu$ .

When the initial distribution is  $\delta_x$  for  $x \in S$ , i.e., the Markov chain is started in  $x$ , we write  $\mathbb{P}_x$  rather than  $\mathbb{P}_{\delta_x}$ . When calculating expectations, we use the notation  $\mathbb{E}_x$  and  $\mathbb{E}_\mu$ . An application of the Monotone Class Theorem (Theorem H.1) yields the following result:

**Proposition 4.25.** Let  $Q$  be a transition kernel and consider the measures  $\mathbb{P}_x$  and  $\mathbb{P}_\mu$  defined above. Then  $x \mapsto \mathbb{E}_x[Z]$  is measurable for every bounded or non-negative  $\mathcal{B}(S)^{\mathbb{N}_0}$ -measurable function  $Z : S^{\mathbb{N}_0} \rightarrow \mathbb{R}$ .

Moreover, for any initial distribution  $\mu$  we have

$$\mathbb{P}_\mu(A) = \int \mathbb{P}_x(A) \mu(dx), \quad \text{for } A \in \mathcal{B}(S)^{\mathbb{N}_0}, \quad (4.28)$$

and, more generally,

$$\mathbb{E}_\mu[Z] = \int_S \mathbb{E}_x[Z] \mu(dx) \quad (4.29)$$

for every bounded and  $\mathcal{B}(S)^{\mathbb{N}_0}$ -measurable function  $Z : S^{\mathbb{N}_0} \rightarrow \mathbb{R}$ .

**Remark 4.26 (One-step and finite-dimensional distributions under  $\mathbb{P}_x$  and  $\mathbb{P}_\mu$ ).**

Let  $(X_n)_{n \geq 0}$  denote the coordinate process on  $S^{\mathbb{N}_0}$  under the measures  $\mathbb{P}_x := \mathbb{P}_{\delta_x}$  and  $\mathbb{P}_\mu$ , as in Theorem 4.24. Then  $X_0 = x$   $\mathbb{P}_x$ -a.s. and  $X_0 \sim \mu$  under  $\mathbb{P}_\mu$ . In particular, it follows from Theorem 4.21 that for any initial distribution  $\mu$  and any  $B \in \mathcal{B}(S)$ ,

$$\mathbb{P}_\mu(X_1 \in B) = \int_S Q(y, B) \mu(dy). \quad (4.30)$$

More generally, for  $n \in \mathbb{N}_0$  and  $B_0, \dots, B_n \in \mathcal{B}(S)$ ,

$$\mathbb{P}_\mu(X_0 \in B_0, \dots, X_n \in B_n) = \int_{B_0} \int_{B_1} \cdots \int_{B_n} Q(y_{n-1}, dy_n) \cdots Q(y_0, dy_1) \mu(dy_0). \quad (4.31)$$

By letting  $\mu = \delta_x$  in (4.30), it follows that for any  $B \in \mathcal{B}(S)$

$$\mathbb{P}_x(X_1 \in B) = Q(x, B). \quad (4.32)$$

Similarly, for  $n \in \mathbb{N}$  and  $B_1, \dots, B_n \in \mathcal{B}(S)$ ,

$$\mathbb{P}_x(X_1 \in B_1, \dots, X_n \in B_n) = \int_{B_1} \cdots \int_{B_n} 1 Q(y_{n-1}, dy_n) \cdots Q(y_1, dy_2) Q(x, dy_1). \quad (4.33)$$

If, in addition,  $S$  is countable and  $p_{ij} = Q(i, \{j\})$ , then writing  $\mathbb{P}_i := \mathbb{P}_{\delta_i}$  we have for  $i, j \in S$ ,

$$\mathbb{P}_i(X_1 = j) = p_{ij}, \quad (4.34)$$

and for  $n \in \mathbb{N}_0$  and  $i_0, \dots, i_n \in S$ ,

$$\mathbb{P}_\mu(X_0 = i_0, \dots, X_n = i_n) = \mu(\{i_0\}) p_{i_0 i_1} \cdots p_{i_{n-1} i_n}. \quad (4.35)$$

In particular, for  $n \in \mathbb{N}$  and  $i, i_1, \dots, i_n \in S$ ,

$$\begin{aligned} \mathbb{P}_i(X_1 = i_1, \dots, X_n = i_n) &= \mathbb{P}_i(X_0 = i, X_1 = i_1, \dots, X_n = i_n) \\ &= \delta_i(\{i\}) p_{ii_1} \cdots p_{i_{n-1} i_n} \\ &= p_{ii_1} \cdots p_{i_{n-1} i_n}. \end{aligned}$$

Moreover, writing  $\mu(i) := \mu(\{i\})$ , we have for  $j \in S$ ,

$$\mathbb{P}_\mu(X_1 = j) = \sum_{i \in S} \mu(i) p_{ij}, \quad (4.36)$$

and for  $n \in \mathbb{N}$  and  $i_1, \dots, i_n \in S$ ,

$$\mathbb{P}_\mu(X_1 = i_1, \dots, X_n = i_n) = \sum_{i_0 \in S} \mu(i_0) p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n}. \quad (4.37)$$

■

*Proof of Theorem 4.21.* Assume (1) and let us prove (2) by induction in  $n$ . For  $n = 0$  the result holds since  $Y_0$  has distribution  $\mu$ , so assume (4.25) holds for some  $n$  and all  $B_0, \dots, B_n \in \mathcal{B}(S)$ . For  $B_0, \dots, B_{n+1} \in \mathcal{B}(S)$ ,

$$\begin{aligned} &\mathbb{P}(Y_0 \in B_0, \dots, Y_{n+1} \in B_{n+1}) \\ &= \mathbb{E}[\mathbb{1}_{\{Y_0 \in B_0, \dots, Y_n \in B_n\}} \mathbb{1}_{\{Y_{n+1} \in B_{n+1}\}}] \\ &= \mathbb{E}[\mathbb{1}_{\{Y_0 \in B_0, \dots, Y_n \in B_n\}} \mathbb{E}[\mathbb{1}_{\{Y_{n+1} \in B_{n+1}\}} \mid \mathcal{F}_n^Y]] \\ &= \mathbb{E}[\mathbb{1}_{\{Y_0 \in B_0, \dots, Y_n \in B_n\}} Q(Y_n, B_{n+1})] \\ &= \int_{B_0} \int_{B_1} \cdots \int_{B_n} Q(y_n, B_{n+1}) Q(y_{n-1}, dy_n) \cdots Q(y_0, dy_1) \mu(dy_0) \\ &= \int_{B_0} \int_{B_1} \cdots \int_{B_n} \int_{B_{n+1}} 1 Q(y_n, dy_{n+1}) Q(y_{n-1}, dy_n) \cdots Q(y_0, dy_1) \mu(dy_0), \end{aligned}$$

where in the second to last equality we used Remark 4.23 with

$$f(y_0, \dots, y_n) = Q(y_n, B_{n+1}) \prod_{j=0}^n \mathbb{1}_{B_j}(y_j).$$

This concludes the proof of (2).

Conversely assume (2) and let us prove (1). First of all, from (4.25) with  $n = 0$  it follows that  $Y_0$  has distribution  $\mu$ . Next we verify that  $\mathbb{P}(Y_{n+1} \in B | \mathcal{F}_n^Y) = Q(Y_n, B)$  for every  $n \in \mathbb{N}_0$  and  $B \in \mathcal{B}(S)$ . Fix such an  $n$  and  $B$ . Since  $Q(Y_n, B)$  is  $\mathcal{F}_n^Y$ -measurable, it suffices to verify that

$$\mathbb{E}[\mathbb{1}_A \mathbb{1}_{\{Y_{n+1} \in B\}}] = \mathbb{E}[\mathbb{1}_A Q(Y_n, B)] \quad (4.38)$$

for  $A \in \mathcal{F}_n^Y$ . As noted in (4.9), there is a  $C \in \mathcal{B}(S^{n+1})$  such that  $A = \{(Y_0, \dots, Y_n) \in C\}$ . An application of Remark 4.23 with  $n$  replaced by  $n + 1$  and

$$f(y_0, \dots, y_{n+1}) = \mathbb{1}_C(y_0, \dots, y_n) \mathbb{1}_B(y_{n+1})$$

yields

$$\begin{aligned} & \mathbb{E}[\mathbb{1}_A \mathbb{1}_{\{Y_{n+1} \in B\}}] \\ &= \iint \cdots \iint \mathbb{1}_C(y_0, \dots, y_n) \mathbb{1}_B(y_{n+1}) Q(y_n, dy_{n+1}) Q(y_{n-1}, dy_n) \cdots Q(y_0, dy_1) \mu(dy_0) \\ &= \iint \cdots \int \mathbb{1}_C(y_0, \dots, y_n) Q(y_n, B) Q(y_{n-1}, dy_n) \cdots Q(y_0, dy_1) \mu(dy_0). \end{aligned}$$

By the same remark with  $f(y_0, \dots, y_n) = Q(y_n, B) \mathbb{1}_C(y_0, \dots, y_n)$  it follows that the above equals  $\mathbb{E}[\mathbb{1}_A Q(Y_n, B)]$ . This concludes the proof of (4.38).  $\square$

*Proof of Corollary 4.22.* Recall that  $Q$  and  $P$  are related as  $Q(i, \{j\}) = p_{ij}$  for  $i, j \in S$ . When  $S$  is countable it suffices to consider  $B_0 = \{i_0\}, \dots, B_n = \{i_n\}$  for  $i_0, \dots, i_n \in S$  in Theorem 4.21. In this case (4.25) reduces to

$$\begin{aligned} & \mathbb{P}(Y_0 = i_0, \dots, Y_n = i_n) \\ &= \int_{\{i_0\}} \int_{\{i_1\}} \cdots \int_{\{i_{n-1}\}} \int_{\{i_n\}} \mathbb{1} Q(y_{n-1}, dy_n) Q(y_{n-2}, dy_{n-1}) \cdots Q(y_0, dy_1) \mu(dy_0) \\ &= \mu(\{i_0\}) Q(i_0, \{i_1\}) \cdots Q(i_{n-1}, \{i_n\}) \\ &= \mu(\{i_0\}) p_{i_0 i_1} \cdots p_{i_{n-1} i_n}. \end{aligned}$$

That is, (4.25) and (4.27) are equivalent.  $\square$

*Proof of Theorem 4.24.* Inspired by Theorem 4.21, for every  $n \in \mathbb{N}_0$  define a probability measure  $\nu_n$  on  $(S^{n+1}, \mathcal{B}(S^{n+1}))$  as

$$\nu_n(A) = \iint \cdots \int \mathbb{1}_A(y_0, \dots, y_n) Q(y_{n-1}, dy_n) \cdots Q(y_0, dy_1) \mu(dy_0)$$

for  $A \in \mathcal{B}(S^{n+1})$ . Then the family  $\{\nu_n : n \in \mathbb{N}_0\}$  is consistent since for  $n \in \mathbb{N}_0$  and  $B_0, \dots, B_n \in \mathcal{B}(S)$ ,

$$\begin{aligned} \nu_{n+1}(B_0 \times \dots \times B_n \times S) &= \int_{B_0} \int_{B_1} \dots \int_{B_n} \int_S Q(y_n, dy_{n+1}) Q(y_{n-1}, dy_n) \dots Q(y_0, dy_1) \mu(dy_0) \\ &= \int_{B_0} \int_{B_1} \dots \int_{B_n} Q(y_n, S) Q(y_{n-1}, dy_n) \dots Q(y_0, dy_1) \mu(dy_0) \\ &= \int_{B_0} \int_{B_1} \dots \int_{B_n} Q(y_{n-1}, dy_n) \dots Q(y_0, dy_1) \mu(dy_0) \\ &= \nu_n(B_0 \times \dots \times B_n). \end{aligned}$$

By Kolmogorov's Consistency Theorem (Theorem 3.5), there exists exactly one probability measure  $\mathbb{P}_\mu$  on  $(S^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$  under which  $(X_0, \dots, X_n)$  has distribution  $\nu_n$  for every  $n \in \mathbb{N}_0$ . By Theorem 4.21,  $(X_n)_{n \geq 0}$  is a Markov chain with initial distribution  $\mu$  under  $\mathbb{P}_\mu$ . By uniqueness of  $\mathbb{P}_\mu$  there can be no other probability measure under which  $(X_n)_{n \geq 0}$  is a Markov chain with initial distribution  $\mu$ .  $\square$

*Proof of Proposition 4.25.* Let  $\mathcal{H}$  be the class of functions  $Z : S^{\mathbb{N}_0} \rightarrow \mathbb{R}$  satisfying (4.29). Precisely,  $Z$  is in  $\mathcal{H}$  if it is bounded,  $\mathcal{B}(S)^{\mathbb{N}_0}$ -measurable, and  $x \mapsto \mathbb{E}_x[Z]$  is measurable with  $\mathbb{E}_\mu[Z] = \int \mathbb{E}_x[Z] \mu(dx)$ . The class  $\mathcal{H}$  satisfies all required conditions in the Monotone Class Theorem.

Let  $\mathcal{A}$  be the collection of sets of the form (3.2). For such a set Theorem 4.21 shows that

$$\begin{aligned} \mathbb{P}_x(X_0 \in B_0, \dots, X_n \in B_n) &= \mathbb{1}_{B_0}(x) \int_{B_1} \dots \int_{B_{n-1}} \int_{B_n} 1 Q(y_{n-1}, dy_n) Q(y_{n-2}, dy_{n-1}) \dots Q(x, dy_1), \quad x \in S. \end{aligned}$$

Hence,  $x \mapsto \mathbb{P}_x(X_0 \in B_0, \dots, X_n \in B_n)$  is measurable, and in addition

$$\begin{aligned} \mathbb{P}_\mu(X_0 \in B_0, \dots, X_n \in B_n) &= \int_{B_0} \int_{B_1} \dots \int_{B_{n-1}} \int_{B_n} 1 Q(y_{n-1}, dy_n) Q(y_{n-2}, dy_{n-1}) \dots Q(y_0, dy_1) \mu(dy_0) \\ &= \int \mathbb{P}_{y_0}(X_0 \in B_0, \dots, X_n \in B_n) \mu(dy_0). \end{aligned}$$

That is, the indicator function of a set as in (3.2) is in  $\mathcal{H}$ . Since  $\mathcal{A}$  is stable under intersection and generates  $\mathcal{B}(S)^{\mathbb{N}_0}$ , the results follows from the Monotone Class Theorem (Theorem H.1).  $\square$

## 4.5 Extending the Markov property – the strong Markov property

Let  $Q$  be a transition kernel and equip  $(S^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$  with the probability measures  $\mathbb{P}_x$  and  $\mathbb{P}_\mu$  as described in Theorem 4.24. For  $n \in \mathbb{N}_0$  define the *shift operator*  $\theta^n : S^{\mathbb{N}_0} \rightarrow S^{\mathbb{N}_0}$  as

$$\theta^n(\omega_0, \omega_1, \dots) = (\omega_n, \omega_{n+1}, \dots). \quad (4.39)$$

That is,  $\theta^n$  removes the first  $n$  coordinates,  $\omega_0, \dots, \omega_{n-1}$ . This operator can be composed with any map from  $S^{\mathbb{N}_0}$ . As an example,  $X_m \circ \theta^n = X_{n+m}$  for  $n, m \in \mathbb{N}_0$ .

For  $n \in \mathbb{N}_0$ ,  $\theta^n$  is  $(\mathcal{B}(S)^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$ -measurable. Indeed, for arbitrary  $m \in \mathbb{N}_0$  and  $B_0, \dots, B_m \in \mathcal{B}(S)$ ,

$$\{\theta^n \in \{X_0 \in B_0, \dots, X_m \in B_m\}\} = \{X_n \in B_0, \dots, X_{n+m} \in B_m\} \in \mathcal{B}(S)^{\mathbb{N}_0}. \quad (4.40)$$

**Theorem 4.27.** *Let  $Q$  be a transition kernel and define the measures  $\mathbb{P}_x$  and  $\mathbb{P}_\mu$  as above.*

*Let  $\mu$  be a probability measure on  $(S, \mathcal{B}(S))$  and  $Z : S^{\mathbb{N}_0} \rightarrow \mathbb{R}$  be  $\mathcal{B}(S)^{\mathbb{N}_0}$ -measurable and bounded or non-negative.*

- (1) *For  $n \in \mathbb{N}_0$ ,  $\mathbb{E}_\mu[Z \circ \theta^n | \mathcal{F}_n^X] = \mathbb{E}_{X_n}[Z]$ .*
- (2) *For every  $(\mathcal{F}_n^X)_{n \geq 0}$ -stopping time  $\tau$ ,  $\mathbb{E}_\mu[Z \circ \theta^\tau | \mathcal{F}_\tau^X] = \mathbb{E}_{X_\tau}[Z]$  on  $\{\tau < \infty\}$ .*

Result (2) above is called the *strong Markov property*; it will be used repeatedly throughout these notes.

**Remark 4.28.** Let us be careful in explaining how the right-hand sides in (1) and (2) are defined. For a given  $Z$ , let  $G : S \rightarrow \mathbb{R}$  be the measurable map defined as  $G(x) = \mathbb{E}_x[Z]$ . Then  $\mathbb{E}_{X_n}[Z]$  is defined as  $\mathbb{E}_{X_n}[Z] = G(X_n)$ , and, similarly,  $\mathbb{E}_{X_\tau}[Z] = G(X_\tau)$  on  $\{\tau < \infty\}$ .

Alternatively, we can use the definition  $\mathbb{E}_{X_n}[Z] = \mathbb{E}_x[Z] \Big|_{x=X_n}$ . ■

**Remark 4.29 (What the strong Markov property says).**

Theorem 4.27 is formulated using shift operators on the canonical path space, which is extremely convenient in proofs. From a more applied point of view, its content can be summarized as follows:

*Given the present state, the conditional distribution of the entire future is the same as if we restarted the Markov chain from that state; i.e., the past influences the future only through the present.*

To make this precise, let  $h$  be a bounded measurable function of the full trajectory and set  $Z = h(X_0, X_1, \dots)$ . Then

$$Z \circ \theta^n = h(X_n, X_{n+1}, \dots),$$

i.e. the shift  $\theta^n$  simply discards the first  $n$  steps. By Theorem 4.27(1),

$$\mathbb{E}_\mu[h(X_n, X_{n+1}, \dots) | \mathcal{F}_n^X] = \mathbb{E}_{X_n}[h(X_0, X_1, \dots)].$$

Equivalently, using regular conditional distributions,

$$\mathbb{E}_\mu[h(X_n, X_{n+1}, \dots) | X_n = x_n, \dots, X_0 = x_0] = \mathbb{E}_{x_n}[h(X_0, X_1, \dots)].$$

Thus, even if we condition on the entire past  $(x_0, \dots, x_n)$ , the conditional distribution of the future depends only on the present state  $x_n$ . In other words: *conditional on  $\mathcal{F}_n^X$ , the process  $(X_{n+k})_{k \geq 0}$  is a Markov chain started from  $X_n$ .*

The genuinely stronger statement is Theorem 4.27(2), where the deterministic time  $n$  is replaced by a stopping time  $\tau$ . On  $\{\tau < \infty\}$ ,

$$\mathbb{E}_\mu[h(X_\tau, X_{\tau+1}, \dots) | \mathcal{F}_\tau^X] = \mathbb{E}_{X_\tau}[h(X_0, X_1, \dots)].$$

Hence we may choose the "restart time" at random, based on the observed past, and still the post- $\tau$  evolution only depends on  $X_\tau$ . In particular, for  $B \in \mathcal{B}(S)$ ,

$$\mathbb{P}(X_{\tau+1} \in B | \mathcal{F}_\tau^X) = Q(X_\tau, B) \quad \text{on } \{\tau < \infty\}.$$

A common example is a hitting time. Let  $A \in \mathcal{B}(S)$  and define

$$\tau_A := \inf\{n \in \mathbb{N}_0 | X_n \in A\}.$$

Then, on  $\{\tau_A < \infty\}$ , the future after the first hit of  $A$  behaves like a fresh chain started from the random point  $X_{\tau_A} \in A$ .

In particular, if  $A = \{a\}$  is a single state and  $\tau_a := \inf\{n \geq 0 : X_n = a\}$ , then on  $\{\tau_a < \infty\}$  we have  $X_{\tau_a} = a$  and hence

$$\mathbb{E}_\mu[h(X_{\tau_a}, X_{\tau_a+1}, \dots) | \mathcal{F}_{\tau_a}^X] = \mathbb{E}_a[h(X_0, X_1, \dots)].$$

For instance, for any  $m \in \mathbb{N}$  and sets  $B_1, \dots, B_m \in \mathcal{B}(S)$ ,

$$\mathbb{P}(X_{\tau_a+1} \in B_1, \dots, X_{\tau_a+m} \in B_m | \mathcal{F}_{\tau_a}^X) = \mathbb{P}_a(X_1 \in B_1, \dots, X_m \in B_m) \quad \text{on } \{\tau_a < \infty\}.$$

In particular,

$$\mathbb{P}(X_{\tau_a+1} \in B | \mathcal{F}_{\tau_a}^X) = Q(a, B) \quad \text{on } \{\tau_a < \infty\}. \quad \blacksquare$$

When  $S$  is countable the above discussion can be made more precise:

**Proposition 4.30.** *Let  $S$  be countable and  $P = \{p_{ij}\}_{i,j \in S}$  denote the transition matrix corresponding to  $Q$  ( $p_{ij} = Q(i, \{j\})$  for all  $i, j$ ). Then for  $\tau$  an  $(\mathcal{F}_n^X)_{n \geq 0}$ -stopping time,  $k \in \mathbb{N}$ ,  $A \in \mathcal{F}_\tau^X$ ,  $i_0, \dots, i_k \in S$ , and an initial distribution  $\mu$ ,*

$$\mathbb{P}_\mu(X_{\tau+1} = i_1, \dots, X_{\tau+k} = i_k | A \cap \{\tau < \infty\} \cap \{X_\tau = i_0\}) = \mathbb{P}_{i_0}(X_1 = i_1, \dots, X_k = i_k). \quad (4.41)$$

*In particular, this is true if  $\tau = n$  for some  $n \in \mathbb{N}_0$ .*

*Proof of Theorem 4.27.* Since  $x \mapsto \mathbb{E}_x[Z]$  is  $(\mathcal{B}(S), \mathcal{B}(\mathbb{R}))$ -measurable,  $\mathbb{E}_{X_n}[Z]$  and  $\mathbb{E}_{X_\tau}[Z] \mathbb{1}_{\{\tau < \infty\}}$  are resp.  $\mathcal{F}_n^X$ -measurable and  $\mathcal{F}_\tau^X$ -measurable. We prove the result in three steps.

*Step 1: Proof of Theorem 4.27(1) in the case where  $Z = \prod_{j=0}^k f_j(X_j)$  for some  $k \in \mathbb{N}_0$ , and  $f_j : S \rightarrow \mathbb{R}$ , for  $j = 0, \dots, k$ , are measurable and bounded.*

In the case  $k = 0$ ,  $Z = f_0(X_0)$ ,  $Z \circ \theta^n = f_0(X_n)$  and  $\mathbb{E}_\mu[f_0(X_n) | \mathcal{F}_n^X] = f_0(X_n)$ . Since  $\mathbb{E}_x[f_0(X_0)] = f_0(x)$  for  $x \in S$ , it follows that

$$\mathbb{E}_\mu[Z \circ \theta^n | \mathcal{F}_n^X] = f_0(X_n) = \mathbb{E}_{X_n}[Z].$$

This concludes the case  $k = 0$ .

Assume the result is true for some  $k \in \mathbb{N}_0$ . Let  $f_j : S \rightarrow \mathbb{R}$ , for  $j = 0, \dots, k+1$ , be measurable and bounded and put  $Z = \prod_{j=0}^{k+1} f_j(X_j)$ , so that  $Z \circ \theta^n = \prod_{j=0}^{k+1} f_j(X_{n+j})$ . By Remark 4.5(1),

$$\begin{aligned} \mathbb{E}_\mu[Z \circ \theta^n | \mathcal{F}_n^X] &= \mathbb{E}_\mu \left[ \prod_{j=0}^{k+1} f_j(X_{n+j}) \middle| \mathcal{F}_n^X \right] \\ &= \mathbb{E}_\mu \left[ \mathbb{E}_\mu \left[ \prod_{j=0}^{k+1} f_j(X_{n+j}) \middle| \mathcal{F}_{n+k}^X \right] \middle| \mathcal{F}_n^X \right] \\ &= \mathbb{E}_\mu \left[ \prod_{j=0}^k f_j(X_{n+j}) \mathbb{E}_\mu \left[ f_{k+1}(X_{n+k+1}) \middle| \mathcal{F}_{n+k}^X \right] \middle| \mathcal{F}_n^X \right] \\ &= \mathbb{E}_\mu \left[ \prod_{j=0}^k f_j(X_{n+j}) (Qf_{k+1})(X_{n+k}) \middle| \mathcal{F}_n^X \right]. \end{aligned}$$

Since the last expression involves the  $k+1$  functions  $f_0, f_1, \dots, f_{k-1}$  and  $x \mapsto f_k(x)(Qf_{k+1})(x)$ , the induction hypothesis yields

$$\mathbb{E}_\mu[Z \circ \theta^n | \mathcal{F}_n^X] = \mathbb{E}_{X_n} \left[ \prod_{j=0}^k f_j(X_j) (Qf_{k+1})(X_k) \right]. \quad (4.42)$$

Similarly, for  $x \in S$ ,

$$\mathbb{E}_x[Z] = \mathbb{E}_x\left[\prod_{j=0}^{k+1} f_j(X_j)\right] = \mathbb{E}_x\left[\mathbb{E}_x\left[\prod_{j=0}^{k+1} f_j(X_j) \mid \mathcal{F}_k^X\right]\right] = \mathbb{E}_x\left[\prod_{j=0}^k f_j(X_j)(Qf_{k+1})(X_k)\right].$$

Inserting this in (4.42) it follows that  $\mathbb{E}_\mu[Z \circ \theta^n \mid \mathcal{F}_n^X] = \mathbb{E}_{X_n}[Z]$ .

*Step 2: Proof of Theorem 4.27(1) in the general case.*

This is a typical application of the Monotone Class Theorem. Let  $\mathcal{A}$  be the collection of sets of the form

$$\{X_0 \in B_0, \dots, X_k \in B_k\} \quad \text{for some } k \in \mathbb{N}_0 \text{ and some } B_0, \dots, B_k \in \mathcal{B}(S). \quad (4.43)$$

Let  $\mathcal{H}$  be the collection of  $Z : S^{\mathbb{N}_0} \rightarrow \mathbb{R}$  satisfying Theorem 4.27(1), that is,

$$\mathcal{H} = \{Z : Z \text{ is bounded and } \mathcal{B}(S)^{\mathbb{N}_0}\text{-measurable, and } \mathbb{E}_\mu[Z \circ \theta^n \mid \mathcal{F}_n^X] = \mathbb{E}_{X_n}[Z]\}.$$

If  $A \in \mathcal{A}$  is as in (4.43), then  $\mathbb{1}_A = \prod_{j=0}^k \mathbb{1}_{B_j}(X_j)$ , and by Step 1,  $\mathbb{1}_A \in \mathcal{H}$ . Moreover, it is easily seen that  $\mathcal{H}$  satisfies conditions (2) and (3) in the Monotone Class Theorem (Theorem H.1). Since  $\mathcal{A}$  is stable under intersection and  $\sigma(\mathcal{A}) = \mathcal{B}(S)^{\mathbb{N}_0}$ , the Monotone Class Theorem shows that every bounded  $\mathcal{B}(S)^{\mathbb{N}_0}$ -measurable function  $Z : S^{\mathbb{N}_0} \rightarrow \mathbb{R}$  is in  $\mathcal{H}$  and hence satisfies Theorem 4.27(1).

*Step 3: Proof of Theorem 4.27(2).*

By definition of conditional expectations we must show that

$$\mathbb{E}_\mu[Z \circ \theta^\tau \mathbb{1}_{A \cap \{\tau < \infty\}}] = \mathbb{E}_\mu[\mathbb{E}_{X_\tau}[Z] \mathbb{1}_{A \cap \{\tau < \infty\}}]$$

for  $A \in \mathcal{F}_\tau^X$ . Note that

$$\begin{aligned} \mathbb{E}_\mu[Z \circ \theta^\tau \mathbb{1}_{A \cap \{\tau < \infty\}}] &= \mathbb{E}_\mu[Z \circ \theta^\tau \mathbb{1}_{A \cap \{\bigcup_{n=0}^{\infty} \{\tau=n\}\}}] = \sum_{n=0}^{\infty} \mathbb{E}_\mu[Z \circ \theta^n \mathbb{1}_{A \cap \{\tau=n\}}], \\ \mathbb{E}_\mu[\mathbb{E}_{X_\tau}[Z] \mathbb{1}_{A \cap \{\tau < \infty\}}] &= \sum_{n=0}^{\infty} \mathbb{E}_\mu[\mathbb{E}_{X_n}[Z] \mathbb{1}_{A \cap \{\tau=n\}}]. \end{aligned}$$

Since  $A \cap \{\tau = n\} \in \mathcal{F}_n^X$  by definition of  $\mathcal{F}_\tau^X$ , (1) shows that for  $n \in \mathbb{N}_0$ ,

$$\mathbb{E}_\mu[Z \circ \theta^n \mathbb{1}_{A \cap \{\tau=n\}}] = \mathbb{E}_\mu[\mathbb{E}_{X_n}[Z] \mathbb{1}_{A \cap \{\tau=n\}}],$$

which implies the result.  $\square$

*Proof of Proposition 4.30.* Using  $Z = \mathbb{1}_{\{X_1=i_1, \dots, X_k=i_k\}}$  and that  $A \cap \{\tau < \infty\} \cap \{X_\tau = i_0\} \in \mathcal{F}_\tau^X$  in Theorem 4.27, it follows that

$$\begin{aligned}
& \mathbb{P}_\mu(\{X_{\tau+1} = i_1, \dots, X_{\tau+k} = i_k\} \cap A \cap \{\tau < \infty\} \cap \{X_\tau = i_0\}) \\
&= \mathbb{E}_\mu \left[ \mathbb{E}_\mu[Z \circ \theta^\tau \mid \mathcal{F}_\tau^X] \mathbb{1}_{A \cap \{\tau < \infty\} \cap \{X_\tau = i_0\}} \right] \\
&= \mathbb{E}_\mu \left[ \mathbb{E}_{i_0}[Z] \mathbb{1}_{A \cap \{\tau < \infty\} \cap \{X_\tau = i_0\}} \right] \\
&= \mathbb{E}_\mu \left[ \mathbb{P}_{i_0}(X_1 = i_1, \dots, X_k = i_k) \mathbb{1}_{A \cap \{\tau < \infty\} \cap \{X_\tau = i_0\}} \right] \\
&= \mathbb{P}_{i_0}(X_1 = i_1, \dots, X_k = i_k) \mathbb{P}_\mu(A \cap \{\tau < \infty\} \cap \{X_\tau = i_0\}),
\end{aligned}$$

which gives the result. □

## Chapman–Kolmogorov and some applications

In the following, we define "products" and "powers" of transition kernels.

**Definition 5.1.** Let  $Q(x, B)$  and  $R(x, B)$ , for  $x \in S$ ,  $B \in \mathcal{B}(S)$ , be transition kernels. Define a new transition kernel  $QR$  as

$$QR(x, B) = \int R(y, B) Q(x, dy), \quad \text{for } x \in S \text{ and } B \in \mathcal{B}(S). \quad (5.1)$$

The reader is encouraged to verify that  $QR$ , in fact, is a transition kernel. By (4.6),

$$(QR)1_B(x) = QR(x, B) = \int \int 1_B(z) R(y, dz) Q(x, dy), \quad \text{for } x \in S \text{ and } B \in \mathcal{B}(S).$$

Thus, by the Standard Proof,

$$(QR)f(x) = \int \int f(z) R(y, dz) Q(x, dy) \quad (5.2)$$

for any  $x \in S$  and  $f : S \rightarrow \mathbb{R}$  bounded and measurable.

**Remark 5.2 (Associativity).**

For  $j = 1, 2, 3$  let  $Q_j(x, B)$ ,  $x \in S$ ,  $B \in \mathcal{B}(S)$ , be a transition kernel. Then for  $x \in S$  and  $B \in \mathcal{B}(S)$ ,

$$Q_1(Q_2 Q_3)(x, B) = \int_S \int_S Q_3(z, B) Q_2(y, dz) Q_1(x, dy) = (Q_1 Q_2) Q_3(x, B). \quad (5.3)$$

Indeed, the first equality is by definition and the second one uses (5.2) with  $f(x) = Q_3(x, B)$ ,  $Q = Q_1$ , and  $R = Q_2$ . More generally, for any  $f : S \rightarrow \mathbb{R}$  measurable and bounded and  $x \in S$ ,

$$Q_1(Q_2 Q_3)f(x) = \int_S \int_S \int_S f(u) Q_3(z, du) Q_2(y, dz) Q_1(x, dy) = (Q_1 Q_2) Q_3 f(x). \quad (5.4)$$

■

Let  $Q(x, B)$ ,  $x \in S$ ,  $B \in \mathcal{B}(S)$ , be a transition kernel. Define  $Q^0$  as the transition kernel given by  $Q^0(x, B) = \delta_x(B)$  for  $x \in S$  and  $B \in \mathcal{B}(S)$ . For  $n \in \mathbb{N}$  define  $Q^n$  as  $Q \cdots Q$  ( $n$  times). By iterating (5.3) and (5.4), it follows that, for  $n \in \mathbb{N}$ ,  $f : S \rightarrow \mathbb{R}$  measurable and bounded, and  $B \in \mathcal{B}(S)$ ,

$$Q^n f(x) = \int \cdots \int f(x_n) Q(x_{n-1}, dx_n) \cdots Q(x, dx_1) \quad (5.5)$$

and

$$Q^n(x, B) = \int_S \cdots \int_S Q(x_{n-1}, B) Q(x_{n-2}, dx_{n-1}) \cdots Q(x, dx_1). \quad (5.6)$$

In particular these results imply

$$Q^{n+m} = Q^m Q^n \quad \text{for } n, m \in \mathbb{N}_0 \quad (5.7)$$

and, more generally,

$$Q^m(Q^n f)(x) = Q^{m+n} f(x) \quad \text{for all } x \in S, n, m \in \mathbb{N}_0, \quad (5.8)$$

and  $f : S \rightarrow \mathbb{R}$  measurable and bounded. As usual, the last identity also holds if  $f$  is measurable and non-negative. Equations (5.7)–(5.8) are referred to as *Chapman–Kolmogorov’s equations*.

Let us also consider the probabilistic interpretation of  $Q^n$ , and for this let  $\mathbb{P}_x$  be the measure corresponding to the initial value  $x$  as defined in Theorem 4.24. Then

$$Q^n(x, B) = \mathbb{P}_x(X_n \in B), \quad \text{for } n \in \mathbb{N}_0, x \in S \text{ and } B \in \mathcal{B}(S). \quad (5.9)$$

That is,  $Q^n(x, B)$  is the probability of being in  $B$  after  $n$  steps when the initial value is  $x$ . To see that this is true, note that for  $n = 0$  it follows by definition of  $Q^0$ . For  $n \in \mathbb{N}$  note that

$$\mathbb{P}_x(X_n \in B) = \mathbb{P}_x(X_1 \in S, \dots, X_{n-1} \in S, X_n \in B),$$

and apply (4.25) and (5.6).

**Remark 5.3 (The countable case).**

Let  $S$  be countable and  $P = \{p_{ij}\}_{i,j \in S}$  be a transition matrix. Recall that the corresponding transition kernel  $Q$  satisfies  $Q(i, \{j\}) = p_{ij}$ .

Put  $P^n = P \cdots P$  ( $n$  times) for  $n \in \mathbb{N}$  and let  $P^0$  be the  $S \times S$  identity matrix. Denote the  $(i, j)$ th entry of  $P^n$  by  $p_{ij}^n$  for  $i, j \in S$ . Then for all  $n \in \mathbb{N}_0$  and  $i, j \in S$ ,  $Q^n(i, \{j\}) = p_{ij}^n$ . This is obvious for  $n = 0$ , so we only need a proof for  $n \in \mathbb{N}$ . By (5.6)

with  $B = \{j\}$  and  $x = i$ ,

$$\begin{aligned}
 Q^n(i, \{j\}) &= \int_S \cdots \int_S \int_S Q(x_{n-1}, \{j\}) Q(x_{n-2}, dx_{n-1}) \cdots Q(i, dx_1) \\
 &= \sum_{x_{n-1} \in S} \cdots \sum_{x_1 \in S} Q(i, \{x_1\}) \cdots Q(x_{n-1}, \{j\}) \\
 &= \sum_{x_{n-1} \in S} \cdots \sum_{x_1 \in S} p_{ix_1} \cdots p_{x_{n-1}j} \\
 &= p_{ij}^n,
 \end{aligned}$$

where the last equality is by definition of matrix multiplication.

Thus, by (5.9),

$$p_{ij}^n = \mathbb{P}_i(X_n = j) \quad \text{for } i, j \in S \text{ and } n \in \mathbb{N}_0. \quad (5.10)$$

See Durrett (2016, Section 1.2) (and in particular Example 1.11), as well as Pedersen (2014, Kapitel 3), for interesting applications of this result. ■

Theorem 4.27(2) is a very applicable version of the strong Markov property which relies on the shift operators on  $S^{\mathbb{N}_0}$ . For general Markov chains (i.e., not defined on  $S^{\mathbb{N}_0}$ ) there are no shift operators, and therefore the strong Markov property has to be reformulated. Items (2) and (3) below show how this can be done.

**Theorem 5.4 (The Markov property for general Markov chains).**

Let  $(Y_n)_{n \geq 0}$  be a Markov chain with transition kernel  $Q$  with respect to  $(\mathcal{F}_n)_{n \geq 0}$  defined on  $(\Omega, \mathcal{F}, \mathbb{P})$ .

(1) For  $m \in \mathbb{N}_0$ ,  $n \in \mathbb{N}$  and  $f : S \rightarrow \mathbb{R}$  bounded and measurable

$$\mathbb{E}[f(Y_{n+m}) | \mathcal{F}_m] = Q^n f(Y_m) \quad a.s. \quad (5.11)$$

In particular,

$$\mathbb{P}(Y_{m+n} \in A | \mathcal{F}_m) = Q^n(Y_m, A) \quad a.s. \text{ for } A \in \mathcal{B}(S).$$

(2) Let  $\tau$  be a stopping time satisfying  $\mathbb{P}(\tau < \infty) = 1$ . Then

$$\mathbb{P}(Y_{\tau+n} \in A | \mathcal{F}_\tau) = Q^n(Y_\tau, A) \quad a.s. \text{ for } n \in \mathbb{N} \text{ and } A \in \mathcal{B}(S).$$

(3) Let  $\tau$  be a stopping time satisfying  $\mathbb{P}(\tau < \infty) = 1$ . Then  $(Y_{\tau+n})_{n \geq 0}$  is a Markov chain with transition kernel  $Q$  with respect to  $(\mathcal{F}_{\tau+n})_{n \geq 0}$ .

*Proof. Item (1):* We use induction in  $n$ . For  $n = 1$  the result holds by definition of a Markov chain. Assume (5.11) is true for some  $n \in \mathbb{N}$ , every  $m \in \mathbb{N}_0$  and every  $f$ , and let us prove it is also true for  $n + 1$ . So let  $m \in \mathbb{N}_0$  and use the tower principle to get

$$\begin{aligned}\mathbb{E}[f(Y_{m+(n+1)}) | \mathcal{F}_m] &= \mathbb{E}[\mathbb{E}[f(Y_{m+(n+1)}) | \mathcal{F}_{m+n}] | \mathcal{F}_m] \\ &= \mathbb{E}[Qf(Y_{m+n}) | \mathcal{F}_m] = Q^n(Qf)(Y_m) = Q^{n+1}f(Y_m),\end{aligned}$$

where the last equality uses (5.8).

*Item (2):* Note that  $Q^n(Y_\tau, A) \mathbb{1}_{\{\tau < \infty\}}$  is  $\mathcal{F}_\tau$ -measurable. Hence we only have to verify the defining equation for conditional expectations:

$$\mathbb{E}[\mathbb{1}_B \mathbb{1}_{\{Y_{\tau+n} \in A\}}] = \mathbb{E}[Q^n(Y_\tau, A) \mathbb{1}_B] \quad \text{for } B \in \mathcal{F}_\tau.$$

This follows as in the proof of Theorem 4.27(2).

*Item (3):* Follows from (2); see Exercise 56. □

## Markov chains with countable state space

In this chapter,  $S$  is countable and  $P = \{p_{ij}\}_{i,j \in S}$  is a transition matrix. Equip  $(S^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$  with the probability measures  $\mathbb{P}_x$  and  $\mathbb{P}_\mu$  as described in Theorem 4.24.

We are particularly interested in the asymptotic behaviour of the chain, i.e., questions like how many times the chain visits a given state. The analysis will be based on the concepts irreducibility, recurrence, and transience. We prove a Law of Large Numbers for Markov chains, which gives information about the fraction of time spent in different states, and for this we need to study so-called stationary distributions. The ergodic theorem, which is related to the Law of Large Numbers, will not be proved here; see, however, Norris (1998, Theorem 1.7.1) and Pedersen (2014).

For chains with a general (i.e., uncountable) state space, it is often possible to prove results similar to the ones we obtain; moreover, using a so-called splitting technique, proofs are often also reduced to arguments very similar to those below; see, e.g., Meyn and Tweedie (2009) and Cappé et al. (2005). For a survey, see also Roberts and Rosenthal (2004). In particular, it is worth mentioning that a Law of Large Numbers can be proved for so-called Harris recurrent Markov chains.

**Remark 6.1.** It is convenient to work on the canonical path space  $S^{\mathbb{N}_0}$ , not least because this gives access to the shift operators. However, all important results below generalize in an obvious way to Markov chains defined on arbitrary probability spaces; this follows since Markov chains with the same initial distribution and transition kernel are versions (see Exercise 33). ■

**Definition 6.2 (Return times, number of visits to a state).**

Let  $i \in S$ . Define the **first return time**  $T_i$  to  $i$  as

$$T_i = \inf\{n \geq 1 : X_n = i\}.$$

Define the  $0$ th return time to  $i$  as  $T_i^0 = 0$ , and for  $n \in \mathbb{N}$  define the  **$n$ th return time** to  $i$  iteratively as  $T_i^1 = T_i$  and

$$T_i^{n+1} = \inf\{k > T_i^n : X_k = i\}.$$

Let  $N_i = \sum_{n=1}^{\infty} \mathbb{1}_{\{X_n=i\}}$  denote the **number of visits to  $i$**  (after time 0).

Note that we have the interpretations:

$$\{T_i^n < \infty\} = \{N_i \geq n\} = \{i \text{ is visited at least } n \text{ times after time } 0\}, \quad \text{for } n \in \mathbb{N}$$

$$\{N_i = \infty\} = \{i \text{ is visited infinitely often}\}$$

$$\{N_i < \infty\} = \{i \text{ is visited finitely many times}\}.$$

Moreover, for  $n \in \mathbb{N}$ ,

$$T_i^{n+1} = T_i^n + T_i \circ \theta^{T_i^n} \quad \text{on } \{T_i^n < \infty\}, \quad (6.1)$$

$$N_i = \mathbb{1}_{\{T_i < \infty\}} (1 + N_i \circ \theta^{T_i}). \quad (6.2)$$

**Remark 6.3.** For each  $n \in \mathbb{N}_0$ , the return time  $T_i^n$  is a stopping time with respect to the natural filtration  $\mathcal{F}_m^X = \sigma(X_0, \dots, X_m)$ , because the event  $\{T_i^n \leq m\}$  depends only on the path up to time  $m$ . Indeed, let  $m \in \mathbb{N}$ . Then  $\{T_i \leq m\} = \bigcup_{k=1}^m \{X_k = i\} \in \mathcal{F}_m^X$ , and, by induction, one can write  $\{T_i^{n+1} \leq m\} = \bigcup_{k=1}^m (\{T_i^n = k\} \cap \bigcup_{\ell=1}^{m-k} \{X_{k+\ell} = i\}) \in \mathcal{F}_m^X$  for  $n \in \mathbb{N}$ . ■

The following result is fundamental.

**Proposition 6.4.** Let  $i \in S$  and  $\mu$  be a probability measure on  $S$ .

(1) We have

$$\mathbb{P}_\mu(T_i^{n+1} < \infty) = \mathbb{P}_\mu(T_i < \infty) [\mathbb{P}_i(T_i < \infty)]^n, \quad n \in \mathbb{N}_0, \quad (6.3)$$

$$\mathbb{E}_\mu[N_i] = \mathbb{P}_\mu(T_i < \infty) (1 + \mathbb{E}_i[N_i]), \quad (6.4)$$

where we put  $0 \cdot \infty = 0$ .

(2) Assume  $\mathbb{P}_\mu(T_i < \infty) = 1$  and  $\mathbb{P}_i(T_i < \infty) = 1$ . Then

(a)  $T_i^n$  is  $\mathbb{P}_\mu$ -a.s. finite for all  $n \in \mathbb{N}$ ,

(b)  $T_i^1, T_i^2 - T_i^1, T_i^3 - T_i^2, \dots$  are independent under  $\mathbb{P}_\mu$ ,

(c) for  $n \in \mathbb{N}$  the distribution of  $T_i^{n+1} - T_i^n$  under  $\mathbb{P}_\mu$  equals that of  $T_i$  under  $\mathbb{P}_i$ .

*Proof. Item (1):* We prove (6.3). For  $n = 0$ , this is obvious, so assume it holds for some  $n \in \mathbb{N}_0$ . Then

$$\begin{aligned}
 \mathbb{P}_\mu(T_i^{n+2} < \infty) &= \mathbb{P}_\mu(T_i^{n+2} < \infty, T_i^{n+1} < \infty) \\
 &= \mathbb{E}_\mu \left[ \mathbb{1}_{\{T_i \circ \theta^{T_i^{n+1}} < \infty\}} \mathbb{1}_{\{T_i^{n+1} < \infty\}} \right] \\
 &= \mathbb{E}_\mu \left[ \mathbb{E}_\mu \left[ \mathbb{1}_{\{T_i \circ \theta^{T_i^{n+1}} < \infty\}} \mid \mathcal{F}_{T_i^{n+1}}^X \right] \mathbb{1}_{\{T_i^{n+1} < \infty\}} \right] \\
 &= \mathbb{E}_\mu \left[ \mathbb{E}_{X_{T_i^{n+1}}} \left[ \mathbb{1}_{\{T_i < \infty\}} \right] \mathbb{1}_{\{T_i^{n+1} < \infty\}} \right] \\
 &= \mathbb{E}_\mu \left[ \mathbb{P}_i(T_i < \infty) \mathbb{1}_{\{T_i^{n+1} < \infty\}} \right] \\
 &= \mathbb{P}_i(T_i < \infty) \mathbb{P}_\mu(T_i^{n+1} < \infty),
 \end{aligned}$$

where we have used (6.1) with  $n$  replaced by  $n + 1$ , that  $T_i^{n+1}$  is  $\mathcal{F}_{T_i^{n+1}}^X$ -measurable, the strong Markov property and that  $X_{T_i^{n+1}} = i$  on  $\{T_i^{n+1} < \infty\}$ . The result follows by the induction hypothesis.

*Item (2):* It follows from (6.3) that  $\mathbb{P}_\mu(T_i^n < \infty) = 1$  for all  $n$ . To conclude the proof, it suffices (check this!) to argue that

$$\mathbb{P}_\mu(T_i \in B_1, \dots, T_i^n - T_i^{n-1} \in B_n) = \mathbb{P}_\mu(T_i \in B_1) \prod_{j=2}^n \mathbb{P}_i(T_i \in B_j) \quad (6.5)$$

for all  $n \in \mathbb{N}$  and  $B_1, \dots, B_n \subseteq \mathbb{N}$ . However, this follows by modification of the proof of (6.3). See Exercise 47.  $\square$

## 6.1 Recurrence, transience and irreducibility

The next definitions are important in describing the behaviour of a Markov chain.

**Definition 6.5.** Consider the following.

- (1) Let  $i, j \in S$ . We say that  $i$  **communicates with**  $j$  (and write  $i \rightarrow j$ ) if  $\mathbb{P}_i(T_j < \infty) > 0$ .
- (2) The transition matrix  $P$  is **irreducible** if  $i \rightarrow j$  for all  $i, j \in S$ .
- (3) A state  $i \in S$  is **recurrent** if  $\mathbb{P}_i(T_i < \infty) = 1$ .

(4) A state  $i \in S$  is **transient** if  $\mathbb{P}_i(T_i < \infty) < 1$ .

**Remark 6.6.** Note that

- $i \rightarrow j$  means that it is possible to visit  $j$  when starting from  $i$ ,
- irreducibility means that it is possible to visit any state no matter where we start,
- $i$  is recurrent if we are certain to return to  $i$  from  $i$ ,
- any state is either recurrent or transient (not both).

Let us discuss how to check these properties.

(1) Often it is rather easy to check for communication and irreducibility. In particular, it is worth mentioning that the following three statements are equivalent for  $i, j \in S$ :

- (a)  $i \rightarrow j$ ,
- (b) there is an  $n \in \mathbb{N}$  such that  $p_{ij}^n = \mathbb{P}_i(X_n = j) > 0$ .
- (c) there is a *path* from  $i$  to  $j$ , that is, there is an  $n \in \mathbb{N}$  and  $i_0, \dots, i_n \in S$  such that  $i_0 = i$ ,  $i_n = j$  and  $p_{i_{k-1}i_k} > 0$  for all  $k = 1, \dots, n$ .

(See Exercise 48). Typically it is straightforward to decide whether there is a path from one state to another.

(2) On the other hand, it may be hard to decide whether a given state is transient or recurrent, but later we give some useful criteria to help us decide.

Let us note the following condition, which is useful in some cases: A state  $j \in S$  is transient if there is a state  $i \in S$  with  $i \neq j$  such that  $j \rightarrow i$  and  $\mathbb{P}_i(T_j < \infty) < 1$ . (That is, (i): it is possible to come from  $j$  to  $i$ , and (ii): starting in  $i$  we are not certain to visit  $j$ ).

To see this, note that

$$\mathbb{P}_j(N_j < \infty) \geq \mathbb{P}_j(T_i < \infty, T_j \circ \theta^{T_i} = \infty) = \mathbb{P}_j(T_i < \infty)\mathbb{P}_i(T_j = \infty). \quad (6.6)$$

If  $j \rightarrow i$  and  $\mathbb{P}_i(T_j < \infty) < 1$  then the right-hand side is positive which means that  $j$  is transient, see Theorem 6.8(3). ■

**Example 6.7 (Gambler's ruin).**

Recall that in this example  $S = \{0, \dots, N\}$  for some integer  $N \in \mathbb{N}$ , and the transition matrix  $P = \{p_{ij}\}_{i,j \in S}$  is

$$\begin{aligned} p_{i,i+1} &= p, & p_{i,i-1} &= 1 - p \text{ for } i = 1, \dots, N-1, \\ p_{00} &= 1, & p_{NN} &= 1, \text{ and } p_{ij} = 0 \text{ otherwise} \end{aligned}$$

where  $p \in (0, 1)$ . We interpret a corresponding Markov chain as modelling the fortune of a gambler cf. Example 4.17.

Since the states  $1, \dots, N-1$  all communicate with 0 (and  $N$ ) but it is impossible to escape from 0 (or  $N$ ), it follows from the criterion in Remark 6.6(2) that  $1, \dots, N-1$  are transient. As we shall see below, transient states will only be visited a finite number of times. Thus, no matter the initial capital, you will always end up being either ruined (i.e., absorbed in 0) or reaching the maximal capital  $N$  (absorption in  $N$ ). ■

The following is the main result in this section.

**Theorem 6.8.** *Let  $i, j \in S$ .*

- (1) *For any initial distribution  $\mu$ ,  $\mathbb{E}_\mu[N_i] = \frac{\mathbb{P}_\mu(T_i < \infty)}{1 - \mathbb{P}_i(T_i < \infty)}$ , where we put  $0/0 = 0$  and  $a/0 = \infty$  for  $a > 0$ .*
- (2) *The following four statements are equivalent:*
  - (a)  *$i$  is recurrent (i.e.,  $\mathbb{P}_i(T_i < \infty) = 1$ ),*
  - (b)  *$\mathbb{P}_i(N_i = \infty) = 1$ ,*
  - (c)  *$\mathbb{E}_i[N_i] = \infty$ ,*
  - (d)  *$\sum_{n=1}^{\infty} p_{ii}^n = \infty$ .*
- (3) *If  $i$  is transient then  $\mathbb{P}_\mu(N_i < \infty) = 1$  and  $\mathbb{E}_\mu[N_i] < \infty$  for every initial distribution  $\mu$ , and  $\sum_{n=1}^{\infty} p_{ji}^n < \infty$  for every  $j \in S$ .*
- (4) *Let  $i$  be recurrent and  $i \rightarrow j$ . Then  $\mathbb{P}_i(N_j = \infty) = 1$ ,  $j$  is recurrent, and  $j \rightarrow i$ .*
- (5) *If  $S$  is finite, there is at least one recurrent state.*

*Proof.* Recall that for every  $n$ ,  $\{N_i \geq n\} = \{T_i^n < \infty\}$ . Also note that by (5.10),

$$\mathbb{E}_j[N_i] = \sum_{n=1}^{\infty} \mathbb{E}_j[\mathbb{1}_{\{X_n=i\}}] = \sum_{n=1}^{\infty} p_{ji}^n \quad (6.7)$$

*Item (1):* If  $\mathbb{P}_\mu(T_i < \infty) = 0$  then  $\mathbb{E}_\mu[N_i] = 0$  and the result follows.

If  $i$  is recurrent then by Proposition 6.4,  $\mathbb{P}_i(N_i \geq n) = 1$  for every  $n$ . That is,  $\mathbb{P}_i(N_i = \infty) = 1$  and hence  $\mathbb{E}_i[N_i] = \infty$ . If additionally  $\mathbb{P}_\mu(T_i < \infty) > 0$  then by (6.4)

$$\infty = \mathbb{E}_\mu[N_i] = \frac{\mathbb{P}_\mu(T_i < \infty)}{1 - \mathbb{P}_i(T_i < \infty)},$$

which is the desired result.

Finally we consider the case  $\mathbb{P}_\mu(T_i < \infty) > 0$  and  $\mathbb{P}_i(T_i < \infty) < 1$  (in particular  $i$  is transient). By Proposition 6.4,

$$\begin{aligned}\mathbb{E}_\mu[N_i] &= \sum_{n=1}^{\infty} \mathbb{P}_\mu(N_i \geq n) \\ &= \sum_{n=1}^{\infty} \mathbb{P}_\mu(T_i < \infty) [\mathbb{P}_i(T_i < \infty)]^{n-1} \\ &= \frac{\mathbb{P}_\mu(T_i < \infty)}{1 - \mathbb{P}_i(T_i < \infty)},\end{aligned}$$

which concludes the proof.

*Item (2):* We saw above that if  $i$  is recurrent then  $\mathbb{P}_i(N_i = \infty) = 1$ . Clearly,  $\mathbb{P}_i(N_i = \infty) = 1$  implies  $\mathbb{E}_i[N_i] = \infty$ . Thus we have "(a) implies (b)" and "(b) implies (c)". By (6.7) with  $j = i$ , (c) and (d) are equivalent. Finally, if (c) is satisfied then by (1) with  $\mu = \delta_i$  it follows that (a) is satisfied as well.

*Item (3):* If  $i$  is transient then by (1),  $\mathbb{E}_\mu[N_i] < \infty$  for every initial distribution  $\mu$ ; with  $\mu = \delta_j$  it follows from (6.7) that  $\sum_{n=1}^{\infty} p_{ji}^n < \infty$  for every  $j$ . Obviously,  $\mathbb{E}_\mu[N_i] < \infty$  implies  $\mathbb{P}_\mu(N_i < \infty) = 1$ .

*Item (4):* Let  $i$  be recurrent and assume  $i \rightarrow j$ . By (2),  $T_i^n$  is  $\mathbb{P}_i$ -a.s. finite for every  $n \in \mathbb{N}$  which means that

$$\mathbb{P}_i(T_j < \infty) = \mathbb{P}_i\left(\bigcup_{n=1}^{\infty} B_n\right), \quad \text{where } B_n = \{\text{there is a } k \text{ in } (T_i^{n-1}, T_i^n] : X_k = j\}. \quad (6.8)$$

By the strong Markov property,  $B_1, B_2, \dots$  are independent under  $\mathbb{P}_i$  with  $\mathbb{P}_i(B_n) = \mathbb{P}_i(B_1)$  for  $n \in \mathbb{N}$ . Since  $\mathbb{P}_i(T_j < \infty) > 0$ , it follows from (6.8) that  $\mathbb{P}_i(B_n) > 0$  for some and hence all  $n \in \mathbb{N}$ . Since  $B_1, B_2, \dots$  are independent and have the same (positive) probability, the Borel-Cantelli Lemma (see, e.g., Thorbjørnsen (2014)) implies  $\mathbb{P}_i(\limsup B_n) = 1$ . That is, with  $\mathbb{P}_i$ -probability one, there are infinitely many  $n$  such that  $j$  is visited between the two time points  $T_i^{n-1}$  and  $T_i^n$ . In particular  $\mathbb{P}_i(N_j = \infty) = 1$ . From (3) it follows that  $j$  is not transient and hence it must be recurrent. Finally, note that  $j \rightarrow i$  since otherwise the fact that  $i \rightarrow j$  would imply that  $i$  is transient by the criterion in Remark 6.6(2).

*Item (5):* Assume  $S$  is finite and fix some  $i \in S$ . If all states  $j \in S$  are transient then  $\sum_{n=1}^{\infty} p_{ij}^n < \infty$  for all  $j$  by (3). But then since  $S$  finite and  $P^n$  is a transition matrix (and hence all row sums are 1),

$$\infty > \sum_{j \in S} \sum_{n=1}^{\infty} p_{ij}^n = \sum_{n=1}^{\infty} \sum_{j \in S} p_{ij}^n = \sum_{n=1}^{\infty} 1 = \infty,$$

which is a contradiction. □

**Definition 6.9.** Let  $P$  be a transition matrix.

- (1) Call  $P$  **recurrent** if it is irreducible and there exists a recurrent state.
- (2) Call  $P$  **transient** if it is irreducible and there exists a transient state.

If  $P$  is irreducible, it is either recurrent or transient. The next result shows that these two cases are mutually exclusive and that we are in the recurrent case when  $S$  is finite.

**Corollary 6.10.** Let  $P$  be irreducible.

- (1) The transition matrix  $P$  is recurrent if and only if every state is recurrent.
- (2) The transition matrix  $P$  is transient if and only if every state is transient.
- (3) The transition matrix  $P$  is either recurrent or transient (not both).
- (4) If  $S$  is finite,  $P$  is recurrent.

*Proof.* If there is a recurrent state then by Theorem 6.8(4) all states are recurrent. Conversely, if some state is transient, there cannot exist a recurrent state by the same result. Thus we have (1) and (2). These results show that  $P$  cannot be both transient and recurrent. Hence we also have (3).

If  $S$  is finite, there is at least one recurrent state by Theorem 6.8(5). Hence  $P$  is recurrent.  $\square$

**Corollary 6.11.** Let  $P$  be irreducible.

- (1) If  $P$  is recurrent then for every  $i \in S$  and initial distribution  $\mu$ ,  $\mathbb{P}_\mu(N_i = \infty) = 1$ .  
That is, no matter the initial distribution, every state is visited infinitely often.
- (2) If  $P$  is transient then for any  $i \in S$  and initial distribution  $\mu$ ,  $\mathbb{P}_\mu(N_i < \infty) = 1$ .  
That is, no matter the initial distribution, any state is only visited finitely often with probability 1.

*Proof.* If  $P$  is transient then each state is only visited finitely many times by Theorem 6.8(3).

If  $P$  is recurrent then  $\mathbb{P}_\mu(N_i = \infty) = 1$  for every  $i$  and  $\mu$ . Indeed, by (4.28),

$$\mathbb{P}_\mu = \int \mathbb{P}_j \mu(dj) = \sum_{j \in S} \mathbb{P}_j \mu(\{j\}).$$

Hence,

$$\mathbb{P}_\mu(N_i = \infty) = \sum_{j \in S} \mathbb{P}_j(N_i = \infty) \mu(\{j\}) = \sum_{j \in S} 1 \mu(\{j\}) = 1$$

by Theorem 6.8(4) since every state  $j$  is recurrent by Corollary 6.10.  $\square$

Finally, let us note that when  $S$  is finite, it is actually often easy to classify states as either transient or recurrent by means of the following.

**Proposition 6.12.** *Let  $S$  be finite,  $P = \{p_{ij}\}_{i,j \in S}$  be a transition matrix and define  $T$  as*

$$T = \{i \in S : \text{there is a } j \in S \text{ such that } i \rightarrow j \text{ and } j \not\rightarrow i\}.$$

*Then  $T$  is precisely the set of transient states.*

See also Theorem 5.11 in Pedersen (2014) for a slightly more general result.

*Proof.* By Remark 6.6(2) all states in  $T$  are transient. Let  $k \in S \setminus T$  and let us show that  $k$  is recurrent. Put

$$[k] = \{i \in S \mid k \rightarrow i\}.$$

Note that

1. if  $i \in [k]$  then  $i \rightarrow k$ . (Otherwise we would have  $k \in T$ ).
2. if  $i, j \in [k]$  then  $i \rightarrow j$ . Indeed,  $i \rightarrow k$  and  $k \rightarrow j$  implies  $i \rightarrow j$ .
3.  $k \in [k]$ . Indeed, take  $i \in [k]$ . Then  $k \rightarrow i$  and  $i \rightarrow k$  which implies  $k \rightarrow k$ .
4. if  $i \in [k]$  and  $p_{ij} > 0$  for some  $j \in S$  then  $j \in [k]$ . Indeed, we have  $k \rightarrow i$  and  $i \rightarrow j$  and hence  $k \rightarrow j$ .

Thus,  $\{p_{ij}\}_{i,j \in [k]}$  is an irreducible transition matrix on a finite state space. By (1) and (4) in Corollary 6.10 all states in  $[k]$  (and in particular  $k$ ) are recurrent.  $\square$

## 6.2 Invariant measures and stationary distributions

Since  $S$  is countable, we may and do represent a measure on  $S$  by its point masses.

**Definition 6.13 (Measures on  $S$ ).**

- (1) A **measure** on  $S$  is a family  $\lambda = (\lambda_i)_{i \in S}$  where  $\lambda_i \in [0, \infty]$  for all  $i \in S$ .
- (2) A measure  $\lambda = (\lambda_i)_{i \in S}$  is **finite** if  $\sum_{i \in S} \lambda_i < \infty$ , and it is **non-zero** if  $\lambda_i > 0$  for some  $i$ .
- (3) A **probability measure** on  $S$  is a measure  $\pi = (\pi_i)_{i \in S}$  satisfying  $\sum_{i \in S} \pi_i = 1$ .

In most cases we consider measures with finite entries, i.e.  $\lambda_i < \infty$  for all  $i$ . A finite measure has finite entries but the converse only applies when  $S$  is finite.

**Definition 6.14.** A measure  $\lambda = (\lambda_i)_{i \in S}$  is **invariant** (for  $P$ ) if it is non-zero, has finite entries, and  $\lambda P = \lambda$ . (Here  $\lambda$  is treated as a row vector). Written out, the last condition means that  $\lambda_i = \sum_{j \in S} \lambda_j p_{ji}$  for every  $i \in S$ .

A probability measure  $\pi = (\pi_i)_{i \in S}$  on  $S$  is called a **stationary distribution** if it is invariant.

**Remark 6.15.** Let  $\lambda = (\lambda_i)_{i \in S}$  be a measure that is invariant for  $P$ . Then

- (1)  $c\lambda$  is invariant for  $P$  for every  $c > 0$ ;
- (2)  $\lambda$  is invariant for  $P^n$  for every  $n \in \mathbb{N}_0$ .

These results are proved in Exercise 64. ■

**Lemma 6.16.** Let  $\pi = (\pi_i)_{i \in S}$  be a probability distribution on  $S$ . Then  $(X_n)_{n \geq 0}$  is stationary under  $\mathbb{P}_\pi$  if and only if  $\pi$  is a stationary distribution.

*Proof.* First assume that  $\pi$  is a stationary distribution. Let  $n, k \in \mathbb{N}_0$  and  $i_0, \dots, i_k \in S$ . By the Markov property as formulated in Proposition 4.30,

$$\begin{aligned} \mathbb{P}_\pi(X_n = i_0, \dots, X_{n+k} = i_k) &= \mathbb{P}_\pi(X_n = i_0, \dots, X_{n+k} = i_k \mid X_n = i_0) \mathbb{P}_\pi(X_n = i_0) \\ &= \mathbb{P}_{i_0}(X_0 = i_0, \dots, X_k = i_k) \mathbb{P}_\pi(X_n = i_0) \\ &= \mathbb{P}_\pi(X_0 = i_0, \dots, X_k = i_k \mid X_0 = i_0) \mathbb{P}_\pi(X_n = i_0). \end{aligned}$$

Moreover, using the Markov property once again, that  $\pi$  is invariant for  $P^n$ , and (5.10), it follows that

$$\begin{aligned} \mathbb{P}_\pi(X_n = i_0) &= \sum_{j \in S} \mathbb{P}_\pi(X_n = i_0 \mid X_0 = j) \mathbb{P}_\pi(X_0 = j) \\ &= \sum_{j \in S} \mathbb{P}_j(X_n = i_0) \pi_j = \sum_{j \in S} \pi_j p_{ji_0}^n = \pi_{i_0} = \mathbb{P}_\pi(X_0 = i_0). \end{aligned}$$

Hence,

$$\begin{aligned} \mathbb{P}_\pi(X_n = i_0, \dots, X_{n+k} = i_k) &= \mathbb{P}_\pi(X_0 = i_0, \dots, X_k = i_k \mid X_0 = i_0) \mathbb{P}_\pi(X_n = i_0) \\ &= \mathbb{P}_\pi(X_0 = i_0, \dots, X_k = i_k \mid X_0 = i_0) \mathbb{P}_\pi(X_0 = i_0) \\ &= \mathbb{P}_\pi(X_0 = i_0, \dots, X_k = i_k). \end{aligned}$$

Thus  $(X_n)_{n \geq 0}$  is stationary under  $\mathbb{P}_\pi$ .

Conversely, assume that  $(X_n)_{n \geq 0}$  is stationary under  $\mathbb{P}_\pi$ . Then for every  $i \in S$ ,

$$\pi_i = \mathbb{P}_\pi(X_0 = i) = \mathbb{P}_\pi(X_1 = i) = \sum_{j \in S} \mathbb{P}_\pi(X_0 = j, X_1 = i) = \sum_{j \in S} \pi_j p_{ji},$$

where we used (4.35) in the last step. Hence  $\pi$  is a stationary distribution. □

The following result ensures the existence of an invariant measure and is shown in Norris (1998, Theorems 1.7.5).

**Theorem 6.17.** *Let  $P$  be recurrent (that is,  $P$  is irreducible and every state is recurrent). Let  $k \in S$ . Put*

$$\gamma_i^k = \mathbb{E}_k \left[ \sum_{n=0}^{T_k-1} \mathbb{1}_{\{X_n=i\}} \right], \quad i \in S. \quad (6.9)$$

*Then  $0 < \gamma_i^k < \infty$  for every  $i \in S$ ,  $\gamma_k^k = 1$ , and  $\gamma^k = (\gamma_i^k)_{i \in S}$  is invariant for  $P$ .*

**Remark 6.18.** In his proof, Norris actually shows that if  $P$  is recurrent then  $\gamma_i^k$  is also given as

$$\gamma_i^k = \mathbb{E}_k \left[ \sum_{n=1}^{T_k} \mathbb{1}_{\{X_n=i\}} \right], \quad i \in S.$$

It is also worth noticing that

$$\sum_{i \in S} \gamma_i^k = \mathbb{E}_k \left[ \sum_{n=0}^{T_k-1} \sum_{i \in S} \mathbb{1}_{\{X_n=i\}} \right] = \mathbb{E}_k \left[ \sum_{n=0}^{T_k-1} 1 \right] = \mathbb{E}_k[T_k]. \quad (6.10)$$

Thus,  $\gamma^k$  is a finite measure if and only if  $\mathbb{E}_k[T_k] < \infty$ . ■

Item (3) in the next result, which is a consequence of Norris (1998, Theorems 1.7.5-1.7.6), shows that, up to scaling,  $\gamma^k$  is the unique invariant measure when  $P$  is recurrent.

**Corollary 6.19.** *Let  $P$  be an irreducible transition matrix.*

- (1) *If  $\lambda = (\lambda_i)_{i \in S}$  is invariant for  $P$  then  $\lambda_i > 0$  for all  $i \in S$ .*
- (2) *Assume  $\lambda = (\lambda_i)_{i \in S}$  is a non-zero measure such that  $\lambda = \lambda P$ . Further assume that there exists an  $i \in S$  with  $\lambda_i < \infty$ . Then  $\lambda_j < \infty$  for all  $j \in S$ . In particular,  $\lambda$  is invariant for  $P$ .<sup>1</sup>*
- (3) *Let  $P$  be recurrent,  $k \in S$  and  $\lambda = (\lambda_i)_{i \in S}$  be invariant for  $P$ . Then  $\lambda = \lambda_k \cdot \gamma^k$ .*

*Proof.* Item (1): Let  $j \in S$  satisfy  $\lambda_j > 0$ . (Such a  $j$  exists since  $\lambda$  is a non-zero measure). Let  $i \in S$ . Since  $P$  is irreducible, (b) in Remark 6.6(1) shows that there exists an  $n \in \mathbb{N}$  such that  $p_{ji}^n > 0$ . Fix such an  $n$ . Since  $\lambda$  is invariant for  $P$ , it is also invariant for  $P^n$ , so in particular

$$\lambda_i = \sum_{k \in S} \lambda_k p_{ki}^n \geq \lambda_j p_{ji}^n > 0.$$

---

<sup>1</sup>The point here is that in order for  $\lambda$  to be invariant, all entries have to be finite.

*Item (2):* Let  $j \in S$ . By irreducibility of  $P$  there exists an  $n \in \mathbb{N}$  such that  $p_{ji}^n > 0$ . Since  $\lambda = \lambda P^n$ , it follows that

$$\lambda_j p_{ji}^n \leq \sum_{k \in S} \lambda_k p_{ki}^n = \lambda_i < \infty$$

and hence  $\lambda_j < \infty$  since  $p_{ji}^n > 0$ .

*Item (3):* Since  $\lambda_k > 0$  by (1),  $\mu$  defined as  $\mu = \lambda/\lambda_k$  is well-defined and invariant for  $P$ . Since  $\mu_k = 1$ , it follows from Norris (1998, Theorem 1.7.6) that  $\mu = \gamma^k$ . Hence  $\lambda = \lambda_k \gamma^k$ .  $\square$

**Definition 6.20.** Let  $P$  be a transition matrix and let  $i \in S$ .

- (1) State  $i$  is **positive recurrent** if  $\mathbb{E}_i[T_i] < \infty$ .
- (2) State  $i$  is **0-recurrent** if  $i$  is recurrent and  $\mathbb{E}_i[T_i] = \infty$ .

**Remark 6.21.** Let  $i \in S$ . Obviously  $\mathbb{E}_i[T_i] < \infty$  implies  $\mathbb{P}_i(T_i < \infty) = 1$ . Thus, if  $i$  is positive recurrent, then it is also recurrent. Conversely, if  $i$  is recurrent, then it is either positive recurrent (namely if  $\mathbb{E}_i[T_i] < \infty$ ) or 0-recurrent (namely if  $\mathbb{E}_i[T_i] = \infty$ ).  $\blacksquare$

**Definition 6.22.** Let  $P$  be a transition matrix.

- (1) Call  $P$  **positive recurrent** if it is irreducible and there exists a positive recurrent state.
- (2) Call  $P$  **0-recurrent** if it is irreducible and there exists a 0-recurrent state.

If  $P$  is recurrent (cf. Definition 6.9), it is either positive recurrent or 0-recurrent by Remark 6.21.

**Theorem 6.23.** Let  $P$  be a transition matrix.

- (1) Let  $P$  be positive recurrent. Then
  - (a) Every invariant measure is finite.
  - (b) All states are positive recurrent.
  - (c) There exists a unique stationary distribution  $\pi = (\pi_j)_{j \in S}$ . Moreover,  $\pi$  has the following representations

$$\pi_j = 1/\mathbb{E}_j[T_j], \quad \text{for } j \in S \tag{6.11}$$

and

$$\pi = \frac{\gamma^j}{\mathbb{E}_j[T_j]} \quad \text{for arbitrary } j \in S. \tag{6.12}$$

- (2) If  $P$  is 0-recurrent then all states are 0-recurrent and there is no stationary distribution.
- (3) If  $P$  is transient then all states are transient and there is no stationary distribution.
- (4) If  $P$  is irreducible then it is either transient, positive recurrent, or 0-recurrent, and these three cases are mutually exclusive.
- (5) If  $S$  is finite and  $P$  is irreducible then  $P$  is positive recurrent.

*Proof.* Item (1): Since  $P$  is positive recurrent, there exists a positive recurrent state, which we denote by  $k$  and fix in the following. By (6.10),  $\sum_{i \in S} \gamma_i^k = \mathbb{E}_k[T_k] < \infty$  and thus  $\gamma^k$  is a finite measure; moreover,  $\gamma^k$  is invariant by Theorem 6.17 since  $P$  is recurrent.

To prove (a), let  $\lambda = (\lambda_i)_{i \in S}$  denote an invariant measure. By Corollary 6.19(3)  $\lambda = \lambda_k \cdot \gamma^k$ , so since  $\gamma^k$  is a finite measure,  $\lambda$  is also finite.

To prove (b), let  $i \in S$ . Since  $P$  is recurrent,  $\gamma^i$  is invariant by Theorem 6.17, so by (a),  $\gamma^i$  is a finite and therefore, by (6.10),  $\mathbb{E}_i[T_i]$  is finite. Hence  $i$  is positive recurrent.

(c): Let  $j \in S$  be fixed but arbitrary. Since  $\gamma^j$  is an invariant measure with  $\sum_{i \in S} \gamma_i^j = \mathbb{E}_j[T_j] < \infty$ ,  $\pi$  given in (6.12) is a stationary distribution. Now let  $\pi$  be an arbitrary stationary distribution. By Corollary 6.19(3),  $\pi = \pi_j \cdot \gamma^j$  and hence, by (6.10),

$$1 = \sum_{i \in S} \pi_i = \pi_j \sum_{i \in S} \gamma_i^j = \pi_j \mathbb{E}_j[T_j].$$

Since  $j$  is arbitrary, it follows that any stationary distribution satisfies (6.11). In particular, the stationary distribution is unique.

*Item (2):* Since  $P$  is 0-recurrent, there exists at least one 0-recurrent state. Since  $P$  is recurrent, all states are recurrent by Corollary 6.10. If one state were positive recurrent, then  $P$  would be positive recurrent (by definition), and (1) would imply that all states are positive recurrent, contradicting that there is a 0-recurrent state. Hence all states are 0-recurrent. Assume there exists a stationary distribution  $\pi = (\pi_j)_{j \in S}$ . Then for an arbitrary  $j \in S$ ,  $\pi = \pi_j \cdot \gamma^j$  by Corollary 6.19(3). Since  $\pi$  is a finite measure, the same applies to  $\gamma^j$ ; from (6.10) it follows that  $\mathbb{E}_j[T_j]$  is finite and hence that  $j$  is positive recurrent, contradicting that all states are 0-recurrent.

*Item (3):* By Corollary 6.10 all states are transient if  $P$  is transient. By Exercise 63 there does not exist a stationary distribution in this case.

*Item (4):* If  $P$  is irreducible, Corollary 6.10 shows that  $P$  is either recurrent or transient and that these cases are mutually exclusive. In the recurrent case  $P$  is,

by definition, either positive recurrent or 0-recurrent, and these two cases are mutually exclusive by (1) and (2).

*Item (5):* If  $S$  is finite and  $P$  is irreducible, it follows from Corollary 6.10 that  $P$  is recurrent. Hence by Theorem 6.17,  $\gamma^j$  is an invariant measure for  $j \in S$ . Since  $S$  is finite,  $\gamma^j$  is a finite measure. Hence  $\gamma^j/\mathbb{E}_j[T_j]$  is a stationary distribution, which by (2) implies that  $P$  is not 0-recurrent; hence  $P$  is positive recurrent.  $\square$

**Remark 6.24.** (1) The case of a 0-recurrent transition matrix is interesting: By Corollary 6.11 every state  $i$  is visited infinitely often no matter the initial distribution. On the other hand, as the previous result shows, the expected return time to any state is infinite!

(2) If  $P$  is irreducible and there exists a stationary distribution then, by (2) and (3) above,  $P$  is neither transient nor 0-recurrent. Therefore  $P$  is positive recurrent and the stationary distribution is unique.

(3) If  $P$  is positive recurrent it is sometimes possible to find the stationary distribution simply by solving  $\pi = \pi P$ . See e.g. Durrett (2016, Example 1.18–1.19) or Pedersen (2014) for the case of a finite state space, and Durrett (2016, Example 1.26) for the case of a birth and death chain.

In some cases one can find the stationary distribution by solving the detailed balance equation as defined below.  $\blacksquare$

**Definition 6.25.** Let  $\pi = (\pi_i)_{i \in S}$  denote a probability measure. We say that  $\pi$  satisfies the *detailed balance equation* if

$$\pi_i p_{ij} = \pi_j p_{ji} \quad (6.13)$$

for all  $i, j \in S$ .

**Proposition 6.26.** Let  $\pi$  be a probability measure on  $S$ . Then  $(X_n)_{n \geq 0}$  is reversible under  $\mathbb{P}_\pi$  if and only if  $\pi$  satisfies the detailed balance equation. In that case,  $\pi$  is a stationary distribution.

In this context, reversibility means that

$$\mathbb{P}_\pi(X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}_\pi(X_0 = i_n, \dots, X_n = i_0) \quad (6.14)$$

for all  $n \in \mathbb{N}$  and  $i_0, \dots, i_n \in S$ .

*Proof.* First assume that  $\pi$  satisfies the detailed balance equation. In Exercise 64 it is shown that then  $\pi$  is a stationary distribution. To verify (6.14), use (4.35) and

(6.13) repeatedly to get

$$\begin{aligned}
 \mathbb{P}_\pi(X_0 = i_0, \dots, X_n = i_n) &= \pi_{i_0} p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n} \\
 &= p_{i_1 i_0} \pi_{i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n} \\
 &= p_{i_1 i_0} p_{i_2 i_1} \pi_{i_2} \cdots p_{i_{n-1} i_n} \\
 &= \dots \\
 &= p_{i_1 i_0} p_{i_2 i_1} \cdots p_{i_n i_{n-1}} \pi_{i_n} \\
 &= \mathbb{P}_\pi(X_0 = i_n, \dots, X_n = i_0).
 \end{aligned}$$

Hence  $(X_n)_{n \geq 0}$  is reversible under  $\mathbb{P}_\pi$ .

Conversely, assume that  $(X_n)_{n \geq 0}$  is reversible under  $\mathbb{P}_\pi$ . Taking  $n = 1$  in (6.14), we obtain for all  $i, j \in S$  that

$$\mathbb{P}_\pi(X_0 = i, X_1 = j) = \mathbb{P}_\pi(X_0 = j, X_1 = i).$$

Using (4.35), this becomes

$$\pi_i p_{ij} = \pi_j p_{ji},$$

so  $\pi$  satisfies the detailed balance equation.  $\square$

**Theorem 6.27 (The Law of Large Numbers).**

Let  $P$  be positive recurrent and  $\pi = (\pi_i)_{i \in S}$  denote the unique stationary distribution. For any function  $f : S \rightarrow \mathbb{R}$  satisfying  $\sum_{i \in S} |f(i)| \pi_i < \infty$  it holds that

$$\frac{1}{n} \sum_{m=1}^n f(X_m) \rightarrow \sum_{i \in S} f(i) \pi_i \quad \mathbb{P}_\mu\text{-a.s.} \quad (6.15)$$

no matter the initial distribution  $\mu$ .

*Proof.* We prove a particular case, namely that  $\frac{1}{n} N_i^n \rightarrow \pi_i$   $\mathbb{P}_\mu$ -a.s. for every  $i \in S$ , where

$$N_i^n = \#\{0 < m \leq n : X_m = i\} \quad (6.16)$$

counts the number of visits to state  $i$  up to time  $n$ . (This result implies (6.15) when  $f$  is bounded, in particular when  $S$  is finite.)

To prove this, let us first note that for  $k \in \mathbb{N}$ ,

$$\frac{1}{k} T_i^k = \frac{1}{k} T_i^1 + \frac{k-1}{k} \frac{1}{k-1} \sum_{j=2}^k (T_i^j - T_i^{j-1}). \quad (6.17)$$

Since, by Proposition 6.4,  $(T_i^j - T_i^{j-1})_{j \geq 2}$  are i.i.d. under  $\mathbb{P}_\mu$  with common distribution equal to the distribution of  $T_i$  under  $\mathbb{P}_i$ , the usual Law of Large Numbers implies that

$$\frac{1}{k} T_i^k \rightarrow_{k \rightarrow \infty} \mathbb{E}_i[T_i] \quad \mathbb{P}_\mu\text{-a.s.}$$

where  $\mathbb{E}_i[T_i]$  is finite since  $P$  is positive recurrent. Since  $N_i^n \rightarrow \infty$   $\mathbb{P}_\mu$ -a.s. it follows that

$$\frac{1}{N_i^n} T_i^{N_i^n}, \frac{1}{N_i^n + 1} T_i^{N_i^n + 1} \rightarrow_{n \rightarrow \infty} \mathbb{E}_i[T_i] \quad \text{for } n \rightarrow \infty \text{ } \mathbb{P}_\mu\text{-a.s.} \quad (6.18)$$

Finally, note that for every  $n \in \mathbb{N}$ ,  $T_i^{N_i^n} \leq n < T_i^{N_i^n + 1}$  and hence

$$\frac{T_i^{N_i^n}}{N_i^n} \leq \frac{n}{N_i^n} \leq \frac{T_i^{N_i^n + 1}}{N_i^n + 1} \frac{N_i^n + 1}{N_i^n}.$$

It follows from (6.18) that  $\frac{n}{N_i^n} \rightarrow_{n \rightarrow \infty} \mathbb{E}_i[T_i]$   $\mathbb{P}_\mu$ -a.s. which is the desired result since  $\pi_i = 1/\mathbb{E}_i[T_i]$ .  $\square$

### 6.3 Random walks on $\mathbb{Z}^d$

In this section, the state space is  $S = \mathbb{Z}^d$  for some  $d \in \mathbb{N}$ .

**Definition 6.28.** A *random walk (on  $\mathbb{Z}^d$ )* is a stochastic process  $(Y_n)_{n \geq 0}$  given by

$$Y_n = Y_0 + \sum_{m=1}^n \epsilon_m, \quad \text{for } n \in \mathbb{N}, \quad (6.19)$$

where  $Y_0, \epsilon_1, \epsilon_2, \dots$  are independent and  $\mathbb{Z}^d$ -valued, and  $(\epsilon_m)_{m \geq 1}$  are i.i.d.

Let  $e_1 = (1, 0, \dots, 0), \dots, e_d = (0, \dots, 0, 1)$  denote the  $d$  standard unit vectors in  $\mathbb{R}^d$ . If  $\mathbb{P}(\epsilon_n = e_j) = \mathbb{P}(\epsilon_n = -e_j) = \frac{1}{2d}$  for  $n \in \mathbb{N}$  and  $j = 1, \dots, d$ , then  $(Y_n)_{n \geq 0}$  is called a **simple symmetric random walk (on  $\mathbb{Z}^d$ )**.

Example 4.16 with  $p = 1/2$  corresponds to the simple symmetric random walk on  $\mathbb{Z}$ .

The representation (6.19) is equivalent to

$$Y_{n+1} = Y_n + \epsilon_{n+1}, \quad \text{for } n \in \mathbb{N}_0, \quad (6.20)$$

which by Example 4.9 shows that a random walk on  $\mathbb{Z}^d$  is a Markov chain with transition matrix  $P = \{p_{ij}\}_{i,j \in \mathbb{Z}^d}$  given by

$$p_{ij} = \mu_{j-i}, \quad \text{for } i, j \in \mathbb{Z}^d, \quad (6.21)$$

where  $(\mu_i)_{i \in \mathbb{Z}^d}$  denotes the common distribution of  $(\epsilon_n)_{n \geq 1}$ ; that is,  $\mu_i = \mathbb{P}(\epsilon_n = i)$  for  $n \in \mathbb{N}$  and  $i \in \mathbb{Z}^d$ .

We are interested in when a random walk is (irreducible and) transient or recurrent. To begin with, note that the counting measure (i.e. the measure  $\lambda = (\lambda_i)_{i \in \mathbb{Z}^d}$  with  $\lambda_i = 1$  for all  $i \in \mathbb{Z}^d$ ) is invariant for  $P$ . By Theorem 6.23, this implies that a random walk cannot be positive recurrent.

**Proposition 6.29.** *The simple random walk on  $\mathbb{Z}$  in Example 4.16 is irreducible; it is transient when  $p \neq 1/2$  and 0-recurrent when  $p = 1/2$ .*

The case  $p = 1/2$  is rather peculiar: Since the chain is irreducible and the states are recurrent, all states will be visited infinitely often; but since the states are 0-recurrent, the expected return times are  $\infty$  cf. also the discussion above. (And more generally the expected time to get from one given state to another is  $\infty$ ). To get some intuition about this, see Pedersen (2014, Eksempel 2.4).

*Proof.* Using e.g. that there is a path (cf. Remark 6.6(1)) between any two states, it is easy to see that the chain is irreducible.

Let  $(Y_n)_{n \geq 0}$  denote a simple random walk on  $\mathbb{Z}$ . Hence, in (6.19)  $(\epsilon_n)_{n \geq 1}$  are i.i.d. with  $\mathbb{P}(\epsilon_n = 1) = 1 - \mathbb{P}(\epsilon_n = -1) = p \in (0, 1)$ . Since  $\mathbb{E}[\epsilon_n] = 2p - 1$ , it follows from the Law of Large Numbers that

$$Y_n \rightarrow \begin{cases} \infty & \text{when } p > 1/2 \\ -\infty & \text{when } p < 1/2 \end{cases}$$

a.s., which implies that any state will only be visited finitely many times a.s. when  $p \neq 1/2$ . Hence the chain is transient in this case.

For  $n$  odd,  $p_{ii}^n = 0$  for all  $i \in \mathbb{Z}$  since a random walk can only return to the initial position in an even number of steps. For  $n \in \mathbb{N}$  and  $i \in \mathbb{Z}$ ,

$$p_{ii}^{2n} = \binom{2n}{n} p^n (1-p)^n. \quad (6.22)$$

since, when starting in  $i$ , the random walk is in  $i$  after  $2n$  steps if and only if it takes precisely  $n$  steps to the right and precisely  $n$  steps to the left among the first  $2n$  steps. (That is, among  $\epsilon_1, \dots, \epsilon_{2n}$  precisely  $n$  are  $+1$  and precisely  $n$  are  $-1$ ). By the approximation (based on Stirling's formula)

$$\binom{2n}{n} \sim \frac{2^{2n}}{\sqrt{\pi n}}$$

it follows that

$$p_{ii}^{2n} \sim \frac{1}{\sqrt{\pi}} \frac{(4p(1-p))^n}{\sqrt{n}}. \quad (6.23)$$

Assume  $p \neq 1/2$ . Since  $4p(1-p) < 1$ , it follows that  $\sum_{n=1}^{\infty} p_{ii}^n < \infty$ , which by Theorem 6.8 gives a second proof of transience.

Assume  $p = 1/2$ . From (6.23) it follows that  $p_{ii}^{2n} \sim \frac{1}{\sqrt{\pi n}}$  which implies  $\sum_{n=1}^{\infty} p_{ii}^n = \infty$ . Hence, by Theorem 6.8 the random walk is recurrent. As noted above, a simple random walk cannot be positive recurrent; hence it is 0-recurrent. For another proof of 0-recurrence, see Pedersen (2014).  $\square$

**Proposition 6.30.** *A simple symmetric random walk on  $\mathbb{Z}^d$  is irreducible; it is recurrent when  $d = 1$  or  $d = 2$ , and it is transient when  $d \geq 3$ .*

*Proof.* It is easily seen that we have an irreducible chain. The case  $d = 1$  was settled above. Fix  $i \in \mathbb{Z}^d$ .

$d = 2$ : By Norris (1998, Example 1.6.2),  $p_{ii}^n = 0$  when  $n$  is odd, and

$$p_{ii}^{2n} = \left( \binom{2n}{n} \left( \frac{1}{2} \right)^{2n} \right)^2 \sim \frac{1}{\pi n}.$$

Thus, by Theorem 6.8 the random walk is recurrent.

$d = 3$ : By Norris (1998, Example 1.6.3),  $p_{ii}^n = 0$  when  $n$  is odd, and

$$p_{ii}^{2n} = \left( \frac{1}{6} \right)^{2n} \sum_{\substack{i,j,k \geq 0 \\ i+j+k=n}} \frac{(2n)!}{(i!j!k!)^2} \sim C n^{-3/2}$$

for some positive constant  $C$ . This implies transience by Theorem 6.8.

$d > 3$ : Durrett (2010) shows how to reduce the case  $d \geq 3$  to  $d = 3$ . We will not give details here.  $\square$

You can find much more about random walks, e.g. in Durrett (2010) and Kallenberg (2002). As an example of an important result, let me mention that if a random walk (on  $\mathbb{Z}^d$ ) has effective dimension at least three, then the states are transient. (The effective dimension is the dimension of the least linear subspace in  $\mathbb{R}^d$  that contains the support of the distribution of  $\epsilon_n$ ).

Let us conclude by giving an alternative characterization of random walks as processes with stationary independent increments. Our interest in such processes stems from that later we will analyze stationary independent increments in continuous time (that is,  $T = \mathbb{R}_+$ ), see Chapter 9. Consider a stochastic process  $(Y_n)_{n \in \mathbb{N}_0}$  with state space  $S = \mathbb{R}^d$  and  $Y_0 = 0$  a.s. This process is said to have *stationary independent increments* if

- (stationary increments)  $Y_n - Y_{n-1} \stackrel{d}{=} Y_1$  for all  $n \in \mathbb{N}$
- (independent increments) for all  $n \in \mathbb{N} \setminus \{1\}$

$Y_1, Y_2 - Y_1, \dots, Y_n - Y_{n-1}$  are independent.

**Proposition 6.31.** *The process  $(Y_n)_{n \in \mathbb{N}_0}$  has stationary independent increments if and only if it is a random walk.*

*Proof.* Assume  $(Y_n)_{n \in \mathbb{N}_0}$  has stationary independent increments. Put  $\epsilon_n = Y_n - Y_{n-1}$  for  $n \in \mathbb{N}$ . Then

$$Y_n \stackrel{\text{a.s.}}{=} Y_n - Y_0 = (Y_n - Y_{n-1}) + \cdots + (Y_1 - Y_0) = \sum_{m=1}^n \epsilon_m \quad \text{for } n \in \mathbb{N}_0.$$

Moreover,  $\epsilon_n = Y_n - Y_{n-1} \stackrel{d}{=} Y_1 \stackrel{\text{a.s.}}{=} \epsilon_1$  for all  $n \in \mathbb{N}$ , and

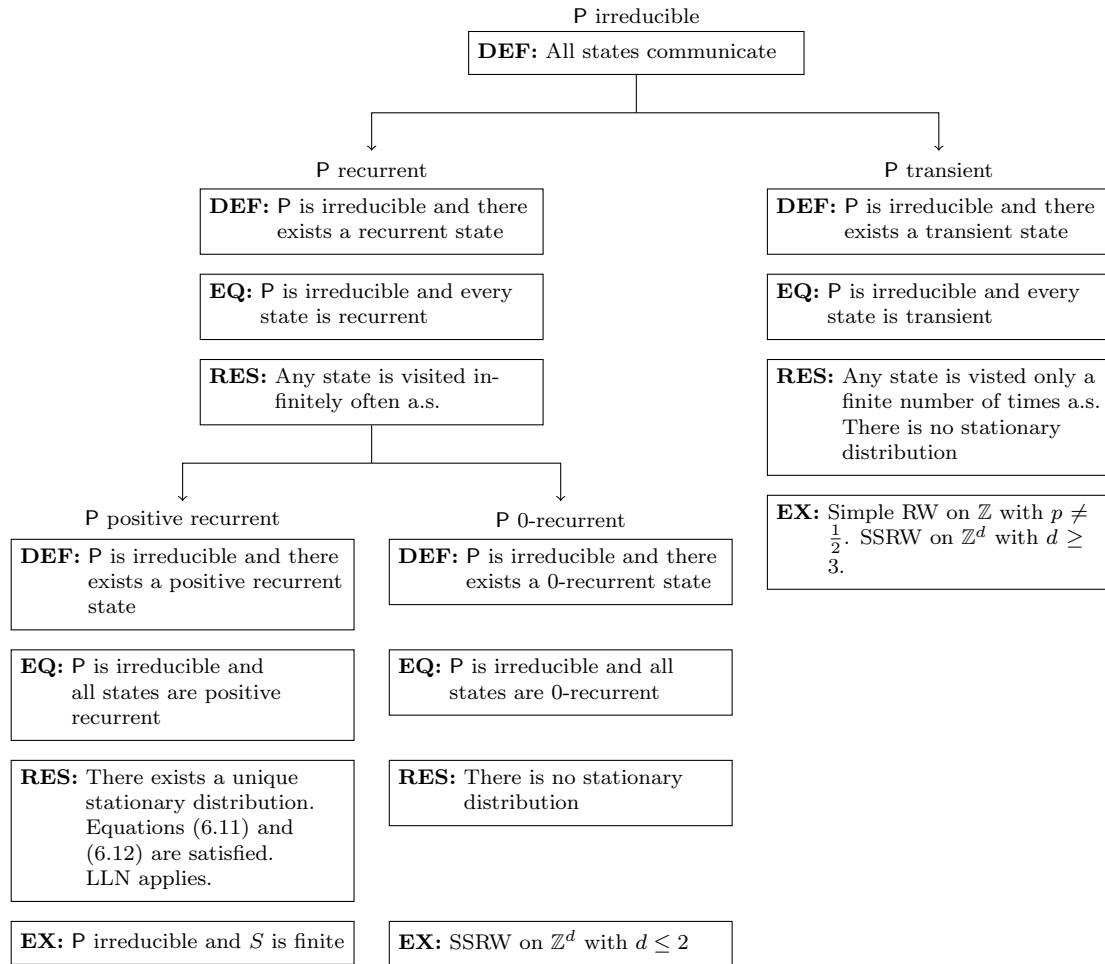
$$\epsilon_1 \stackrel{\text{a.s.}}{=} Y_1, \epsilon_2 = Y_2 - Y_1, \dots, \epsilon_n = Y_n - Y_{n-1} \text{ are independent}$$

for all  $n \in \mathbb{N} \setminus \{1\}$ . That is,  $(\epsilon_n)_{n \in \mathbb{N}}$  is i.i.d. and hence  $(Y_n)_{n \in \mathbb{N}_0}$  is a random walk.

The converse follows similarly. □

## 6.4 Concluding remarks

In this chapter, we have studied irreducible Markov chains in great detail, and Figure 6.1 gives an overview of the most important results.



**Figure 6.1:** An overview of the most important results for irreducible Markov chains. Important examples are mentioned as well. Abbreviations: DEF is the Definition, EQ is an equivalent characterization, RES are important results, and EX lists some examples. Finally, RW is short for Random Walk, SSRW is short for Simple Symmetric Random Walk, and LLN is the Law of Large Numbers.

# Markov processes

Recall that, to define *Markov chains* (i.e., stochastic processes possessing the Markov property where the index set is  $\mathbb{N}_0$ ), we need to specify a transition kernel  $Q(x, A)$ ,  $x \in S, A \in \mathcal{B}(S)$ , where  $Q(x, A)$  is interpreted as the probability of being in  $A$  “one step ahead” if the current state is  $x$ . As shown in Chapter 5,  $Q^n(x, A)$ , for  $n \in \mathbb{N}_0$  and  $x \in S$ , is the probability of being in  $A$  after  $n$  steps if the current state is  $x$ . In addition, we have *Chapman-Kolmogorov’s equation*:

$$Q^{n+m} = Q^n Q^m \quad \text{for } n, m \in \mathbb{N}_0. \quad (7.1)$$

In this chapter we study *Markov processes*, i.e., processes possessing the Markov property where the index set is  $\mathbb{R}_+$ . For  $t \geq 0$ ,  $P_t(x, A)$ ,  $x \in S, A \in \mathcal{B}(S)$ , will be used to model the probability of being in  $A$   $t$  time units ahead if the current state is  $x$ . As Chapman-Kolmogorov’s equation suggests, the transition kernels for different values of  $t$  must be closely related. This motivates the following.

**Definition 7.1.** A family  $(P_t)_{t \geq 0}$  of transition kernels is a **transition semigroup** if

- (1)  $P_0(x, A) = \delta_x(A)$  for all  $x \in S$  and  $A \in \mathcal{B}(S)$ ;
- (2)  $P_{t+s} = P_t P_s$  for all  $s, t \geq 0$ .

A *filtered probability space*  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$  is a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  equipped with a filtration  $(\mathcal{F}_t)_{t \geq 0}$ , i.e., an increasing family of  $\sigma$ -fields in  $\mathcal{F}$ . A stochastic process  $(Y_t)_{t \geq 0}$  defined on  $(\Omega, \mathcal{F}, \mathbb{P})$  is *adapted* if  $Y_t$  is  $\mathcal{F}_t$ -measurable for all  $t \geq 0$ . See also Appendix C.

**Definition 7.2.** Let  $(P_t)_{t \geq 0}$  be a transition semigroup and  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$  denote a filtered probability space.

An  $(\mathcal{F}_t)_{t \geq 0}$ -adapted  $S$ -valued stochastic process  $(Y_t)_{t \geq 0}$  is called a **Markov process with transition semigroup**  $(P_t)_{t \geq 0}$  **with respect to**  $(\mathcal{F}_t)_{t \geq 0}$  if

$$\mathbb{E}[f(Y_{t+s}) | \mathcal{F}_s] = P_t f(Y_s) \quad \text{for all } t, s \geq 0 \quad (7.2)$$

and every  $f : S \rightarrow \mathbb{R}$  bounded and measurable.

If  $\mathcal{F}_t = \mathcal{F}_t^Y$  for all  $t \geq 0$  above, then  $(Y_t)_{t \geq 0}$  is also simply called a Markov process with transition semigroup  $(P_t)_{t \geq 0}$ .

The probability measure  $\mu$  on  $(S, \mathcal{B}(S))$  defined as  $\mu(A) = \mathbb{P}(Y_0 \in A)$  for  $A \in \mathcal{B}(S)$  is **the initial distribution**.

Here, the natural filtration is defined just as in discrete time; that is,  $\mathcal{F}_t^Y = \sigma(Y_s | s \leq t)$  for  $t \geq 0$ .

Arguments as in Remark 4.5 show that (7.2) is equivalent to

$$\mathbb{P}(Y_{t+s} \in A | \mathcal{F}_s) = P_t(Y_s, A) \quad \text{for all } t, s \geq 0 \text{ and all } A \in \mathcal{B}(S). \quad (7.3)$$

Moreover, by arguments as in Theorem 4.21 it follows that  $(Y_t)_{t \geq 0}$  is a Markov process with transition semigroup  $(P_t)_{t \geq 0}$  and initial distribution  $\mu$  if and only if, for arbitrary  $n \in \mathbb{N}$ ,  $0 \leq t_1 < \dots < t_n$ , and  $B_1, \dots, B_n \in \mathcal{B}(S)$ ,

$$\begin{aligned} \mathbb{P}(Y_{t_1} \in B_1, \dots, Y_{t_n} \in B_n) \\ = \int_S \int_{B_1} \dots \int_{B_n} P_{t_n - t_{n-1}}(y_{n-1}, dy_n) \dots P_{t_1 - t_0}(y_0, dy_1) \mu(dy_0), \end{aligned} \quad (7.4)$$

where we put  $t_0 = 0$ . To make sure that such Markov processes exist, one has to rely on Kolmogorov's Consistency Theorem (cf. Theorem 3.5), and the next lemma ensures that this theorem is applicable.

**Lemma 7.3.** For  $n \in \mathbb{N}$  and  $0 \leq t_1 < \dots < t_n$  define a probability measure  $\nu(t_1, \dots, t_n)$  on  $(S^n, \mathcal{B}(S^n))$  as

$$\begin{aligned} \nu(t_1, \dots, t_n)(A) \\ = \int \int \dots \int \mathbb{1}_A(y_1, \dots, y_n) P_{t_n - t_{n-1}}(y_{n-1}, dy_n) \dots P_{t_1 - t_0}(y_0, dy_1) \mu(dy_0) \end{aligned} \quad (7.5)$$

for  $A \in \mathcal{B}(S^n)$ . Here we put  $t_0 = 0$ .

Then, for  $n \geq 2$ ,  $j = 1, \dots, n$  and  $B_1, \dots, B_{j-1}, B_{j+1}, \dots, B_n \in \mathcal{B}(S)$ ,

$$\begin{aligned} \nu(t_1, \dots, t_{j-1}, t_j, t_{j+1}, \dots, t_n)(B_1 \times \dots \times B_{j-1} \times S \times B_{j+1} \times \dots \times B_n) \\ = \nu(t_1, \dots, t_{j-1}, t_{j+1}, \dots, t_n)(B_1 \times \dots \times B_{j-1} \times B_{j+1} \times \dots \times B_n). \end{aligned} \quad (7.6)$$

*Proof.* The case  $j = n$  is left to the reader (see Exercise 81), so in the following assume  $j < n$ . By Definition 7.1(2) and the standard proof,

$$P_{t_{j+1}-t_{j-1}}f(y_{j-1}) = \iint f(y_{j+1})P_{t_{j+1}-t_j}(y_j, dy_{j+1})P_{t_j-t_{j-1}}(y_{j-1}, dy_j) \quad (7.7)$$

for  $f : S \rightarrow \mathbb{R}$  measurable and bounded and  $y_{j-1} \in S$ . Defining  $f$  as

$$f(y_{j+1}) = \mathbb{1}_{B_{j+1}}(y_{j+1}) \int_{B_{j+2}} \cdots \int_{B_n} \mathbb{1}_{P_{t_n-t_{n-1}}(y_{n-1}, dy_n) \cdots P_{t_{j+2}-t_{j+1}}(y_{j+1}, dy_{j+2}),$$

the left-hand side of (7.6) equals

$$\begin{aligned} & \int_S \int_{B_1} \cdots \int_{B_{j-1}} \int_S \int_S f(y_{j+1}) \\ & P_{t_{j+1}-t_j}(y_j, dy_{j+1})P_{t_j-t_{j-1}}(y_{j-1}, dy_j)P_{t_{j-1}-t_{j-2}}(y_{j-2}, dy_{j-1}) \cdots P_{t_1-t_0}(y_0, dy_1)\mu(dy_0). \end{aligned}$$

By (7.7), this is also equal to

$$\begin{aligned} & \int_S \int_{B_1} \cdots \int_{B_{j-1}} \int_S f(y_{j+1})P_{t_{j+1}-t_{j-1}}(y_{j-1}, dy_{j+1})P_{t_{j-1}-t_{j-2}}(y_{j-2}, dy_{j-1}) \\ & \cdots P_{t_1-t_0}(y_0, dy_1)\mu(dy_0), \end{aligned}$$

which is the right-hand side of (7.6).  $\square$

By Kolmogorov's Consistency Theorem there exists a stochastic process with finite-dimensional marginal distributions given by (7.5). As in Section 4.4, this implies the following result, where  $(X_t)_{t \geq 0}$  denotes the coordinate process on  $S^{[0, \infty)}$  and, for  $s \geq 0$ ,  $\theta^s : S^{[0, \infty)} \rightarrow S^{[0, \infty)}$  is the shift operator defined as  $\theta^s((\omega_t)_{t \geq 0}) = (\omega_{s+t})_{t \geq 0}$ .

**Theorem 7.4.** *Let  $(P_t)_{t \geq 0}$  be a transition semigroup and  $\mu$  denote a probability measure on  $(S, \mathcal{B}(S))$ .*

- (1) *There exists a unique probability measure  $\mathbb{P}_\mu$  on  $(S^{[0, \infty)}, \mathcal{B}(S)^{[0, \infty)})$  under which  $(X_t)_{t \geq 0}$  is a Markov process with transition semigroup  $(P_t)_{t \geq 0}$  and initial distribution  $\mu$ . When  $\mu = \delta_x$  for  $x \in S$  we use the notation  $\mathbb{P}_x$ .*
- (2) *For  $A \in \mathcal{B}(S)^{[0, \infty)}$ ,  $x \mapsto \mathbb{P}_x(A)$  is measurable, and  $\mathbb{P}_\mu(A) = \int \mathbb{P}_x(A) \mu(dx)$ .*
- (3) *Let  $Z : S^{[0, \infty)} \rightarrow \mathbb{R}$  be bounded and  $\mathcal{B}(S)^{[0, \infty)}$ -measurable. Then, for every  $s \geq 0$ ,  $\mathbb{E}_\mu[Z \circ \theta^s | \mathcal{F}_s^X] = \mathbb{E}_{X_s}[Z]$ .*

The observant reader will note that there is no strong Markov property formulated here, and there are in fact examples of Markov processes not possessing this

property (search online for examples). However, the strong Markov property can be verified by imposing additional regularity conditions on  $P_t(x, \cdot)$ , such as the so-called *Feller property*, see e.g., Revuz and Yor (1999). Later we verify the strong Markov property for special classes of processes.

**Remark 7.5.** Let  $S$  be countable. Then we often specify the point probabilities  $p_t(x, y) = P_t(x, \{y\})$  for  $x, y \in S$  and  $t \geq 0$  rather than  $P_t(x, A)$  for  $A \in \mathcal{B}(S)$ . Let  $p_t$  be the  $S \times S$  matrix  $p_t = \{p_t(x, y)\}_{x, y \in S}$ . In this case the two conditions in Definition 7.1 can be reformulated as

$$p_0 = I \text{ (the } S \times S \text{ identity matrix) and } p_{s+t} = p_s p_t \text{ for } s, t \geq 0. \quad (7.8)$$

Here,  $p_s p_t$  is a matrix product. In this case we also call  $(p_t)_{t \geq 0}$  a transition semigroup. ■

**Example 7.6.** Although there are many (interesting) examples of Markov processes, the transition semigroup is often more or less unknown. Therefore, Markov processes are usually not constructed by first specifying the transition semigroup as indicated in Theorem 7.4. We will return to this important point in Chapter 12.

The following are some examples of transition semigroups.

1. In the CIR model (cf. Example 4.13) we have  $S = (0, \infty)$  and there are three parameters  $\alpha, \gamma, \eta > 0$  satisfying  $2\alpha\gamma > \eta^2$ . For  $x, t > 0$ ,  $P_t(x, \cdot)$  has density  $P_t(x, dy) = p_t(x, y)dy$ , where

$$p_t(x, y) = \frac{e^{\alpha t}}{2c} \left( \frac{ye^{\alpha t}}{x} \right)^{\nu/2} \exp\left(-\frac{x + ye^{\alpha t}}{2c}\right) I_\nu\left(\frac{\sqrt{xye^{\alpha t}}}{c}\right), \quad x, y > 0 \quad (7.9)$$

$c = \frac{\eta^2}{4\alpha}(e^{\alpha t} - 1)$ ,  $\nu = \frac{2\alpha\gamma}{\eta^2} - 1$ , and  $I_\nu$  is the modified Bessel function of the first kind of order  $\nu$ . (It is not obvious that  $(P_t)_{t \geq 0}$  is a transition semigroup – but it is).

2. The so-called Gaussian Ornstein-Uhlenbeck semigroup is defined in Exercise 75.
3. Let  $S = \{1, 2\}$  and  $q \in (0, 1)$ . Then  $(p_t)_{t \geq 0}$ , defined by

$$p_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad p_t = \begin{pmatrix} q & 1-q \\ q & 1-q \end{pmatrix} \text{ for } t > 0, \quad (7.10)$$

is a transition semigroup. See Exercise 74.

4. The semigroup corresponding to a Lévy process is studied in Chapter 9.

For Markov jump processes (see Chapter 12), the transition semigroup may be known only implicitly. ■

## Stopping times and related concepts

In this chapter, we collect some of the basic definitions and results related to stopping times in a continuous time framework.

Two stochastic processes  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  with the same state space and index set  $\mathbb{R}_+$  are said to be *versions* (or to *have the same distribution*) if  $(Y_{t_1}, \dots, Y_{t_n}) \stackrel{d}{=} (Y'_{t_1}, \dots, Y'_{t_n})$  for every  $n \in \mathbb{N}$  and every sequence  $0 \leq t_1 < \dots < t_n$ . (Equivalently, this holds for any finite collection of time points; the ordering is irrelevant.)

**Definition 8.1.** Let  $(\Omega, \mathcal{F}, \mathbb{P})$  denote a probability space on which  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are stochastic processes with state space  $S$ .

- (1)  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are **modifications** if  $\mathbb{P}(Y_t = Y'_t) = 1$  for every  $t \geq 0$ ;
- (2)  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are **indistinguishable** if the set

$$\{\omega \in \Omega : \text{there is a } t \geq 0 \text{ such that } Y_t(\omega) \neq Y'_t(\omega)\} \quad (8.1)$$

is a null set.

**Remark 8.2.** For stochastic processes indexed e.g. by  $\mathbb{N}_0$ , indistinguishability and modifications can be defined in a similar way. But since  $\mathbb{N}_0$  is countable, these two concepts coincide in that case. ■

**Example 8.3.** Let  $\tau$  denote an absolutely continuous random variable with state space  $[0, \infty)$  and put  $Y_t = \mathbb{1}_{\{\tau=t\}}$  and  $Y'_t = 0$  for  $t \geq 0$ . Then  $\mathbb{P}(Y_t = 0) = 1$  for every  $t \geq 0$ , which means that  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are modifications; however, they are not indistinguishable since the set (8.1) is equal to  $\Omega$ . ■

The next result clarifies the relations between the concepts introduced above.

**Proposition 8.4.** *We have the following.*

- (1) *If  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are indistinguishable, then they are modifications.*
- (2) *Assume  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are modifications. If the maps  $t \mapsto Y_t(\omega)$  and  $t \mapsto Y'_t(\omega)$  are right-continuous for almost every  $\omega$ , then  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are indistinguishable.*
- (3) *Assume  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are modifications. If the maps  $t \mapsto Y_t(\omega)$  and  $t \mapsto Y'_t(\omega)$  are left-continuous for almost every  $\omega$ , then  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are indistinguishable.*
- (4) *If  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are modifications, then they are versions.*

*Proof.* Item (1): For  $s \geq 0$ ,

$$\{\omega \in \Omega : Y_s(\omega) \neq Y'_s(\omega)\} \subseteq \{\omega \in \Omega : \text{there is a } t \geq 0 \text{ such that } Y_t(\omega) \neq Y'_t(\omega)\}.$$

Since the right-hand side is a null set, so is the left-hand side, which means that  $(Y_t)_{t \geq 0}$  and  $(Y'_t)_{t \geq 0}$  are modifications.

Item (2): Choose a set  $A \in \mathcal{F}$  with  $\mathbb{P}(A) = 1$  such that the maps  $t \mapsto Y_t(\omega)$  and  $t \mapsto Y'_t(\omega)$  are right-continuous for  $\omega \in A$ . Then choose a set  $B \in \mathcal{F}$  with  $\mathbb{P}(B) = 1$  such that  $Y_t(\omega) = Y'_t(\omega)$  for every  $t \in \mathbb{Q}_+$  and every  $\omega \in B$ . For  $\omega \in A \cap B$ ,

$$Y_t(\omega) = \lim_{s > t, s \in \mathbb{Q}, s \rightarrow t} Y_s(\omega) = \lim_{s > t, s \in \mathbb{Q}, s \rightarrow t} Y'_s(\omega) = Y'_t(\omega), \quad t \geq 0.$$

Thus, the set (8.1) is contained in  $(A \cap B)^c$ , which is a null set.

Item (4): See Exercise 80. □

Given a filtration  $(\mathcal{F}_t)_{t \geq 0}$ , put

$$\mathcal{F}_{t+} = \bigcap_{s > t} \mathcal{F}_s \quad \text{for } t \geq 0.$$

The filtration is called *right-continuous* if  $\mathcal{F}_t = \mathcal{F}_{t+}$  for  $t \geq 0$ .

**Definition 8.5.** *Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration. A map  $\tau : \Omega \rightarrow [0, \infty]$  is a **stopping time** if  $\{\tau \leq t\} \in \mathcal{F}_t$  for  $t \geq 0$ , and an **extended stopping time** if  $\{\tau < t\} \in \mathcal{F}_t$  for  $t \geq 0$ .*

If necessary, we emphasize the filtration by saying that  $\tau$  is a stopping time (or an extended stopping time) with respect to  $(\mathcal{F}_t)_{t \geq 0}$ . For  $A \in \mathcal{B}(S)$  and an adapted stochastic process  $(Y_t)_{t \geq 0}$  define

$$\tau_A = \inf\{t > 0 : Y_t \in A\} \quad \text{and} \quad \sigma_A = \inf\{t \geq 0 : Y_t \in A\}.$$

If one is willing to impose some rather mild regularity conditions on the filtered probability space (e.g., the so-called *usual conditions*), then  $\sigma_A$  and  $\tau_A$  are always stopping times if the process has e.g. right- or left-continuous paths. The proof of this result is, however, surprisingly hard, see, e.g., Jacod and Shiryaev (2003). The next result gives regularity conditions ensuring that  $\sigma_A$  and  $\tau_A$  are (extended) stopping times without restrictions on the probability space.

**Proposition 8.6.** *Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration.*

- (1)  *$\tau$  is an extended stopping time with respect to  $(\mathcal{F}_t)_{t \geq 0}$  if and only if it is a stopping time with respect to  $(\mathcal{F}_{t+})_{t \geq 0}$ .*
- (2) *If  $\tau$  is a stopping time with respect to  $(\mathcal{F}_t)_{t \geq 0}$ , then it is also an extended stopping time with respect to  $(\mathcal{F}_t)_{t \geq 0}$ .*
- (3) *Let  $(Y_t)_{t \geq 0}$  be an adapted process where **all** paths are right-continuous or **all** paths are left-continuous. If  $U \subseteq S$  is open, then  $\tau_U$  and  $\sigma_U$  are extended stopping times.*
- (4) *Let  $(Y_t)_{t \geq 0}$  be an adapted process where **all** paths are continuous. If  $F \subseteq S$  is closed, then  $\sigma_F$  is a stopping time.*

Item (1) shows that if  $(\mathcal{F}_t)_{t \geq 0}$  is right-continuous then  $\tau$  is a stopping time if and only if it is an extended stopping time.

*Proof. Item (1):* Assume  $\tau$  is an extended stopping time with respect to  $(\mathcal{F}_t)_{t \geq 0}$ . To infer that  $\tau$  is a stopping time with respect to  $(\mathcal{F}_{t+})_{t \geq 0}$ , it suffices to show that, for  $t < s$ , the event  $\{\tau \leq t\}$  is in  $\mathcal{F}_s$ . But for such an  $s$ ,  $\{\tau \leq t\} = \bigcap \{\tau < t + 1/n\}$ , where the intersection is over  $n \in \mathbb{N}$  satisfying  $t + 1/n < s$ . Moreover, if  $t + 1/n < s$ , then  $\{\tau < t + 1/n\} \in \mathcal{F}_{t+1/n} \subseteq \mathcal{F}_s$ .

Conversely, assume  $\tau$  is a stopping time with respect to  $(\mathcal{F}_{t+})_{t \geq 0}$ . Since, for  $t > 0$ ,

$$\{\tau < t\} = \bigcup_{n=1}^{\infty} \{\tau \leq t - 1/n\}$$

and  $\{\tau \leq t - 1/n\} \in \mathcal{F}_{(t-1/n)+} \subseteq \mathcal{F}_t$ , it follows that  $\{\tau < t\} \in \mathcal{F}_t$ .

*Item (2):* If  $\tau$  is a stopping time with respect to  $(\mathcal{F}_t)_{t \geq 0}$  then it is also a stopping time with respect to  $(\mathcal{F}_{t+})_{t \geq 0}$ , so the result follows from item (1).

*Item (3):* For  $t > 0$ , since  $U$  is open and all paths are either right- or left-continuous, we have

$$\begin{aligned}\{\tau_U < t\} &= \bigcup_{s \in (0, t) \cap \mathbb{Q}} \{Y_s \in U\} \in \mathcal{F}_t \\ \{\sigma_U < t\} &= \{Y_0 \in U\} \cup \bigcup_{s \in (0, t) \cap \mathbb{Q}} \{Y_s \in U\} \in \mathcal{F}_t.\end{aligned}$$

*Item (4):* Let  $d$  denote the metric on  $S$  and for  $x \in S$  let  $d(x, F) = \inf\{d(x, y) : y \in F\}$  denote the distance to  $F$ . The function  $x \mapsto d(x, F)$  is continuous; in particular  $s \mapsto d(Y_s, F)$  is a (random) continuous function. By closedness of  $F$ ,  $d(x, F)$  is zero if and only if  $x \in F$ . From all this follows that  $\sigma_F$  is a stopping time since

$$\{\sigma_F \leq t\} = \left\{ \inf_{s \in [0, t]} d(Y_s, F) = 0 \right\} = \left\{ \inf_{s \in [0, t] \cap \mathbb{Q}_+} d(Y_s, F) = 0 \right\} \in \mathcal{F}_t, \quad t \geq 0.$$

The second equality above is by continuity of  $s \mapsto d(Y_s, F)$ . To see that the first equality is true, note that if  $\sigma_F$  is less than or equal to  $t$  then there exists a (random) sequence  $(t_n)$  such that  $d(Y_{t_n}, F) = 0$ ,  $t_n \geq \sigma_F$  and  $t_n \rightarrow \sigma_F$ . By right continuity of  $s \mapsto d(Y_s, F)$ ,  $d(Y_{\sigma_F}, F) = 0$  and in particular  $\inf_{s \in [0, t]} d(Y_s, F) = 0$ . Conversely, if the latter holds, there is, by continuity, an  $s \in [0, t]$  such that  $d(Y_s, F) = 0$ . Hence,  $\sigma_F \leq s \leq t$ .  $\square$

**Proposition 8.7.** *Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration.*

- (1) *If  $\sigma$  and  $\tau$  are stopping times, then so are  $\sigma + \tau$ ,  $\sigma \wedge \tau$  and  $\sigma \vee \tau$ .*
- (2) *If  $(\tau_n)_{n \geq 1}$  is a sequence of stopping times, then  $\sum_{n=1}^{\infty} \tau_n$  and  $\sup_{n \in \mathbb{N}} \tau_n$  are stopping times as well.*

There is a similar result for extended stopping times. Moreover,  $\inf_{n \in \mathbb{N}} \tau_n$  is an extended stopping time if  $\tau_n$  is an extended stopping time for  $n \in \mathbb{N}$ .

*Proof.* Let us prove that if  $\sigma$  and  $\tau$  are stopping times then so is  $\sigma + \tau$ . For this, note that for  $t \geq 0$ ,

$$\{\sigma + \tau > t\} = \{\sigma > t\} \cup \{\tau > t\} \bigcup \left[ \bigcup_{r, s \in [0, t] \cap \mathbb{Q}} \{\sigma > s\} \cap \{\tau > r\} \right],$$

where the union on the right-hand side is over  $r, s \in [0, t] \cap \mathbb{Q}$  satisfying  $r + s > t$ . Since, for such  $s$  and  $r$ ,  $\{\sigma > s\} \in \mathcal{F}_s \subseteq \mathcal{F}_t$  and  $\{\tau > r\} \in \mathcal{F}_r \subseteq \mathcal{F}_t$ , we see that  $\{\sigma + \tau > t\} \in \mathcal{F}_t$ , which means that  $\sigma + \tau$  is a stopping time.

For the remaining assertions, see Exercise 82.  $\square$

For a stopping time  $\tau$  with respect to  $(\mathcal{F}_t)_{t \geq 0}$  define the  $\sigma$ -field  $\mathcal{F}_\tau$  of events prior to time  $\tau$  as

$$\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for } t \geq 0\}.$$

For an extended stopping time  $\tau$  the  $\sigma$ -field  $\mathcal{F}_{\tau+}$  is defined in the same way with  $\mathcal{F}_t$  replaced by  $\mathcal{F}_{t+}$ . Another way of representing  $\mathcal{F}_{\tau+}$  is as

$$\mathcal{F}_{\tau+} = \{A \in \mathcal{F} : A \cap \{\tau < t\} \in \mathcal{F}_t \text{ for } t \geq 0\}.$$

The following summarizes some basic properties.

**Proposition 8.8.** *Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration.*

- (1) *If  $\tau$  is a stopping time then  $\mathcal{F}_\tau$  is a  $\sigma$ -field and  $\tau$  is  $\mathcal{F}_\tau$ -measurable. If  $\tau$  is an extended stopping time then  $\mathcal{F}_{\tau+}$  is a  $\sigma$ -field and  $\tau$  is  $\mathcal{F}_{\tau+}$ -measurable.*
- (2) *Let  $\sigma$  and  $\tau$  be stopping times. Then  $\mathcal{F}_{\sigma \wedge \tau} = \mathcal{F}_\sigma \cap \mathcal{F}_\tau$ . Moreover, the events  $\{\tau < \sigma\}$ ,  $\{\tau \leq \sigma\}$  and  $\{\tau = \sigma\}$  are in  $\mathcal{F}_\sigma \cap \mathcal{F}_\tau$ .*
- (3) *Let  $\sigma$  and  $\tau$  be stopping times satisfying  $\sigma(\omega) \leq \tau(\omega)$  for every  $\omega \in \Omega$ . Then  $\mathcal{F}_\sigma \subseteq \mathcal{F}_\tau$ .*

Using that extended stopping times are stopping times with respect to  $(\mathcal{F}_{t+})_{t \geq 0}$  it follows that there are results similar to (2) and (3) for extended stopping times.

*Proof.* The proof of (1), (3) and the last part of (2) is left to the reader as an exercise (see Exercise 84), so let us concentrate on the first part of (2).

Let  $A \in \mathcal{F}_{\sigma \wedge \tau}$ . For  $t \geq 0$ ,

$$A \cap \{\sigma \leq t\} = [A \cap \{\sigma \wedge \tau \leq t\}] \cap \{\sigma \leq t\}.$$

Since  $A \in \mathcal{F}_{\sigma \wedge \tau}$ ,  $A \cap \{\sigma \wedge \tau \leq t\} \in \mathcal{F}_t$ . And since  $\sigma$  is a stopping time,  $\{\sigma \leq t\} \in \mathcal{F}_t$ . Hence,  $A \cap \{\sigma \leq t\} \in \mathcal{F}_t$  for every  $t \geq 0$ , which implies that  $A \in \mathcal{F}_\sigma$ . Similar arguments show that  $A \in \mathcal{F}_\tau$  and hence  $A \in \mathcal{F}_\sigma \cap \mathcal{F}_\tau$ .

Conversely, let  $A \in \mathcal{F}_\sigma \cap \mathcal{F}_\tau$ . Since  $A \cap \{\sigma \wedge \tau \leq t\} = [A \cap \{\sigma \leq t\}] \cup [A \cap \{\tau \leq t\}] \in \mathcal{F}_t$  for  $t \geq 0$  we conclude that  $A \in \mathcal{F}_{\sigma \wedge \tau}$ .  $\square$

Next we consider some questions related to the measurability of  $Y_\tau$ , i.e., looking at a stochastic process at random time points. Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration. For a stochastic process  $(Y_t)_{t \geq 0}$  and  $t \geq 0$  define the map  $\phi_t^Y : [0, t] \times \Omega \rightarrow S$  as  $\phi_t^Y(s, \omega) = Y_s(\omega)$ .

**Definition 8.9.** *The process  $(Y_t)_{t \geq 0}$  is **progressively measurable** if for every  $t \geq 0$ , the map  $\phi_t^Y$  is  $(\mathcal{B}([0, t]) \times \mathcal{F}_t, \mathcal{B}(S))$ -measurable.*

If  $\tau$  is a map from  $\Omega$  to  $[0, \infty]$ , define  $Y_\tau$  on  $\{\tau < \infty\}$  in the obvious way as  $Y_\tau(\omega) = Y_{\tau(\omega)}(\omega)$ . An extended stopping time  $\tau$  is *finite* if  $\tau(\omega) < \infty$  for every  $\omega \in \Omega$ .

**Proposition 8.10.** *Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration.*

- (1) *A progressively measurable process  $(Y_t)_{t \geq 0}$  is adapted.*
- (2) *Assume  $(Y_t)_{t \geq 0}$  is adapted and that all paths are right-continuous or all paths are left-continuous. Then  $(Y_t)_{t \geq 0}$  is progressively measurable.*
- (3) *Let  $(Y_t)_{t \geq 0}$  be progressively measurable and  $\tau$  be an extended stopping time. Then  $(Y_{\tau \wedge t})_{t \geq 0}$  is adapted.*
- (4) *Let  $(Y_t)_{t \geq 0}$  be progressively measurable. Then  $Y_\tau$  is  $\mathcal{F}_\tau$ -measurable if  $\tau$  is a finite stopping time and  $\mathcal{F}_{\tau+}$ -measurable if  $\tau$  is a finite extended stopping time.*
- (5) *Let the state space be  $S = \mathbb{R}^d$  for some  $d \in \mathbb{N}$  and  $(Y_t)_{t \geq 0}$  be progressively measurable. Then  $Y_\tau \mathbb{1}_{\{\tau < \infty\}}$  is  $\mathcal{F}_\tau$ -measurable if  $\tau$  is a stopping time and  $\mathcal{F}_{\tau+}$ -measurable if  $\tau$  is an extended stopping time.*

**Remark 8.11.** Let the state space be  $S = \mathbb{R}^d$  for some  $d \in \mathbb{N}$  and let  $(Y_t)_{t \geq 0}$  be progressively measurable. If  $\lim_{t \rightarrow \infty} Y_t$  exists a.s. we can, e.g., define  $Y_\tau$  as

$$Y_\tau(\omega) = \begin{cases} \lim_{n \rightarrow \infty} Y_{\tau(\omega) \wedge n}(\omega) & \text{if the limit exists,} \\ 0 & \text{otherwise.} \end{cases}$$

That is, if  $\tau(\omega)$  is finite, then  $Y_\tau(\omega)$  equals  $Y_{\tau(\omega)}(\omega)$ , and if  $\tau(\omega) = \infty$  and the limit  $\lim_{t \rightarrow \infty} Y_t(\omega)$  exists then  $Y_\tau(\omega)$  equals  $\lim_{t \rightarrow \infty} Y_t(\omega)$ .

In this case  $Y_\tau$  is  $\mathcal{F}_\tau$ -measurable if  $\tau$  is a stopping time and  $\mathcal{F}_{\tau+}$ -measurable if  $\tau$  is an extended stopping time. Let us prove this when  $\tau$  is a stopping time. Then  $Y_{\tau \wedge n}$  is  $\mathcal{F}_\tau$ -measurable by Proposition 8.10(4) since  $\mathcal{F}_{\tau \wedge n} \subseteq \mathcal{F}_\tau$  for  $n \in \mathbb{N}$  (why?). By going to the limit, we conclude that  $Y_\tau$  is  $\mathcal{F}_\tau$ -measurable as well. ■

*Proof of Proposition 8.10. Item (1):* Let  $t \geq 0$ . Since  $(s, \omega) \mapsto \phi_t^Y(s, \omega) = Y_s(\omega)$  is  $(\mathcal{B}([0, t]) \times \mathcal{F}_t, \mathcal{B}(S))$ -measurable, it follows that for arbitrary  $s \in [0, t]$ ,  $\omega \mapsto Y_s(\omega)$  is  $(\mathcal{F}_t, \mathcal{B}(S))$ -measurable. By choosing  $s = t$  we conclude that  $\omega \mapsto Y_t(\omega)$  is  $(\mathcal{F}_t, \mathcal{B}(S))$ -measurable. Thus,  $(Y_t)_{t \geq 0}$  is adapted.

*Item (2):* Assume  $(Y_t)_{t \geq 0}$  is adapted and right-continuous. Let  $t \geq 0$ . For  $n \in \mathbb{N}$  define  $\phi_t^{Y,n} : [0, t] \times \Omega \rightarrow S$  as

$$\phi_t^{Y,n}(s, \omega) = \begin{cases} Y_0(\omega) & \text{for } s = 0, \\ Y_{\frac{it}{n}}(\omega) & \text{for } s \in (\frac{(i-1)t}{n}, \frac{it}{n}] \text{ and } i = 1, \dots, n. \end{cases}$$

That is, in the lower case  $(0, t]$  is divided into the intervals  $(0, \frac{t}{n}]$ ,  $(\frac{t}{n}, \frac{2t}{n}]$ ,  $\dots$ ,  $(\frac{(n-1)t}{n}, t]$ , and on each interval  $\phi_t^{Y,n}$  is set to the value of  $(Y_s)_{s \geq 0}$  in the right end-point. Then  $\phi_t^{Y,n}$  is  $(\mathcal{B}([0, t]) \times \mathcal{F}_t, \mathcal{B}(S))$ -measurable since for  $A \in \mathcal{B}(S)$ ,

$$\begin{aligned} & \{(s, \omega) \in [0, t] \times \Omega : \phi_t^{Y,n}(s, \omega) \in A\} \\ &= [\{0\} \times \{Y_0 \in A\}] \cup \bigcup_{i=1}^n \left[ \left( \frac{(i-1)t}{n}, \frac{it}{n} \right] \times \{Y_{\frac{it}{n}} \in A\} \right] \in \mathcal{B}([0, t]) \times \mathcal{F}_t. \end{aligned}$$

Moreover, by right-continuity of  $s \mapsto Y_s(\omega)$ ,  $\phi_t^Y(s, \omega) = \lim_{n \rightarrow \infty} \phi_t^{Y,n}(s, \omega)$  for  $(s, \omega) \in [0, t] \times \Omega$ , which implies that  $\phi_t^Y$  is  $(\mathcal{B}([0, t]) \times \mathcal{F}_t, \mathcal{B}(S))$ -measurable as well. Hence,  $(Y_t)_{t \geq 0}$  is progressively measurable.

*Item (3):* Let  $t \geq 0$ . The map

$$\Omega \ni \omega \mapsto (\tau(\omega) \wedge t, \omega) \in [0, t] \times \Omega$$

is  $(\mathcal{F}_t, \mathcal{B}([0, t]) \times \mathcal{F}_t)$ -measurable. By composing with  $\phi_t^Y$  it follows that

$$\Omega \ni \omega \mapsto \phi_t^Y(\tau(\omega) \wedge t, \omega) = Y_{\tau(\omega) \wedge t}(\omega)$$

is  $(\mathcal{F}_t, \mathcal{B}(S))$ -measurable.

*Item (4):* We must show that for  $A \in \mathcal{B}(S)$  the set  $\{Y_\tau \in A\}$  is in  $\mathcal{F}_\tau$  if  $\tau$  is a stopping time and in  $\mathcal{F}_{\tau+}$  if  $\tau$  is an extended stopping time.

Assume  $\tau$  is a stopping time and let  $t \geq 0$ . Then

$$\{Y_\tau \in A\} \cap \{\tau \leq t\} = \{Y_{\tau \wedge t} \in A\} \cap \{\tau \leq t\} \in \mathcal{F}_t,$$

where the inclusion follows since  $\{\tau \leq t\}$  and  $\{Y_{\tau \wedge t} \in A\}$  are in  $\mathcal{F}_t$  because  $\tau$  is a stopping time and because of item (3). Thus,  $\{Y_\tau \in A\}$  is in  $\mathcal{F}_\tau$ .

Replacing the event  $\{\tau \leq t\}$  by  $\{\tau < t\}$  above, it follows that  $\{Y_\tau \in A\}$  is in  $\mathcal{F}_{\tau+}$  if  $\tau$  is an extended stopping time.

*Item (5):* Follows in the same way as above. □

**Remark 8.12.** Let  $(\mathcal{F}_t)_{t \geq 0}$  denote a filtration and  $\tau$  be an extended stopping time. For  $n \in \mathbb{N}$  define  $\tau_n$  as

$$\tau_n = \begin{cases} \frac{k}{2^n} & \text{on } \{\tau \in [\frac{k-1}{2^n}, \frac{k}{2^n})\} \text{ for } k \in \mathbb{N}, \\ \infty & \text{on } \{\tau = \infty\}. \end{cases}$$

Call  $(\tau_n)_{n \geq 1}$  the discrete approximation of  $\tau$ .

Concretely,  $\tau_1$  and  $\tau_2$  are given by

$$\tau_1 = \begin{cases} \frac{1}{2} & \text{if } \tau \in [0, \frac{1}{2}), \\ 1 & \text{if } \tau \in [\frac{1}{2}, 1), \\ \frac{3}{2} & \text{if } \tau \in [1, \frac{3}{2}), \\ \vdots & \end{cases} \quad \tau_2 = \begin{cases} \frac{1}{4} & \text{if } \tau \in [0, \frac{1}{4}), \\ \frac{1}{2} & \text{if } \tau \in [\frac{1}{4}, \frac{1}{2}), \\ \frac{3}{4} & \text{if } \tau \in [\frac{1}{2}, \frac{3}{4}), \\ \vdots & \end{cases}$$

We then have the following properties (see Exercise 85):

- (1) For  $n \in \mathbb{N}$ ,  $\tau_n$  is a stopping time.
- (2) For  $n \in \mathbb{N}$ ,  $\{\tau < \infty\} = \{\tau_n < \infty\}$ .
- (3) For  $n \in \mathbb{N}$ ,  $\tau_n > \tau$  on  $\{\tau < \infty\} = \{\tau_n < \infty\}$ .
- (4)  $\tau_1 \geq \tau_2 \geq \dots$  and  $\tau_n \rightarrow \tau$  pointwise.
- (5) For  $A \in \mathcal{F}_{\tau+}$  and  $k, n \in \mathbb{N}$ ,

$$A \cap \{\tau_n = \frac{k}{2^n}\} = A \cap \{\tau \in [\frac{k-1}{2^n}, \frac{k}{2^n})\} \in \mathcal{F}_{\frac{k}{2^n}}.$$

- (6) For  $n \in \mathbb{N}$ ,

$$\mathcal{F}_{\tau_n} = \{B \in \mathcal{F} : B \cap \{\tau_n = \frac{k}{2^n}\} \in \mathcal{F}_{\frac{k}{2^n}} \text{ for all } k \in \mathbb{N}\}.$$

- (7)  $\mathcal{F}_{\tau+} = \bigcap_{n=1}^{\infty} \mathcal{F}_{\tau_n}$ .

■

# Lévy processes

In this chapter, we study the continuous time analogue of random walks – namely so-called Lévy processes. Standard references for the theory of Lévy processes include Sato (2013), Bertoin (1996), Cont and Tankov (2004), Applebaum (2004), and Kyprianou (2014).

## 9.1 Definitions

We begin by giving two definitions of Lévy processes and then study how they are related.

**Definition 9.1.** Let  $d \in \mathbb{N}$ . An  $\mathbb{R}^d$ -valued stochastic process  $(Y_t)_{t \geq 0}$  has

- (1) **stationary increments** if  $Y_t - Y_s \stackrel{d}{=} Y_{t-s}$  for  $0 \leq s \leq t$ ;
- (2) **independent increments** if for arbitrary  $n \in \mathbb{N}$  and  $0 \leq t_1 < \dots < t_n$ ,  $Y_{t_1}, Y_{t_2} - Y_{t_1}, \dots, Y_{t_n} - Y_{t_{n-1}}$  are independent.

The process has **stationary independent increments** if both conditions are satisfied.

By taking  $s = t$  in Definition 9.1(1) it follows that if  $(Y_t)_{t \geq 0}$  has stationary increments then  $Y_0 = 0$  a.s.

**Definition 9.2.** Let  $d \in \mathbb{N}$ . An  $\mathbb{R}^d$ -valued stochastic process  $(Y_t)_{t \geq 0}$  is a ( $d$ -dimensional) **Lévy process** if

- (1)  $Y_0 = 0$  a.s.
- (2)  $t \mapsto Y_t(\omega)$  is càdlàg for any  $\omega \in \Omega$ .
- (3) The process has stationary independent increments.

As noted above, condition (1) is implied by (3).

If  $d = 1$  we also speak of a real-valued Lévy process. Many authors only require the càdlàg property in Definition 9.2(2) to hold for  $\mathbb{P}$ -almost all  $\omega$ . Assuming the càdlàg property for all  $\omega$  is convenient, since, with respect to the natural filtration  $(\mathcal{F}_t^Y)_{t \geq 0}$ , the process is adapted. Then Proposition 8.10 yields progressive measurability and hence measurability of  $Y_\tau$  for (extended) stopping times  $\tau$ .

At this point the reader is strongly encouraged to consider some examples of Lévy processes, see, e.g., Section 9.3. Also recall that in discrete time a process with stationary independent increments is a random walk cf. Proposition 6.31.

**Remark 9.3.** A Lévy process  $(Y_t)_{t \geq 0}$  is continuous in probability in the sense that if  $t_n \rightarrow t$  then  $Y_{t_n} \rightarrow Y_t$  in probability (see Exercise 93). In particular, for every fixed  $t > 0$ ,  $\Delta Y_t = 0$  a.s.

The reader should not confuse the last part with the statement that  $t \mapsto Y_t$  is a.s. continuous because the latter is only true in very particular cases, see Section 9.3 and Section 9.5. ■

**Remark 9.4.** Let  $(Y_t)_{t \geq 0}$  be a Lévy process and let  $\Delta > 0$  be arbitrary. Then the discrete time process  $(Y_{n\Delta})_{n \geq 0}$  is a random walk. Indeed, decompose  $Y_{n\Delta}$  as

$$Y_{n\Delta} = Y_0 + \sum_{k=1}^n [Y_{k\Delta} - Y_{(k-1)\Delta}] \quad \text{for } n \in \mathbb{N},$$

and note that by the stationary independent increments,  $(Y_{n\Delta} - Y_{(n-1)\Delta})_{n \geq 1}$  is an i.i.d. sequence of random vectors all having the same distribution as  $Y_\Delta$ . ■

We can also define Lévy processes with respect to a filtration:

**Definition 9.5.** Let  $d \in \mathbb{N}$  and  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration. An  $\mathbb{R}^d$ -valued stochastic process  $(Y_t)_{t \geq 0}$  is an  $(\mathcal{F}_t)_{t \geq 0}$ -**Lévy process** if

- (1)  $Y_0 = 0$  a.s.
- (2)  $t \mapsto Y_t(\omega)$  is càdlàg for any  $\omega \in \Omega$ .
- (3) The process has stationary increments.
- (4)  $(Y_t)_{t \geq 0}$  is  $(\mathcal{F}_t)_{t \geq 0}$ -adapted.
- (5) For  $0 \leq s < t$ ,  $Y_t - Y_s$  is independent of  $\mathcal{F}_s$ .

If  $(Y_t)_{t \geq 0}$  is an  $(\mathcal{F}_t)_{t \geq 0}$ -Lévy process then it is also an  $(\mathcal{F}_t^Y)_{t \geq 0}$ -Lévy process.

The two definitions above are related as follows:

**Proposition 9.6.** *Let  $(Y_t)_{t \geq 0}$  be an  $\mathbb{R}^d$ -valued stochastic process and  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration.*

- (1) *If  $(Y_t)_{t \geq 0}$  has independent increments, then it satisfies condition (5) in Definition 9.5 with  $\mathcal{F}_s = \mathcal{F}_s^Y$ .*
- (2) *Assume  $(Y_t)_{t \geq 0}$  satisfies conditions (4) and (5) in Definition 9.5 and that  $Y_0 = 0$  a.s. Then  $(Y_t)_{t \geq 0}$  has independent increments, and  $\mathcal{F}_0$  is independent of  $(Y_t)_{t \geq 0}$ .*
- (3)  *$(Y_t)_{t \geq 0}$  is a Lévy process (in the sense of Definition 9.2) if and only if it is an  $(\mathcal{F}_t^Y)_{t \geq 0}$ -Lévy process (in the sense of Definition 9.5).*

*Proof.* Item (1): Let  $0 \leq s < t$ . Let  $n \in \mathbb{N}$  and  $0 \leq t_1 < \dots < t_n = s < t_{n+1} = t$ . By independent increments,

$$Y_{t_1}, Y_{t_2} - Y_{t_1}, \dots, Y_{t_n} - Y_{t_{n-1}}, Y_{t_{n+1}} - Y_{t_n}$$

are independent. Since

$$\begin{pmatrix} Y_{t_1} \\ Y_{t_2} \\ \vdots \\ Y_{t_n} \end{pmatrix} = \begin{pmatrix} Y_{t_1} \\ Y_{t_1} + [Y_{t_2} - Y_{t_1}] \\ \vdots \\ Y_{t_1} + \sum_{k=2}^n [Y_{t_k} - Y_{t_{k-1}}] \end{pmatrix},$$

$Y_t - Y_s$  is independent of  $(Y_{t_1}, \dots, Y_{t_n})$  as well. Put differently,  $Y_t - Y_s$  is independent of every event of the form

$$\bigcap_{k=1}^n \{Y_{t_k} \in A_k\},$$

where  $n \in \mathbb{N}$ ,  $0 \leq t_1 < \dots < t_n = s$  and  $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R}^d)$ . Since the collection of such events is stable under intersection and generates  $\mathcal{F}_s^Y$ , it follows that  $Y_t - Y_s$  is independent of  $\mathcal{F}_s^Y$ . That is, condition (5) in Definition 9.5 is satisfied with  $\mathcal{F}_s = \mathcal{F}_s^Y$ .

Item (2): Let  $n \in \mathbb{N}$ ,  $0 \leq t_1 < \dots < t_n$  and  $U$  be an  $\mathbb{R}$ -valued  $\mathcal{F}_0$ -measurable random variable. Put  $t_0 = 0$ . Below, we show that

$$U, Y_{t_1} - Y_{t_0}, Y_{t_2} - Y_{t_1}, \dots, Y_{t_n} - Y_{t_{n-1}} \text{ are independent.} \quad (9.1)$$

Since  $Y_{t_1} = Y_{t_1} - Y_{t_0}$  a.s., this in particular implies that  $(Y_t)_{t \geq 0}$  has independent increments. Moreover, by adding the increments, it also follows that  $U$  is independent of  $(Y_{t_1}, \dots, Y_{t_n})$  for every  $n \in \mathbb{N}$  and  $0 \leq t_1 < \dots < t_n$ . Hence,  $U$  is independent of  $\sigma(Y_t | t \geq 0)$ . Taking  $U = \mathbb{1}_A$  for  $A \in \mathcal{F}_0$ , we see that  $\mathcal{F}_0$  is independent of  $\sigma(Y_t | t \geq 0)$ .

To prove (9.1), let  $\theta \in \mathbb{R}$  and  $\theta_1, \dots, \theta_n \in \mathbb{R}^d$ . Then,

$$\begin{aligned}
& \mathbb{E}[e^{i\theta U} \prod_{m=1}^n e^{i\langle \theta_m, Y_{t_m} - Y_{t_{m-1}} \rangle}] \\
&= \mathbb{E}\left[\mathbb{E}\left[e^{i\theta U} \prod_{m=1}^n e^{i\langle \theta_m, Y_{t_m} - Y_{t_{m-1}} \rangle} \mid \mathcal{F}_{t_{n-1}}\right]\right] \\
&= \mathbb{E}\left[e^{i\theta U} \prod_{m=1}^{n-1} e^{i\langle \theta_m, Y_{t_m} - Y_{t_{m-1}} \rangle} \mathbb{E}[e^{i\langle \theta_n, Y_{t_n} - Y_{t_{n-1}} \rangle} \mid \mathcal{F}_{t_{n-1}}]\right] \\
&= \mathbb{E}\left[e^{i\theta U} \prod_{m=1}^{n-1} e^{i\langle \theta_m, Y_{t_m} - Y_{t_{m-1}} \rangle}\right] \mathbb{E}[e^{i\langle \theta_n, Y_{t_n} - Y_{t_{n-1}} \rangle}],
\end{aligned}$$

where we use the assumption that  $Y_{t_n} - Y_{t_{n-1}}$  is independent of  $\mathcal{F}_{t_{n-1}}$  in the last step. Repeating these calculations where we successively condition on  $\mathcal{F}_{t_{n-2}}, \dots, \mathcal{F}_{t_0} = \mathcal{F}_0$ , we arrive at

$$\mathbb{E}[e^{i\theta U} \prod_{m=1}^n e^{i\langle \theta_m, Y_{t_m} - Y_{t_{m-1}} \rangle}] = \mathbb{E}[e^{i\theta U}] \prod_{m=1}^n \mathbb{E}[e^{i\langle \theta_m, Y_{t_m} - Y_{t_{m-1}} \rangle}],$$

which is equivalent to (9.1).

*Item (3):* Assume  $(Y_t)_{t \geq 0}$  is a Lévy process. Conditions (1)–(3) in Definition 9.5 are satisfied by Definition 9.2, and obviously  $(Y_t)_{t \geq 0}$  is  $(\mathcal{F}_t^Y)_{t \geq 0}$ -adapted. Condition (5) in Definition 9.5 is satisfied by item (1) above.

Conversely, if  $(Y_t)_{t \geq 0}$  is an  $(\mathcal{F}_t^Y)_{t \geq 0}$ -Lévy process then by Definition 9.5,  $(Y_t)_{t \geq 0}$  has stationary increments and conditions (1)–(2) in Definition 9.2 are satisfied. The independent increments follow from item (2) above.  $\square$

## 9.2 The Markov property and convolution semigroups

This section establishes a one-to-one correspondence between Lévy processes and so-called *regular convolution semigroups*. In doing so, we also establish the Markov property of Lévy processes.

**Definition 9.7.** Let  $d \in \mathbb{N}$ . A family  $(\nu_t)_{t \geq 0}$  of probability measures on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  is called a **regular convolution semigroup** (on  $\mathbb{R}^d$ ) if

- (1)  $\nu_0 = \delta_0$  and  $\nu_{t+s} = \nu_t * \nu_s$  for  $s, t \geq 0$ .
- (2) If  $t_n \geq 0$  and  $t_n \rightarrow 0$  then  $\nu_{t_n} \xrightarrow{w} \nu_0 = \delta_0$ .

Since  $\widehat{\delta_0}(z) = 1$  for  $z \in \mathbb{R}^d$ , Definition 9.7(2) is equivalent to that  $\widehat{\nu_{t_n}}(z) \rightarrow 1$  as  $n \rightarrow \infty$  for every  $z \in \mathbb{R}^d$ . Moreover, the term *regular* is related to the second property above, which is a kind of continuity in  $t$ . It can be sharpened to the following:

**Lemma 9.8.** *Let  $(\nu_t)_{t \geq 0}$  be a regular convolution semigroup. If  $t_n, t \geq 0$  and  $t_n \rightarrow t$  then  $\nu_{t_n} \xrightarrow{\sim} \nu_t$ .*

*Proof.* It suffices to consider the two cases (a):  $t_n \geq t$  for all  $n$ , and (b):  $t_n \leq t$  for all  $n$ .

In case (a), use the decomposition  $\nu_{t_n} = \nu_t * \nu_{t_n-t}$ . Since  $t_n - t \rightarrow 0$  it follows that  $\nu_{t_n-t} \xrightarrow{\sim} \delta_0$  and hence  $\nu_{t_n} = \nu_t * \nu_{t_n-t} \xrightarrow{\sim} \nu_t * \delta_0 = \nu_t$ .

In case (b),  $\nu_t = \nu_{t_n} * \nu_{t-t_n}$ . Since  $\nu_{t-t_n} \xrightarrow{\sim} \delta_0$ ,  $\widehat{\nu_{t-t_n}}(z) \rightarrow 1$  for  $z \in \mathbb{R}^d$ . Since  $\widehat{\nu_t}(z) = \widehat{\nu_{t_n}}(z)\widehat{\nu_{t-t_n}}(z)$ , it follows that  $\widehat{\nu_{t_n}}(z) \rightarrow \widehat{\nu_t}(z)$  for  $z \in \mathbb{R}^d$ . Thus,  $\nu_{t_n} \xrightarrow{\sim} \nu_t$ .  $\square$

The next theorem shows that a Lévy process induces a regular convolution semigroup.

**Theorem 9.9.** *Let  $(Y_t)_{t \geq 0}$  be a Lévy process and define  $\nu_t$  as  $\nu_t = \mathbb{P}_{Y_t}$  for  $t \geq 0$ . Then  $(\nu_t)_{t \geq 0}$  is a regular convolution semigroup.*

*Proof.* Since  $Y_0 = 0$  a.s.,  $\nu_0 = \delta_0$ . Let  $s, t \geq 0$ . Since  $Y_{t+s} = [Y_{t+s} - Y_t] + Y_t$  and  $Y_{t+s} - Y_t$  is independent of  $Y_t$  by the independent increments,

$$\nu_{t+s} = \mathbb{P}_{Y_{t+s}} = \mathbb{P}_{Y_{t+s}-Y_t} * \mathbb{P}_{Y_t} = \mathbb{P}_{Y_{t+s}-Y_t} * \nu_t.$$

By the stationary increments,  $Y_{t+s} - Y_t \stackrel{d}{=} Y_s$  and hence  $\mathbb{P}_{Y_{t+s}-Y_t} = \mathbb{P}_{Y_s} = \nu_s$ . That is,  $\nu_{t+s} = \nu_t * \nu_s$ .

By right-continuity of the paths, if  $t_n \rightarrow 0$  then  $Y_{t_n} \rightarrow Y_0$  pointwise, which implies  $Y_{t_n} \xrightarrow{\sim} Y_0$ , i.e.,  $\nu_{t_n} = \mathbb{P}_{Y_{t_n}} \xrightarrow{\sim} \mathbb{P}_{Y_0} = \nu_0$ .  $\square$

Let us show that, conversely, a regular convolution semigroup induces a Lévy process. If  $(\nu_t)_{t \geq 0}$  is a regular convolution semigroup on  $\mathbb{R}^d$ , define  $P_t(x, A)$ , for  $t \geq 0$ ,  $x \in \mathbb{R}^d$  and  $B \in \mathcal{B}(\mathbb{R}^d)$ , as

$$P_t(x, B) = \nu_t(B - x) = \int_{\mathbb{R}^d} \mathbb{1}_B(x + y) \nu_t(dy). \quad (9.2)$$

The following result, together with Theorem 9.9, establishes a one-to-one correspondence between Lévy processes and regular convolution semigroups. Moreover, it shows that Lévy processes are Markov processes with transition semigroup  $(P_t)_{t \geq 0}$ .

**Theorem 9.10.** *Let  $(\nu_t)_{t \geq 0}$  be a regular convolution semigroup and define  $(P_t)_{t \geq 0}$  by (9.2). We then have the following.*

- (1)  $(P_t)_{t \geq 0}$  is a transition semigroup.
- (2) Let  $(Y_t)_{t \geq 0}$  be an  $\mathbb{R}^d$ -valued stochastic process with  $Y_0 = 0$  a.s. Then  $(Y_t)_{t \geq 0}$  is a Markov process with transition semigroup  $(P_t)_{t \geq 0}$  if and only if  $(Y_t)_{t \geq 0}$  has stationary independent increments and  $Y_t \sim \nu_t$  for  $t \geq 0$ .
- (3) There exists a Lévy process  $(Y_t)_{t \geq 0}$  such that  $Y_t \sim \nu_t$  for  $t \geq 0$ .

*Proof.* Item (1): By the standard proof and (9.2),

$$\int f(y) P_t(x, dy) = \int f(y+x) \nu_t(dy) \quad (9.3)$$

for any  $t \geq 0$  and  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  measurable and bounded. Thus, for  $s, t \geq 0$ ,  $x \in \mathbb{R}^d$ , and  $A \in \mathcal{B}(\mathbb{R}^d)$ ,

$$\begin{aligned} P_t P_s(x, A) &= \int P_s(y, A) P_t(x, dy) = \int \nu_s(A-y) P_t(x, dy) \\ &= \int \nu_s(A-y-x) \nu_t(dy) = \nu_t * \nu_s(A-x) \\ &= \nu_{t+s}(A-x) = P_{s+t}(x, A), \end{aligned}$$

where the first equality is by definition of  $P_t P_s$ , the third equality is by (9.3) with  $f(y) = P_s(y, A) = \nu_s(A-y)$  and the fifth equality is due to  $(\nu_t)_{t \geq 0}$  being a convolution semigroup. Since  $\nu_0 = \delta_0$ , (9.2) shows that  $P_0(x, \cdot) = \delta_x(\cdot)$ . Thus,  $(P_t)_{t \geq 0}$  is a transition semigroup.

Item (2): Let  $(Y_t)_{t \geq 0}$  be a Markov process with transition semigroup  $(P_t)_{t \geq 0}$ . By (9.3),

$$\mathbb{E}[f(Y_{s+t}) | \mathcal{F}_s^Y] = P_t f(Y_s) = \int f(y + Y_s) \nu_t(dy)$$

for all  $s, t \geq 0$  and  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  measurable and bounded. Thus,

$$\mathbb{E}[e^{i\langle \theta, Y_{t+s} \rangle} | \mathcal{F}_s^Y] = \int e^{i\langle \theta, y + Y_s \rangle} \nu_t(dy) = e^{i\langle \theta, Y_s \rangle} \int e^{i\langle \theta, y \rangle} \nu_t(dy)$$

for  $s, t \geq 0$  and  $\theta \in \mathbb{R}^d$ . Since  $Y_s$  is  $\mathcal{F}_s^Y$ -measurable it follows that

$$\mathbb{E}[e^{i\langle \theta, Y_{t+s} - Y_s \rangle} | \mathcal{F}_s^Y] = \int e^{i\langle \theta, y \rangle} \nu_t(dy).$$

By Exercise 70,  $Y_{t+s} - Y_s$  is independent of  $\mathcal{F}_s^Y$ , so by Proposition 9.6(2),  $(Y_t)_{t \geq 0}$  has independent increments. The same exercise shows that  $Y_{t+s} - Y_s \sim \nu_t$ . In particular,  $Y_{t+s} - Y_s \sim \nu_t \sim Y_t$  since  $Y_0 = 0$ , so  $(Y_t)_{t \geq 0}$  has stationary increments.

Conversely, let  $(Y_t)_{t \geq 0}$  have stationary independent increments and satisfy  $Y_t \sim \nu_t$  for  $t \geq 0$ . Let  $s, t \geq 0$  and  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be measurable and bounded. By Proposition 9.6(1),  $Y_{s+t} - Y_s$  is independent of  $\mathcal{F}_s^Y$ . Hence

$$\begin{aligned} \mathbb{E}[f(Y_{s+t}) | \mathcal{F}_s^Y] &= \mathbb{E}[f(Y_s + (Y_{s+t} - Y_s)) | \mathcal{F}_s^Y] = \mathbb{E}[f(x + (Y_{s+t} - Y_s)) | \mathcal{F}_s^Y]_{x=Y_s} \\ &= \int_{\mathbb{R}^d} f(x+y) \mathbb{P}_{Y_{s+t}-Y_s}(dy) \Big|_{x=Y_s} = \int_{\mathbb{R}^d} f(x+y) \nu_t(dy) \Big|_{x=Y_s} \\ &= P_t f(Y_s), \end{aligned}$$

where the second equality is by the useful rule Theorem E.7, the fourth equality is by the stationary increments and the definition of  $\nu_t$ , and the final equality is by (9.3). Thus  $(Y_t)_{t \geq 0}$  is a Markov process with transition semigroup  $(P_t)_{t \geq 0}$ .

*Item (3):* By items (1)-(2) and Theorem 7.4 there exists a process  $(Z_t)_{t \geq 0}$  with stationary independent increments satisfying  $Z_0 = 0$  a.s. and  $Z_t \sim \nu_t$  for all  $t \geq 0$ . In addition,  $(Z_t)_{t \geq 0}$  is continuous in probability by Exercise 105.

A process like  $(Z_t)_{t \geq 0}$  with stationary independent increments which is continuous in probability and starts at 0 a.s. is sometimes called a *Lévy process in law*. By Sato (2013, Theorem 11.5) any Lévy process in law has a càdlàg modification. That is, there is a modification  $(Y_t)_{t \geq 0}$  of  $(Z_t)_{t \geq 0}$  satisfying that  $t \mapsto Y_t(\omega)$  is càdlàg for every  $\omega \in \Omega$ . It follows that  $(Y_t)_{t \geq 0}$  is a Lévy process with  $Y_t \sim \nu_t$  for every  $t \geq 0$ .

Later we give a direct proof of the existence of the càdlàg modification  $(Y_t)_{t \geq 0}$  in some special cases. However, let us sketch a general argument, covering at least the case when  $Z_t$  is integrable. So, assume  $d = 1$  and that  $Z_t$  is integrable for all  $t > 0$ . For simplicity also assume  $\mathbb{E}[Z_t] = 0$  for  $t \geq 0$ . By Exercise 104,  $(Z_t)_{t \geq 0}$  is then an  $(\mathcal{F}_t^Z)_{t \geq 0}$ -martingale and is, as argued above, also continuous in probability. By the *Martingale Modification Theorem* (see e.g., Graversen (2011) or Rogers and Williams (2000) cf. also Theorem 13.1) such a process has a càdlàg modification.  $\square$

**Example 9.11 (A Gaussian convolution semigroup and Brownian motion).**

Let the state space be  $S = \mathbb{R}$ . Let  $\sigma^2 > 0$  and  $\mu \in \mathbb{R}$ , and define

$$\nu_t = N(\mu t, \sigma^2 t), \quad t \geq 0. \quad (9.4)$$

By Exercise 94,  $(\nu_t)_{t \geq 0}$  is a regular convolution semigroup.

Recall that the corresponding transition semigroup  $(P_t)$  is given by (9.2). Thus,

for  $t > 0$ ,  $x \in \mathbb{R}$  and  $B \in \mathcal{B}(\mathbb{R})$ ,

$$\begin{aligned} P_t(x, B) &= \nu_t(B - x) \\ &= N(\mu t, \sigma^2 t)(B - x) \\ &= N(x + \mu t, \sigma^2 t)(B) \\ &= \int_B \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left(-\frac{(y - x - \mu t)^2}{2\sigma^2 t}\right) dy. \end{aligned}$$

The interpretation is that, if we are in state  $x$ , then after  $t$  time units we are in  $B$  with probability  $P_t(x, B) = N(x + \mu t, \sigma^2 t)(B)$ .

The corresponding Lévy process is a Brownian motion with drift  $\mu$  and parameter  $\sigma^2$ . As we discuss in Chapter 10, it can be written as  $X_t = \mu t + \sigma B_t$ , where  $(B_t)_{t \geq 0}$  is a standard Brownian motion. ■

**Example 9.12 (A Poisson convolution semigroup).**

Let  $\lambda > 0$  and define

$$\nu_t = \text{po}(\lambda t), \quad t \geq 0.$$

(Here,  $\text{po}(0)$  is set to  $\delta_0$ .) Then for  $m, n \in \mathbb{N}_0$  with  $m \leq n$  and  $t > 0$ ,

$$P_t(m, \{n\}) = \nu_t(\{n - m\}) = e^{-\lambda t} \frac{(\lambda t)^{n-m}}{(n-m)!}.$$

The corresponding Lévy process is a Poisson process with parameter  $\lambda$ ; see Chapter 11. ■

## 9.3 Examples and simulation

Theorem 9.10 tells us that if  $(\nu_t)_{t \geq 0}$  is a regular convolution semigroup then there is a Lévy process  $(Y_t)_{t \geq 0}$  satisfying  $Y_t \sim \nu_t$  for  $t \geq 0$ . We say that  $(Y_t)_{t \geq 0}$  is a Lévy process with regular convolution semigroup  $(\nu_t)_{t \geq 0}$ . By Exercise 91, the distribution of  $(Y_t)_{t \geq 0}$  is uniquely determined in the sense that if  $(Z_t)_{t \geq 0}$  is another Lévy process with the same regular convolution semigroup then  $(Y_t)_{t \geq 0}$  and  $(Z_t)_{t \geq 0}$  are versions.

In this section we discuss simulation of Lévy processes and give some examples.

**Remark 9.13 (Simulation).**

Let  $(\nu_t)_{t \geq 0}$  be a regular convolution semigroup. We can simulate a Lévy process having this semigroup at discrete time-points in the following way:

First choose a step size  $\Delta > 0$  (usually  $\Delta$  is very small) and  $n \in \mathbb{N}$ , the number of observations to be generated. Then simulate  $Y_0, Y_\Delta, \dots, Y_{n\Delta}$  as follows:

- (1) Put  $Y_0 = 0$ ;
- (2) Let  $k = 0, 1, \dots, n-1$ . Assuming  $Y_0, \dots, Y_{k\Delta}$  have already been simulated, simulate  $Y_{(k+1)\Delta} - Y_{k\Delta}$  independent of  $Y_0, \dots, Y_{k\Delta}$  such that  $Y_{(k+1)\Delta} - Y_{k\Delta} \sim \nu_\Delta$ . Put  $Y_{(k+1)\Delta} = Y_{k\Delta} + [Y_{(k+1)\Delta} - Y_{k\Delta}]$ . ■

Although the above may seem straightforward, the problem is that in many interesting cases the semigroup  $(\nu_t)_{t \geq 0}$  is not known in explicit form, which makes the method in Remark 9.13 impossible; see Section 9.5.

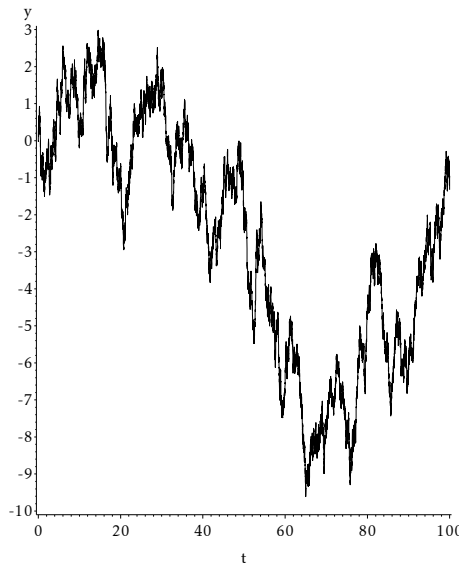
In the next examples the approach sketched in Remark 9.13 is feasible. Let  $d = 1$ .

**Example 9.14 (A Gaussian convolution semigroup).**

Let  $\sigma^2 > 0$  and define

$$\nu_t = N(0, \sigma^2 t), \quad t \geq 0. \quad (9.5)$$

By Exercise 94,  $(\nu_t)_{t \geq 0}$  is a regular convolution semigroup. A Lévy process  $(Y_t)_{t \geq 0}$  having this regular convolution semigroup has stationary independent increments and  $Y_t - Y_s \sim N(0, \sigma^2(t-s))$  for  $0 \leq s \leq t$ . Figure 9.1 presents a simulation of  $(Y_t)_{t \geq 0}$  with  $\sigma^2 = 1$  where  $\Delta = 0.01$  and  $n\Delta = 100$ . The figure indicates that the path  $t \mapsto Y_t$  is continuous but not too smooth. In Chapter 10 we show that almost all paths of  $(Y_t)_{t \geq 0}$  are continuous but not differentiable at any point. ■



**Figure 9.1:** A simulation of a Lévy process with convolution semigroup (9.5).

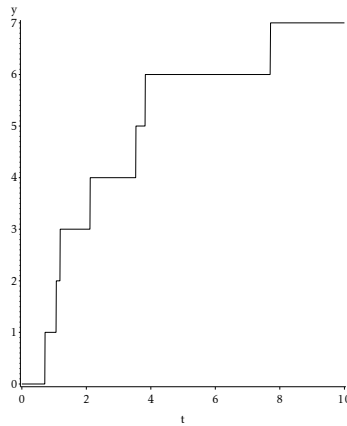
**Example 9.15 (A Poisson convolution semigroup).**

Let  $\lambda > 0$  and define

$$\nu_t = \text{po}(\lambda t), \quad t \geq 0. \quad (9.6)$$

■

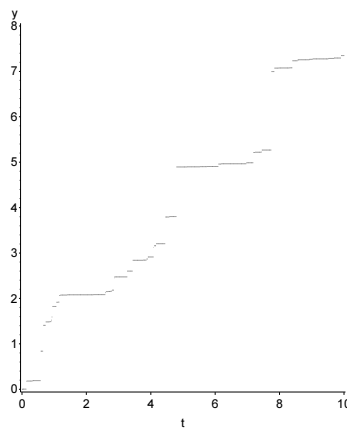
(Here,  $\text{po}(0)$  is set to  $\delta_0$ .) By Exercise 94,  $(\nu_t)_{t \geq 0}$  is a regular convolution semigroup. Figure 9.2 presents a simulation of the corresponding Lévy process with  $\lambda = 1$  when  $\Delta = 0.01$  and  $n\Delta = 10$ . It seems that  $t \mapsto Y_t$  is integer-valued and non-decreasing and that all jumps are of size +1. In Chapter 11 we show that this is indeed the case and, conversely, that any Lévy process having these properties has convolution semigroup given by (9.6).



**Figure 9.2:** A simulation of a Lévy process with convolution semigroup (9.6).

**Example 9.16 (A Gamma convolution semigroup).**

Let  $\alpha, \beta > 0$ , and let  $\nu_t = \Gamma(\alpha t, \beta)$  for  $t > 0$  and  $\nu_0 = \delta_0$ . It is readily seen that  $(\nu_t)_{t \geq 0}$  is a regular convolution semigroup. A simulation of the corresponding Lévy process with  $\alpha = \beta = 1$  is plotted in Figure 9.3. It can be shown that in this case, almost surely,  $t \mapsto Y_t$  is non-decreasing and there are infinitely many jumps on any non-empty open interval of  $(0, \infty)$ .



**Figure 9.3:** A simulation of a Lévy process with convolution semigroup given in Example 9.16.

## 9.4 The strong Markov property

In this section,  $(Y_t)_{t \geq 0}$  denotes an  $(\mathcal{F}_t)_{t \geq 0}$ -Lévy process. For an extended stopping time  $\tau$  define  $(D_\tau Y_t)_{t \geq 0}$  as  $D_\tau Y_t = (Y_{\tau+t} - Y_\tau) \mathbb{1}_{\{\tau < \infty\}}$  for  $t \geq 0$ . By Proposition 8.8(3) and Proposition 8.10(5),  $D_\tau Y_t$  is  $\mathcal{F}_{(\tau+t)+}$ -measurable for  $t \geq 0$ .

### Theorem 9.17 (The strong Markov property).

Let  $(Y_t)_{t \geq 0}$  be an  $(\mathcal{F}_t)_{t \geq 0}$ -Lévy process and  $\tau$  be an extended stopping time with respect to  $(\mathcal{F}_t)_{t \geq 0}$ . Assume  $\tau$  is a.s. finite. Then

- (1)  $(D_\tau Y_t)_{t \geq 0}$  is independent of  $\mathcal{F}_{\tau+}$ ;
- (2)  $(D_\tau Y_t)_{t \geq 0}$  is a Lévy process;
- (3)  $(D_\tau Y_t)_{t \geq 0}$  and  $(Y_t)_{t \geq 0}$  are versions.

**Remark 9.18.** Let  $r \in [0, \infty)$  and put  $\tau = r$ . Then  $\tau$  is an extended stopping time and  $\mathcal{F}_{\tau+} = \mathcal{F}_+$ . In this case the theorem says that  $(Y_{r+t} - Y_r)_{t \geq 0}$  is a Lévy process independent of  $\mathcal{F}_{r+}$ , and that  $(Y_t)_{t \geq 0}$  and  $(Y_{r+t} - Y_r)_{t \geq 0}$  are versions. ■

We follow the proof of Graversen (2011, Theorem 6.3) closely.

*Proof of Theorem 9.17.* First of all we note that for  $\omega \in \Omega$ ,  $D_\tau Y_0(\omega) = 0$ , and that  $t \mapsto D_\tau Y_t(\omega) = (Y_{\tau(\omega)+t}(\omega) - Y_{\tau(\omega)}(\omega)) \cdot \mathbb{1}_{\{\tau < \infty\}}(\omega)$  is càdlàg.

Let us show that

$$\mathbb{E}[e^{i\langle \theta, D_\tau Y_t - D_\tau Y_s \rangle} \mathbb{1}_A] = \mathbb{P}(A) \mathbb{E}[e^{i\langle \theta, Y_{t-s} \rangle}] \quad \text{for } \theta \in \mathbb{R}^d, 0 \leq s < t \text{ and } A \in \mathcal{F}_{(\tau+s)+}. \quad (9.7)$$

Put  $\sigma = \tau + s$  and note that  $D_\tau Y_t - D_\tau Y_s = D_\sigma Y_{t-s}$ . Let  $(\sigma_n)_{n \geq 1}$  denote the discrete approximation of  $\sigma$ , cf. Remark 8.12. By right continuity of the paths,

$$D_\sigma Y_{t-s} = (Y_{\sigma+t-s} - Y_\sigma) \mathbb{1}_{\{\sigma < \infty\}} = \lim_{n \rightarrow \infty} (Y_{\sigma_n+t-s} - Y_{\sigma_n}) \mathbb{1}_{\{\sigma_n < \infty\}} \quad \text{for every } \omega \in \Omega. \quad (9.8)$$

Thus, since  $\sigma_n$  and  $\sigma$  are a.s. finite,

$$\mathbb{E}[e^{i\langle \theta, D_\tau Y_t - D_\tau Y_s \rangle} \mathbb{1}_A] = \mathbb{E}[e^{i\langle \theta, D_\sigma Y_{t-s} \rangle} \mathbb{1}_{A \cap \{\sigma < \infty\}}] = \lim_{n \rightarrow \infty} \mathbb{E}[e^{i\langle \theta, Y_{\sigma_n+t-s} - Y_{\sigma_n} \rangle} \mathbb{1}_{A \cap \{\sigma_n < \infty\}}],$$

where the last equality uses (9.8) and the Dominated Convergence Theorem. For  $n \in \mathbb{N}$ ,

$$\begin{aligned}
& \mathbb{E}[e^{i\langle \theta, Y_{\sigma_n+t-s} - Y_{\sigma_n} \rangle} \mathbb{1}_{A \cap \{\sigma_n < \infty\}}] \\
&= \mathbb{E}\left[\sum_{k=1}^{\infty} e^{i\langle \theta, Y_{\frac{k}{2^n}+t-s} - Y_{\frac{k}{2^n}} \rangle} \mathbb{1}_{A \cap \{\sigma_n = \frac{k}{2^n}\}}\right] \\
&= \sum_{k=1}^{\infty} \mathbb{E}[e^{i\langle \theta, Y_{\frac{k}{2^n}+t-s} - Y_{\frac{k}{2^n}} \rangle} \mathbb{1}_{A \cap \{\sigma_n = \frac{k}{2^n}\}}] \\
&= \sum_{k=1}^{\infty} \mathbb{P}(A \cap \{\sigma_n = \frac{k}{2^n}\}) \mathbb{E}[e^{i\langle \theta, Y_{\frac{k}{2^n}+t-s} - Y_{\frac{k}{2^n}} \rangle}] \\
&= \sum_{k=1}^{\infty} \mathbb{P}(A \cap \{\sigma_n = \frac{k}{2^n}\}) \mathbb{E}[e^{i\langle \theta, Y_{t-s} \rangle}] \\
&= \mathbb{E}[e^{i\langle \theta, Y_{t-s} \rangle}] \mathbb{P}(A),
\end{aligned}$$

where the second equality uses the Dominated Convergence Theorem and the third that  $A \cap \{\sigma_n = \frac{k}{2^n}\} \in \mathcal{F}_{k/2^n}$  (see Remark 8.12) and that the latter  $\sigma$ -field is independent of  $Y_{\frac{k}{2^n}+t-s} - Y_{\frac{k}{2^n}}$ . This concludes the proof of (9.7).

By (9.7) with  $A = \Omega$ , it follows that  $D_\tau Y_t - D_\tau Y_s \stackrel{d}{=} Y_{t-s}$ . Likewise, with  $s = 0$  and  $t$  replaced by  $t - s$  it follows that  $D_\tau Y_{t-s} \stackrel{d}{=} Y_{t-s}$ . That is,

$$D_\tau Y_t - D_\tau Y_s \stackrel{d}{=} Y_{t-s} \stackrel{d}{=} D_\tau Y_{t-s} \quad \text{for } 0 \leq s \leq t, \quad (9.9)$$

so  $(D_\tau Y_t)_{t \geq 0}$  has stationary increments.

By (9.7), (9.9) and Exercise 69,  $(D_\tau Y_t)_{t \geq 0}$  and the filtration  $(\mathcal{F}_{(\tau+t)_+})_{t \geq 0}$  satisfies condition (5) in Definition 9.5.

In conclusion, we have verified that  $(D_\tau Y_t)_{t \geq 0}$  is an  $(\mathcal{F}_{(\tau+t)_+})_{t \geq 0}$ -Lévy process, so in particular it is a Lévy process. By Proposition 9.6(2),  $(D_\tau Y_t)_{t \geq 0}$  is independent of  $\mathcal{F}_{\tau+}$ . By (9.9) and Exercise 91,  $(D_\tau Y_t)_{t \geq 0}$  and  $(Y_t)_{t \geq 0}$  are versions.  $\square$

When  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$  is a filtered probability space, let  $\mathcal{N} = \{N \in \mathcal{F} : \mathbb{P}(N) = 0\}$  and  $\overline{\mathcal{F}}_t = \sigma(\mathcal{F}_t \cup \mathcal{N})$  for  $t \geq 0$ . That is,  $\overline{\mathcal{F}}_t$  is the least  $\sigma$ -field containing  $\mathcal{F}_t$  and all measurable null sets. When  $\mathcal{F}_t = \mathcal{F}_t^Y$  for some stochastic process  $(Y_t)_{t \geq 0}$  we write  $\overline{\mathcal{F}}_t^Y$  rather than  $\overline{\mathcal{F}}_t$ .

**Corollary 9.19.** *Let  $(Y_t)_{t \geq 0}$  denote a  $d$ -dimensional Lévy process.*

- (1) *The filtration  $(\overline{\mathcal{F}}_t^Y)_{t \geq 0}$  is right-continuous.*
- (2) *(Blumenthal's 0-1 law) If  $A \in \mathcal{F}_{0+}^Y$  then  $\mathbb{P}(A) \in \{0, 1\}$ .*

*Proof.* For the proof of item (1) see Exercise 89. In item (2) use Theorem 9.17 with  $\tau = 0$  to infer that  $\mathcal{F}_{0+}^Y$  is independent of  $(Y_t - Y_0)_{t \geq 0}$ . Since  $Y_0 = 0$  a.s.,  $\mathcal{F}_{0+}^Y$  is independent of  $(Y_t)_{t \geq 0}$  as well. In particular  $\mathcal{F}_{0+}^Y$  is independent of itself, which implies the result.  $\square$

## 9.5 Further remarks

In this section, we discuss the structure of Lévy processes in further detail, focusing particularly on the relations to the important class of *infinitely divisible distributions*. The exposition will be brief, and I refer to the course *Lévy processes* for more information. See also the references given at the beginning of the chapter.

**Definition 9.20.** A probability measure  $\mu$  on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  is called an **infinitely divisible distribution** if for every  $n \in \mathbb{N}$  there is a probability measure  $\mu_n$  on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  such that  $\mu = \mu_n * \dots * \mu_n$  ( $n$  times).

By (Sato, 2013, Lemma 7.5), if  $\mu$  is infinitely divisible, then  $\widehat{\mu}(z) \neq 0$  for every  $z \in \mathbb{R}^d$ . Moreover, as  $\widehat{\mu}$  is a characteristic function, the map  $z \mapsto \widehat{\mu}(z)$  is continuous and  $\widehat{\mu}(0) = 1$ . These properties make it possible to define a logarithm of the (complex-valued) function  $\widehat{\mu}$ :

**Proposition 9.21.** Let  $\mu$  be an infinitely divisible distribution.

- (1) There exists a unique continuous function from  $\mathbb{R}^d$  to  $\mathbb{C}$ , denoted by  $\log \widehat{\mu}$ , satisfying  $\log \widehat{\mu}(0) = 0$  and  $\exp(\log \widehat{\mu}(z)) = \widehat{\mu}(z)$  for  $z \in \mathbb{R}^d$ .
- (2) The probability measure  $\mu_n$  in Definition 9.20 is uniquely determined as

$$\widehat{\mu}_n(z) = \exp\left[\frac{1}{n} \log \widehat{\mu}(z)\right], \quad z \in \mathbb{R}^d.$$

For the proof of this result, see Sato (2013, Lemma 7.5-7.6). Let us study the relations between infinitely divisible distributions and Lévy processes, keeping in mind that the latter can be represented by regular convolution semigroups.

**Proposition 9.22.** Let  $(\nu_t)_{t \geq 0}$  be a regular convolution semigroup on  $\mathbb{R}^d$ . Then  $\nu_t$  is infinitely divisible for every  $t \geq 0$ .

In particular, whenever  $(Y_t)_{t \geq 0}$  is a Lévy process, the distribution  $\mathbb{P}_{Y_t}$  is infinitely divisible for  $t \geq 0$ .

To prove the first part of this result, use that, for  $n \in \mathbb{N}$ ,  $\nu_t = \nu_{\frac{t}{n}} * \dots * \nu_{\frac{t}{n}}$  ( $n$  times). In the second part recall from Theorem 9.9 that  $(\mathbb{P}_{Y_t})_{t \geq 0}$  is a regular convolution semigroup.

Next we give a famous characterization of the class of infinitely divisible distributions, called the Lévy-Khintchine representation:

**Theorem 9.23 (Lévy-Khintchine).**

If  $\mu$  is infinitely divisible then there exists a unique triplet  $(\Sigma, \nu, \gamma)$ , called *the characteristic triplet of  $\mu$* , such that

$$\widehat{\mu}(z) = \exp\left[-\frac{1}{2}z\Sigma z^\top + i\langle\gamma, z\rangle + \int_{\mathbb{R}^d} g(z, x) \nu(dx)\right], \quad z \in \mathbb{R}^d, \quad (9.10)$$

where  $\top$  denotes the transpose,  $D = \{x \in \mathbb{R}^d : \|x\| \leq 1\}$  and

$$g(z, x) = e^{i\langle z, x \rangle} - 1 - i\langle z, x \rangle \mathbb{1}_D(x).$$

Moreover,  $\Sigma$  is a  $d \times d$  non-negative definite matrix, and  $\nu$  is a Lévy measure on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ , i.e.,  $\nu(\{0\}) = 0$  and  $\int_{\mathbb{R}^d} (1 \wedge \|x\|^2) \nu(dx) < \infty$ . Finally,  $\gamma \in \mathbb{R}^d$  is a constant. In (9.10),  $z$  is treated as a row vector.

Conversely, given such a characteristic triplet there exists an infinitely divisible distribution  $\mu$  on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  such that (9.10) holds.

Using  $\log \widehat{\mu}$ , Equation (9.10) can be written as

$$\log \widehat{\mu}(z) = -\frac{1}{2}z\Sigma z^\top + i\langle\gamma, z\rangle + \int_{\mathbb{R}^d} g(z, x) \nu(dx), \quad z \in \mathbb{R}^d.$$

This follows from the fact that the right-hand side is continuous as a function of  $z$  and is equal to 0 for  $z = 0$ .

We have seen that there is a one-to-one correspondence between regular convolution semigroups and Lévy processes, and that these induce an infinitely divisible distribution, as described in Proposition 9.22. There is a converse:

**Theorem 9.24.** *Let  $\mu$  be an infinitely divisible distribution with characteristic triplet  $(\Sigma, \nu, \gamma)$ . Then there exists a unique regular convolution semigroup  $(\nu_t)_{t \geq 0}$  satisfying  $\nu_1 = \mu$ . The characteristic function of  $\nu_t$  is given by*

$$\widehat{\nu}_t(z) = \exp\left[t\left(-\frac{1}{2}z\Sigma z^\top + i\langle\gamma, z\rangle + \int_{\mathbb{R}^d} g(z, x) \nu(dx)\right)\right], \quad z \in \mathbb{R}^d. \quad (9.11)$$

*Proof.* For  $t \geq 0$  there is a (unique) probability measure  $\nu_t$  on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  with characteristic function (9.11). This follows from Theorem 9.23 since  $(t\Sigma, t\nu, t\gamma)$  is a characteristic triplet. From (9.11) it is immediate that  $\widehat{\nu}_{t+s} = \widehat{\nu}_t \widehat{\nu}_s$  for  $s, t \geq 0$  and that  $\widehat{\nu}_{t_n}(z) \rightarrow 1$  as  $n \rightarrow \infty$  for every  $z \in \mathbb{R}^d$  and  $t_n \rightarrow 0$ , so  $(\nu_t)_{t \geq 0}$  is a regular convolution semigroup satisfying  $\nu_1 = \mu$ .

Let  $(\widetilde{\nu}_t)_{t \geq 0}$  denote a regular convolution semigroup with  $\widetilde{\nu}_1 = \mu$ . For  $n \in \mathbb{N}$ ,  $\widetilde{\nu}_{\frac{1}{n}} * \cdots * \widetilde{\nu}_{\frac{1}{n}} = \widetilde{\nu}_1 = \mu$  (where there are  $n$  convolutions). Thus, by Proposition 9.21(2),  $\widetilde{\nu}_{\frac{1}{n}} = \nu_{\frac{1}{n}}$  for  $n \in \mathbb{N}$ . Since  $\widetilde{\nu}_{\frac{m}{n}} = \widetilde{\nu}_{\frac{1}{n}} * \cdots * \widetilde{\nu}_{\frac{1}{n}}$  ( $m$  times) for  $m, n \in \mathbb{N}$  we see that  $\widetilde{\nu}_t = \nu_t$  for every  $t \in \mathbb{Q}_+$ . Finally, by Lemma 9.8,  $\widetilde{\nu}_t = \nu_t$  for  $t \geq 0$ .  $\square$

Another way of expressing this result is by saying that given an infinitely divisible distribution  $\mu$  there exists a unique (in distribution) Lévy process  $(Y_t)_{t \geq 0}$  such that  $Y_1 \sim \mu$ . The distribution of  $Y_t$  for  $t > 0$  is then  $\nu_t$ , where the characteristic function of  $\nu_t$  is given by (9.11).

To sum up, a Lévy process  $(Y_t)_{t \geq 0}$  can be specified by

- the convolution semigroup  $(\nu_t)_{t \geq 0}$ ;
- an infinitely divisible  $\mu$  and then defining the convolution semigroup by (9.11);
- a characteristic triplet  $(\Sigma, \nu, \gamma)$  and then defining the convolution semigroup by (9.11).

An interesting question is the probabilistic role of the Lévy measure  $\nu$ . This will be studied in detail in the course *Lévy processes*, so let me just mention a few properties:

- $(Y_t)_{t \geq 0}$  has a.s. continuous paths if and only if  $\nu = 0$ ;
- On any non-empty, open, bounded interval of  $(0, \infty)$ ,  $(Y_t)_{t \geq 0}$  has a.s. finitely many jumps if and only if  $\nu$  is a finite measure.
- For  $t > 0$ ,  $\sum_{0 < s \leq t} \|\Delta Y_s\| < \infty$  a.s. (i.e. the jumps converge absolutely) if and only if  $\int_{\mathbb{R}^d} (1 \wedge \|x\|) \nu(dx) < \infty$ .
- In general,  $\sum_{0 < s \leq t} \|\Delta Y_s\|^2 < \infty$  a.s.

Finally, let me emphasize that, if one starts by specifying the characteristic triplet  $(\Sigma, \nu, \gamma)$ , the problem is that, although the characteristic function is given by (9.11), the probability measure  $\nu_t$  is generally not explicitly known. As a consequence, simulation and analysis of  $(Y_t)_{t \geq 0}$  is often a non-trivial task.

# Brownian Motion

This chapter concerns one of the most important Lévy processes, namely Brownian motion.

## 10.1 Definitions and the strong Markov property

**Definition 10.1.** Let  $\sigma^2 > 0$  and  $\mu \in \mathbb{R}$ . A real stochastic process  $(Y_t)_{t \geq 0}$  is called a **Brownian motion with parameter  $\sigma^2$  and drift  $\mu$**  if

- (1)  $Y_0(\omega) = 0$  for almost all  $\omega \in \Omega$ .
- (2)  $t \mapsto Y_t(\omega)$  is continuous for every  $\omega \in \Omega$ .
- (3)  $Y_t - Y_s \sim N(\mu(t-s), \sigma^2(t-s))$  for all  $0 \leq s < t$ .
- (4)  $(Y_t)_{t \geq 0}$  has independent increments.

When  $\mu = 0$  we also call  $(Y_t)_{t \geq 0}$  a **Brownian motion with parameter  $\sigma^2$** . If in addition  $\sigma^2 = 1$  then we speak of a **standard Brownian motion**.

In the literature, Brownian motions are also called *Wiener processes*.

A typical path of a Brownian motion (without drift) is depicted in Figure 9.1.

**Remark 10.2.** Assuming condition (1), condition (3) in Definition 10.1 is equivalent to the following

$$Y_t \sim N(\mu t, \sigma^2 t) \text{ for all } t \geq 0 \text{ and } (Y_t)_{t \geq 0} \text{ has stationary increments.} \quad (3^*)$$

In particular, this shows that a Brownian motion is a Lévy process with regular convolution semigroup  $\nu_t = N(\mu t, \sigma^2 t)$ . Conversely, a Lévy process satisfying (2) and (3\*) is a Brownian motion with drift  $\mu$  and parameter  $\sigma^2$ .

To see that (3) and (3\*) are equivalent under (1), first assume that (3) holds. Then (1) and (3) with  $s = 0$  imply  $Y_t = Y_t - Y_0 \sim N(\mu t, \sigma^2 t)$ , where the equality holds a.s. Moreover, for  $0 \leq s < t$ ,

$$Y_t - Y_s \sim N(\mu(t-s), \sigma^2(t-s)) \sim Y_{t-s}$$

so  $(Y_t)_{t \geq 0}$  has stationary increments. Conversely, assume (3\*). Then for  $0 \leq s < t$ ,  $Y_t - Y_s \stackrel{d}{=} Y_{t-s} \sim N(\mu(t-s), \sigma^2(t-s))$ , which is exactly (3). ■

**Remark 10.3.** Brownian motions without drift will often be denoted by  $(B_t)_{t \geq 0}$ .

If  $(B_t)_{t \geq 0}$  is a standard Brownian motion then  $(Y_t)_{t \geq 0}$  defined as  $Y_t = \mu t + \sigma B_t$  is a Brownian motion with parameter  $\sigma^2 > 0$  and drift  $\mu$ . Conversely, if  $(Y_t)_{t \geq 0}$  is a Brownian motion with parameter  $\sigma^2 > 0$  and drift  $\mu$  then  $(B_t)_{t \geq 0}$  defined as  $B_t = (Y_t - \mu t)/\sigma$  is a standard Brownian motion.

Let us give another characterization of Brownian motions.

**Proposition 10.4.** Let  $\sigma^2 > 0$  and  $\mu \in \mathbb{R}$ . A real stochastic process  $(Y_t)_{t \geq 0}$  satisfies (1), (3) and (4) in Definition 10.1 if and only if  $(Y_t)_{t \geq 0}$  is a Gaussian process with  $\mathbb{E}[Y_t] = \mu t$  and  $\text{Cov}(Y_s, Y_t) = \sigma^2(s \wedge t)$  for all  $s, t \geq 0$ .

Recall that  $(Y_t)_{t \geq 0}$  is Gaussian if  $(Y_{t_1}, \dots, Y_{t_n})^\top$  follows an  $n$ -variate normal distribution for every  $n \in \mathbb{N}$  and  $t_1, \dots, t_n \geq 0$ . (It suffices even to consider the case  $0 \leq t_1 \leq \dots \leq t_n$ .)

*Proof.* The proof relies on basic properties of multivariate normal distributions, see Chapter F.

First assume that  $(Y_t)_{t \geq 0}$  satisfies (1), (3) and (4). Let  $n \in \mathbb{N}$  and  $0 \leq t_1 \leq \dots \leq t_n$ . By assumption,  $Y_{t_1}, Y_{t_2} - Y_{t_1}, \dots, Y_{t_n} - Y_{t_{n-1}}$  are independent Gaussian random variables. Since

$$\begin{pmatrix} Y_{t_1} \\ Y_{t_2} \\ \vdots \\ Y_{t_n} \end{pmatrix} = \begin{pmatrix} Y_{t_1} \\ Y_{t_1} + [Y_{t_2} - Y_{t_1}] \\ \vdots \\ Y_{t_1} + \sum_{j=2}^n [Y_{t_j} - Y_{t_{j-1}}] \end{pmatrix},$$

the left-hand side follows an  $n$ -variate normal distribution. That is,  $(Y_t)_{t \geq 0}$  is Gaussian. By (3\*),  $\mathbb{E}[Y_t] = \mu t$  for  $t \geq 0$ . Since  $Y_t - Y_s$  is independent of  $Y_s$  for  $0 \leq s \leq t$ ,

$$\text{Cov}(Y_s, Y_t) = \text{Cov}(Y_s, Y_s) + \text{Cov}(Y_s, Y_t - Y_s) = \text{Var}(Y_s) = \sigma^2 s = \sigma^2(s \wedge t).$$

Conversely, assume  $(Y_t)_{t \geq 0}$  is a Gaussian process satisfying  $\mathbb{E}[Y_t] = \mu t$  and  $\text{Cov}(Y_s, Y_t) = \sigma^2(s \wedge t)$  for  $s, t \geq 0$ . Then  $\text{Var}(Y_0) = \text{Cov}(Y_0, Y_0) = 0$ , so  $Y_0$  is a.s. constant. Since  $\mathbb{E}[Y_0] = 0$ , it follows that  $Y_0 = 0$  a.s., i.e. (1).

To verify (3), fix  $0 \leq s < t$ . Then  $\mathbb{E}[Y_t - Y_s] = \mu(t - s)$  and

$$\text{Var}(Y_t - Y_s) = \text{Var}(Y_t) + \text{Var}(Y_s) - 2\text{Cov}(Y_s, Y_t) = \sigma^2(t + s - 2s) = \sigma^2(t - s).$$

By Gaussianity,  $(Y_s, Y_t)^\top$  is bivariate normal; therefore, by linearity,  $Y_t - Y_s$  is normal. Since we have already calculated the mean and variance, it follows that  $Y_t - Y_s \sim N(\mu(t - s), \sigma^2(t - s))$ ; that is, we have (3).

Finally, let us verify (4), the independent increments. Let  $n \in \mathbb{N}$  and  $0 = t_0 \leq t_1 \leq \dots \leq t_n$ . Then

$$(Y_{t_1} - Y_{t_0}, Y_{t_2} - Y_{t_1}, \dots, Y_{t_n} - Y_{t_{n-1}})^\top \stackrel{\text{a.s.}}{=} (Y_{t_1}, Y_{t_2} - Y_{t_1}, \dots, Y_{t_n} - Y_{t_{n-1}})^\top \quad (10.1)$$

follows an  $n$ -variate normal distribution since it is a linear function of the vector  $(Y_{t_0}, Y_{t_1}, \dots, Y_{t_n})^\top$ , which follows an  $(n + 1)$ -variate normal distribution by Gaussianity. To verify independent increments, it hence suffices to check that all covariances of the variables in (10.1) are 0. Let  $i, j = 1, \dots, n$  with  $i < j$ . Then

$$\begin{aligned} & \text{Cov}(Y_{t_i} - Y_{t_{i-1}}, Y_{t_j} - Y_{t_{j-1}}) \\ &= \text{Cov}(Y_{t_i}, Y_{t_j}) - \text{Cov}(Y_{t_i}, Y_{t_{j-1}}) - \text{Cov}(Y_{t_{i-1}}, Y_{t_j}) + \text{Cov}(Y_{t_{i-1}}, Y_{t_{j-1}}) \\ &= \sigma^2(t_i - t_i - t_{i-1} + t_{i-1}) = 0. \end{aligned}$$

This completes the proof.  $\square$

Brownian motions with respect to a filtration are defined as follows:

**Definition 10.5.** Let  $\sigma^2 > 0$ ,  $\mu \in \mathbb{R}$  and  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$  denote a filtered probability space. A real stochastic process  $(Y_t)_{t \geq 0}$  is called an  $(\mathcal{F}_t)_{t \geq 0}$ -**Brownian motion with parameter  $\sigma^2$  and drift  $\mu$**  if

- (1)  $Y_0(\omega) = 0$  for almost all  $\omega \in \Omega$ .
- (2)  $t \mapsto Y_t(\omega)$  is continuous for every  $\omega \in \Omega$ .
- (3)  $Y_t - Y_s \sim N(\mu(t - s), \sigma^2(t - s))$  for  $0 \leq s < t$ .
- (4)  $(Y_t)_{t \geq 0}$  is  $(\mathcal{F}_t)_{t \geq 0}$ -adapted.
- (5)  $Y_t - Y_s$  is independent of  $\mathcal{F}_s$  for  $0 \leq s \leq t$ .

When  $\mu = 0$  and  $\sigma^2 = 1$ , we also call  $(Y_t)_{t \geq 0}$  a *standard  $(\mathcal{F}_t)_{t \geq 0}$ -Brownian motion*.

**Remark 10.6.** As argued in Remark 10.2, condition (3) is equivalent to (3\*) on page 85 under condition (1).

From Proposition 9.6(3) it follows that  $(Y_t)_{t \geq 0}$  is a Brownian motion with parameter  $\sigma^2$  and drift  $\mu$  if and only if it is an  $(\mathcal{F}_t^Y)_{t \geq 0}$ -Brownian motion with parameter  $\sigma^2$  and drift  $\mu$ . ■

For Brownian motions, the strong Markov property for Lévy processes (see Theorem 9.17) simplifies as follows.

**Theorem 10.7.** *Let  $(Y_t)_{t \geq 0}$  be an  $(\mathcal{F}_t)_{t \geq 0}$ -Brownian motion with drift  $\mu$  and parameter  $\sigma^2$ . Let  $\tau$  be an extended stopping time with respect to  $(\mathcal{F}_t)_{t \geq 0}$  and assume  $\tau$  is a.s. finite. Then  $(D_\tau Y_t)_{t \geq 0}$  is a Brownian motion with drift  $\mu$  and parameter  $\sigma^2$  which is independent of  $\mathcal{F}_{\tau+}$ .*

Recall that  $D_\tau Y_t = (Y_{\tau+t} - Y_\tau) \mathbb{1}_{\{\tau < \infty\}}$  for  $t \geq 0$ .

## 10.2 Existence and properties

For simplicity, we concentrate in this section on Brownian motions without drift; however, using Remark 10.3 most results can be translated to the case with drift.

**Theorem 10.8.** *Let  $\sigma^2 > 0$ . There exists a Brownian motion  $(B_t)_{t \geq 0}$  with parameter  $\sigma^2$ . We have the following properties.*

- (1) *Almost surely,  $t \mapsto B_t$  is locally Hölder continuous with exponent  $\gamma$  for every  $\gamma \in (0, 1/2)$  and  $t \mapsto B_t$  is not locally Hölder continuous with exponent  $\gamma \geq 1/2$ . Moreover, almost surely,  $t \mapsto B_t$  is not differentiable at any point  $t \in (0, \infty)$ .*
- (2) *Let  $T_{0+} = \inf\{t \geq 0 : B_t > 0\}$  and  $T_{0-} = \inf\{t \geq 0 : B_t < 0\}$ . Then*

$$\mathbb{P}(T_{0+} = 0 = T_{0-}) = 1.$$

- (3) *For  $\omega \in \Omega$  let  $H(\omega) = \{t \geq 0 : B_t(\omega) = 0\}$ . Then for almost every  $\omega$ ,  $H(\omega)$  is a closed subset of  $\mathbb{R}_+$  with no isolated points and is of Lebesgue measure 0. Moreover,  $H(\omega)$  is uncountable for almost every  $\omega$ .*

In the proof of (1) we actually construct a specific Brownian motion satisfying that  $t \mapsto B_t(\omega)$  is locally Hölder continuous with exponent  $\gamma$  for every  $\gamma \in (0, 1/2)$  and every  $\omega \in \Omega$ . Using this it is then easily seen that an arbitrary Brownian motion satisfies (1).

Item (2) tells us that in any non-empty open neighborhood of  $t = 0$  the path  $t \mapsto B_t$  is infinitely often above and below 0 almost surely; in particular, level  $B_t = 0$  is hit infinitely often.

*Proof.* Let  $\nu_t = N(0, \sigma^2 t)$  for  $t \geq 0$  which gives a regular convolution semigroup by Exercise 94. By Theorem 9.10(1)–(2) and Theorem 7.4 there exists a process  $(Y_t)_{t \geq 0}$  with stationary independent increments such that  $Y_t \sim N(0, \sigma^2 t)$  for  $t \geq 0$ . The existence of such a process is also granted by Proposition 10.4 and Exercise 17.

Since  $Y_t - Y_s \sim N(0, \sigma^2(t-s))$  for  $0 \leq s \leq t$ , there exists, for  $n \in \mathbb{N}$ , a constant  $C_n$  (not depending on  $s$  and  $t$ ) such that  $\mathbb{E}[|Y_t - Y_s|^{2n}] = C_n \sigma^{2n} |t-s|^n$  for  $s, t \geq 0$ . By Kolmogorov-Chentsov's Theorem, see Theorem 14.3 (cf. also Karatzas and Shreve (1991, Theorem 2.8)), with  $\alpha = 2n$  and  $\beta = n-1$  for  $n \in \mathbb{N} \setminus \{1\}$ , there exists, for every  $n \in \mathbb{N} \setminus \{1\}$ , a modification  $(B_t^{(n)})_{t \geq 0}$  of  $(Y_t)_{t \geq 0}$  satisfying that  $t \mapsto B_t^{(n)}(\omega)$  is locally Hölder continuous with exponent  $\gamma$  for every  $\gamma \in (0, (n-1)/(2n))$  and  $\omega \in \Omega$ . In particular  $(B_t^{(n)})_{t \geq 0}$  and  $(B_t^{(m)})_{t \geq 0}$  are modifications and hence indistinguishable (why?) for arbitrary  $n, m \in \mathbb{N} \setminus \{1\}$ . Choose  $N \in \mathcal{F}$  with  $\mathbb{P}(N) = 0$  such that  $B_t^{(n)}(\omega) = B_t^{(2)}(\omega)$  for every  $t \geq 0$ ,  $n \in \mathbb{N} \setminus \{1\}$ , and  $\omega \in N^c$ . Put

$$B_t(\omega) = \begin{cases} B_t^{(2)}(\omega) & \text{for } \omega \in N^c \text{ and } t \geq 0 \\ 0 & \text{for } \omega \in N. \end{cases}$$

Then  $(B_t)_{t \geq 0}$  is a continuous modification of  $(Y_t)_{t \geq 0}$  and, by letting  $n \rightarrow \infty$  above, it follows that every path of  $(B_t)_{t \geq 0}$  is locally Hölder continuous with exponent  $\gamma$  for every  $\gamma \in (0, 1/2)$ .

It is shown in Durrett (2010) and Schilling and Partzsch (2014) that, almost surely,  $t \mapsto B_t$  is not differentiable at any point and not locally Hölder continuous with exponent  $\gamma \geq 1/2$ .

*Item (2):* Note that for  $t > 0$ ,  $\mathbb{P}(T_{0+} < t) \geq \mathbb{P}(B_t > 0) = 1/2$ . Thus,  $\mathbb{P}(T_{0+} = 0) = \lim_{t \downarrow 0} \mathbb{P}(T_{0+} < t) \geq 1/2$ . Since  $\{T_{0+} = 0\} \in \mathcal{F}_{0+}^B$  (why?), Blumenthal's 0-1 law (Corollary 9.19) implies  $\mathbb{P}(T_{0+} = 0) = 1$ . Similarly,  $\mathbb{P}(T_{0-} = 0) = 1$ .

*Item (3):* Since the paths  $t \mapsto B_t$  are continuous, arguments as in the proof of Proposition 8.10(2) show that the map  $\mathbb{R}_+ \times \Omega \ni (t, \omega) \mapsto B_t(\omega)$  is  $(\mathcal{B}(\mathbb{R}_+) \times \mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable. From this follows that the set  $A = \{(t, \omega) : B_t(\omega) = 0\}$  is in  $\mathcal{B}(\mathbb{R}_+) \times \mathcal{F}$ . Moreover,

$$\int \left[ \int \mathbb{1}_{H(\omega)}(t) dt \right] \mathbb{P}(d\omega) = \iint \mathbb{1}_A(t, \omega) \mathbb{P}(d\omega) dt = \int \mathbb{P}(B_t = 0) dt = 0$$

and so  $H(\omega)$  is a Lebesgue null set for almost all  $\omega$ . By continuity of  $t \mapsto B_t$  the set  $H(\omega)$  is closed. Finally, to see that  $H(\omega)$  has no isolated points, let  $\tau_s = \inf\{t > s : B_t = 0\}$  for  $s \in \mathbb{Q}_+$ . By the strong Markov property and item (2),  $B_{\tau_s+t} = 0$  for infinitely many  $t \in (0, \epsilon)$  for any  $\epsilon > 0$  almost surely on  $\{\tau_s < \infty\}$ . An isolated point  $t$  in  $H(\omega)$  is of the form  $t = \tau_s$  for some  $s \in \mathbb{Q}_+$  with  $s < t$ . Consequently,  $H(\omega)$  can only have isolated points on a null set of  $\omega$ 's. The last part of (3) follows from the fact that a closed non-empty set with no isolated points is uncountable.  $\square$

**Theorem 10.9.** Let  $(Y_t)_{t \geq 0}$  be a real Lévy process such that  $Y_t \sim N(0, \sigma^2 t)$  for every  $t \geq 0$ , where  $\sigma^2 > 0$ . Then, almost surely,  $t \mapsto Y_t$  is continuous.

**Remark 10.10.** Let the setting be as in Theorem 10.9. Choose a null set  $N \in \mathcal{F}$  so that  $t \mapsto Y_t(\omega)$  is continuous for  $\omega \in N^c$ . Defining  $(B_t)_{t \geq 0}$  as

$$B_t(\omega) = \begin{cases} Y_t(\omega) & \text{for } \omega \in N^c \text{ and } t \geq 0, \\ 0 & \text{for } \omega \in N \text{ and } t \geq 0, \end{cases}$$

we see that  $(B_t)_{t \geq 0}$  is a continuous process indistinguishable from  $(Y_t)_{t \geq 0}$ . Thus,  $(B_t)_{t \geq 0}$  is a Brownian motion. ■

*Proof.* Let  $(Y_t)_{t \geq 0}$  be defined on  $(\Omega, \mathcal{F}, \mathbb{P})$ . By Exercise 91,  $(Y_t)_{t \geq 0}$  is a version of a Brownian motion  $(B_t)_{t \geq 0}$  with parameter  $\sigma^2$ . Note that

$$\begin{aligned} & \{t \mapsto Y_t \text{ is continuous}\} \\ &= \bigcap_{m=1}^{\infty} \{t \mapsto Y_t \text{ is continuous on } [0, m]\} \\ &= \bigcap_{m=1}^{\infty} \{t \mapsto Y_t \text{ is uniformly continuous on } [0, m]\} \\ &= \bigcap_{m=1}^{\infty} \left\{ \forall k \geq 1 \exists n \geq 1 \text{ such that } |Y_t - Y_s| \leq \frac{1}{k} \forall t, s \in [0, m] \text{ s.t. } |t - s| \leq \frac{1}{n} \right\} \\ &= \bigcap_{m=1}^{\infty} \left\{ \forall k \geq 1 \exists n \geq 1 \text{ such that } |Y_t - Y_s| \leq \frac{1}{k} \forall t, s \in [0, m] \cap \mathbb{Q} \text{ s.t. } |t - s| \leq \frac{1}{n} \right\} \\ &= \bigcap_{m=1}^{\infty} \bigcap_{k=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{\substack{t, s \in [0, m] \cap \mathbb{Q} \\ |t-s| \leq \frac{1}{n}}} \{|Y_t - Y_s| \leq \frac{1}{k}\}, \end{aligned} \tag{10.2}$$

where the next to last equality uses that  $t \mapsto Y_t(\omega)$  is càdlàg. This shows that  $\{t \mapsto Y_t \text{ is continuous}\}$  is an event (i.e., is in  $\mathcal{F}$ ). Since  $(Y_t)_{t \geq 0}$  and  $(B_t)_{t \geq 0}$  are versions, (10.2) shows that the probability that  $t \mapsto Y_t$  is continuous is equal to the probability that  $t \mapsto B_t$  is continuous, which is 1. □

**Theorem 10.11.** Let  $(Y_t)_{t \geq 0}$  be a real Lévy process satisfying that  $t \mapsto Y_t(\omega)$  is continuous for  $\omega \in \Omega$ .

Then either there is a constant  $\mu \in \mathbb{R}$  such that  $(Y_t)_{t \geq 0}$  is indistinguishable from the deterministic process  $(\mu t)_{t \geq 0}$  or there exist constants  $\mu \in \mathbb{R}$  and  $\sigma^2 > 0$  such that  $(Y_t)_{t \geq 0}$  is a Brownian motion with parameter  $\sigma^2$  and drift  $\mu$ .

*Proof.* See Graversen (2011, Theorem 7.2). □

**Theorem 10.12.** *Let  $(B_t)_{t \geq 0}$  denote a Brownian motion with parameter  $\sigma^2 > 0$ .*

- (1) *For  $a \geq 0$  the process  $(B_{t+a} - B_a)_{t \geq 0}$  is a Brownian motion with parameter  $\sigma^2$ .*
- (2) *Define  $(\widetilde{B}_t)_{t \geq 0}$  as  $\widetilde{B}_0 = 0$  and  $\widetilde{B}_t = tB_{1/t}$  for  $t > 0$ . Then  $(\widetilde{B}_t)_{t \geq 0}$  is indistinguishable from a Brownian motion with parameter  $\sigma^2$ .*

The reason that the process  $(\widetilde{B}_t)_{t \geq 0}$  is only indistinguishable from a Brownian motion (i.e., it is not necessarily a Brownian motion itself) is that the path  $t \mapsto \widetilde{B}_t$  is continuous at  $t = 0$  only on a set of probability 1.

*Proof.* See Exercise 107. □

Schilling and Partzsch (2014) is a very readable reference on (additional) properties and constructions of the Brownian motion.

# Poisson processes

In this chapter we study *Poisson processes*. We give the basic definitions, show that these processes exist and derive various properties and characterizations.

Some proofs are based on Graversen (2011).

## 11.1 Definitions, existence and the strong Markov property

**Definition 11.1.** Let  $\lambda > 0$ . A stochastic process  $(N_t)_{t \geq 0}$  is called a **Poisson process** with parameter  $\lambda$  if

- (1)  $N_0(\omega) = 0$  for every  $\omega \in \Omega$ .
- (2)  $t \mapsto N_t(\omega)$  is càdlàg, non-decreasing and  $\mathbb{N}_0$ -valued for every  $\omega \in \Omega$ .
- (3)  $N_t - N_s \sim \text{po}(\lambda(t-s))$  for  $0 \leq s < t$ .
- (4)  $(N_t)_{t \geq 0}$  has independent increments.

A typical path of a Poisson process is depicted in Figure 9.2.

**Remark 11.2.** Assuming (1), condition (3) above is equivalent to the following:

$$N_t \sim \text{po}(\lambda t) \text{ for } t > 0, \text{ and } (N_t)_{t \geq 0} \text{ has stationary increments.} \quad (3^*)$$

(This follows by arguments as in Remark 10.2). In particular, a Poisson process is a Lévy process whose regular convolution semigroup is  $\nu_t = \text{po}(\lambda t)$ .

Conversely, let  $(N_t)_{t \geq 0}$  be a real Lévy process satisfying (1)–(2) and  $N_t \sim \text{po}(\lambda t)$  for  $t > 0$ . Then (3\*) holds and (4) holds since  $(N_t)_{t \geq 0}$  is a Lévy process, so  $(N_t)_{t \geq 0}$  is a Poisson process with parameter  $\lambda$ . ■

Poisson processes with respect to a filtration are defined as follows:

**Definition 11.3.** Let  $\lambda > 0$  and  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$  denote a filtered probability space. A stochastic process  $(N_t)_{t \geq 0}$  is called an  $(\mathcal{F}_t)_{t \geq 0}$ -**Poisson process with parameter  $\lambda$**  if

- (1)  $N_0(\omega) = 0$  for every  $\omega \in \Omega$ .
- (2)  $t \mapsto N_t(\omega)$  is càdlàg, non-decreasing and  $\mathbb{N}_0$ -valued for every  $\omega \in \Omega$ .
- (3)  $N_t - N_s \sim \text{po}(\lambda(t-s))$  for  $0 \leq s < t$ .
- (4)  $(N_t)_{t \geq 0}$  is  $(\mathcal{F}_t)_{t \geq 0}$ -adapted and  $N_t - N_s$  is independent of  $\mathcal{F}_s$  for  $0 \leq s \leq t$ .

**Remark 11.4.** As seen above, condition (3) is equivalent to (3\*) under (1).

From Proposition 9.6(3) it follows that  $(N_t)_{t \geq 0}$  is a Poisson process with parameter  $\lambda$  if and only if it is an  $(\mathcal{F}_t^N)_{t \geq 0}$ -Poisson process with parameter  $\lambda$ . ■

**Proposition 11.5.** Let  $\lambda > 0$ . There exists a Poisson process with parameter  $\lambda$ .

*Proof.* Let  $(\nu_t)_{t \geq 0}$  be the regular convolution semigroup on  $\mathbb{R}$  defined by  $\nu_t = \text{po}(\lambda t)$  for  $t \geq 0$ . By Theorem 9.10(1)–(2) and Theorem 7.4 there exists a process  $(Z_t)_{t \geq 0}$  with stationary independent increments such that  $Z_t - Z_s \sim \text{po}(\lambda(t-s))$  for  $0 \leq s \leq t$  and  $Z_0 = 0$  a.s.

Since Poisson distributions are supported on  $\mathbb{N}_0$ , we have  $Z_q \in \mathbb{N}_0$  a.s. for every  $q \geq 0$ , and for  $q < r$  we have  $Z_r - Z_q \in \mathbb{N}_0$  a.s., hence  $Z_q \leq Z_r$  a.s. Because  $\mathbb{Q}_+$  is countable, there exists a null set  $A_0 \in \mathcal{F}$  such that for every  $\omega \in A_0^c$  we have  $Z_0(\omega) = 0$  and the map  $\mathbb{Q}_+ \ni q \mapsto Z_q(\omega)$  is non-decreasing and  $\mathbb{N}_0$ -valued.

Define, for  $t \geq 0$  and  $\omega \in \Omega$ ,

$$N_t(\omega) = \begin{cases} 0, & \omega \in A_0, \\ \inf\{Z_q(\omega) : q \in \mathbb{Q}_+, q > t\}, & \omega \in A_0^c. \end{cases}$$

Note that  $t \mapsto N_t(\omega)$  is non-decreasing for every  $\omega$ , since  $s > t$  implies  $\{q \in \mathbb{Q}_+ \mid q > s\} \subseteq \{q \in \mathbb{Q}_+ \mid q > t\}$ .

Fix  $\omega \in A_0^c$  and  $t \geq 0$ . Choose  $(q_n)_{n \geq 1} \subset \mathbb{Q}_+$  such that  $q_n > t$  and  $q_n \downarrow t$ . Then  $Z_{q_n}(\omega)$  is non-increasing and  $\mathbb{N}_0$ -valued, hence eventually constant, and

$$\inf_{n \geq 1} Z_{q_n}(\omega) = \inf\{Z_q(\omega) \mid q \in \mathbb{Q}_+, q > t\} = N_t(\omega),$$

because for any  $q > t$  there exists  $n$  with  $q_n < q$ , hence  $Z_{q_n}(\omega) \leq Z_q(\omega)$ . Thus there exists  $n_0$  such that  $Z_{q_{n_0}}(\omega) = N_t(\omega)$ . Put  $\varepsilon = q_{n_0} - t > 0$ . Then for all  $q \in (t, t + \varepsilon) \cap \mathbb{Q}_+$  we have  $q < q_{n_0}$ , hence  $Z_q(\omega) \leq Z_{q_{n_0}}(\omega) = N_t(\omega)$ , while by definition  $Z_q(\omega) \geq N_t(\omega)$ . Therefore

$$Z_q(\omega) = N_t(\omega) \quad \text{for all } q \in (t, t + \varepsilon) \cap \mathbb{Q}_+. \quad (11.1)$$

In particular  $N_t(\omega) \in \mathbb{N}_0$ . Moreover, for  $s \in [t, t + \varepsilon)$  we have  $N_s(\omega) = N_t(\omega)$  (since the infimum defining  $N_s$  sees rationals in  $(s, t + \varepsilon)$  where  $Z_q(\omega) = N_t(\omega)$ ), so  $t \mapsto N_t(\omega)$  is right-continuous on  $A_0^c$ . Since it is non-decreasing, it has left limits; hence  $t \mapsto N_t(\omega)$  is càdlàg for every  $\omega \in \Omega$ .

We now show that  $(N_t)_{t \geq 0}$  is a modification of  $(Z_t)_{t \geq 0}$ . Fix  $t \geq 0$  and choose a deterministic sequence  $(q_n)_{n \geq 1} \subset \mathbb{Q}_+$  with  $q_n > t$  and  $q_n \downarrow t$ . By continuity in probability of  $(Z_t)_{t \geq 0}$  we have  $Z_{q_n} \rightarrow Z_t$  in probability. On the other hand, for  $\omega \in A_0^c$  the above argument gives  $Z_{q_n}(\omega) \downarrow N_t(\omega)$ , so  $Z_{q_n} \rightarrow N_t$  in probability. By uniqueness of limits in probability it follows that  $N_t = Z_t$  a.s. for every  $t \geq 0$ .

Since  $(N_t)_{t \geq 0}$  and  $(Z_t)_{t \geq 0}$  are modifications, and  $(Z_t)_{t \geq 0}$  satisfies the properties Definition 11.1(3)–(4), the same is true for  $(N_t)_{t \geq 0}$ .

Finally, set  $A_1 = \{\omega \in \Omega \mid N_0(\omega) \neq 0\}$ . Since  $N_0 = Z_0$  a.s. and  $Z_0 = 0$  a.s., we have  $\mathbb{P}(A_1) = 0$ . Re-define  $N_t(\omega) = 0$  for all  $t \geq 0$  and  $\omega \in A_0 \cup A_1$ . Then Definition 11.1(1)–(2) hold for every  $\omega \in \Omega$ , and the finite-dimensional distributions are unchanged.

We have shown that  $(N_t)_{t \geq 0}$  is a Poisson process with parameter  $\lambda$ .  $\square$

The strong Markov property (see Theorem 9.17) can be stated as follows.

**Theorem 11.6.** *Let  $(N_t)_{t \geq 0}$  denote an  $(\mathcal{F}_t)_{t \geq 0}$ -Poisson process with parameter  $\lambda$ , and let  $\tau$  be an a.s. finite extended stopping time with respect to  $(\mathcal{F}_t)_{t \geq 0}$ . Then  $(D_\tau N_t)_{t \geq 0}$  is a Poisson process with parameter  $\lambda$  and is independent of  $\mathcal{F}_{\tau+}$ .*

Recall that  $D_\tau N_t$  is defined as  $(N_{\tau+t} - N_\tau) \mathbb{1}_{\{\tau < \infty\}}$  for  $t \geq 0$ .

## 11.2 Properties

In this section we derive various properties and characterizations of Poisson processes.

First, we note that sums of independent Poisson processes are Poisson processes.

**Proposition 11.7.** *For  $j = 1, 2$  let  $(N_t^{(j)})_{t \geq 0}$  be a Poisson process with parameters  $\lambda_j$ . Assume these processes are independent. Then  $(N_t^{(1)} + N_t^{(2)})_{t \geq 0}$  is a Poisson process with parameter  $\lambda_1 + \lambda_2$ .*

*Proof.* By Exercise 103,  $(N_t^{(1)} + N_t^{(2)})_{t \geq 0}$  is a Lévy process. In addition, as  $(N_t^{(j)})_{t \geq 0}$  satisfies Definition 11.1(1)–(2) for  $j = 1, 2$ , so does  $(N_t^{(1)} + N_t^{(2)})_{t \geq 0}$ . Finally,  $N_t^{(1)} + N_t^{(2)} \sim \text{po}(\lambda_1 + \lambda_2)$  for  $t \geq 0$ , so the sum is a Poisson process by Remark 11.2.  $\square$

Next we are interested in the jumps of a Poisson process, and looking at the simulation in Figure 9.2 one gets the suspicion that the jump sizes are exactly equal to +1. This motivates the following.

**Definition 11.8.** A stochastic process  $(N_t)_{t \geq 0}$  is called a **counting process** if

- (1)  $N_0(\omega) = 0$  for  $\omega \in \Omega$ .
- (2)  $t \mapsto N_t(\omega)$  is càdlàg, non-decreasing and  $\mathbb{N}_0$ -valued for  $\omega \in \Omega$ .
- (3)  $\Delta N_t \in \{0, 1\}$  for every  $t > 0$  a.s.

If  $(\mathcal{F}_t)_{t \geq 0}$  is a filtration then  $(N_t)_{t \geq 0}$  is called an  $(\mathcal{F}_t)_{t \geq 0}$ -**counting process** if it is  $(\mathcal{F}_t)_{t \geq 0}$ -adapted and a counting process.

The first two conditions above are recognized from Definition 11.1 and the third one simply says that, almost surely, all jumps are of size +1.

**Proposition 11.9.** A Poisson process is a counting process.

*Proof.* Let  $\lambda > 0$  be the parameter. We will use that

$$\{\text{there is a } t > 0 \text{ such that } \Delta N_t \geq 2\} = \bigcup_{n=1}^{\infty} \bigcap_{k=1}^{\infty} \bigcup_{i=1}^{n2^k} \{N_{\frac{i}{2^k}} - N_{\frac{i-1}{2^k}} \geq 2\}.$$

We must show that the (ugly) right-hand side has probability 0, and it suffices to show that  $\mathbb{P}(\bigcap_{k=1}^{\infty} \bigcup_{i=1}^{n2^k} \{N_{\frac{i}{2^k}} - N_{\frac{i-1}{2^k}} \geq 2\}) = 0$  for all  $n \in \mathbb{N}$ . Since

$$\mathbb{P}\left(\bigcap_{k=1}^{\infty} \bigcup_{i=1}^{n2^k} \{N_{\frac{i}{2^k}} - N_{\frac{i-1}{2^k}} \geq 2\}\right) \leq \mathbb{P}\left(\bigcup_{i=1}^{n2^k} \{N_{\frac{i}{2^k}} - N_{\frac{i-1}{2^k}} \geq 2\}\right) \quad \text{for } k \in \mathbb{N},$$

it suffices to show that the right-hand side converges to 0 as  $k \rightarrow \infty$  for all  $n \in \mathbb{N}$ . By sub-additivity of  $\mathbb{P}$  and the stationary increments,

$$\mathbb{P}\left(\bigcup_{i=1}^{n2^k} \{N_{\frac{i}{2^k}} - N_{\frac{i-1}{2^k}} \geq 2\}\right) \leq \sum_{i=1}^{n2^k} \mathbb{P}(N_{\frac{i}{2^k}} - N_{\frac{i-1}{2^k}} \geq 2) = n2^k \mathbb{P}(N_{\frac{1}{2^k}} \geq 2).$$

Since  $N_{\frac{1}{2^k}} \sim \text{po}(\frac{\lambda}{2^k})$ ,

$$n2^k \mathbb{P}(N_{\frac{1}{2^k}} \geq 2) = n2^k (1 - e^{-\frac{\lambda}{2^k}} - \frac{\lambda}{2^k} e^{-\frac{\lambda}{2^k}}) \xrightarrow{k \rightarrow \infty} 0,$$

which concludes the proof.  $\square$

**Remark 11.10.** The above shows that a Poisson process has jumps of size +1. In addition, if  $(N_t)_{t \geq 0}$  is a real Lévy process such that  $N_t \sim \text{po}(\lambda t)$  for every  $t \geq 0$  then, with probability one, the path  $t \mapsto N_t$  is non-decreasing and  $\mathbb{N}_0$ -valued, and all jumps are of size +1 a.s.

To see this, note that since the Poisson distribution is concentrated on  $\mathbb{N}_0$  the map  $\mathbb{Q}_+ \ni t \mapsto N_t(\omega)$  is  $\mathbb{N}_0$ -valued and non-decreasing for almost every  $\omega \in \Omega$ . Thus, by the càdlàg property,  $\mathbb{R}_+ \ni t \mapsto N_t(\omega)$  is  $\mathbb{N}_0$ -valued and non-decreasing for almost every  $\omega \in \Omega$  as well. The proof of Proposition 11.9 shows that all jumps are +1 a.s. ■

**Theorem 11.11.** *Let  $(N_t)_{t \geq 0}$  be a Lévy process and a counting process. Then either  $\mathbb{P}(N_t = 0) = 1$  for  $t \geq 0$  or there exists a  $\lambda > 0$  such that  $(N_t)_{t \geq 0}$  is a Poisson process with parameter  $\lambda$ .*

In the case when  $N_t = 0$  a.s. for  $t \geq 0$ , the process  $(N_t)_{t \geq 0}$  is, by right-continuity, indistinguishable from the process that is constantly equal to 0.

*Proof.* For  $s, t \geq 0$  we have

$$\begin{aligned} \mathbb{P}(N_{t+s} = 0) &= \mathbb{P}(N_{s+t} = 0, N_t = 0) = \mathbb{P}(N_{s+t} - N_t = 0, N_t = 0) \\ &= \mathbb{P}(N_{s+t} - N_t = 0) \mathbb{P}(N_t = 0) = \mathbb{P}(N_s = 0) \mathbb{P}(N_t = 0), \end{aligned}$$

where we used the independent increments in the third equality, and the stationary increments in the fourth equality.

Put  $\phi(t) = \mathbb{P}(N_t = 0)$  for  $t \geq 0$ . By the above,  $\phi(t+s) = \phi(t)\phi(s)$  for all  $s, t \geq 0$ . In addition,  $\phi(0) = \mathbb{P}(N_0 = 0) = 1$  and  $t \mapsto \phi(t)$  is non-increasing (since  $s \leq t$  implies  $\phi(t) = \mathbb{P}(N_t = 0) \leq \mathbb{P}(N_s = 0) = \phi(s)$  by the non-decreasing paths). Moreover, since  $\phi(\frac{1}{m}) \rightarrow \phi(0) = 1$  as  $m \rightarrow \infty$  (why?), there exists an  $m$  such that  $\phi(\frac{1}{m}) > 0$ , and this implies that  $\phi(t) > 0$  for every  $t \geq 0$ . By Exercise 117 with  $f(t) = \log \phi(t)$  there exists a  $\lambda \in [0, \infty)$  such that  $\phi(t) = e^{-\lambda t}$  for all  $t \geq 0$ . When  $\lambda = 0$  we have  $\phi(t) = \mathbb{P}(N_t = 0) = 1$  for every  $t \geq 0$ , so assume  $\lambda > 0$  in the following.

We show that  $(N_t)_{t \geq 0}$  is a Poisson process with parameter  $\lambda$ . The conditions in Definition 11.1(1)–(2) are satisfied since  $(N_t)_{t \geq 0}$  is a counting process; and since  $(N_t)_{t \geq 0}$  is a Lévy process, it has stationary independent increments. Thus, it suffices to show that  $N_t \sim \text{po}(\lambda t)$  for  $t > 0$  by Remark 11.2.

Let  $t > 0$  be fixed in the following. Since the paths are integer-valued and

non-decreasing and  $N_0 = 0$ ,

$$\begin{aligned}
N_t &= N_t - N_0 = \sum_{i=1}^n (N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t}) \\
&= \sum_{i=1}^n (N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t}) \mathbb{1}_{\{N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} \geq 1\}} \\
&= \sum_{i=1}^n \mathbb{1}_{\{N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} \geq 1\}} + \sum_{i=1}^n (N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} - 1) \mathbb{1}_{\{N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} \geq 1\}} \\
&= \sum_{i=1}^n \mathbb{1}_{\{N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} \geq 1\}} + \sum_{i=1}^n (N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} - 1) \mathbb{1}_{\{N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} \geq 2\}}. \tag{11.2}
\end{aligned}$$

Since all jumps are of size +1 a.s., for almost all  $\omega$  there exists an  $M(\omega) \in \mathbb{N}$  such that for  $n \geq M(\omega)$ ,  $N_{\frac{i}{n}t}(\omega) - N_{\frac{i-1}{n}t}(\omega) \leq 1$  for  $i = 1, \dots, n$ . Thus, almost surely, the last term in (11.2) is 0 for  $n$  large enough.

Note that

$$\mathbb{P}(N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} \geq 1) = \mathbb{P}(N_{\frac{1}{n}t} \geq 1) = \left(1 - \phi\left(\frac{t}{n}\right)\right) = 1 - e^{-\lambda \frac{t}{n}},$$

where the first equality is by the stationary increments. Moreover,

$$n(1 - e^{-\lambda \frac{t}{n}}) = \frac{1 - e^{-\lambda \frac{t}{n}}}{\lambda \frac{t}{n}} \lambda t \rightarrow \lambda t \quad \text{for } n \rightarrow \infty.$$

Since  $\sum_{i=1}^n \mathbb{1}_{\{N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} \geq 1\}} \sim \text{Bin}(n, 1 - e^{-\lambda \frac{t}{n}})$ , it follows that  $\sum_{i=1}^n \mathbb{1}_{\{N_{\frac{i}{n}t} - N_{\frac{i-1}{n}t} \geq 1\}}$  converges in distribution to  $\text{po}(\lambda t)$ . (See Exercise 136).

Summing up, the first term in (11.2) converges to  $\text{po}(\lambda t)$  and the second term converges to 0 almost surely. It follows that  $N_t \sim \text{po}(\lambda t)$ .  $\square$

Next we are interested in the distribution of the *holding times*, i.e., how much time a Poisson process spends in a state  $n \in \mathbb{N}_0$  before it jumps to the next state  $n+1$ . Let us give precise definitions.

**Remark 11.12 (Jump times and holding times).**

Let  $(N_t)_{t \geq 0}$  be a stochastic process satisfying

- (1)  $N_0(\omega) = 0$  for  $\omega \in \Omega$ ;
- (2)  $t \mapsto N_t(\omega)$  is càdlàg, non-decreasing and  $\mathbb{N}_0$ -valued for  $\omega \in \Omega$ .

For  $n \in \mathbb{N}_0$  define the *nth jump time* as

$$T_n = \inf\{t > 0 : N_t \geq n\}.$$

We see that  $0 = T_0 < T_1 \leq T_2 \leq \dots$  and  $T_n \rightarrow \infty$  as  $n \rightarrow \infty$  pointwise.

For  $n \in \mathbb{N}$  define the  $n$ th holding time as

$$H_n = \begin{cases} T_n - T_{n-1} & \text{if } T_{n-1} < \infty, \\ 0 & \text{if } T_{n-1} = \infty. \end{cases}$$

Obviously, the holding times are determined by the jump times. Conversely the holding times determine  $T_n$  for every  $n \in \mathbb{N}$  since

$$T_n = H_1 + \dots + H_n \quad \text{for } n \in \mathbb{N}. \quad (11.3)$$

The jump times  $T_n$  are stopping times with respect to  $(\mathcal{F}_t^N)_{t \geq 0}$  since

$$\{T_n \leq t\} = \{N_t \geq n\} \quad \text{for } n \in \mathbb{N}_0 \text{ and } t \geq 0. \quad (11.4)$$

Furthermore, the jump times and the holding times determine  $(N_t)_{t \geq 0}$  uniquely; for instance,  $N_t$  can be calculated from the jump times as  $N_t = \sum_{n=1}^{\infty} \mathbb{1}_{[T_n, \infty)}(t)$ .

Generally, inconvenient things may happen. For instance, if  $(N_t)_{t \geq 0}$  jumps from 0 to 2 without visiting 1 then  $T_1 = T_2$ , and then the second holding time  $H_2$  is  $H_2 = 0$ ; similarly if, e.g.,  $N_t$  never reaches state 7 then  $T_n$  equals  $\infty$  for  $n \geq 7$ . (In this case  $H_n$  is put to 0 for  $n \geq 7$ .) We are mainly interested in Poisson processes, and we know from Proposition 11.9 and Exercise 11.4 that, almost surely, all jumps of a Poisson process are +1 and that the process converges to  $\infty$ . Thus, we consider an  $\omega \in \Omega$  such that

$$\Delta N_t(\omega) \in \{0, 1\} \text{ for all } t \geq 0 \quad \text{and} \quad N_t(\omega) \rightarrow \infty \text{ as } t \rightarrow \infty. \quad (11.5)$$

In this case  $0 = T_0(\omega) < T_1(\omega) < T_2(\omega) < \dots$  and  $H_n(\omega) > 0$  for all  $n \in \mathbb{N}$ . Moreover,

$$H_n(\omega) = \inf\{t > 0 : D_{T_{n-1}} N_t(\omega) \geq 1\} \quad \text{for } n \in \mathbb{N}. \quad (11.6)$$

Here, the process  $(D_{T_{n-1}} N_t)_{t \geq 0}$  is defined in Section 9.4. In words, (11.6) says that  $n$ th holding time  $H_n$  is equal to the first holding time (or jump time) for the process  $(D_{T_{n-1}} N_t)_{t \geq 0}$ . ■

For  $\lambda > 0$  let  $\text{Exp}(\lambda)$  denote the exponential distribution with mean  $1/\lambda$ .

**Theorem 11.13.** *Let  $(N_t)_{t \geq 0}$  denote a Poisson process with parameter  $\lambda > 0$ . Then the holding times  $(H_n)_{n \geq 1}$  are i.i.d. and  $\text{Exp}(\lambda)$ -distributed.*

**Remark 11.14.** This result gives an alternative way of simulating a Poisson process; namely, one should simply simulate i.i.d. holding times. Since exponential distributions are easy to simulate, this method is usually preferred over the approach in Example 9.15.

By the above theorem and (11.3),  $T_n$  has a gamma distribution for  $n \in \mathbb{N}$ . ■

*Proof.* We prove by induction in  $n$  that  $H_1, \dots, H_n$  are i.i.d.  $\text{Exp}(\lambda)$ -distributed. By (11.4) and the fact that  $H_1 = T_1$ ,

$$\mathbb{P}(H_1 \leq t) = \mathbb{P}(N_t \geq 1) = 1 - e^{-\lambda t} \quad \text{for } t \geq 0,$$

so  $H_1 \sim \text{Exp}(\lambda)$ . Assuming we have proved that  $H_1, \dots, H_n$  are i.i.d. and  $\text{Exp}(\lambda)$ -distributed, we have to show that  $H_{n+1}$  is independent of  $(H_1, \dots, H_n)$  and  $\text{Exp}(\lambda)$ -distributed.

By the strong Markov property (Theorem 11.6) the process  $(D_{T_n} N_t)_{t \geq 0}$  is a Poisson process with parameter  $\lambda$  independent of  $\mathcal{F}_{T_n+}^N$ , and consequently the first holding time of  $(D_{T_n} N_t)_{t \geq 0}$  is independent of  $\mathcal{F}_{T_n+}^N$  and  $\text{Exp}(\lambda)$ -distributed. By the last part of Remark 11.12, the first holding time of  $(D_{T_n} N_t)_{t \geq 0}$  is a.s. equal to  $H_{n+1}$ . Thus,  $H_{n+1}$  is independent of  $\mathcal{F}_{T_n+}^N$  and  $\text{Exp}(\lambda)$ -distributed. Since  $T_1, \dots, T_{n-1}$  are all less than or equal to  $T_n$ , it follows that  $(T_1, \dots, T_n)$  is  $\mathcal{F}_{T_n+}^N$ -measurable. Since  $H_1, \dots, H_n$  are determined by  $(T_1, \dots, T_n)$ , the vector  $(H_1, \dots, H_n)$  is  $\mathcal{F}_{T_n+}^N$ -measurable as well. That is,  $(H_1, \dots, H_n)$  is independent of  $H_{n+1}$ .  $\square$

**Corollary 11.15.** *Let  $(N_t)_{t \geq 0}$  denote a Poisson process with parameter  $\lambda > 0$ .*

- (1) *For  $n \in \mathbb{N}$  the vector  $(T_1, \dots, T_n)$  is absolutely continuous with respect to Lebesgue measure on  $\mathbb{R}^n$  with density  $f_{(T_1, \dots, T_n)}$  given by*

$$f_{(T_1, \dots, T_n)}(v_1, \dots, v_n) = \begin{cases} \lambda^n e^{-\lambda v_n} & \text{for } 0 < v_1 < \dots < v_n, \\ 0 & \text{otherwise.} \end{cases}$$

- (2) *For  $n \in \mathbb{N}$  and  $t > 0$  the conditional distribution  $(T_1, \dots, T_n) \mid N_t = n$  is absolutely continuous with respect to the Lebesgue measure on  $\mathbb{R}^n$  with density  $f_{(T_1, \dots, T_n) \mid N_t = n}(v_1, \dots, v_n)$  given by*

$$f_{(T_1, \dots, T_n) \mid N_t = n}(v_1, \dots, v_n) = \begin{cases} \frac{n!}{t^n} & \text{for } 0 < v_1 < \dots < v_n < t, \\ 0 & \text{otherwise.} \end{cases}$$

The density in the last equation above is that of the order statistics of  $n$  i.i.d.  $\text{Uniform}(0, t)$ -distributed random variables. This gives yet another way of simulating a Poisson process – how?

*Proof. Item (1):* By Theorem 11.13 the density of  $(H_1, \dots, H_n)$  is given by

$$\begin{aligned} f_{(H_1, \dots, H_n)}(u_1, \dots, u_n) &= \prod_{j=1}^n f_{H_j}(u_j) \\ &= \begin{cases} \prod_{j=1}^n \lambda e^{-\lambda u_j} & \text{for } u_j > 0 \text{ for } j = 1, \dots, n, \\ 0 & \text{otherwise,} \end{cases} \\ &= \begin{cases} \lambda^n e^{-\lambda(u_1 + \dots + u_n)} & \text{for } u_j > 0 \text{ for } j = 1, \dots, n, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Since  $H_1 = T_1, T_2 = H_1 + H_2, \dots, T_n = H_1 + \dots + H_n$ , we have

$$\begin{pmatrix} T_1 \\ \vdots \\ T_n \end{pmatrix} = \mathbf{T} \begin{pmatrix} H_1 \\ \vdots \\ H_n \end{pmatrix},$$

where  $\mathbf{T}$  is the lower triangular  $n \times n$  matrix having all entries in the lower triangle equal to 1. In particular, the determinant of  $\mathbf{T}$  is 1. Thus,

$$\begin{aligned} f_{(T_1, \dots, T_n)}(v_1, \dots, v_n) &= f_{(H_1, \dots, H_n)}(v_1, v_2 - v_1, \dots, v_n - v_{n-1}) \\ &= \begin{cases} \lambda^n e^{-\lambda(v_1 + (v_2 - v_1) + \dots + (v_n - v_{n-1}))} & \text{for } v_j - v_{j-1} > 0, j = 1, \dots, n, \\ 0 & \text{otherwise,} \end{cases} \\ &= \begin{cases} \lambda^n e^{-\lambda v_n} & \text{for } 0 < v_1 < \dots < v_n, \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

where we put  $v_0 = 0$ .

*Item (2):* Put  $K_t^n = \{(v_1, \dots, v_n) \in \mathbb{R}^n : 0 < v_1 < \dots < v_n < t\}$ . Since

$$\{N_t = n\} = \{T_n \leq t\} \cap \{T_{n+1} > t\}, \quad (11.7)$$

we see that  $\mathbb{P}((T_1, \dots, T_n) \in K_t^n \mid N_t = n) = 1$ . We must show that

$$\mathbb{P}((T_1, \dots, T_n) \in A \mid N_t = n) = \int_A \frac{n!}{t^n} \mathbb{1}_{K_t^n}(v) \text{Leb}_n(dv) \quad (11.8)$$

for  $A \in \mathcal{B}(\mathbb{R}^n)$ , where  $\text{Leb}_n$  denotes Lebesgue measure on  $\mathbb{R}^n$ . It suffices to consider

$$A = [0, t_1] \times \dots \times [0, t_n], \quad \text{where } 0 \leq t_1 \leq t_2 \leq \dots \leq t_n \leq t.$$

Using this  $A$  and (11.7), it follows that

$$\begin{aligned} \mathbb{P}(\{(T_1, \dots, T_n) \in A\} \cap \{N_t = n\}) \\ &= \mathbb{P}(0 \leq T_1 \leq t_1, \dots, 0 \leq T_n \leq t_n, T_{n+1} > t) \\ &= \mathbb{P}(0 \leq T_1 \leq t_1, \dots, 0 \leq T_n \leq t_n) - \mathbb{P}(0 \leq T_1 \leq t_1, \dots, 0 \leq T_n \leq t_n, 0 \leq T_{n+1} \leq t). \end{aligned}$$

The probabilities in the last expression can be calculated by integrating the joint densities given in item (1), resulting in

$$\begin{aligned} \mathbb{P}(\{(T_1, \dots, T_n) \in A\} \cap \{N_t = n\}) \\ &= \int_0^{t_1} \int_{u_1}^{t_2} \dots \int_{u_{n-1}}^{t_n} \lambda^n e^{-\lambda u_n} du_n \dots du_1 - \int_0^{t_1} \dots \int_{u_{n-1}}^{t_n} \int_{u_n}^t \lambda^{n+1} e^{-\lambda u_{n+1}} du_{n+1} \dots du_1 \\ &= \int_0^{t_1} \int_{u_1}^{t_2} \dots \int_{u_{n-1}}^{t_n} \lambda^n (e^{-\lambda u_n} - (e^{-\lambda u_n} - e^{-\lambda t})) du_n \dots du_2 du_1 \\ &= \lambda^n e^{-\lambda t} \int_0^{t_1} \dots \int_0^{t_n} \mathbb{1}_{K_t^n}(u) du_n \dots du_1. \end{aligned}$$

Since  $\mathbb{P}(N_t = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$ , the result in (11.8) follows.  $\square$

## 11.3 Extensions

In this section we consider various extensions of Poisson processes.

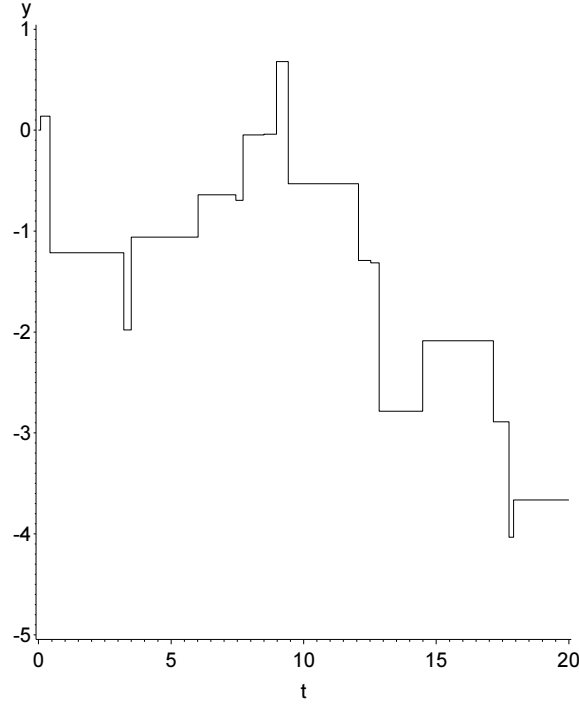
We have seen that a Poisson process has i.i.d. exponentially distributed holding times and that, almost surely, all jump sizes are +1. One way of generalizing this is by allowing the jump sizes to be random. If the jump sizes are i.i.d., the resulting process is called a *compound Poisson process*:

**Definition 11.16.** Let  $d \in \mathbb{N}$ ,  $\lambda > 0$ , and  $(N_t)_{t \geq 0}$  denote a Poisson process with parameter  $\lambda$ . Let  $(Z_n)_{n \geq 1}$  denote an i.i.d. sequence of  $\mathbb{R}^d$ -valued random vectors that are independent of  $(N_t)_{t \geq 0}$ . Define the  $\mathbb{R}^d$ -valued process  $(Y_t)_{t \geq 0}$  as

$$Y_t = \sum_{j=1}^{N_t} Z_j \quad \text{for } t \geq 0, \quad (11.9)$$

where we use the convention  $\sum_{j=1}^0 = 0$ . The process  $(Y_t)_{t \geq 0}$  is called a **compound Poisson process**.

Figure 11.1 displays a simulated path of a compound Poisson process in the case  $\lambda = 1$ ,  $d = 1$  and  $Z_n \sim N(0, 1)$ .



**Figure 11.1:** A simulation of a compound Poisson process in the case  $\lambda = 1$ ,  $d = 1$  and  $Z_n \sim N(0, 1)$ .

**Proposition 11.17.** *The compound Poisson process defined in (11.9) is a Lévy process and the characteristic function of  $Y_t$  is given by*

$$\mathbb{E}[e^{i\langle \theta, Y_t \rangle}] = \exp(-\lambda t(1 - \phi(\theta))) \quad \text{for } t \geq 0 \text{ and } \theta \in \mathbb{R}^d, \quad (11.10)$$

where  $\phi(\theta) = \mathbb{E}[e^{i\langle \theta, Z_n \rangle}]$  is the characteristic function of the  $Z_n$ 's.

*Proof.* Since  $N_0(\omega) = 0$  and  $t \mapsto N_t(\omega)$  is càdlàg for  $\omega \in \Omega$ ,  $(Y_t)_{t \geq 0}$  satisfies (1) and (2) in Definition 9.2. To verify (3) it suffices to show that for  $n \in \mathbb{N}$ ,  $\theta_1, \dots, \theta_n \in \mathbb{R}^d$  and  $0 = t_0 < t_1 < \dots < t_n$  we have

$$\mathbb{E}[\exp(\sum_{j=1}^n i\langle \theta_j, Y_{t_j} - Y_{t_{j-1}} \rangle)] = \prod_{j=1}^n \exp(-\lambda(t_j - t_{j-1})(1 - \phi(\theta_j))). \quad (11.11)$$

This equation is verified in Exercise 115.  $\square$

Next we consider *thinning of a Poisson process*, which in a sense is the opposite of summing Poisson processes. Let  $(N_t)_{t \geq 0}$  denote a Poisson process with parameter  $\lambda > 0$ . Think of the jumps as some kind of event occurring. Now assume that there are  $K$  types of events (for some  $K \geq 2$ ) and that each event is of the  $k$ th type with probability  $p_k$  for  $k = 1, \dots, K$ , where  $p_k \in (0, 1)$  and  $\sum_{k=1}^K p_k = 1$ . Moreover, assume that the jumps are classified according to type independently. An example could

be that an event is a vehicle passing a road. Then  $k = 1$  could be *car*,  $k = 2$  could be *truck* and  $k = 3$  could be *other*.

Let  $N_t^k$  denote the total number of events of type  $k$  which have occurred on  $[0, t]$  for  $t \geq 0$  and  $k = 1, \dots, K$ . Then in particular  $N_t = \sum_{k=1}^K N_t^k$ .

**Theorem 11.18 (Thinning).**

For  $k = 1, \dots, K$ ,  $(N_t^k)_{t \geq 0}$  is a Poisson process with parameter  $\lambda p_k$ . Moreover, the processes  $(N_t^1)_{t \geq 0}, \dots, (N_t^K)_{t \geq 0}$  are independent.

*Proof.* First we give a precise definition of  $(N_t^1, \dots, N_t^K)$ . Put

$$e_1 = (1, 0, \dots, 0) \in \mathbb{R}^K, \dots, e_K = (0, \dots, 0, 1) \in \mathbb{R}^K$$

and let  $(Z_j)_{j \geq 1}$  denote i.i.d. random vectors with state space  $\{e_1, \dots, e_K\}$  such that

$$\mathbb{P}(Z_j = e_k) = p_k \quad \text{for } k = 1, \dots, K \text{ and } j \in \mathbb{N}.$$

Also assume  $(Z_j)_{j \geq 1}$  is independent of  $(N_t)_{t \geq 0}$ . The interpretation is that  $Z_j = e_k$  corresponds to that the  $j$ th event is of type  $k$ . Thus,  $(N_t^1, \dots, N_t^K)$  is given as

$$(N_t^1, \dots, N_t^K) = \sum_{j=1}^{N_t} Z_j \quad \text{for } t \geq 0.$$

Let  $Y_t$  denote the random sum on the right-hand side.

Since  $(Y_t)_{t \geq 0}$  is a compound Poisson process and  $Z_j$  has state space  $\{e_1, \dots, e_K\}$  it follows that  $N_0^k(\omega) = 0$  and  $t \mapsto N_t^k(\omega)$  is càdlàg, non-decreasing and  $\mathbb{N}_0$ -valued for all  $\omega$  and  $k = 1, \dots, K$ .

Since a compound Poisson process is a Lévy process,  $(Y_t)_{t \geq 0}$  has stationary independent increments; these properties are inherited to the coordinates  $(N_t^k)_{t \geq 0}$  for  $k = 1, \dots, K$ .

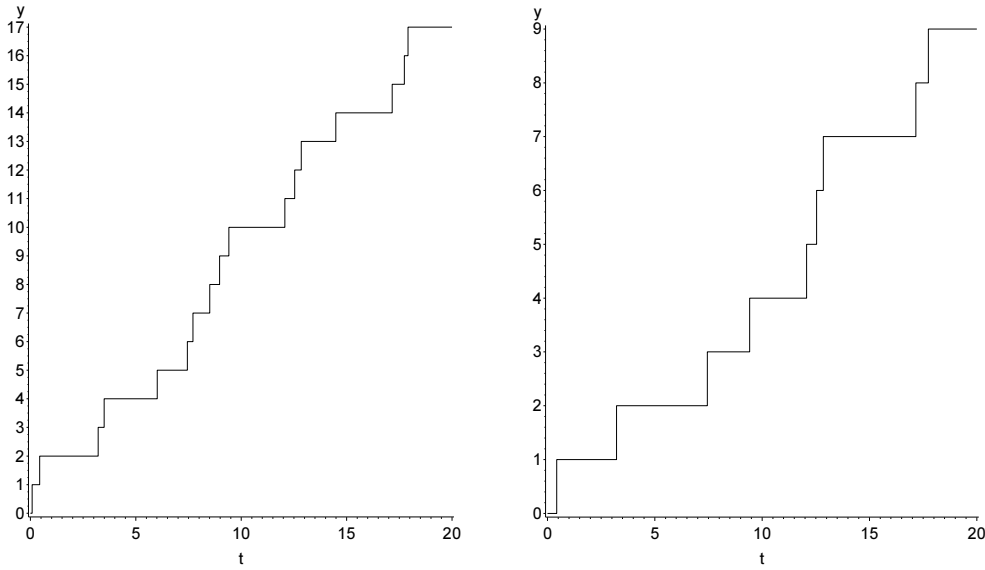
Let  $t > 0$  and  $\theta = (\theta_1, \dots, \theta_K) \in \mathbb{R}^K$ . Put  $\phi(\theta) = \mathbb{E}[e^{i\langle \theta, Z_j \rangle}]$  and note that since  $Z_j$  has state space  $\{e_1, \dots, e_K\}$ ,

$$\phi(\theta) = \sum_{k=1}^K e^{i\langle \theta, e_k \rangle} \mathbb{P}(Z_j = e_k) = \sum_{k=1}^K e^{i\theta_k} p_k.$$

By (11.10),

$$\mathbb{E}[e^{i\langle \theta, Y_t \rangle}] = \exp(-\lambda t(1 - \phi(\theta))) = \prod_{k=1}^K \exp(-\lambda t p_k(1 - e^{i\theta_k})).$$

This shows that  $N_t^1, \dots, N_t^K$  are independent and  $N_t^k \sim \text{po}(\lambda t p_k)$  for  $k = 1, \dots, K$ . Finally, using that  $(Y_t)_{t \geq 0}$  has independent increments it follows that the processes  $(N_t^1)_{t \geq 0}, \dots, (N_t^K)_{t \geq 0}$  are independent.  $\square$



**Figure 11.2:** Left: A simulation of a Poisson process  $(N_t)_{t \geq 0}$  with parameter  $\lambda = 1$ . Right: A simulation of the corresponding thinned process  $(N_t^1)_{t \geq 0}$  with  $p_1 = 1/2$ .

Finally, we introduce nonhomogeneous Poisson processes.

**Example 11.19 (Nonhomogeneous Poisson processes).**

Let  $\lambda : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  be a continuous function. Following Durrett (2016, p. 85), a real-valued stochastic process  $(M_t)_{t \geq 0}$  is called a *nonhomogeneous Poisson process* with rate  $\lambda(t)$ ,  $t \geq 0$ , if

1.  $M_0(\omega) = 0$  for  $\omega \in \Omega$ ;
2.  $t \mapsto M_t(\omega)$  is non-decreasing, càdlàg and  $\mathbb{N}_0$ -valued;
3.  $(M_t)_{t \geq 0}$  has independent increments and  $M_t - M_s \sim \text{po}(\int_s^t \lambda(u) du)$  for  $0 \leq s \leq t$ .

(With the convention that  $\text{po}(0)$  is the distribution concentrated in 0).

Nonhomogeneous Poisson processes can be constructed from homogeneous ones in the following way. For simplicity assume that  $\int_0^\infty \lambda(u) du = \infty$  and let  $(N_t)_{t \geq 0}$  be a Poisson process with parameter 1. Put  $\Lambda(t) = \int_0^t \lambda(u) du$  and define  $M_t = N_{\Lambda(t)}$  for  $t \geq 0$ . In Exercise 119 you are asked to show the following:

- (a)  $(M_t)_{t \geq 0}$  is a nonhomogeneous Poisson process with rate  $\lambda(t)$ ,  $t \geq 0$ .
- (b) Let  $(H_n)_{n \geq 1}$  denote the holding times of  $(N_t)_{t \geq 0}$  and  $(K_n)_{n \geq 1}$  be the holding times of  $(M_t)_{t \geq 0}$ . Note that  $H_1 = \Lambda(K_1)$  and  $H_n = \Lambda(K_1 + \dots + K_n) - \Lambda(K_1 + \dots + K_{n-1})$  for  $n \geq 2$ .

Using that  $(H_n)_{n \geq 1}$  are i.i.d. and  $\text{Exp}(1)$ -distributed it follows that the density of  $(K_1, \dots, K_n)$  (for  $n \in \mathbb{N}$ ) is

$$f_{(K_1, \dots, K_n)}(v_1, \dots, v_n) = e^{-\Lambda(v_1 + \dots + v_n)} \prod_{j=1}^n \lambda(v_1 + \dots + v_j) \quad \text{for } v_1, \dots, v_n > 0.$$

Point (b) shows, in particular, that, in general, the holding times of a nonhomogeneous Poisson process are *not* independent and they *do not* have an exponential distribution. ■

## Markov jump processes

So far we have always constructed Markov processes by specifying the transition semigroup and then appealing to Theorem 7.4. In this chapter we study an important alternative approach where the idea is to specify the *infinitesimal behavior* of the process (in some sense). Let us first consider two examples.

### Example 12.1 (Stochastic differential equations).

Let the state space be  $S = \mathbb{R}$  and let  $b, \sigma : \mathbb{R} \rightarrow \mathbb{R}$  be functions that are sufficiently regular. Consider the *stochastic differential equation* (SDE)

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t$$

where  $(B_t)_{t \geq 0}$  is a standard Brownian motion. We will not go into details with the exact meaning of this equation, but loosely speaking, for  $dt$  close to 0, the infinitesimal behavior is

$$X_{t+dt} \approx X_t + b(X_t)dt + \sigma(X_t)[B_{t+dt} - B_t].$$

If we think of  $t$  as present time and condition on the past at time  $t$ , the conditional distribution of  $X_{t+dt}$  is approximately  $N(X_t + b(X_t)dt, \sigma^2(X_t)dt)$  which only depends on  $X_t$ . In other words, it is likely that a solution  $(X_t)_{t \geq 0}$  of the SDE is Markov.

The above can be made precise and it turns out that there is a result of the following type:

*Assume the SDE has a unique solution (in some sense) no matter the initial condition. Then any solution is a strong Markov process and the corresponding transition semigroup is*

$$P_t(x, A) = \mathbb{P}(X_t^x \in A) \quad \text{for } t \geq 0, x \in \mathbb{R} \text{ and } A \in \mathcal{B}(\mathbb{R}),$$

where  $X_t^x$  denotes the solution with initial condition  $x$ .

The CIR process (see Example 4.13) is one of the cases where  $P_t(x, dy)$  is explicitly known; indeed, it has a density given by (4.23).

In other words, we can construct Markov processes by solving stochastic differential equations. The downside, however, is that in most cases, just as for ordinary differential equations, solutions to stochastic differential equations are not known in any explicit form and in particular the transition semigroup is not known explicitly. ■

**Example 12.2 (The Poisson process as a Markov process).**

Let  $(N_t)_{t \geq 0}$  denote a Poisson process with parameter  $\lambda > 0$  which is a Lévy process and in particular a Markov process. The state space is  $S = \mathbb{N}_0$ . Let  $(P_t)_{t \geq 0}$  denote the transition semigroup.

Since  $S$  is discrete, it is typically more convenient to work with the point probabilities  $p_t(n, m) = P_t(n, \{m\})$  (defined for  $t \geq 0$  and  $n, m \in S$ ) than with  $(P_t)_{t \geq 0}$ . Note that  $p_t(n, m)$  is the probability of being in state  $m$   $t$  time units ahead given that the current state is  $n$ . For  $t > 0$ ,

$$p_t(n, m) = \begin{cases} e^{-\lambda t} \frac{(\lambda t)^{m-n}}{(m-n)!} & \text{for } n \leq m \\ 0 & \text{for } m < n. \end{cases} \quad (12.1)$$

Indeed, the case  $m < n$  follows from the non-decreasing paths, and for  $n \leq m$  we can use that  $p_t(n, m) = \mathbb{P}(N_{t+s} = m \mid N_s = n)$  (for an arbitrary  $s > 0$ ) and the useful rule to infer that

$$\begin{aligned} p_t(n, m) &= \mathbb{P}(N_{t+s} = m \mid N_s = n) = \mathbb{P}(N_{t+s} - N_s + n = m \mid N_s = n) \\ &= \mathbb{P}(N_{t+s} - N_s = m - n) = e^{-\lambda t} \frac{(\lambda t)^{m-n}}{(m-n)!}. \end{aligned}$$

By Taylor expansion of (12.1) it follows that

$$p_t(n, m) = \begin{cases} e^{-\lambda t} = (1 - \lambda t) + o(t) & \text{if } m = n \\ e^{-\lambda t} \lambda t = \lambda t + o(t) & \text{if } m = n + 1 \\ e^{-\lambda t} \frac{(\lambda t)^{m-n}}{(m-n)!} = o(t) & \text{if } m \geq n + 2. \end{cases}$$

As usual,  $o(t)$  denotes a term satisfying  $o(t)/t \rightarrow 0$  as  $t \rightarrow 0$ . The interpretation of this is that if current state is  $n$  then in a small interval of length  $t$  the process will

$$\begin{cases} \text{stay in } n \text{ with a probability close to } 1 - \lambda t \\ \text{jump one up (to } n + 1) \text{ with a probability close to } \lambda t \\ \text{jump at least twice with probability close to } 0. \end{cases}$$

This is an example of a specification of the *infinitesimal behavior* of a Markov jump process. ■

From now on let  $S$  be a non-empty countable state space. As always, we choose a nice metric on  $S$  such that  $\mathcal{B}(S)$  is the set of all subsets of  $S$ .

## 12.1 Definitions and basic properties

Consider a measurable space  $(\Omega, \mathcal{F})$  on which an  $S$ -valued process  $(X_t)_{t \geq 0}$  is defined. We will always assume that  $X_t : \Omega \rightarrow S$  is measurable. Moreover, we impose the assumption that  $t \mapsto X_t(\omega)$  is càdlàg and piecewise constant and that, on every bounded interval  $[0, m]$ , it has only finitely many jumps for every  $\omega \in \Omega$ . Finally, we assume that  $\mathbb{P}_i$ , for  $i \in S$ , is a probability measure on  $(\Omega, \mathcal{F})$  satisfying  $\mathbb{P}_i(X_0 = i) = 1$ , that is,  $\mathbb{P}_i$  is a probability measure corresponding to the initial condition  $X_0 = i$  a.s.

For a probability measure  $\mu$  on  $(S, \mathcal{B}(S))$  define the probability measure  $\mathbb{P}_\mu$  on  $(\Omega, \mathcal{F})$  as  $\mathbb{P}_\mu = \sum_{i \in S} \mu(\{i\})\mathbb{P}_i$ , which means that  $X_0 \sim \mu$  under  $\mathbb{P}_\mu$ .

**Definition 12.3.** Let  $(\mathcal{F}_t)_{t \geq 0}$  denote a filtration. The process  $(X_t)_{t \geq 0}$  is called a **Markov jump process** with respect to  $(\mathcal{F}_t)_{t \geq 0}$  if it is adapted and

$$\mathbb{P}_\mu(X_{s+t} = j \mid \mathcal{F}_s) = \mathbb{P}_{X_s}(X_t = j) \quad (12.2)$$

for all  $s, t \geq 0$  and  $j \in S$  and every initial distribution  $\mu$  on  $(S, \mathcal{B}(S))$ .

On the right-hand side of (12.2),  $\mathbb{P}_{X_s}(X_t = j)$  means that we calculate  $\mathbb{P}_i(X_t = j)$  for  $i = X_s \in S$ . It suffices to check (12.2) for  $\mathbb{P}_\mu = \mathbb{P}_k$  for  $k \in S$ .

When  $\mathcal{F}_t = \mathcal{F}_t^X$  for every  $t \geq 0$ , we speak as usual of a Markov jump process without explicitly mentioning the filtration. In this case, (12.2) can be written alternatively as

$$\mathbb{P}_\mu(X_{s+t} = j \mid X_s = i, X_{s_n} = i_n, \dots, X_{s_1} = i_1) = \mathbb{P}_i(X_t = j) \quad (12.3)$$

for all  $s, t \geq 0$ ,  $i, j \in S$ ,  $n \in \mathbb{N}$ ,  $0 \leq s_1 \leq s_2 \leq \dots \leq s_n \leq s$  and every initial distribution  $\mu$  on  $(S, \mathcal{B}(S))$ .

**Remark 12.4.** If  $(X_t)_{t \geq 0}$  is a Markov jump process with respect to  $(\mathcal{F}_t)_{t \geq 0}$  then it is also a Markov jump process with respect to  $(\mathcal{F}_{t+})_{t \geq 0}$ ; see Exercise 120. (For this to hold it is crucial that  $(X_t)_{t \geq 0}$  is càdlàg.)

For simplicity, we will therefore always assume that  $(\mathcal{F}_t)_{t \geq 0}$  is right-continuous. In particular this implies that  $\tau$  is a stopping time if and only if it is an extended stopping time. ■

In the remainder of this section assume that  $(X_t)_{t \geq 0}$  is a Markov jump process with respect to  $(\mathcal{F}_t)_{t \geq 0}$ . Put  $p_t(i, j) = \mathbb{P}_i(X_t = j)$  for  $i, j \in S$  and  $t \geq 0$ , and let  $p_t = \{p_t(i, j)\}_{i, j \in S}$ . Note that  $p_t(i, j)$  is the probability of being in state  $j$   $t$  time units ahead given that the current state is  $i$ .

**Proposition 12.5 (Chapman–Kolmogorov).**

We have

$$p_0 = I \text{ (the } S \times S \text{ identity matrix) and } p_{s+t} = p_s p_t \text{ for } s, t \geq 0. \quad (12.4)$$

*Proof.* Obviously,

$$p_0(i, j) = \mathbb{P}_i(X_0 = j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

For  $i, k \in S$  and  $s, t \geq 0$ ,

$$\begin{aligned} p_{s+t}(i, k) &= \mathbb{P}_i(X_{s+t} = k) = \sum_{j \in S} \mathbb{P}_i(X_s = j) \mathbb{P}_i(X_{s+t} = k | X_s = j) \\ &= \sum_{j \in S} \mathbb{P}_i(X_s = j) \mathbb{P}_j(X_t = k) = \sum_{j \in S} p_s(i, j) p_t(j, k), \end{aligned}$$

where we note that the last term is the  $(i, k)$ th entry of  $p_s p_t$ .  $\square$

Proposition 12.5 shows that a Markov jump process is a Markov process in the sense of Definition 7.2; put differently  $(p_t)_{t \geq 0}$  is a transition semigroup in the sense of Remark 7.5. In particular the finite-dimensional marginals can be calculated as in (7.4) (where the integrals are sums since the state space is countable). The converse statement, that any transition semigroup on a countable state space, corresponds to a Markov jump process, is *not* true; (7.10) is a simple example illustrating this. Roughly speaking, continuity of  $t \mapsto p_t$  is needed for a transition semigroup  $(p_t)_{t \geq 0}$  to correspond to a Markov jump process.

**Theorem 12.6 (The strong Markov Property).**

Let  $\tau$  be a stopping time. The process  $(X_{\tau+t})_{t \geq 0}$  is a Markov jump process with respect to  $(\mathcal{F}_{\tau+t})_{t \geq 0}$  on  $\{\tau < \infty\}$ .

More precisely,

$$\mathbb{P}_\mu(X_{\tau+s+t} = j | \mathcal{F}_{\tau+s}) = \mathbb{P}_{X_{\tau+s}}(X_t = j) = p_t(X_{\tau+s}, j) \quad \text{on } \{\tau < \infty\} \quad (12.5)$$

for all  $s, t \geq 0$  and  $j \in S$  and every initial distribution  $\mu$  on  $(S, \mathcal{B}(S))$ .

As a generalization of (12.5) we also have

$$\mathbb{P}_\mu((X_{\tau+t})_{t \geq 0} \in A | \mathcal{F}_\tau) = \mathbb{P}_{X_\tau}((X_t)_{t \geq 0} \in A) \quad (12.6)$$

on  $\{\tau < \infty\}$  for "nice" sets of paths  $A$ . (Compare Theorem 4.27 and Theorem 9.17).

We shall not give a proof of the Strong Markov Property here (although it is, as we shall see, very important). Let us just note that the proof follows using the same approach as in the proof of Theorem 9.17.

For  $n \in \mathbb{N}_0$  define the  $n$ th jump time  $T_n$  as  $T_0 = 0$ ,

$$T_1 = \inf\{t > 0 : X_t \neq X_0\}$$

and

$$T_{n+1} = \inf\{t > T_n : X_t \neq X_{T_n}\},$$

where as always  $\inf \emptyset = \infty$ . The associated *holding times*  $(H_n)_{n \geq 1}$  are defined as  $H_1 = T_1$  and for  $n \geq 2$ ,

$$H_n = \begin{cases} T_n - T_{n-1} & \text{if } T_{n-1} < \infty \\ 0 & \text{if } T_{n-1} = \infty \end{cases}. \quad (12.7)$$

Obviously, the holding times are determined by the jump times but the converse is also true since  $T_n = H_1 + \dots + H_n$  for  $n \in \mathbb{N}$ .

Define the associated *jump chain*  $(Y_n)_{n \geq 0}$  as  $Y_0 = X_0 = X_{T_0}$  and recursively,

$$Y_n = \begin{cases} X_{T_n} & \text{if } T_n < \infty \\ Y_{n-1} & \text{if } T_n = \infty. \end{cases}$$

By definition, the jump times  $(T_n)_{n \geq 0}$  (equivalently: the holding times  $(H_n)_{n \geq 1}$ ) and the jump chain  $(Y_n)_{n \geq 0}$  are determined by  $(X_t)_{t \geq 0}$ . Conversely,  $(X_t)_{t \geq 0}$  is also fully determined by  $(T_n)_{n \geq 0}$  (equivalently:  $(H_n)_{n \geq 1}$ ) and  $(Y_n)_{n \geq 0}$ . (To realize this, the reader is encouraged to plot possible paths of  $(X_t)_{t \geq 0}$ ).

In the following we give a complete description of the jump times  $(T_n)_{n \geq 0}$  and the jump chain  $(Y_n)_{n \geq 0}$ , and hence also of  $(X_t)_{t \geq 0}$ .

**Proposition 12.7.** *Let  $i \in S$ . There is a number  $\lambda(i) \in [0, \infty)$  such that  $\mathbb{P}_i(T_1 > t) = \exp(-\lambda(i)t)$  for every  $t \geq 0$ .*

*In other words,  $T_1 \sim \text{Exp}(\lambda(i))$  under  $\mathbb{P}_i$  if  $\lambda(i) > 0$  and  $T_1 = \infty$   $\mathbb{P}_i$ -a.s. if  $\lambda(i) = 0$ .*

*Proof.* For  $s, t \geq 0$ ,

$$\mathbb{P}_i(T_1 > s+t) = \mathbb{P}_i(T_1 > s+t \mid T_1 > s) \mathbb{P}_i(T_1 > s) = \mathbb{P}_i(T_1 > t) \mathbb{P}_i(T_1 > s),$$

where the second equality uses that  $\{T_1 > s\} \in \mathcal{F}_s^X$  and the Markov property. Since the function  $t \mapsto \mathbb{P}_i(T_1 > t)$  is right-continuous and non-increasing and  $\mathbb{P}_i(T_1 > 0) = 1$  by the right-continuous paths, the result follows.  $\square$

When  $\lambda(i) = 0$  for some state  $i \in S$  the Markov jump process can not escape from  $i$  once it is in this state. Call  $i$  *absorbing* in this case.

Put  $p_{ij} = \mathbb{P}_i(Y_1 = j)$  for  $i, j \in S$  and let  $P = \{p_{ij}\}_{i,j \in S}$  be the corresponding transition matrix. Note that for  $i \in S$ ,

$$p_{ii} = \begin{cases} 1 & \text{if } \lambda(i) = 0 \\ 0 & \text{if } \lambda(i) > 0. \end{cases}$$

Let

$$q_{ij} = \begin{cases} \lambda(i)p_{ij} & \text{for } i \neq j \\ -\lambda(i) & \text{for } i = j \end{cases} \quad (12.8)$$

and put  $Q = \{q_{ij}\}_{i,j \in S}$ . We have  $q_{ij} \geq 0$  for  $i \neq j$ ,  $q_{ii} \leq 0$  and  $\sum_{j \in S} q_{ij} = 0$  for every  $i \in S$ .

For future reference, note that  $P$  and the  $\lambda(i)$ s are uniquely determined by  $Q$ . Indeed,

$$\lambda(i) = -q_{ii} \quad \text{for } i \in S \quad (12.9)$$

and

$$p_{ij} = \begin{cases} q_{ij}/\lambda(i) & \text{if } i \neq j \text{ and } \lambda(i) > 0 \\ 0 & \text{if } i = j \text{ and } \lambda(i) > 0 \\ 0 & \text{if } i \neq j \text{ and } \lambda(i) = 0 \\ 1 & \text{if } i = j \text{ and } \lambda(i) = 0. \end{cases} \quad (12.10)$$

**Theorem 12.8.** Let  $\mu$  be a probability measure on  $(S, \mathcal{B}(S))$ .

- (1) The process  $(Y_n)_{n \geq 0}$  is a Markov chain with transition matrix  $P$  and initial distribution  $\mu$  with respect to  $(\mathcal{F}_{T_n})_{n \geq 0}$  under  $\mathbb{P}_\mu$ .
- (2) For  $n \in \mathbb{N}_0$  and  $t \geq 0$ ,  $\mathbb{P}(H_{n+1} > t \mid \mathcal{F}_{T_n}) = \exp(-\lambda(Y_n)t)$  on  $\{T_n < \infty\}$  under  $\mathbb{P}_\mu$ .
- (3) The  $S \times S$  matrix-valued map  $\mathbb{R}_+ \ni t \mapsto p_t$  is continuously differentiable componentwise. In addition,  $p'_0(i, j) = q_{ij}$  for every  $i, j \in S$ . That is, written in matrix notation,  $p'_0 = Q$ .

Moreover,  $t \mapsto p_t$  solves Kolmogorov's backward equations

$$p'_t = Qp_t \quad \text{for } t \geq 0 \quad (12.11)$$

and Kolmogorov's forward equations

$$p'_t = p_t Q \quad \text{for } t \geq 0. \quad (12.12)$$

- (4) When  $S$  is finite,  $p_t = e^{Qt}$  for  $t \geq 0$ . In addition, in this case  $p_t$  is the only solution of Kolmogorov's equations satisfying  $p_0 = I$ .

For a uniqueness result related to Kolmogorov's equations when  $S$  is infinite, see Theorems 2.8.3 and 2.8.6 in Norris (1998).

*Proof. Item (1):* We must show that for  $n \in \mathbb{N}_0$  and  $j \in S$ ,  $\mathbb{P}_\mu(Y_{n+1} = j | \mathcal{F}_{T_n}) = p_{Y_n j}$ . This is true by definition on  $\{T_n = \infty\}$ . On  $\{T_n < \infty\}$  apply the Strong Markov Property to conclude that

$$\mathbb{P}_\mu(Y_{n+1} = j | \mathcal{F}_{T_n}) = \mathbb{P}_{X_{T_n}}(X_{T_1} = j) = \mathbb{P}_{X_{T_n}}(Y_1 = j) = p_{Y_n j}.$$

(The observant reader notes that we actually use the Strong Markov Property as indicated in (12.6).)

*Item (2):* Follows from Strong Markov Property and Proposition 12.7.

*Item (3):* We shall not prove this result; see e.g. Norris (1998). It should be emphasized, however, that it relies very much on the fact that we have assumed  $t \mapsto X_t$  is càdlàg and  $S$ -valued. Without these assumptions a process satisfying the Markov property (12.3) does not give rise to a result like (3). See (7.10) for an example.

*Item (4):* When  $S$  is finite,  $t \mapsto e^{Qt}$  solves Kolmogorov's forward and backward equations.  $\square$

**Example 12.9.** Let  $(N_t)_{t \geq 0}$  be a Poisson process with parameter  $\lambda$  independent of  $(Y_n)_{n \geq 0}$  which is a (discrete time) Markov chain with one-step transition probabilities  $u(i, j) = \mathbb{P}_i(Y_1 = j)$  and  $n$ -step transition probabilities  $u^n(i, j) = \mathbb{P}_i(Y_n = j)$  for  $i, j \in S$ .

Define

$$X_t = Y_{N_t} \quad \text{for } t \geq 0.$$

Then  $(X_t)_{t \geq 0}$  is a Markov jump process and, by the useful rule,

$$p_t(i, j) = e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} u^n(i, j) \quad \text{for } i, j \in S \text{ and } t \geq 0. \quad (12.13)$$

See Exercise 121.  $\blacksquare$

## 12.2 Specifying the process via $Q$

In the previous section the starting point was a Markov jump process. But as discussed in the beginning of this chapter in practice it is often more natural to start by specifying the "infinitesimal behavior". We pursue this approach in the following.

**Definition 12.10.** An  $S \times S$  matrix  $Q = \{q_{ij}\}_{i,j \in S}$  is called a  **$Q$ -matrix** if  $q_{ij} \in [0, \infty)$  for  $i \neq j$ ,  $q_{ii} \in (-\infty, 0]$  and  $q_{ii} = -\sum_{j \neq i} q_{ij}$  for  $i \in S$ .

The matrix  $Q$  defined by (12.8) is a fundamental example of a  $Q$ -matrix.

Let  $Q$  denote a  $Q$ -matrix. The idea is that we want to define a Markov jump process  $(X_t)_{t \geq 0}$  such that the corresponding transition semigroup  $(p_t)_{t \geq 0}$  satisfies  $Q = p'_0$ . This corresponds to assuming

$$p_t(i, j) = q_{ij}t + o(t) \quad \text{for } i \neq j \quad (12.14)$$

$$p_t(i, i) = 1 + q_{ii}t + o(t) = 1 - \left( \sum_{j \neq i} q_{ij} \right) t + o(t) \quad \text{for } i \in S. \quad (12.15)$$

The interpretation of (12.14) is that for  $t \approx 0$  the probability of going from state  $i$  to state  $j \neq i$  in an interval of length  $t$  should be close to  $q_{ij}t$ ; i.e.,  $q_{ij}$  models the rate at which we go from  $i$  to  $j$ . Similarly, (12.15) tells us that the probability of not leaving  $i$  in a small interval should be close to  $1 + q_{ii}t$ .

Define  $\lambda(i)$  for  $i \in S$  by (12.9) and the transition matrix  $P = \{p_{ij}\}_{i,j \in S}$  by (12.10). Proposition 12.7 and Theorem 12.8 tell us how the requested Markov jump process evolves if it exists; indeed, the jump chain must be a Markov chain and holding times must have an exponential distribution with a parameter determined by current state. Guided by this observation we pursue the following idea. For simplicity assume  $\lambda(i) > 0$  for all  $i \in S$ . (The case  $\lambda(i) = 0$  corresponds to an absorbing state, and the construction below is easily modified to include this case).

Consider a measurable space  $(\Omega, \mathcal{F})$  equipped with a probability measure  $\mathbb{P}_\mu$  for every probability measure  $\mu$  on  $(S, \mathcal{B}(S))$ . Assume  $(Y_n)_{n \geq 0}$  is a Markov chain with state space  $S$ , transition matrix  $P$  and initial distribution  $\mu$  under  $\mathbb{P}_\mu$ . In addition, let  $(U_n)_{n \geq 1}$  be an i.i.d. sequence of exponential distributions with parameter 1 independent of  $(Y_n)_{n \geq 0}$  under every  $\mathbb{P}_\mu$ . As usual, we use the notation  $\mathbb{P}_i$  when  $\mu = \delta_i$  for  $i \in S$ .

Define  $H_n$  for  $n \in \mathbb{N}$  as  $H_n = U_n / \lambda(Y_{n-1})$ . Observe that conditioned on  $(Y_n)_{n \geq 0}$ ,  $H_n$  has an  $\text{Exp}(\lambda(Y_{n-1}))$  distribution which is in the spirit of Theorem 12.8. Define  $T_n = H_1 + \cdots + H_n$  for  $n \in \mathbb{N}$  and  $T_0 = 0$ . Finally define a prospective Markov jump process  $X_t$  as

$$X_t = Y_{n-1} \quad \text{for } t \in [T_{n-1}, T_n) \text{ and } n \in \mathbb{N}. \quad (12.16)$$

That is,

$$X_t = \begin{cases} Y_0 & \text{for } t \in [0, T_1) \\ Y_1 & \text{for } t \in [T_1, T_2) \\ Y_2 & \text{for } t \in [T_2, T_3) \\ \vdots & \end{cases}.$$

Note that  $t \mapsto X_t$  is piecewise constant, and by the Markov property of  $(Y_n)_{n \geq 0}$  and since the exponential distribution is memoryless, it is likely that  $X_t$  is a Markov jump process. There is a problem, however, namely that it may be that the process *explodes* i.e. jumps infinitely many times in finite time. To define this more rigorously put  $T_\infty = \lim_{n \rightarrow \infty} T_n$ . We say that  $Q$  is *explosive* if there exists an initial distribution  $\mu$  such that  $\mathbb{P}_\mu(T_\infty < \infty) > 0$ . Equivalently,  $Q$  is explosive if there exists an  $i \in S$  such that  $\mathbb{P}_i(T_\infty < \infty) > 0$ .

When  $Q$  is non-explosive, (12.16) specifies  $X_t$  for every  $t \geq 0$  (at least  $\mathbb{P}_\mu$ -a.s. for every initial distribution  $\mu$ ) and we have the following result.

**Theorem 12.11.** *Let  $Q$  be non-explosive. Then  $(X_t)_{t \geq 0}$  defined by (12.16) is a Markov jump process. In addition, defining the  $S \times S$  matrix  $p_t = \{p_t(i, j)\}_{i, j \in S}$  for  $t \geq 0$  as  $p_t(i, j) = \mathbb{P}_i(X_t = j)$ , we have  $p'_0 = Q$ .*

In particular the properties in Theorem 12.8 hold when  $Q$  is non-explosive.

The tricky part of Theorem 12.11 is the assumption that  $Q$  is non-explosive, which is often quite hard to check. *Reuter's criterion* (see Nielsen (2009) or Norris (1998, Corollary 2.7.3)) provides a necessary and sufficient condition for non-explosiveness. This criterion may be hard to apply in general but see Nielsen (2009) for the case of birth and death processes. Let us delimit ourselves to the following useful result.

**Theorem 12.12.** *The  $Q$ -matrix  $Q$  is non-explosive if  $S$  is finite or more generally  $\sup_{i \in S} (-q_{ii}) < \infty$ . When  $S$  is finite, the corresponding transition semigroup  $(p_t)_{t \geq 0}$  is  $p_t = e^{Qt}$  for  $t \geq 0$ .*

## 12.3 What happens after explosion?

When the  $Q$ -matrix  $Q$  is explosive the construction (12.16) leads to explosion under some initial distribution, and in this case there are different ways to continue the process after the explosion time  $T_\infty$ . In the so-called *minimal construction* the idea is to absorb the Markov process in a *coffin state*  $\dagger$ , i.e. a state not contained in  $S$ , at time  $T_\infty$ . Precisely, we complete the construction in (12.16) by putting  $X_t$  equal to  $\dagger$  from time  $T_\infty$ . In this case  $(X_t)_{t \geq 0}$  is a Markov jump process with state space defined as  $S^\dagger = S \cup \{\dagger\}$ . See Norris (1998) (Section 2.9) for examples of non-minimal Markov jump processes and references to the literature.

## 12.4 Asymptotics

The asymptotic behavior (ergodic theorems and Laws of Large Numbers) is very well described in Norris (1998), Sections 3.4–3.6.

## Continuous time martingale results

In the following, consider the index set  $T = [0, \infty)$  and a filtered probability space  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$  on which all stochastic processes are defined.

The first result is one of many versions of the Martingale Modification Theorem. See e.g. Rogers and Williams (2000) or Graversen (2011) for further results.

**Theorem 13.1.** *Any  $(\mathcal{F}_t)_{t \geq 0}$ -martingale  $(M_t)_{t \geq 0}$  which is continuous in probability has a càdlàg modification. That is, there exists a modification  $(\tilde{M}_t)_{t \geq 0}$  of  $(M_t)_{t \geq 0}$  for which  $t \mapsto \tilde{M}_t(\omega)$  is càdlàg for every  $\omega \in \Omega$ .*

*Proof.* See Graversen (2011, p. 78). □

Next, we consider a result which is closely related to the Optional Sampling Theorem. See the above references for more on this.

**Theorem 13.2.** *Let  $(M_t)_{t \geq 0}$  denote an  $(\mathcal{F}_t)_{t \geq 0}$ -martingale for which  $t \mapsto M_t(\omega)$  is right-continuous for every  $\omega \in \Omega$ . Then for any extended stopping time  $\tau$ ,  $(M_{t \wedge \tau})_{t \geq 0}$  is an  $(\mathcal{F}_t)_{t \geq 0}$ -martingale as well.*

The pathwise right-continuity assumption can be relaxed: it suffices to assume right-continuity a.s. if  $(M_t)$  is progressively measurable.

*Proof.* The process  $(M_{t \wedge \tau})_{t \geq 0}$  is  $(\mathcal{F}_t)_{t \geq 0}$ -adapted. (Why?).

Let  $(\tau_n)_{n \geq 1}$  denote the discrete approximation of  $\tau$  and for  $n \in \mathbb{N}$  let  $D_n = \{\frac{k}{2^n} : k \in \mathbb{N}_0\}$ . The process  $(M_t)_{t \in D_n}$  is an  $(\mathcal{F}_t)_{t \in D_n}$ -martingale and  $\tau_n$  is an  $(\mathcal{F}_t)_{t \in D_n}$ -stopping time so by discrete time martingale theory it is well known that the process  $(M_{t \wedge \tau_n})_{t \in D_n}$  is an  $(\mathcal{F}_t)_{t \in D_n}$ -martingale.

For  $t \in [0, \infty)$  and  $n \in \mathbb{N}$  let  $t_n$  denote the smallest number in  $D_n$  with  $t_n \geq t$ . Then  $t_n \downarrow t$  and  $0 \leq t_n - t \leq 2^{-n}$ . Put

$$T_n = t_n \wedge \tau_n.$$

Then  $(T_n)_{n \geq 1}$  is a non-increasing sequence of bounded stopping times and  $T_n \downarrow t \wedge \tau$ . By right-continuity of  $t \mapsto M_t(\omega)$ , we therefore have

$$M_{T_n} \rightarrow M_{t \wedge \tau} \quad \text{pointwise.}$$

Moreover, for  $m \geq n$  we have  $T_m \leq T_n$  and  $T_m, T_n \leq t_1$ , so by the discrete-time Optional Sampling Theorem applied to the martingale  $(M_u)_{u \in D_m}$  and the  $D_m$ -valued stopping times  $T_m \leq T_n$ ,

$$\mathbb{E}[M_{T_n} | \mathcal{F}_{T_m}] = M_{T_m}.$$

Hence  $(M_{T_n})_{n \geq 1}$  is a backwards martingale with respect to the decreasing filtration  $(\mathcal{F}_{T_n})_{n \geq 1}$ . By the backwards Martingale Convergence Theorem it follows that  $M_{T_n} \rightarrow M_{t \wedge \tau}$  in  $L^1(\mathbb{P})$ . In particular,  $M_{t \wedge \tau} \in L^1(\mathbb{P})$ .

Finally, let  $0 \leq s < t$  and choose  $s_n$  and  $t_n$  as above. Then

$$\mathbb{E}[M_{t \wedge \tau} | \mathcal{F}_{s+}] = \lim_{n \rightarrow \infty} \mathbb{E}[M_{t \wedge \tau} | \mathcal{F}_{s_n}] = \lim_{n \rightarrow \infty} \mathbb{E}[M_{t_n \wedge \tau_n} | \mathcal{F}_{s_n}] = \lim_{n \rightarrow \infty} M_{s_n \wedge \tau_n} = M_{s \wedge \tau}$$

in  $L^1(\mathbb{P})$ , where the second equality follows because  $M_{t_n \wedge \tau_n} \rightarrow M_{t \wedge \tau}$  in  $L^1(\mathbb{P})$  and the third follows since  $(M_{t \wedge \tau_n})_{t \in D_n}$  is an  $(\mathcal{F}_t)_{t \in D_n}$ -martingale. This concludes the proof.  $\square$

# Kolmogorov-Chentsov's Theorem

For  $n \in \mathbb{N}_0$  let  $D_n = \{\frac{k}{2^n} : k = 0, \dots, 2^n\}$ . Note that

$$D_0 \subseteq D_1 \subseteq D_2 \subseteq \dots \quad (14.1)$$

and put  $D = \cup_{n=0}^{\infty} D_n$ .

**Remark 14.1.** For  $t \in [0, 1]$  and  $n \in \mathbb{N}_0$  let  $t_n$  denote the unique number in  $D_n$  such that  $t \in [t_n, t_n + \frac{1}{2^n})$ . We then have the following properties:

- (1) For every  $n \in \mathbb{N}_0$  either  $t_{n+1} = t_n$  or  $t_{n+1} = t_n + \frac{1}{2^{n+1}}$ . Indeed, the first case corresponds to  $t \in [t_n, t_n + \frac{1}{2^{n+1}})$  and the second to  $t \in [t_n + \frac{1}{2^{n+1}}, t_n + \frac{1}{2^n})$ .
- (2) For every  $t \in [0, 1]$  we have  $t_n \rightarrow t$  as  $n \rightarrow \infty$ .
- (3) For  $t \in D$  there is an  $N \in \mathbb{N}_0$  (depending on  $t$ ) such that  $t_n = t$  for  $n \geq N$ . Indeed, choose  $N$  such that  $t \in D_N$  and use (14.1). ■

**Lemma 14.2.** Let  $f : D \rightarrow \mathbb{R}$  and assume there are constants  $c \in [0, \infty)$  and  $\gamma > 0$  such that  $|f(t) - f(s)| \leq c|t - s|^\gamma$  for all  $s, t \in D$ . Then there exists a unique function  $g : [0, 1] \rightarrow \mathbb{R}$  extending  $f$  in the sense that  $f(t) = g(t)$  for all  $t \in D$  and  $|g(t) - g(s)| \leq c|t - s|^\gamma$  for all  $s, t \in [0, 1]$ .

*Proof.* For  $t \in [0, 1]$  let  $(t_n)_{n \in \mathbb{N}_0}$  be the approximating sequence in Remark 14.1. Note that  $(f(t_n))_{n \in \mathbb{N}_0}$  is Cauchy in  $\mathbb{R}$  since for all  $n, m \in \mathbb{N}_0$ ,  $|f(t_n) - f(t_m)| \leq c|t_n - t_m|^\gamma$ . Let  $g(t) = \lim_{n \rightarrow \infty} f(t_n)$ . Since  $t_n = t$  for  $n$  large when  $t \in D$  it follows that  $f(t) = g(t)$  in this case.

For  $s, t \in [0, 1]$  we have

$$|g(t) - g(s)| = \lim_{n \rightarrow \infty} |f(t_n) - f(s_n)| \leq \lim_{n \rightarrow \infty} c|t_n - s_n|^\gamma = c|t - s|^\gamma.$$

As for the uniqueness, let  $h : [0, 1] \rightarrow \mathbb{R}$  be continuous and satisfy  $h(t) = f(t)$  for  $t \in D$ . Then, for arbitrary  $t \in [0, 1]$ ,  $g(t) = \lim_{n \rightarrow \infty} f(t_n) = \lim_{n \rightarrow \infty} h(t_n) = h(t)$  by continuity; thus  $g = h$ .  $\square$

**Theorem 14.3.** *Let  $(Y_t)_{t \in [0, 1]}$  be a real stochastic process defined on a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ . Assume there exist constants  $c, \alpha, \beta > 0$  such that*

$$\mathbb{E}[|Y_t - Y_s|^\alpha] \leq c|t - s|^{1+\beta} \quad \text{for all } s, t \in [0, 1]. \quad (14.2)$$

*Then there exists a modification  $(X_t)_{t \in [0, 1]}$  of  $(Y_t)_{t \in [0, 1]}$  such that  $t \mapsto X_t(\omega)$  is Hölder continuous with exponent  $\gamma$  for every  $\gamma \in (0, \frac{\beta}{\alpha})$  and every  $\omega \in \Omega$ .*

**Remark 14.4.** (1) In the proof we actually show a little less than stated above, namely that for a *fixed*  $\gamma \in (0, \frac{\beta}{\alpha})$  there exists a modification  $(X_t)_{t \in [0, 1]}$  satisfying that  $t \mapsto X_t(\omega)$  is Hölder continuous with exponent  $\gamma$ . How can we fix this?

(2) Assume  $(Y_t)_{t \geq 0}$  is a real stochastic process satisfying the inequality in (14.2) for *all*  $s, t \geq 0$  and not only  $s, t \in [0, 1]$ . Then there exists a modification  $(X_t)_{t \geq 0}$  of  $(Y_t)_{t \geq 0}$  satisfying that  $t \mapsto X_t(\omega)$  is locally Hölder continuous with exponent  $\gamma$  for every  $\gamma \in (0, \frac{\beta}{\alpha})$  and all  $\omega \in \Omega$ . How does this follow from the above?

(3) There are many – quite similar – proofs of this result available. Here we follow the argument given by Professor Baudoin ([click here](#)).  $\blacksquare$

*Proof.* Fix  $\gamma \in (0, \frac{\beta}{\alpha})$ . For  $n \in \mathbb{N}_0$  we have

$$\begin{aligned}
& \mathbb{P}(\max_{1 \leq k \leq 2^n} |Y_{\frac{k}{2^n}} - Y_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}) \\
&= \mathbb{P}\left(\bigcup_{k=1}^{2^n} \{|Y_{\frac{k}{2^n}} - Y_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}\}\right) \\
&\leq \sum_{k=1}^{2^n} \mathbb{P}(|Y_{\frac{k}{2^n}} - Y_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}) \\
&= \sum_{k=1}^{2^n} \mathbb{P}(|Y_{\frac{k}{2^n}} - Y_{\frac{k-1}{2^n}}|^\alpha \geq 2^{-\gamma \alpha n}) \\
&\leq \sum_{k=1}^{2^n} \frac{\mathbb{E}[|Y_{\frac{k}{2^n}} - Y_{\frac{k-1}{2^n}}|^\alpha]}{2^{-\gamma \alpha n}} \\
&\leq c 2^n \frac{\left(\frac{1}{2^n}\right)^{1+\beta}}{2^{-\gamma \alpha n}} = c 2^{-n(\beta-\gamma\alpha)},
\end{aligned}$$

where we used Markov's inequality in the next to last inequality and (14.2) in the last one. Thus, since  $\gamma\alpha < \beta$ ,

$$\sum_{n=0}^{\infty} \mathbb{P}(\max_{1 \leq k \leq 2^n} |Y_{\frac{k}{2^n}} - Y_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}) \leq c \sum_{n=0}^{\infty} 2^{-n(\beta-\gamma\alpha)} < \infty,$$

which by the Borel-Cantelli Lemma implies

$$\mathbb{P}(\limsup_{n \rightarrow \infty} \{\max_{1 \leq k \leq 2^n} |Y_{\frac{k}{2^n}} - Y_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}\}) = 0.$$

Choose  $NS \in \mathcal{F}$  with  $\mathbb{P}(NS) = 0$  such that for all  $\omega \in (NS)^c$  there exists an  $N(\omega) \in \mathbb{N}_0$  such that

$$\max_{1 \leq k \leq 2^n} |Y_{\frac{k}{2^n}}(\omega) - Y_{\frac{k-1}{2^n}}(\omega)| \leq 2^{-\gamma n} \quad \text{for all } n \geq N(\omega). \quad (14.3)$$

Fix  $\omega \in (NS)^c$ . We show that there exists  $c(\omega) \in (0, \infty)$  such that

$$|Y_t(\omega) - Y_s(\omega)| \leq c(\omega)|t - s|^\gamma \quad \text{for all } s, t \in D. \quad (14.4)$$

Define

$$K(\omega) = \sup_{n \in \mathbb{N}_0} 2^{\gamma n} \max_{1 \leq k \leq 2^n} |Y_{\frac{k}{2^n}}(\omega) - Y_{\frac{k-1}{2^n}}(\omega)|.$$

Then  $K(\omega) < \infty$  since (14.3) gives

$$2^{\gamma n} \max_{1 \leq k \leq 2^n} |Y_{\frac{k}{2^n}}(\omega) - Y_{\frac{k-1}{2^n}}(\omega)| \leq 1 \quad \text{for all } n \geq N(\omega),$$

and for  $n < N(\omega)$  there are only finitely many terms. Consequently, for every  $n \in \mathbb{N}_0$ ,

$$\max_{1 \leq k \leq 2^n} |Y_{\frac{k}{2^n}}(\omega) - Y_{\frac{k-1}{2^n}}(\omega)| \leq K(\omega) 2^{-\gamma n}. \quad (14.5)$$

Let  $s, t \in D$  with  $s < t$  and consider the corresponding approximating sequences  $(t_n)_{n \in \mathbb{N}_0}$  and  $(s_n)_{n \in \mathbb{N}_0}$ . Choose an  $n \in \mathbb{N}_0$  such that

$$\frac{1}{2^{n+1}} \leq t - s \leq \frac{1}{2^n}. \quad (14.6)$$

Then

$$t_n - s_n \leq (t - s) + (s - s_n) < \frac{2}{2^n}$$

which implies  $t_n - s_n \leq 2^{-n}$  since  $t_n - s_n$  is a non-negative multiple of  $2^{-n}$  and  $t_n - s_n < 2 \cdot 2^{-n}$ .

Choose  $M \in \mathbb{N}_0$  such that  $s, t \in D_M$ . Then  $s_M = s$  and  $t_M = t$  by Remark 14.1(3). Moreover, since  $t - s > 0$  is a multiple of  $2^{-M}$ , we have  $2^{-M} \leq t - s$ , and hence by (14.6) we get  $n \leq M$ . Then

$$\begin{aligned} Y_t(\omega) - Y_s(\omega) &= Y_{t_M}(\omega) - Y_{s_M}(\omega) \\ &= Y_{t_n}(\omega) - Y_{s_n}(\omega) + \sum_{j=n}^{M-1} (Y_{t_{j+1}}(\omega) - Y_{t_j}(\omega)) - \sum_{j=n}^{M-1} (Y_{s_{j+1}}(\omega) - Y_{s_j}(\omega)). \end{aligned}$$

Since  $t_n$  and  $s_n$  are either equal or neighbouring points in  $D_n$ , and  $t_{j+1}$  and  $t_j$  are either equal or neighbouring points in  $D_{j+1}$  (and similarly with  $s_{j+1}, s_j$ ) by (1) in Remark 14.1, (14.5) and (14.6) imply

$$\begin{aligned} |Y_t(\omega) - Y_s(\omega)| &\leq K(\omega) \left( 2^{-\gamma n} + 2 \sum_{j=n}^{\infty} 2^{-\gamma(j+1)} \right) \\ &\leq 2K(\omega) \sum_{j=n}^{\infty} 2^{-\gamma j} \\ &= 2K(\omega) \frac{2^{-\gamma n}}{1 - 2^{-\gamma}} = (2^{-(n+1)})^\gamma \frac{2^{\gamma+1} K(\omega)}{1 - 2^{-\gamma}} \leq (t - s)^\gamma \frac{2^{\gamma+1} K(\omega)}{1 - 2^{-\gamma}}, \end{aligned}$$

which concludes the proof of (14.4).

For  $\omega \in NS$ , let  $X_t(\omega) = 0$  for all  $t \in [0, 1]$ ; obviously this path is Hölder continuous with exponent  $\gamma$ . For  $\omega \in (NS)^c$  let  $(X_t(\omega))_{t \in [0, 1]}$  be the extension of  $(Y_t(\omega))_{t \in D}$  in Lemma 14.2. Since the latter satisfies (14.4),  $(X_t(\omega))_{t \in [0, 1]}$  is Hölder continuous with exponent  $\gamma$  by Lemma 14.2.

Since  $NS$  is a null set,  $X_t = Y_t$  a.s. for  $t \in D$ . Next let  $t \in [0, 1]$ . We have  $Y_{t_n} \rightarrow Y_t$  in probability by (14.2). At the same time  $X_{t_n} \rightarrow X_t$  pointwise since  $(X_u)_{u \in [0, 1]}$  has continuous paths. Hence  $X_t = Y_t$  a.s. That is,  $(X_t)_{t \in [0, 1]}$  is a modification of  $(Y_t)_{t \in [0, 1]}$ .  $\square$



## A

# Characteristic functions

Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and let  $X = (X_1, \dots, X_n)^\top : \Omega \rightarrow \mathbb{R}^n$  be a random vector. Denote the distribution (push-forward measure) of  $X$  by

$$\mathbb{P}_X(B) = \mathbb{P}(X \in B), \quad \text{for } B \in \mathcal{B}(\mathbb{R}^n).$$

## A.1 Definition and uniqueness

**Definition A.1 (Characteristic function).**

The *characteristic function* of  $X$  is the map  $\phi_X : \mathbb{R}^n \rightarrow \mathbb{C}$  defined by

$$\phi_X(\theta) = \mathbb{E}[e^{i\langle \theta, X \rangle}], \quad \text{for } \theta \in \mathbb{R}^n.$$

Note that the characteristic function is well-defined since

$$|e^{i\langle \theta, X \rangle}| = 1, \quad \text{for all } \theta \in \mathbb{R}^n.$$

By the transformation theorem,

$$\phi_X(\theta) = \int_{\Omega} e^{i\langle \theta, X(\omega) \rangle} \mathbb{P}(d\omega) = \int_{\mathbb{R}^n} e^{i\langle \theta, x \rangle} \mathbb{P}_X(dx), \quad \text{for } \theta \in \mathbb{R}^n.$$

Recall that for random vectors  $X, Y : \Omega \rightarrow \mathbb{R}^n$  we write  $X \stackrel{d}{=} Y$  if

$$\mathbb{P}(X \in A) = \mathbb{P}(Y \in A), \quad \text{for all } A \in \mathcal{B}(\mathbb{R}^n).$$

**Theorem A.2 (Characteristic functions determine the distribution).**

Let  $X, Y : \Omega \rightarrow \mathbb{R}^n$  be random vectors. Then

$$X \stackrel{d}{=} Y \quad \text{if and only if} \quad \phi_X = \phi_Y.$$

*Proof.* Assume first that  $X \stackrel{d}{=} Y$ . Then  $\mathbb{P}_X = \mathbb{P}_Y$ , and hence for every  $\theta \in \mathbb{R}^n$ ,

$$\phi_X(\theta) = \int_{\mathbb{R}^n} e^{i\langle \theta, x \rangle} \mathbb{P}_X(dx) = \int_{\mathbb{R}^n} e^{i\langle \theta, x \rangle} \mathbb{P}_Y(dx) = \phi_Y(\theta).$$

This is the easy direction.

Conversely, the implication  $\phi_X = \phi_Y \Rightarrow X \stackrel{d}{=} Y$  is nontrivial; a proof is given in a course in *Advanced Probability*.  $\square$

## A.2 How to compute characteristic functions

### Proposition A.3 (Discrete case).

Assume that  $X$  has a countable state space  $\{x_k\}_{k \in I} \subseteq \mathbb{R}^n$ , and write  $p_k = \mathbb{P}(X = x_k)$ . Then, for  $\theta \in \mathbb{R}^n$ ,

$$\phi_X(\theta) = \sum_{k \in I} e^{i\langle \theta, x_k \rangle} p_k.$$

### Proposition A.4 (Absolutely continuous case).

Assume that  $X$  is absolutely continuous with respect to Lebesgue measure on  $\mathbb{R}^n$ , with density  $f_X$ . Then, for  $\theta \in \mathbb{R}^n$ ,

$$\phi_X(\theta) = \int_{\mathbb{R}^n} e^{i\langle \theta, x \rangle} f_X(x) dx.$$

## A.3 Characteristic functions and independence

Suppose that  $n_1, n_2 \in \mathbb{N}$  with  $n_1 + n_2 = n$ , and decompose  $X$  as

$$X = \begin{pmatrix} X^1 \\ X^2 \end{pmatrix},$$

where  $X^1 : \Omega \rightarrow \mathbb{R}^{n_1}$  and  $X^2 : \Omega \rightarrow \mathbb{R}^{n_2}$ . For  $\theta \in \mathbb{R}^n$  we write

$$\theta = \begin{pmatrix} \theta^1 \\ \theta^2 \end{pmatrix},$$

where  $\theta^1 \in \mathbb{R}^{n_1}$  and  $\theta^2 \in \mathbb{R}^{n_2}$ . With this notation,

$$\langle \theta, X \rangle = \left\langle \begin{pmatrix} \theta^1 \\ \theta^2 \end{pmatrix}, \begin{pmatrix} X^1 \\ X^2 \end{pmatrix} \right\rangle = \langle \theta^1, X^1 \rangle + \langle \theta^2, X^2 \rangle.$$

**Proposition A.5 (Independence and factorization).**

The random vectors  $X^1$  and  $X^2$  are independent if and only if

$$\phi_X(\theta) = \phi_{X^1}(\theta^1) \phi_{X^2}(\theta^2), \quad \text{for all } \theta = \begin{pmatrix} \theta^1 \\ \theta^2 \end{pmatrix} \text{ with } \theta^1 \in \mathbb{R}^{n_1} \text{ and } \theta^2 \in \mathbb{R}^{n_2}.$$

*Proof.* Assume first that  $X^1$  and  $X^2$  are independent. Let  $\theta = \begin{pmatrix} \theta^1 \\ \theta^2 \end{pmatrix}$  with  $\theta^1 \in \mathbb{R}^{n_1}$  and  $\theta^2 \in \mathbb{R}^{n_2}$ . Then

$$\begin{aligned} \phi_X(\theta) &= \mathbb{E}[e^{i\langle \theta, X \rangle}] = \mathbb{E}[\exp(i(\langle \theta^1, X^1 \rangle + \langle \theta^2, X^2 \rangle))] \\ &= \mathbb{E}[\exp(i\langle \theta^1, X^1 \rangle) \exp(i\langle \theta^2, X^2 \rangle)] \\ &= \mathbb{E}[\exp(i\langle \theta^1, X^1 \rangle)] \mathbb{E}[\exp(i\langle \theta^2, X^2 \rangle)] \\ &= \phi_{X^1}(\theta^1) \phi_{X^2}(\theta^2), \end{aligned}$$

which is the desired factorization.

Conversely, assume that for every  $\theta = \begin{pmatrix} \theta^1 \\ \theta^2 \end{pmatrix}$  with  $\theta^1 \in \mathbb{R}^{n_1}$  and  $\theta^2 \in \mathbb{R}^{n_2}$  we have

$$\phi_X(\theta) = \phi_{X^1}(\theta^1) \phi_{X^2}(\theta^2).$$

Let  $Y^1 \stackrel{d}{=} X^1$  and  $Y^2 \stackrel{d}{=} X^2$  be defined on some probability space and assume that  $Y^1$  and  $Y^2$  are independent. Let  $Y = ((Y^1)^\top, (Y^2)^\top)^\top$ . Then, for any  $\theta = \begin{pmatrix} \theta^1 \\ \theta^2 \end{pmatrix}$ ,

$$\begin{aligned} \phi_Y(\theta) &= \mathbb{E}[e^{i\langle \theta, Y \rangle}] = \mathbb{E}[\exp(i(\langle \theta^1, Y^1 \rangle + \langle \theta^2, Y^2 \rangle))] \\ &= \mathbb{E}[\exp(i\langle \theta^1, Y^1 \rangle)] \mathbb{E}[\exp(i\langle \theta^2, Y^2 \rangle)] \\ &= \phi_{Y^1}(\theta^1) \phi_{Y^2}(\theta^2) = \phi_{X^1}(\theta^1) \phi_{X^2}(\theta^2) = \phi_X(\theta). \end{aligned}$$

By Theorem A.2 we conclude that  $X \stackrel{d}{=} Y$ . Since  $Y^1$  and  $Y^2$  are independent, it follows that  $X^1$  and  $X^2$  are independent.  $\square$

## B

## Convolutions and characteristic functions

Let  $\mu$  and  $\nu$  denote two probability measures on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  for some  $d \in \mathbb{N}$ . The *convolution of  $\mu$  and  $\nu$*  is the probability measure  $\mu * \nu$  on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  defined as

$$\mu * \nu(B) = \iint \mathbb{1}_B(x+y) \nu(dx) \mu(dy) = \int \nu(B-y) \mu(dy), \quad \text{for } B \in \mathcal{B}(\mathbb{R}^d), \quad (\text{B.1})$$

where  $B-y = \{x \in \mathbb{R}^d \mid x+y \in B\}$ . Note that  $\mu * \nu = \nu * \mu$ .

The probabilistic interpretation of a convolution is that if  $X, Y$  are independent  $\mathbb{R}^d$ -valued random vectors such that  $X \sim \mu$  and  $Y \sim \nu$  then  $X + Y \sim \mu * \nu$ . Thus, some well-known results about the distribution of a sum of independent random variables can be formulated as

$$\begin{aligned} N(\mu_1, \sigma_1^2) * N(\mu_2, \sigma_2^2) &= N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) && \text{for } \sigma_1^2, \sigma_2^2 > 0 \text{ and } \mu_1, \mu_2 \in \mathbb{R}, \\ \Gamma(\alpha_1, \beta) * \Gamma(\alpha_2, \beta) &= \Gamma(\alpha_1 + \alpha_2, \beta) && \text{for } \alpha_1, \alpha_2, \beta > 0, \\ \text{po}(\lambda_1) * \text{po}(\lambda_2) &= \text{po}(\lambda_1 + \lambda_2) && \text{for } \lambda_1, \lambda_2 > 0. \end{aligned}$$

The *characteristic function*  $\widehat{\mu}$  of a probability measure  $\mu$  on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  is defined as

$$\widehat{\mu}(z) = \int_{\mathbb{R}^d} e^{i\langle z, x \rangle} \mu(dx), \quad z \in \mathbb{R}^d. \quad (\text{B.2})$$

For probability measures  $\mu$  and  $\nu$  as above we then have

$$\widehat{\mu * \nu}(z) = \widehat{\mu}(z) \widehat{\nu}(z), \quad z \in \mathbb{R}^d. \quad (\text{B.3})$$

Indeed, starting with  $f = \mathbb{1}_B$  for  $B \in \mathcal{B}(\mathbb{R}^d)$  in (B.1) and then using the Standard Proof it follows that for an arbitrary measurable bounded function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  (or

$f : \mathbb{R}^d \rightarrow \mathbb{C}$ ,

$$\int_{\mathbb{R}^d} f(u) (\mu * \nu)(du) = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x+y) \nu(dx) \mu(dy).$$

Choosing  $f(u) = e^{i\langle z, u \rangle}$  yields (B.3).

## C

# Information and filtrations

Consider a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ . In the following we will not use probabilities, only measurability.

We model *information* as a sub- $\sigma$ -field  $\mathcal{G} \subseteq \mathcal{F}$ . The interpretation is that based on the information  $\mathcal{G}$  we know whether any event  $A \in \mathcal{G}$  has occurred or not. However, we do not know whether an event  $A \in \mathcal{F} \setminus \mathcal{G}$  has occurred. Recall that a map  $Z : \Omega \rightarrow S$  (where  $S$  is a metric space) is  $\mathcal{G}$ -measurable if  $\{Z \in B\} \in \mathcal{G}$  for any  $B \in \mathcal{B}(S)$ . Thus, if  $Z$  is  $\mathcal{G}$ -measurable, we can think of it as being known based on the information  $\mathcal{G}$ .

**Example C.1.** If  $\mathcal{G} = \{\emptyset, \Omega\}$ , we only know that  $\emptyset$  does not occur and  $\Omega$  occurs. Since both cases are trivial, we have essentially no information in this case. Moreover,  $Z$  is  $\mathcal{G}$ -measurable if and only if  $Z$  is constant.

We get a little more information by looking at  $X : \Omega \rightarrow \{0, 1\}$  and letting  $\mathcal{G} = \{\emptyset, \{X = 0\}, \{X = 1\}, \Omega\}$ . In this case the information available is the value of  $X$ , and  $Z$  is  $\mathcal{G}$ -measurable if and only if  $Z$  is a function of  $X$ . ■

In applications it is often important to model how information develops over time, and for this we use *filtrations*. Let us start by looking at the case  $T = \mathbb{N}_0$ .

**Definition C.2 (The case  $T = \mathbb{N}_0$ ).**

A **filtration** is an increasing family  $(\mathcal{F}_n)_{n \geq 0}$  of sub  $\sigma$ -fields in  $\mathcal{F}$ .

That is, a family of  $\sigma$ -fields  $(\mathcal{F}_n)_{n \geq 0}$  is a filtration if

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}. \quad (\text{C.1})$$

Thinking of  $\mathcal{F}_n$  as the information at time  $n$ , (C.1) reflects that we get more and more information as time goes by.

**Definition C.3 (The case  $T = \mathbb{N}_0$ ).**

Let  $S$  be a metric space. Consider an  $S$ -valued stochastic process  $(Y_n)_{n \geq 0}$  and a filtration  $(\mathcal{F}_n)_{n \geq 0}$ . We say that  $(Y_n)_{n \geq 0}$  is **adapted** (or  $(\mathcal{F}_n)_{n \geq 0}$ -adapted) if  $Y_n$  is  $\mathcal{F}_n$ -measurable for every  $n \in \mathbb{N}_0$ .

Based on the discussion of information above, if  $(Y_n)_{n \geq 0}$  is adapted, we can think of  $Y_n$  as being known at time  $n$  for any  $n \in \mathbb{N}_0$ .

The natural filtration  $(\mathcal{F}_n^Y)_{n \geq 0}$  is defined as

$$\mathcal{F}_n^Y = \sigma(Y_0, \dots, Y_n), \quad \text{for } n \in \mathbb{N}_0.$$

**Remark C.4 (The case  $T = \mathbb{N}_0$ ).**

Let  $(\mathcal{F}_n)_{n \geq 0}$  be a filtration and  $(Y_n)_{n \geq 0}$  be a stochastic process. The following statements are equivalent

- $(Y_n)_{n \geq 0}$  is  $(\mathcal{F}_n)_{n \geq 0}$ -adapted;
- $Y_j$  is  $\mathcal{F}_n$ -measurable for all  $j = 0, \dots, n$  and  $n \in \mathbb{N}_0$ ;
- $\mathcal{F}_n^Y \subseteq \mathcal{F}_n$  for all  $n \in \mathbb{N}_0$ . ■

In the continuous time case  $T = \mathbb{R}_+$  we have the following.

**Definition C.5 (The case  $T = \mathbb{R}_+$ ).**

A filtration  $(\mathcal{F}_t)_{t \geq 0}$  is a family of  $\sigma$ -fields satisfying  $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$  for all  $0 \leq s \leq t$ .

Let  $S$  be a metric space, let  $(Y_t)_{t \geq 0}$  be an  $S$ -valued stochastic process, and let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration. We say that  $(Y_t)_{t \geq 0}$  is adapted if  $Y_t$  is  $\mathcal{F}_t$ -measurable for all  $t \geq 0$ .

The natural filtration  $(\mathcal{F}_t^Y)_{t \geq 0}$  generated by  $(Y_t)_{t \geq 0}$  is defined as

$$\mathcal{F}_t^Y = \sigma(Y_s \mid s \leq t), \quad t \geq 0.$$

**Remark C.6 (The case  $T = \mathbb{R}_+$ ).**

Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration and  $(Y_t)_{t \geq 0}$  be a stochastic process. The following statements are equivalent

- $(Y_t)_{t \geq 0}$  is  $(\mathcal{F}_t)_{t \geq 0}$ -adapted;
- $Y_s$  is  $\mathcal{F}_t$ -measurable for all  $0 \leq s \leq t$  and all  $t \geq 0$ ;
- $\mathcal{F}_t^Y \subseteq \mathcal{F}_t$  for all  $t \geq 0$ . ■

## D

## Stopping times in discrete time

Consider a filtered probability space  $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n \geq 0}, \mathbb{P})$ .

A map  $\tau : \Omega \rightarrow \mathbb{N}_0 \cup \{\infty\}$  is called a stopping time (or a stopping time with respect to  $(\mathcal{F}_n)_{n \geq 0}$ , or an  $(\mathcal{F}_n)_{n \geq 0}$ -stopping time) if

$$\{\tau = n\} \in \mathcal{F}_n \quad \text{for every } n \in \mathbb{N}_0.$$

In particular, a stopping time is a random variable (i.e.  $\mathcal{F}$ -measurable).

**Remark D.1.** In a course in Advanced Probability, it is shown that  $\tau : \Omega \rightarrow \mathbb{N}_0 \cup \{\infty\}$  is a stopping time if and only if

$$\{\tau \leq n\} \in \mathcal{F}_n \quad \text{for every } n \in \mathbb{N}_0. \quad \blacksquare$$

When a stopping time  $\tau$  is given, the associated stopping time  $\sigma$ -field is defined as

$$\mathcal{F}_\tau = \left\{ A \in \mathcal{F} : A \cap \{\tau = n\} \in \mathcal{F}_n \text{ for every } n \in \mathbb{N}_0 \right\}.$$

**Example D.2.** Let  $(Y_n)_{n \geq 0}$  be an adapted process with state space  $S$  and let

$$\tau = \inf\{n \in \mathbb{N}_0 : Y_n \in B\},$$

for some  $B \in \mathcal{B}(S)$  (set  $\inf \emptyset = \infty$ ). Then  $\tau$  is a stopping time, since for every  $n \in \mathbb{N}_0$ ,

$$\{\tau = n\} = \{Y_0 \notin B, \dots, Y_{n-1} \notin B, Y_n \in B\} \in \mathcal{F}_n.$$

Similarly,

$$\tau = \inf\{n \in \mathbb{N} : Y_n \in B\}$$

is a stopping time. \blacksquare

## E

## Conditional expectations and probabilities

In the following let  $(\Omega, \mathcal{F}, \mathbb{P})$  denote a fixed probability space and let  $S$  denote a Polish state space.

Recall that for  $A, B \in \mathcal{F}$  we define

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \quad \text{if } \mathbb{P}(B) > 0. \quad (\text{E.1})$$

In the case  $\mathbb{P}(B) = 0$  we let  $\mathbb{P}(A | B)$  be an arbitrary number in the interval  $[0, 1]$ . More generally, if  $Z$  is an  $\mathcal{F}$ -measurable random variable,  $Z : \Omega \rightarrow \mathbb{R}$ , which is either integrable or non-negative, define

$$\mathbb{E}[Z | B] = \frac{\mathbb{E}[Z \mathbf{1}_B]}{\mathbb{P}(B)} \quad \text{if } \mathbb{P}(B) > 0. \quad (\text{E.2})$$

(That is,  $\mathbb{E}[Z | B]$  is the mean of  $Z$  under the probability measure  $A \mapsto \mathbb{P}(A | B)$ ). If  $\mathbb{P}(B) = 0$ , we let  $\mathbb{E}[Z | B]$  be an arbitrary number in  $\mathbb{R}$  if  $Z$  is integrable and an arbitrary non-negative number if  $Z$  is non-negative.

For any sub  $\sigma$ -field  $\mathcal{G} \subseteq \mathcal{F}$  it is possible to define  $\mathbb{P}(A | \mathcal{G})$  and  $\mathbb{E}[Z | \mathcal{G}]$  in a sensible way; see Thorbjørnsen (2014). Recall the following basic definitions and existence results.

**Remark E.1.** Let  $\mathcal{G} \subseteq \mathcal{F}$  be a sub  $\sigma$ -field.

- (1) For any  $\mathcal{F}$ -measurable integrable random variable  $Z : \Omega \rightarrow \mathbb{R}$  there exists a random variable  $\mathbb{E}[Z | \mathcal{G}]$  satisfying the following two conditions (i) and (ii):
  - (i)  $\mathbb{E}[Z | \mathcal{G}] : \Omega \rightarrow \mathbb{R}$  is  $\mathcal{G}$ -measurable and integrable.
  - (ii) For every  $B \in \mathcal{G}$ ,  $\mathbb{E}[Z | \mathcal{G}]$  satisfies the *defining equation*:

$$\mathbb{E}[Z \mathbf{1}_B] = \mathbb{E}[\mathbb{E}[Z | \mathcal{G}] \mathbf{1}_B]. \quad (\text{E.3})$$

Moreover,  $\mathbb{E}[Z | \mathcal{G}]$  is a.s. uniquely determined by (i) and (ii).

- (2) For any  $\mathcal{F}$ -measurable non-negative random variable  $Z : \Omega \rightarrow [0, \infty)$  there exists a random variable  $\mathbb{E}[Z | \mathcal{G}]$  such that

- (i)  $\mathbb{E}[Z | \mathcal{G}](\omega) \in [0, \infty]$  for all  $\omega$  and  $\mathbb{E}[Z | \mathcal{G}]$  is  $\mathcal{G}$ -measurable;
- (ii)  $\mathbb{E}[Z | \mathcal{G}]$  satisfies the defining equation (E.3) for every  $B \in \mathcal{G}$ .

Moreover,  $\mathbb{E}[Z | \mathcal{G}]$  is a.s. uniquely determined by (i) and (ii).

- (3) Exercise 25 shows that in the square-integrable case,  $\mathbb{E}[Z | \mathcal{G}]$  is actually the unique  $\mathcal{G}$ -measurable random variable minimizing the  $L^2(\mathbb{P})$ -distance to  $Z$ .
- (4) For  $A \in \mathcal{F}$  let

$$\mathbb{P}(A | \mathcal{G}) = \mathbb{E}[\mathbb{1}_A | \mathcal{G}].$$

- (5) Let  $Y : \Omega \rightarrow S$  be a random variable. In this case we use the notation

$$\mathbb{E}[Z | Y] = \mathbb{E}[Z | \sigma(Y)]. \quad (\text{E.4})$$

Thorbjørnsen (2014, Opgave 4.15) shows that  $\mathbb{E}[Z | Y]$  is a measurable function of  $Y$ . ■

In (E.4),  $Y$  may have state space  $\mathbb{R}^n$ , which means that  $Y$  is of the form  $Y = (Y_1, \dots, Y_n)$  where  $Y_1, \dots, Y_n$  are real random variables.

We need some ways of calculating conditional expectations. First, a simple example is given, followed by several recipes for calculating conditional expectations in concrete cases.

**Example E.2.** Consider a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ , and let  $\mathcal{G} = \{\emptyset, A, A^c, \Omega\}$  for an  $A \in \mathcal{F}$  with  $\mathbb{P}(A) \in (0, 1)$ . Let  $Z : \Omega \rightarrow \mathbb{R}$  be  $\mathcal{F}$ -measurable and integrable or non-negative. Then

$$\mathbb{E}[Z | \mathcal{G}] = \mathbb{1}_A \mathbb{E}[Z | A] + \mathbb{1}_{A^c} \mathbb{E}[Z | A^c].$$

For a proof, see Exercise 24. ■

**Theorem E.3.** Let  $Z$  be either non-negative or integrable and let  $Y : \Omega \rightarrow S$ . Then  $\mathbb{E}[Z | Y]$  can be calculated as

$$\mathbb{E}[Z | Y] = \mathbb{E}[Z | Y = y] \Big|_{y=Y}. \quad (\text{E.5})$$

By specializing to  $Z$  being an indicator function we get a way of calculating conditional probabilities as well.

In the above theorem  $\mathbb{E}[\cdot | Y = y]$  generally indicates that we are working under a so-called regular conditional probability, which in general may be hard to determine. In the following we specialize to discrete and absolutely continuous random variables.

**Corollary E.4.** *Let  $Z$  and  $Y$  be random variables where  $Y$  has a countable state space  $S$ . Assume  $Z$  is either integrable or non-negative. Then*

$$\mathbb{E}[Z | Y] = \mathbb{E}[Z | Y = y] \text{ on } \{Y = y\} \text{ for every } y \in S \text{ with } \mathbb{P}(Y = y) > 0,$$

where the right-hand side is defined as in (E.2) with  $B = \{Y = y\}$ . Equivalently,

$$\mathbb{E}[Z | Y] = \sum_{y \in S} \mathbb{1}_{\{Y=y\}} \mathbb{E}[Z | Y = y]. \quad (\text{E.6})$$

To show this result, note that the right-hand side of (E.6) is  $\sigma(Y)$ -measurable, and therefore it is enough to verify the defining equation (E.3). This is done essentially as in Example E.2.

**Example E.5.** Let  $X$  and  $Y$  be independent such that  $X \sim N(0, 1)$  and  $Y$  has state space  $\{0, 1\}$ . Define  $Z$  as

$$Z = \begin{cases} X & \text{if } Y = 0 \\ X^2 & \text{if } Y = 1. \end{cases}$$

Then

$$\mathbb{E}[Z | Y = y] = \begin{cases} \mathbb{E}[X] = 0 & \text{if } y = 0 \\ \mathbb{E}[X^2] = 1 & \text{if } y = 1 \end{cases}$$

and hence, by (E.6),

$$\mathbb{E}[Z | Y] = 0 \cdot \mathbb{1}_{\{Y=0\}} + 1 \cdot \mathbb{1}_{\{Y=1\}}. \quad \blacksquare$$

**Corollary E.6.** *Let  $(X, Y)$  be absolutely continuous with respect to a measure  $\mu \times \nu$  with density  $(x, y) \mapsto f_{X,Y}(x, y)$ . Define the conditional density  $f_{X|Y}(x|y)$  as*

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \quad \text{for } y \text{ with } f_Y(y) > 0,$$

where  $f_Y$  denotes the marginal density of  $Y$ , that is,  $f_Y(y) = \int f_{X,Y}(x, y) \mu(dx)$ . Let  $\phi(X, Y)$  be a measurable real-valued function of  $(X, Y)$  which is either non-negative or satisfies that  $\phi(X, Y)$  is integrable. Then

$$\mathbb{E}[\phi(X, Y) | Y] = \mathbb{E}[\phi(X, Y) | Y = y] \Big|_{y=Y}, \quad (\text{E.7})$$

where  $\mathbb{E}[\phi(X, Y) | Y = y]$  is given as

$$\mathbb{E}[\phi(X, Y) | Y = y] = \int \phi(x, y) f_{X|Y}(x|y) \mu(dx).$$

Once again, in order to show this result, it is not too difficult to verify the defining equation (E.3).

**Theorem E.7 (The useful rule).**

Let  $\mathcal{G}$  be a  $\sigma$ -field and  $U, V$  be random variables taking values in  $S$ . Assume  $U$  is independent of  $\mathcal{G}$  and  $V$  is  $\mathcal{G}$ -measurable. Let  $H : S^2 \rightarrow \mathbb{R}$  be measurable and assume  $H$  is non-negative or  $H(U, V)$  is integrable. Then

$$\mathbb{E}[H(U, V) | \mathcal{G}] = \mathbb{E}[H(U, v)] \Big|_{v=V}. \quad (\text{E.8})$$

**Remark E.8.** We have only stated the useful rule when  $U$  and  $V$  have the same state space but it generalizes in the obvious way to different state spaces.

A proof of Theorem E.7 is given in Exercise 26. In the proof it may be useful to note that (E.8) can also be written as

$$\mathbb{E}[H(U, V) | \mathcal{G}] = \tilde{H}(V),$$

where  $\tilde{H} : S \rightarrow \mathbb{R}$  is defined as  $\tilde{H}(v) = \mathbb{E}[H(U, v)]$  for  $v \in S$ . ■

**Example E.9.** Let  $U$  and  $V$  be independent such that  $U \stackrel{d}{=} V \sim N(0, 1)$ . Then  $\mathbb{P}(U \leq V | V) = \Phi(V)$ , where  $\Phi$  denotes the CDF of the standard normal distribution.

To see this, first note that  $\mathbb{P}(U \leq v) = \Phi(v)$  for  $v \in \mathbb{R}$ . Hence by the (very!) useful rule with  $\mathcal{G} = \sigma(V)$ ,

$$\mathbb{P}(U \leq V | V) = \mathbb{P}(U \leq v) \Big|_{v=V} = \Phi(V). \quad \blacksquare$$

Let us list some of the most useful rules for calculating with conditional expectations. (See Thorbjørnsen (2014) for proofs and further results.)

**Theorem E.10.** Let  $\mathcal{G} \subseteq \mathcal{F}$  be a sub  $\sigma$ -field and  $Z : \Omega \rightarrow \mathbb{R}$  be an  $\mathcal{F}$ -measurable random variable that is either integrable or non-negative.

- (1) If  $\mathcal{G} = \{\emptyset, \Omega\}$  then  $\mathbb{E}[Z | \mathcal{G}] = \mathbb{E}[Z]$ .
- (2) (The tower principle). If  $\mathcal{H} \subseteq \mathcal{G}$  is a sub  $\sigma$ -field then  $\mathbb{E}[\mathbb{E}[Z | \mathcal{G}] | \mathcal{H}] = \mathbb{E}[Z | \mathcal{H}]$ .
- (3) If  $Z$  is  $\mathcal{G}$ -measurable then  $\mathbb{E}[Z | \mathcal{G}] = Z$ .
- (4) We have  $\mathbb{E}[Z] = \mathbb{E}[\mathbb{E}[Z | \mathcal{G}]]$ .
- (5) If  $U : \Omega \rightarrow \mathbb{R}$  is  $\mathcal{G}$ -measurable and bounded and  $Z$  is integrable then  $\mathbb{E}[UZ | \mathcal{G}] = U\mathbb{E}[Z | \mathcal{G}]$ .
- (6) If  $U : \Omega \rightarrow [0, \infty)$  is  $\mathcal{G}$ -measurable and  $Z$  is non-negative then  $\mathbb{E}[UZ | \mathcal{G}] = U\mathbb{E}[Z | \mathcal{G}]$ .

- (7) If  $Z_j : \Omega \rightarrow \mathbb{R}$ ,  $j = 1, 2$ , are  $\mathcal{F}$ -measurable and both are either non-negative or integrable then  $\mathbb{E}[Z_1 + Z_2 | \mathcal{G}] = \mathbb{E}[Z_1 | \mathcal{G}] + \mathbb{E}[Z_2 | \mathcal{G}]$ .
- (8) If  $Z$  and  $\mathcal{G}$  are independent then  $\mathbb{E}[Z | \mathcal{G}] = \mathbb{E}[Z]$ .

## F

## The multivariate normal distribution

Let  $n \in \mathbb{N}$ . Recall that for a real  $n \times n$  matrix  $A$ , we say that  $A$  is *non-negative definite* (also *positive semidefinite*) if  $A$  is symmetric ( $A^\top = A$ ) and

$$x^\top A x \geq 0, \quad \text{for all } x \in \mathbb{R}^n.$$

Likewise,  $A$  is *positive definite* if  $A$  is symmetric and

$$x^\top A x > 0, \quad \text{for all } x \in \mathbb{R}^n \setminus \{0\}.$$

Also recall that a symmetric  $n \times n$  matrix  $A$  has  $n$  real eigenvalues. Then  $A$  is non-negative definite if and only if all of these are non-negative, and  $A$  is positive definite if and only if all of these are positive. In particular, if  $A$  is positive definite, then it is invertible.

**Definition F.1.** A random vector  $X = (X_1, \dots, X_n)^\top$  with state space  $S = \mathbb{R}^n$  follows an  $n$ -variate normal distribution if the characteristic function of  $X$  is

$$\mathbb{E}[e^{i\langle \theta, X \rangle}] = \exp \left[ i\langle \theta, \mu \rangle - \frac{1}{2} \theta^\top \Sigma \theta \right], \quad \text{for } \theta \in \mathbb{R}^n \quad (\text{F.1})$$

for some  $\mu = (\mu_1, \dots, \mu_n)^\top \in \mathbb{R}^n$  and some  $n \times n$  non-negative definite matrix  $\Sigma = \{\sigma_{ij}\}_{i,j=1}^n$ .

Recall that the distribution of  $X$  is uniquely determined by the characteristic function. In particular the distribution of  $X$  is completely determined by  $\mu$  and  $\Sigma$ . We use the notation  $X \sim N_n(\mu, \Sigma)$  when the characteristic function of  $X$  is given by (F.1). Example F.3 below shows that we also have existence of  $N_n(\mu, \Sigma)$ .

**Example F.2 (The case  $n = 1$ ).**

Let  $n = 1$ . Then  $\Sigma$  is a  $1 \times 1$  matrix, i.e. essentially a number, which is non-negative since  $\Sigma$  is non-negative definite. Hence we write  $\Sigma = \{\sigma^2\}$  where  $\sigma^2 \geq 0$ . Also use the usual notation  $N(\mu, \sigma^2)$  as an alternative to  $N_1(\mu, \Sigma)$ .

In the case  $\sigma^2 = 0$ ,  $X \sim N(\mu, 0)$  if

$$\mathbb{E}[e^{i\theta X}] = \exp(i\mu\theta), \quad \theta \in \mathbb{R},$$

which is the characteristic function of the constant  $\mu$ . So in other words,  $X \sim N(\mu, 0)$  if and only if  $\mathbb{P}(X = \mu) = 1$ .

When  $\sigma^2 > 0$ ,  $X \sim N(\mu, \sigma^2)$  if and only if  $X$  is absolutely continuous with density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad x \in \mathbb{R}. \quad \blacksquare$$

**Example F.3 (Existence and construction).**

In the following we show how to construct an  $N_n(\mu, \Sigma)$ -distributed random vector from i.i.d. standard normals.

Let  $\mu \in \mathbb{R}^n$  and  $\Sigma$  be an  $n \times n$  non-negative definite matrix, and let  $U_1, \dots, U_n$  be i.i.d. with  $U_j \sim N(0, 1)$ . Then

$$U := (U_1, \dots, U_n)^\top \sim N_n(0, I_n), \quad (\text{F.2})$$

where  $I_n$  denotes the  $n \times n$  identity matrix. (To verify (F.2) calculate the characteristic function or apply (5) in Theorem F.5 below).

Let  $\Sigma^{1/2}$  denote the unique non-negative definite square root of  $\Sigma$ ; that is,  $\Sigma^{1/2}$  is  $n \times n$  non-negative definite and  $\Sigma = \Sigma^{1/2} \Sigma^{1/2}$ . Then  $X$  defined as  $X = \mu + \Sigma^{1/2} U$  satisfies

$$X \sim N_n(\mu, \Sigma).$$

Indeed, this follows from (F.3) below. ■

**Remark F.4 (Notation).**

Consider random vectors  $X = (X_1, \dots, X_n)^\top$  and  $Y = (Y_1, \dots, Y_d)^\top$  in  $\mathbb{R}^n$  resp.  $\mathbb{R}^d$ . Assume all random variables are square integrable. Then we let  $\mathbb{E}[X]$  denote the vector  $(\mathbb{E}[X_1], \dots, \mathbb{E}[X_n])^\top$  and  $\text{Cov}(X, Y)$  be the  $n \times d$  matrix whose  $(i, j)$ th element is  $\text{Cov}(X_i, Y_j)$ . Let  $\text{Var}(X) = \text{Cov}(X, X)$ . Note that  $\text{Cov}(Y, X) = \text{Cov}(X, Y)^\top$ .

We say that  $X$  and  $Y$  are uncorrelated if  $\text{Cov}(X, Y) = 0$  (where the right-hand side is of dimension  $n \times d$ ). ■

The following result is essentially proved in Exercise 5 and 6.

**Theorem F.5 (Properties of the multivariate normal distribution).**

- (1) If  $X \sim N_n(\mu, \Sigma)$  then  $\mu = \mathbb{E}[X]$  and  $\Sigma = \text{Cov}(X)$ .
- (2) If  $X \sim N_n(\mu, \{\sigma_{ij}\}_{i,j=1}^n)$  then  $X_i \sim N(\mu_i, \sigma_{ii})$  for  $i = 1, \dots, n$ .
- (3) Let  $X \sim N_n(\mu, \Sigma)$ ,  $d \in \mathbb{N}$ ,  $a \in \mathbb{R}^d$  and  $B$  be a  $d \times n$  matrix with real entries. Then

$$a + BX \sim N_d(a + B\mu, B\Sigma B^\top). \quad (\text{F.3})$$

- (4) In a multivariate normal distribution, a collection of components is independent if and only if all pairwise covariances are zero (equivalently, the corresponding covariance matrix is diagonal).

We make this precise in the case of two components. Assume  $X \sim N_n(\mu, \Sigma)$  and decompose  $X$  as

$$X = \begin{pmatrix} X^{(1)} \\ X^{(2)} \end{pmatrix}, \quad (\text{F.4})$$

where  $X^{(i)}$  is  $n_i$ -dimensional with  $n_i \in \mathbb{N}$  and  $n_1 + n_2 = n$ . Then  $X^{(1)}$  and  $X^{(2)}$  are independent if and only if  $\text{Cov}(X^{(1)}, X^{(2)}) = 0$ .

- (5) Assume  $X^{(i)} \sim N_{n_i}(\mu^{(i)}, \Sigma_{ii})$  for  $i = 1, 2$ , and that  $X^{(1)}$  and  $X^{(2)}$  are independent. Then

$$\begin{pmatrix} X^{(1)} \\ X^{(2)} \end{pmatrix} \sim N_{n_1+n_2} \left( \begin{pmatrix} \mu^{(1)} \\ \mu^{(2)} \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix} \right),$$

where the 0's are of appropriate dimensions. There is a similar result for more than two components.

- (6) If  $X \sim N_n(\mu, \Sigma)$  then  $X$  has a density with respect to the Lebesgue measure on  $\mathbb{R}^n$  if and only if  $\Sigma$  is positive definite. In that case, the density is

$$f_X(x) = \frac{1}{(2\pi)^{n/2} \det(\Sigma)^{1/2}} \exp \left( -\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right), \quad x \in \mathbb{R}^n.$$

**Remark F.6.** If  $X^{(i)} \sim N_{n_i}(\mu^{(i)}, \Sigma_{ii})$  for  $i = 1, 2$  then generally the vector  $X$  in (F.4) does not follow a multivariate normal distribution – unless e.g.  $X^{(1)}$  and  $X^{(2)}$  are independent. An example is given in Exercise 4. ■

Let us also note that the class of  $n$ -variate normal distributions is closed under convergence in distribution. Indeed, this is proved in Thorbjørnsen (2014, opgave 5.11) in the case  $n = 1$ , and it is not too difficult to reduce the general case to  $n = 1$  by the Cramér-Wold device.

## G

## The càdlàg property

A map  $f : [0, \infty) \rightarrow S$  is called càdlàg (French: *continue à droite, limites à gauche*) if it is right-continuous on  $[0, \infty)$  and has left limits on  $(0, \infty)$ . In other words, the *left limit*  $f(t-)$ , defined as  $f(t-) = \lim_{s \uparrow t} f(s)$ , exists for every  $t > 0$ . Similarly, the *right limit*,  $f(t+) = \lim_{s \downarrow t} f(s)$ , exists and is equal to  $f(t)$  for all  $t \geq 0$ .

For example, a distribution function  $F : \mathbb{R} \rightarrow [0, 1]$  is non-decreasing and càdlàg when restricted to  $[0, \infty)$ .

Let  $S$  be a subset of  $\mathbb{R}^d$  for some  $d \in \mathbb{N}$ . Then, for  $t > 0$  we define *the jump* at time  $t$  as  $\Delta f(t) = f(t) - f(t-)$ . We have the following properties.

**Proposition G.1.** *Let  $f$  be càdlàg and  $S \subseteq \mathbb{R}^d$  for some  $d \in \mathbb{N}$ .*

- (1) *The map  $t \mapsto f(t)$  is locally bounded; that is, for every  $T \in (0, \infty)$ ,  $\sup_{t \leq T} \|f(t)\| < \infty$ .*
- (2) *For every  $T \in (0, \infty)$  and  $\epsilon > 0$ , the set of  $t \in (0, T]$  for which  $\|\Delta f(t)\| \geq \epsilon$  is finite. Consequently, the set*

$$\{t > 0 : \Delta f(t) \neq 0\}$$

*is countable.*

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## The Monotone Class Theorem

**Theorem H.1 (The Monotone Class Theorem).**

Let  $\Omega$  be a set. Assume  $\mathcal{A}$  is a collection of subsets of  $\Omega$  which is stable under intersection. Also assume  $\Omega \in \mathcal{A}$ .

Let  $\mathcal{H}$  be a collection of real-valued functions that satisfies:

- (1) If  $A \in \mathcal{A}$  then  $\mathbb{1}_A \in \mathcal{H}$ .
- (2) If  $f, g \in \mathcal{H}$  then  $f + g \in \mathcal{H}$  and  $cf \in \mathcal{H}$  for  $c \in \mathbb{R}$ .
- (3) If  $0 \leq f_1 \leq f_2 \leq \dots$  with  $f_n \in \mathcal{H}$  for all  $n$ , and if  $f := \lim_{n \rightarrow \infty} f_n$  is bounded, then  $f \in \mathcal{H}$ .

Then  $\mathcal{H}$  contains every bounded function that is measurable with respect to  $\sigma(\mathcal{A})$ .

The particular case when  $\mathcal{A}$  is a  $\sigma$ -field (i.e.,  $\mathcal{A} = \sigma(\mathcal{A})$ ) is often referred to as *the standard proof*.

For an even more general formulation of the Monotone Class Theorem, see e.g., Graversen (2011).

# Exercises

## Exercise 1.

Consider a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  and let  $S$  be a metric space.

- (1) Let  $A_1, A_2, A_3 \in \mathcal{F}$ . What does it mean that  $A_1, A_2$  and  $A_3$  are independent?

*Hint:* Pairwise independence is not enough...

- (2) Consider  $n$  sub- $\sigma$ -fields  $\mathcal{G}_1, \dots, \mathcal{G}_n \subseteq \mathcal{F}$ .

(a) What does it mean that  $\mathcal{G}_1, \dots, \mathcal{G}_n$  are independent? You may want to recall different characterizations.

(b) Let  $n = 3$ . Assume  $\mathbb{P}(A_1 \cap A_2 \cap A_3) = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3)$  for all  $A_i \in \mathcal{G}_i$ ,  $i = 1, 2, 3$ . Are  $\mathcal{G}_1, \mathcal{G}_2$ , and  $\mathcal{G}_3$  necessarily independent?

(c) Let  $n = 3$ . Assume  $\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\mathbb{P}(A_2)$ ,  $\mathbb{P}(A_1 \cap A_3) = \mathbb{P}(A_1)\mathbb{P}(A_3)$ , and  $\mathbb{P}(A_2 \cap A_3) = \mathbb{P}(A_2)\mathbb{P}(A_3)$  for all  $A_i \in \mathcal{G}_i$ ,  $i = 1, 2, 3$ . Are  $\mathcal{G}_1, \mathcal{G}_2$ , and  $\mathcal{G}_3$  necessarily independent?

- (3) What does it mean that  $n$  random variables  $X_1, \dots, X_n$  with state space  $S$  are independent?

Characterize independence:

(a) in terms of the PMF when  $X_1, \dots, X_n$  are discrete

(b) in terms of the density when  $X_1, \dots, X_n$  are absolutely continuous

(c) in terms of the CDF when  $X_1, \dots, X_n$  are real random variables.

- (4) Assume  $S = \mathbb{R}$  and that  $X_1$  and  $X_2$  are square-integrable random variables. Does independence of  $X_1$  and  $X_2$  imply that they are uncorrelated? Does the converse hold?

**Exercise 2.**

Let  $X$  and  $Y$  be independent and assume  $X \sim Y \sim N(0, 1)$ .

- (1) Determine the distribution of  $X + Y$  and of  $X - Y$ .
- (2) Are  $X + Y$  and  $X - Y$  independent?

**Exercise 3.**

Let  $X \sim N_1(\mu, \sigma^2)$  where  $\mu \in \mathbb{R}$  and  $\sigma^2 \geq 0$ .

- (1) Does  $(X, X)^\top$  follow a bivariate normal distribution?
- (2) Is  $(X, X)^\top$  absolutely continuous with respect to the Lebesgue measure on  $\mathbb{R}^2$ ?

**Exercise 4.**

Let  $X$  and  $B$  be independent such that  $X \sim N(0, 1)$  and  $\mathbb{P}(B = 0) = \mathbb{P}(B = 1) = 1/2$ . Put

$$Y = X \mathbb{1}_{\{B=0\}} - X \mathbb{1}_{\{B=1\}}.$$

- (1) Show that  $Y \sim N(0, 1)$ .  
*Hint:* Show that  $\mathbb{P}(Y \leq y) = \Phi(y)$  for every  $y \in \mathbb{R}$ , where  $\Phi$  denotes the CDF of a standard normal distribution.
- (2) Show that  $\mathbb{P}(X + Y = 0) = 1/2$ .
- (3) Does  $(X, Y)^\top$  follow a bivariate normal distribution?
- (4) Calculate  $\text{Cov}(Y, B)$ .

**Exercise 5.**

The purpose of this exercise is to show Theorem F.5(4)-(5) – although only in a particular case in order not to get carried away by notation.

- (1) Assume  $X_i \sim N_1(\mu_i, \sigma_{ii})$  for  $i = 1, 2$  and that  $X_1$  and  $X_2$  are independent. Determine the characteristic function of  $(X_1, X_2)^\top$ . Deduce that the joint distribution is  $(X_1, X_2)^\top \sim N_2(\mu, \Sigma)$ , where

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{22} \end{pmatrix}. \quad (\text{H.1})$$

- (2) Assume  $(Y_1, Y_2)^\top \sim N_2(\mu, \Sigma)$ , where  $\mu$  and  $\Sigma$  are as in (H.1). Show that  $Y_i \sim N_1(\mu_i, \sigma_{ii})$  for  $i = 1, 2$  and that  $Y_1$  and  $Y_2$  are independent.

**Exercise 6.**

- (1) Prove properties (3) and (2) in Theorem F.5.

- (2) Read Example F.3. Use the representation of  $X$  in this example to show (6) in Theorem F.5.

*Hint:* Note the following. If an  $\mathbb{R}^n$ -valued random vector  $X$  is a.s. contained in an affine subset of dimension strictly less than  $n$  then  $X$  cannot be absolutely continuous with respect to the Lebesgue measure on  $\mathbb{R}^n$ .

**Exercise 7.**

A stochastic process  $(Y_n)_{n \geq 0}$  is called *reversible* if  $(Y_0, \dots, Y_n) \stackrel{d}{=} (Y_n, \dots, Y_0)$  for every  $n \in \mathbb{N}_0$ . Recall that  $(Y_n)_{n \geq 0}$  is stationary if  $(Y_0, \dots, Y_n) \stackrel{d}{=} (Y_k, \dots, Y_{k+n})$  for all  $k, n \in \mathbb{N}_0$ . In the following let  $(Y_n^k)_{n \geq 0}$  (for  $k \in \mathbb{N}_0$ ) be the stochastic process defined as  $Y_n^k = Y_{k+n}$  for  $n \in \mathbb{N}_0$ .

- (1) Argue that  $(Y_n)_{n \geq 0}$  is stationary if and only if  $Y \stackrel{d}{=} Y^k$  for every  $k \in \mathbb{N}_0$ .

- (2) Show that  $(Y_n)_{n \geq 0}$  is stationary if and only if  $Y \stackrel{d}{=} Y^1$ .

*Hint:* Your solution may involve the shift operators in (4.39).

- (3) Show that if  $(Y_n)_{n \geq 0}$  is reversible then it is also stationary.

**Exercise 8.**

The following relies on Definition 2.2 and Definition 2.4.

- (1) Let  $(\epsilon_n)_{n \in \mathbb{Z}}$  be real random variables that are i.i.d. and assume  $\epsilon_n \in L^2(\mathbb{P})$  with  $\mathbb{E}[\epsilon_n] = 0$  for  $n \in \mathbb{Z}$ . Argue that  $(\epsilon_n)_{n \in \mathbb{Z}}$  is a white noise.
- (2) Let  $(X_n)_{n \in \mathbb{Z}}$  be a stationary process with  $X_n \in L^2(\mathbb{P})$  for  $n \in \mathbb{Z}$ . Show that  $(X_n)_{n \in \mathbb{Z}}$  is weakly stationary.

**Exercise 9.**

Let  $(\epsilon_n)_{n \in \mathbb{Z}}$  denote a white noise and  $(\alpha(j))_{j \in \mathbb{N}_0}$  be a real sequence such that  $\sum_{j=0}^{\infty} \alpha(j)^2 < \infty$ .

- (1) Argue that  $X_n$  in (2.2) exists in  $L^2(\mathbb{P})$  for  $n \in \mathbb{Z}$ .
- (2) Show that  $\mathbb{E}[X_n] = 0$  for every  $n \in \mathbb{Z}$  and that (2.3) is satisfied. (That is,  $(X_n)_{n \in \mathbb{Z}}$  is weakly stationary.)
- (3) Argue that  $(X_n)_{n \in \mathbb{Z}}$  is stationary if  $(\epsilon_n)_{n \in \mathbb{Z}}$  is i.i.d.

**Exercise 10.**

Prove the three results in Example 2.6.

**Exercise 11.**

Let the situation be as in Example 2.7 with  $p = q = 1$ . Assume  $\alpha_1 + \beta_1 < 1$ ,  $\alpha_0 > 0$  and  $\alpha_1, \beta_1 \geq 0$ . Define  $\sigma_n^2$  by (2.12).

- (1) Show that  $\mathbb{E}[\sigma_n^2] = \frac{\alpha_0}{1-\alpha_1-\beta_1}$ .
- (2) Show that  $(X_n)_{n \in \mathbb{Z}}$  in (2.10) is a white noise.

**Exercise 12.**

Let  $(Y_n)_{n \geq 0}$  and  $(Z_n)_{n \geq 0}$  be real stochastic processes that are versions. Show that  $Y_n \rightarrow 7$  a.s. as  $n \rightarrow \infty$  if and only if  $Z_n \rightarrow 7$  a.s. as  $n \rightarrow \infty$ .

**Exercise 13.**

Let  $(\epsilon_n)_{n \geq 0}$  denote i.i.d. random variables such that  $\epsilon_n \sim N(0, 1)$  for  $n \in \mathbb{N}_0$ .

- (1) Show that  $(\epsilon_n)_{n \geq 0}$  is a Gaussian and stationary process.
- (2) Let  $X_n = \epsilon_n + \epsilon_{n+1}$  for  $n \in \mathbb{N}_0$ . Is  $(X_n)_{n \geq 0}$  a Gaussian process? Is it stationary?
- (3) Let  $Y_n = X_n + X_0$  for  $n \in \mathbb{N}_0$ . Is  $(Y_n)_{n \geq 0}$  a Gaussian process? Is it stationary?
- (4) Are  $(X_n)_{n \geq 0}$  and  $(\sqrt{2}\epsilon_n)_{n \geq 0}$  versions?
- (5) Let  $Z_n = \epsilon_0$  for  $n \in \mathbb{N}_0$ . Is  $(Z_n)_{n \geq 0}$  a Gaussian process? Is it stationary?

**Exercise 14.**

Prove the results in point 4. of Example 2.3.

**Exercise 15.**

Let  $(Y_n)_{n \geq 0}$  and  $(Z_n)_{n \geq 0}$  denote real stochastic processes.

- (1) Assume  $Y \stackrel{d}{=} Z$ . Show that  $Y_n \stackrel{d}{=} Z_n$  for every  $n \in \mathbb{N}_0$ .
- (2) Assume  $Y \stackrel{d}{=} Z$ . Show that if  $(Y_n)_{n \geq 0}$  is i.i.d. then so is  $(Z_n)_{n \geq 0}$ .
- (3) Assume  $Y \stackrel{d}{=} Z$ . Show that if  $(Y_n)_{n \geq 0}$  is stationary then so is  $(Z_n)_{n \geq 0}$ .
- (4) Assume  $Y \stackrel{d}{=} Z$  and let  $r > 0$ . Show that if  $Y_n \in L^r(\mathbb{P})$  for every  $n \in \mathbb{N}_0$  then  $Z_n \in L^r(\mathbb{P})$  for every  $n \in \mathbb{N}_0$ .
- (5) Assume  $Y \stackrel{d}{=} Z$  and  $Y_n, Z_n \in L^2(\mathbb{P})$  for every  $n \in \mathbb{N}_0$ . Show that  $\mathbb{E}[Y_n] = \mathbb{E}[Z_n]$  and  $\text{Cov}(Y_n, Y_m) = \text{Cov}(Z_n, Z_m)$  for all  $n, m \in \mathbb{N}_0$ .
- (6) Assume  $(Y_n)_{n \geq 0}$  and  $(Z_n)_{n \geq 0}$  are i.i.d. and  $Y_0 \stackrel{d}{=} Z_0$ . Show that  $Y \stackrel{d}{=} Z$ .

**Exercise 16.**

Let  $d \in \mathbb{N}$  and  $\mu$  denote your favorite probability measure on  $S = \mathbb{R}^d$ . (Thus, when  $d = 1$ ,  $\mu$  could e.g. be a normal, Poisson or gamma distribution).

- (1) Show that there exists a probability measure on  $(S^{\mathbb{N}_0}, \mathcal{B}(S)^{\mathbb{N}_0})$  under which  $(X_n)_{n \geq 0}$  is i.i.d. with  $X_n \sim \mu$  for all  $n \in \mathbb{N}_0$ .
- (2) Argue that this probability measure is unique.

**Exercise 17.**

In the following we consider real processes.

- (1) Let  $n \in \mathbb{N}$  and consider real numbers  $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$ . Put  $t_0 = 0$ . Let  $Z_1, \dots, Z_n$  be independent random variables such that  $Z_j \sim N(0, t_j - t_{j-1})$  for  $j = 1, \dots, n$ . Determine the distribution of the  $n$ -dimensional vector  $(Z_1, Z_1 + Z_2, \dots, Z_1 + \dots + Z_n)$ .
- (2) Show that there exists a Gaussian process  $(Y_t)_{t \geq 0}$  (with index set  $T = \mathbb{R}_+$ ) satisfying  $\mathbb{E}[Y_t] = 0$  and  $\text{Cov}(Y_s, Y_t) = s \wedge t$  for  $s, t \geq 0$ .
- (3) Let  $(Y_t)_{t \geq 0}$  be as above. Determine the distribution of  $Y_t - Y_s$  for  $0 \leq s \leq t$  and show that  $Y_t - Y_s$  and  $Y_v - Y_u$  are independent for  $0 \leq s \leq t \leq u \leq v$ .

**Exercise 18.**

Show the result in Example 3.3.

**Exercise 19.**

In the setting of Chapter 3  $\mathcal{B}(S)^{\mathbb{N}_0}$  is the least  $\sigma$ -field containing all events of the form (3.1).

Argue that  $\mathcal{B}(S)^{\mathbb{N}_0}$  is also the least  $\sigma$ -field containing all events of the form (3.2). Explain in a few words why it is often preferable to use (3.2) rather than (3.1).

**Exercise 20.**

Let  $(Y_n)_{n \geq 0}$  and  $(Y'_n)_{n \geq 0}$  be stochastic processes. Is it true that these are versions if  $Y_n \stackrel{d}{=} Y'_n$  for all  $n \in \mathbb{N}_0$ ? Prove it if you think the answer is “yes” and provide a simple counterexample if you think the answer is “no”.

**Exercise 21.**

Let  $R, T$  be non-empty index sets, and let  $(Y_t)_{t \in T}$  and  $(Z_r)_{r \in R}$  be stochastic processes indexed by  $T$  resp.  $R$  defined on the same probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ . For simplicity assume the two processes have the same state space  $S$ . Let  $\mathcal{F}^Y$  be the  $\sigma$ -field generated by  $Y_t, t \in T$ , and  $\mathcal{F}^Z$  be the  $\sigma$ -field generated by  $Z_r, r \in R$ .

The two processes are called independent if  $\mathcal{F}^Y$  and  $\mathcal{F}^Z$  are independent.

Show that  $(Y_t)_{t \in T}$  and  $(Z_r)_{r \in R}$  are independent if and only if

$$\begin{aligned} \mathbb{P}(Y_{t_1} \in A_1, \dots, Y_{t_n} \in A_n, Z_{r_1} \in B_1, \dots, Z_{r_m} \in B_m) \\ = \mathbb{P}(Y_{t_1} \in A_1, \dots, Y_{t_n} \in A_n) \cdot \mathbb{P}(Z_{r_1} \in B_1, \dots, Z_{r_m} \in B_m) \end{aligned}$$

for all  $n, m \in \mathbb{N}$ ,  $t_1, \dots, t_n \in T$ ,  $r_1, \dots, r_m \in R$  and  $A_1, \dots, A_n, B_1, \dots, B_m \in \mathcal{B}(S)$ .

**Exercise 22.**

Verify equation (4.16).

**Exercise 23.**

Verify the result in (4.9).

**Exercise 24.**

- (1) Show the existence and uniqueness result in Remark E.1(2).

*Hint:* You may rely on Remark E.1(1).

- (2) Show the results in Example E.2.

**Exercise 25.**

Let  $Z : \Omega \rightarrow \mathbb{R}$  denote a square integrable random variable defined on some probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ .

- (1) Show that for any  $a \in \mathbb{R}$ ,  $\mathbb{E}[(Z - a)^2] \geq \mathbb{E}[(Z - \mathbb{E}Z)^2]$  with equality if and only if  $a = \mathbb{E}Z$ . (That is,  $\mathbb{E}Z$  is the unique constant which minimizes the  $L^2(\mathbb{P})$ -distance to  $Z$ ).

*Hint:* Use the decomposition  $Z - a = (Z - \mathbb{E}Z) + (\mathbb{E}Z - a)$ .

- (2) Let  $\mathcal{G} \subset \mathcal{F}$  denote a  $\sigma$ -field. Then  $\mathbb{E}[Z | \mathcal{G}]$  is square integrable as well. Show that for any  $\mathcal{G}$ -measurable square integrable random variable  $U$ ,

$$\mathbb{E}[(Z - U)^2] \geq \mathbb{E}[(Z - \mathbb{E}[Z | \mathcal{G}])^2]$$

with equality if and only if  $U = \mathbb{E}[Z | \mathcal{G}]$  (a.s.). (That is,  $\mathbb{E}[Z | \mathcal{G}]$  is the a.s. unique square-integrable  $\mathcal{G}$ -measurable random variable which minimizes the  $L^2(\mathbb{P})$ -distance to  $Z$ ).

**Exercise 26.**

Prove the useful rule Theorem E.7 cf. also Remark E.8.

*Hint:* First consider  $H(u, v) = \mathbb{1}_{A \times B}(u, v)$  where  $A, B \in \mathcal{B}(S)$ .

**Exercise 27.**

Let  $Y_1$  and  $Y_2$  be independent real random variables such that  $Y_j \sim \text{Exp}(\lambda_j)$  (where  $\lambda_j > 0$ ) for  $j = 1, 2$ .

Determine the (conditional) probabilities and expectations below.

- (1)  $\mathbb{P}(Y_1 > Y_2 | Y_2)$
- (2)  $\mathbb{P}(Y_1 > Y_2)$
- (3)  $\mathbb{E}[Y_1 \cdot Y_2 | Y_2]$
- (4)  $\text{Var}(3Y_1 - 4Y_2 | Y_2)$
- (5)  $\text{Var}(3Y_1 - 4Y_2)$
- (6)  $\mathbb{P}(Y_1 > Y_2 | Y_1, Y_2)$

(7)  $\text{Var}(Y_1 \cdot Y_2 | Y_2)$ .

**Exercise 28.**

Let  $(X, Y)$  be a random vector where

$$\mathbb{P}(X = x, Y = y) = \begin{cases} \frac{1}{4} & \text{if } y = 0 \text{ and } x = 0 \\ \frac{1}{4} & \text{if } y = 1 \text{ and } x = 0 \\ \frac{1}{4} & \text{if } y = 1 \text{ and } x = 1 \\ \frac{1}{4} e^{-\lambda} \frac{\lambda^x}{x!} & \text{if } y = 2 \text{ and } x \in \mathbb{N}_0 \end{cases}$$

and  $\lambda > 0$ .

- (1) Determine the conditional distribution of  $X$  given  $Y = y$  for  $y = 0, 1, 2$ .
- (2) Are  $X$  and  $Y$  independent?
- (3) Determine  $\mathbb{E}[X | Y]$ .

**Exercise 29.**

Let  $(X, Y, Z)$  be an absolutely continuous random vector taking values in  $\mathbb{R}^3$  with joint density

$$f_{X,Y,Z}(x, y, z) = \begin{cases} 2 & \text{if } x, y, z \in (0, 1) \text{ and } x \leq y \\ 0 & \text{otherwise.} \end{cases}$$

- (1) Are  $(X, Y)$  and  $Z$  independent?
- (2) Determine  $\mathbb{E}[X | Y]$ ,  $\mathbb{E}[XY | Y]$ ,  $\mathbb{E}[XZ | Y]$ , and  $\mathbb{E}[Z | X]$ .

**Exercise 30.**

- (1) In the following,  $Y$  is a random variable with state space  $S = \mathbb{Z} \setminus \{0\}$ . Let  $p(y) = \mathbb{P}(Y = y)$  for  $y \in S$  and for simplicity assume  $p(y) > 0$  for all  $y \in S$ . Put  $Z = |Y|$ .

Determine  $\mathbb{P}(Y = Z | Z = z)$  for  $z \in \mathbb{N}$  in terms of  $p(z)$  and  $p(-z)$ . Deduce that

$$\mathbb{P}(Y = Z | Z) = \frac{p(Z)}{p(Z) + p(-Z)}.$$

- (2) Let  $Y$  be a real random variable which is absolutely continuous with density  $f$ . Assume  $f(y) > 0$  for every  $y \in \mathbb{R}$ .

(a) Show that

$$\mathbb{P}(Y = Z, Z \in B) = \mathbb{E} \left[ \frac{f(Z)}{f(Z) + f(-Z)} \mathbf{1}_B(Z) \right] \quad \text{for } B \in \mathcal{B}(\mathbb{R}_+).$$

*Hint:* Show that the right-hand side equals  $\mathbb{P}(Y \in B)$  and note that this is also equal to the left-hand side.

(b) Determine  $\mathbb{P}(Y = Z \mid Z)$ .

**Exercise 31 (A discretely observed Black-Scholes model).**

Consider i.i.d. random variables  $Y_1, Y_2, \dots$  such that  $Y_n \sim N(\mu, \sigma^2)$  for  $n \in \mathbb{N}$ , where  $\mu \in \mathbb{R}$  and  $\sigma^2 > 0$ . Let  $S_0 > 0$  be fixed and define  $(S_n)_{n \geq 1}$  as

$$S_n = S_0 e^{Y_1 + \dots + Y_n} \quad \text{for } n \in \mathbb{N}.$$

Also define

$$\mathcal{F}_n = \begin{cases} \{\emptyset, \Omega\} & \text{for } n = 0 \\ \sigma(Y_1, \dots, Y_n) & \text{for } n \in \mathbb{N}. \end{cases}$$

(1) Determine  $\mathbb{P}(S_{n+1} > S_0)$  and  $\mathbb{P}(S_{n+1} > S_0 \mid \mathcal{F}_n)$  for  $n \in \mathbb{N}_0$ .

(2) Determine  $\mathbb{E}[S_{n+1} \mid \mathcal{F}_n]$  and  $\text{Var}(S_{n+1} \mid \mathcal{F}_n)$  for  $n \in \mathbb{N}_0$ .

**Exercise 32.**

Consider i.i.d. real square integrable random variables  $Y_1, Y_2, \dots$  such that  $\mathbb{E}[Y_n] = 0$  for all  $n$ . Put  $\sigma^2 = \text{Var}(Y_n)$ , let  $S_0 = 0$  and define  $(S_n)_{n \geq 1}$  and  $(M_n)_{n \geq 0}$  as

$$\begin{aligned} S_n &= Y_1 + \dots + Y_n & \text{for } n \in \mathbb{N} \\ M_n &= S_n^2 - n\sigma^2 & \text{for } n \in \mathbb{N}_0. \end{aligned}$$

Also define

$$\mathcal{F}_n = \begin{cases} \{\emptyset, \Omega\} & \text{for } n = 0 \\ \sigma(Y_1, \dots, Y_n) & \text{for } n \in \mathbb{N}. \end{cases}$$

(1) Show that  $(S_n)_{n \geq 0}$  and  $(M_n)_{n \geq 0}$  are martingales. That is,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n \quad \text{and} \quad \mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \quad \text{for } n \in \mathbb{N}_0.$$

**Exercise 33.**

Let  $(Y_n)_{n \geq 0}$  and  $(Z_n)_{n \geq 0}$  be stochastic processes that are versions and let  $Q$  be a transition kernel on  $S$ . Argue that if  $(Y_n)_{n \geq 0}$  is a Markov chain with transition kernel  $Q$  then so is  $(Z_n)_{n \geq 0}$ .

**Exercise 34.**

Let the state space be  $S = \mathbb{R}$  and consider the transition kernel  $Q(x, \cdot) = N(0, 1 + x^2)$  for  $x \in \mathbb{R}$ . That is,

$$Q(x, B) = \int_B \frac{1}{\sqrt{2\pi(1+x^2)}} \exp\left(-\frac{y^2}{2(1+x^2)}\right) dy, \quad \text{for } x \in \mathbb{R}, B \in \mathcal{B}(\mathbb{R}). \quad (\text{H.2})$$

(1) Determine  $(Qf)(x)$  for  $x \in \mathbb{R}$  when  $f(u) = u^2$  for all  $u$ .

- (2) Let  $(Y_n)_{n \geq 0}$  denote a Markov chain with transition kernel  $Q$  and  $Y_0 = 0$ . Determine  $\mathbb{E}[Y_n^2 | \mathcal{F}_{n-1}^Y]$  and  $\mathbb{E}[Y_n^2]$  for  $n \in \mathbb{N}$ .
- (3) Simulate the process.

**Exercise 35.**

Let the state space be  $S = \mathbb{R}$  and consider the transition kernel  $Q(x, \cdot) = N(0, 1 + x^2)$  for  $x \in \mathbb{R}$ . That is,  $Q(x, B)$  is given as in (H.2).

- (1) Determine  $\mathbb{P}_x(X_1 > 0, X_2 < 0)$  for  $x \in \mathbb{R}$ .
- (2) Use this to show that, regardless of the initial distribution  $\mu$ ,

$$\mathbb{P}_\mu(X_{n+1} > 0, X_{n+2} < 0 | \mathcal{F}_n^X) = \frac{1}{4}$$

for every  $n \in \mathbb{N}_0$ .

**Exercise 36.**

A fair die is thrown repeatedly. For  $n \in \mathbb{N}_0$  put  $Y_n = 1$  if the  $n$ th and  $(n-1)$ st throws both yield a six, and put  $Y_n = 0$  otherwise. (For convenience the die is thrown for the first time at time  $-1$ ).

Calculate  $\mathbb{P}(Y_2 = 0 | Y_1 = 0, Y_0 = 0)$  and  $\mathbb{P}(Y_2 = 0 | Y_1 = 0, Y_0 = 1)$ . Is  $(Y_n)_{n \geq 0}$  a Markov chain?

**Exercise 37.**

A ticket office has two computers to manage reservations. Since it takes some time to switch computers, only one computer will be in use on a given day. During a given day, a computer in use breaks down with probability  $p$ . In this case, the computer is sent out for repair after closing time; moreover, the next day the office shifts to the other computer if this is not under repair. It takes two days to repair a computer and the computer is returned at closing time.

Let  $(X_n, Y_n)$  denote the situation after closing time on the  $n$ th day, where  $X_n$  denotes the number of computers that are not under repair and  $Y_n$  is 1 if a computer has been under repair for one day, and 0 otherwise. Assume  $((X_n, Y_n))_{n \geq 0}$  is a Markov chain with state space

$$S = \{(2, 0), (1, 0), (0, 1), (1, 1)\}.$$

- (1) Determine the transition matrix.
- (2) Assume the office is in state  $(1, 1)$ . Determine the probability that the office will visit this state again within the next three days.

**Exercise 38.**

A fair die is thrown repeatedly. Argue that the two processes described in (1) and (2) below are Markov chains. Determine the state spaces and the transition matrices as well. In case (1) you may e.g. choose to start your chain in an auxiliary state " $\infty$ " (corresponding to "never seen a six"), and in (2) you may e.g. start in "0".

- (1) "Number of throws since last six".
- (2) "Total number of pips (Danish: øjne) seen when the die has been thrown  $n$  times".
- (3) Assume the die is thrown infinitely often. How often do you think the chain described in (1) visits state  $10^{90}$ ? (No mathematics needed here - just guess).

**Exercise 39.**

Consider the state space  $S = \{1, 2\}$  and the transition matrix  $P$  given by

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix},$$

where  $a, b \in [0, 1]$ . Let  $\mathbb{P}_1$  denote the probability measure corresponding to the Markov chain starting in 1. That is,  $\mathbb{P}_1(X_0 = 1) = 1$ . Define

$$T_1 = \inf\{n \in \mathbb{N} : X_n = 1\} \quad (\text{where } \inf \emptyset = \infty).$$

- (1) Determine  $\mathbb{P}_1(T_1 = n)$  for  $n \in \mathbb{N}$ .
- (2) Determine the set of values  $(a, b)$  for which  $\mathbb{P}_1(T_1 < \infty) = 1$ .

**Exercise 40.**

Let  $S$  be countable and  $P = \{p_{ij}\}_{i,j \in S}$  denote a transition matrix.

- (1) Show that if  $(Y_n)_{n \geq 0}$  is a Markov chain with transition matrix  $P$  with respect to a filtration  $(\mathcal{F}_n)_{n \geq 0}$  then  $(Y_{n+m})_{n \geq 0}$  is a Markov chain with transition matrix  $P$  for every  $m \in \mathbb{N}_0$ .
- (2) Let  $i \in S$  and assume  $p_{ii} = 1$ . Show that if  $(Y_n)_{n \geq 0}$  is a Markov chain with transition matrix  $P$  then

$$\mathbb{P}(Y_m = i) = \mathbb{P}(Y_{m+n} = i \text{ for all } n \in \mathbb{N}_0) \quad \text{for all } m \in \mathbb{N}_0.$$

Explain what this result says.

**Exercise 41.**

Let  $S = \{1, 2, 3, 4, 5\}$  and consider the transition matrix

$$P = \begin{pmatrix} 0.2 & 0.2 & 0.3 & 0.3 & 0 \\ 0 & 0.2 & 0.8 & 0 & 0 \\ 0 & 0.8 & 0.2 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 0.8 \\ 0 & 0 & 0 & 0.8 & 0.2 \end{pmatrix}.$$

- (1) Determine the transient and recurrent states.
- (2) Describe how a Markov chain with transition matrix  $P$  develops if it starts in state  $i$  for  $i = 1, \dots, 5$ .
- (3) Assume a Markov chain starts in 1. Determine the probability that it will eventually visit state 5.

**Exercise 42.**

Let  $S$  be countable. For  $k = 1, 2$  let  $(Y_n^k)_{n \geq 0}$  denote a Markov chain with transition matrix  $P = \{p_{ij}^k\}_{i,j \in S}$ . Assume  $(Y_n^1)_{n \geq 0}$  and  $(Y_n^2)_{n \geq 0}$  are independent. Show that  $(Y_n^1, Y_n^2)_{n \geq 0}$  is a Markov chain with state space  $S^2$  and determine the state space.

**Exercise 43.**

Let  $(Y_n)_{n \geq 0}$  be a branching process cf. Example 4.19. Assume  $\mathbb{P}(\xi_1 \neq 1) > 0$ . Find all the transient states. Argue that, no matter the initial distribution, with probability one either  $Y_n$  is eventually absorbed in 0 or  $Y_n \rightarrow \infty$ .

**Exercise 44.**

Let  $S = \{1, 2\}$  and consider the transition matrix  $P$  given by

$$P = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}.$$

Let  $\tau$  and  $\sigma$  be the stopping times with respect to  $(\mathcal{F}_n^X)_{n \geq 0}$  defined as

$$\tau = \inf\{n \in \mathbb{N} : X_n = 1\} \text{ and } \sigma = \inf\{n \in \mathbb{N} : X_{n-1} = X_n\}.$$

It can be shown that, no matter the initial distribution  $\mu$ ,  $\mathbb{P}_\mu(\tau < \infty) = \mathbb{P}_\mu(\sigma < \infty) = 1$ . You may take this for granted in the following.

- (1) Determine  $\mathbb{P}_\mu(X_{\tau+1} = X_{\tau+2} = 1 \mid \mathcal{F}_\tau^X)$  and  $\mathbb{P}_\mu(X_{\tau+1} = X_{\tau+2} = 1)$ .
- (2) Argue that

$$\mathbb{P}_\mu(X_{\sigma+1} = X_{\sigma+2} = 1 \mid \mathcal{F}_\sigma^X) = \frac{1}{9} \mathbb{1}_{\{X_\sigma=1\}} + \frac{1}{12} \mathbb{1}_{\{X_\sigma=2\}}.$$

**Exercise 45.**

In the following  $(Y_n)_{n \geq 0}$  denotes a Markov chain with transition kernel  $Q$  with respect to a filtration  $(\mathcal{F}_n)_{n \geq 0}$ .

- (1) For  $n \in \mathbb{N}_0$  let  $f_n : S \rightarrow \mathbb{R}_+$  be measurable. Assume  $f_0(Y_0) \in L^1(\mathbb{P})$  and  $Qf_{n+1} = f_n$  for all  $n \in \mathbb{N}_0$ . Show that  $(f_n(Y_n))_{n \geq 0}$  is a martingale with respect to  $(\mathcal{F}_n)_{n \geq 0}$ .
- (2) Let  $f : S \rightarrow \mathbb{R}_+$  be measurable, bounded and satisfy  $Qf = f$ . Argue that  $(f(Y_n))_{n \geq 0}$  is a martingale with respect to  $(\mathcal{F}_n)_{n \geq 0}$ .

In the following let  $S = \{0, 1, 2\}$ . Thus, there is a transition matrix corresponding to  $Q$  and as usual we denote it by

$$P = \begin{pmatrix} p_{00} & p_{01} & p_{02} \\ p_{10} & p_{11} & p_{12} \\ p_{20} & p_{21} & p_{22} \end{pmatrix}.$$

We are interested in knowing when  $(Y_n)_{n \geq 0}$  is a martingale so in the following let

$$f(i) = i \quad \text{for } i \in S.$$

- (3) Show that if  $Qf = f$  then  $p_{00} = p_{22} = 1$ . (That is, states 0 and 2 are absorbing.)
- (4) Give concrete examples of transition matrices  $P$  where  $(Y_n)_{n \geq 0}$  is a martingale with respect to  $(\mathcal{F}_n)_{n \geq 0}$ . Avoid cases where all transition probabilities are 0 or 1.

**Exercise 46.**

Consider *Gambler's ruin* cf. Example 4.17. In this case  $1, \dots, N-1$  are transient states and therefore the chain will be absorbed in 0 or  $N$  sooner or later. That is,  $X_\infty = \lim_{n \rightarrow \infty} X_n$  exists  $\mathbb{P}_\mu$ -a.s. and equals

$$X_\infty = 0 \mathbb{1}_{\{\text{absorption in } 0\}} + N \mathbb{1}_{\{\text{absorption in } N\}} \quad \mathbb{P}_\mu - a.s.$$

no matter the initial distribution  $\mu$ . You may use this in the following. Besides, you will probably need parts of Exercise 45 to solve this exercise.

Let  $q = 1 - p$ .

- (1) Assume  $p = q = \frac{1}{2}$ .
  - (a) Show that  $(X_n)_{n \geq 0}$  is a martingale with respect to  $(\mathcal{F}_n^X)_{n \geq 0}$  under  $\mathbb{P}_\mu$  no matter the initial distribution  $\mu$ .
  - (b) Use (a) to determine the probability of ruin, i.e.  $\mathbb{P}_x(\text{absorption in } 0)$  for every  $x \in S = \{0, \dots, N\}$ .

(2) Assume  $p > q$ .

- (a) Show that  $((\frac{q}{p})^{X_n})_{n \geq 0}$  is a martingale with respect to  $(\mathcal{F}_n^X)_{n \geq 0}$  under  $\mathbb{P}_\mu$  no matter the initial distribution  $\mu$ .
- (b) Use (a) to determine the probability of ruin, i.e.  $\mathbb{P}_x(\text{absorption in } 0)$  for every  $x \in S = \{0, \dots, N\}$ .

**Exercise 47.**

Consider Proposition 6.4.

- (1) Prove (6.4).
- (2) Prove Proposition 6.4(2).

**Exercise 48.**

Prove that (a)–(c) in Remark 6.6(1) are equivalent.

**Exercise 49.**

Consider the proof of Theorem 6.8(4).

- (1) Explain why the events  $B_1, B_2, \dots$  are independent and all have the same probability under  $\mathbb{P}_i$ .
- (2) Prove this result.

**Exercise 50.**

Let  $S$  be countable and  $P$  be a transition matrix.

- (1) Let  $i, j \in S$  with  $i \neq j$ . Show that if  $i \rightarrow j$  then  $\mathbb{P}_i(T_j < T_i) > 0$ .
- (2) Show that if  $P$  is positive recurrent then  $\mathbb{E}_i[T_j] < \infty$  for all  $i, j \in S$ .

**Exercise 51.**

Consider the Markov chain in case (1) of Exercise 38.

- (1) Is this an irreducible Markov chain?
- (2) Remove auxiliary state  $\infty$ . Is the resulting Markov chain irreducible?
- (3) Calculate  $\mathbb{P}_0(T_0 < \infty)$ . Is state 0 recurrent?
- (4) How often is state  $10^{90}$  visited? (No guessing needed here - just mathematics!)

**Exercise 52.**

Let  $(Y_n)_{n \geq 0}$  denote an i.i.d. sequence with common distribution  $\mu$ . That is,  $\mu(B) = \mathbb{P}(Y_n \in B)$  for  $B \in \mathcal{B}(S)$ .

- (1) Show that  $(Y_n)_{n \geq 0}$  is a Markov chain and determine the transition kernel.

- (2) Assume  $S$  is countable. Put  $p_j = \mathbb{P}(Y_n = j)$  for  $j \in S$  and  $n \in \mathbb{N}$ . Determine the transition matrix.
- (3) Assume  $S$  is countable. Show that state  $j$  is recurrent if and only if  $p_j > 0$ .

**Exercise 53.**

Let  $S = \{1, 2, 3\}$  and consider the transition matrix

$$P = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0 & 0.3 & 0.7 \\ 0 & 0.2 & 0.8 \end{pmatrix}.$$

Let  $\mu$  denote a probability measure on  $S$ .

- (1) Determine the transient and recurrent states.
- (2) Argue that  $T_3^n < \infty$  for all  $n \in \mathbb{N}$   $\mathbb{P}_\mu$ -a.s.
- (3) Let  $Z_n = X_{T_3^n+1}$  for  $n \in \mathbb{N}$ . Argue that if  $T_3^n(\omega_0, \omega_1, \dots) < \infty$  then  $Z_n(\omega_0, \omega_1, \dots) = X_1 \circ \theta^{T_3^n}(\omega_0, \omega_1, \dots)$ .
- (4) Let  $Z_n = X_{T_3^n+1}$  for  $n \in \mathbb{N}$ . Argue that  $(Z_n)_{n \geq 1}$  is an i.i.d. sequence under  $\mathbb{P}_\mu$  and determine the common distribution.

**Exercise 54.**

Consider the Markov chain in Exercise 38, case (1). Recall that the state space is  $S = \mathbb{N}_0 \cup \{\infty\}$ .

- (1) Determine  $\mathbb{E}_0[T_0]$ .
- (2) Argue that

$$\mathbb{E}_i[T_i] \begin{cases} < \infty & \text{if } i \in \mathbb{N}_0 \\ = \infty & \text{if } i = \infty. \end{cases}$$

- (3) Show that there exists a unique stationary distribution and determine this distribution.  
You may use (cf. Exercise 63) that if  $\pi$  is a stationary distribution and  $j$  is transient then  $\pi_j = 0$ .
- (4) Assume the die is thrown many times. On average, how often will we experience that we have not seen a six in the last three throws?

**Exercise 55.**

- (1) The proof of Proposition 6.4 is a very good illustration of how to apply the strong Markov property. In lay terms (“proof by hand-waving”), explain why Proposition 6.4 is true.

- (2) Prove Proposition 6.4.

**Exercise 56.**

Use Theorem 5.4(2) to prove Theorem 5.4(3).

**Exercise 57 (Renewal processes – an example).**

Let  $S = \mathbb{N}$  and for  $i \in \mathbb{N}$  let  $q_i \in (0, 1)$ . Assume  $\sum_{i=1}^{\infty} q_i = 1$ . Define the transition matrix  $P = \{p_{ij}\}_{i,j \in \mathbb{N}}$  as  $p_{1i} = q_i$  for  $i \in \mathbb{N}$  and  $p_{i,i-1} = 1$  for  $i \in \mathbb{N} \setminus \{1\}$ ; all other transition probabilities are 0.

In several of the questions below, it is a good idea to think about the relations between  $X_1$  and  $T_1$  under  $\mathbb{P}_1$ .

- (1) Describe how a Markov chain with transition matrix  $P$  develops.
- (2) Argue that  $P$  is irreducible and recurrent.
- (3) Show that  $P$  is positive recurrent if and only if  $\sum_{i=1}^{\infty} i q_i < \infty$ .
- (4) Assume  $\sum_{i=1}^{\infty} i q_i < \infty$ . Show that the unique stationary distribution is given as

$$\pi_i = \frac{\sum_{j=i}^{\infty} q_j}{\sum_{j=1}^{\infty} j q_j}, \quad i \in \mathbb{N}.$$

**Exercise 58.**

- (1) Let  $S = \{1, 2, 3\}$  and consider the transition matrix

$$P = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0 & 0.2 & 0.8 \\ 0 & 0.8 & 0.2 \end{pmatrix}.$$

Argue that  $N_i^n/n$  converges  $\mathbb{P}_j$ -a.s. for all  $i, j \in S$  and determine the limit for all  $(i, j)$ . (See (6.16) for the definition of  $N_i^n$ ).

- (2) Let  $S = \{1, 2, 3, 4\}$  and consider the transition matrix

$$P = \begin{pmatrix} 0.2 & 0.8 & 0 & 0 \\ 0.8 & 0.2 & 0 & 0 \\ 0 & 0 & 0.2 & 0.8 \\ 0 & 0 & 0.8 & 0.2 \end{pmatrix}.$$

Determine the expected return time  $\mathbb{E}_i[T_i]$  for all  $i \in S$ .

**Exercise 59.**

Let  $S$  be *finite* and  $P = \{p_{ij}\}_{i,j \in S}$  denote an *irreducible* transition matrix. Let  $i \in S$ .

- (1) Argue that there exists a number  $n \in \mathbb{N}$  and a number  $\gamma \in (0, 1)$  such that  $\mathbb{P}_j(T_i > n) \leq \gamma$  for all  $j \in S$ .
- (2) Show that for all  $m \in \mathbb{N}$   $\mathbb{P}_i(T_i > mn) \leq \gamma^m$ .

**Exercise 60.**

As always let  $S$  denote a Polish state space and  $\nu$  denote a  $\sigma$ -finite measure on  $(S, \mathcal{B}(S))$ . Let  $Q(x, B)$ ,  $x \in S$ ,  $B \in \mathcal{B}(S)$ , denote a transition kernel. Assume there exists a measurable function  $p : S^2 \rightarrow [0, \infty)$  such that

$$Q(x, B) = \int_B p(x, y) \nu(dy) \quad \text{for all } x \in S \text{ and } B \in \mathcal{B}(S).$$

That is,  $Q(x, \cdot)$  is absolutely continuous with respect to  $\nu$  with density  $y \mapsto p(x, y)$ .

- (1) Show that for  $x \in S$  and  $n \in \mathbb{N}$  the vector  $(X_1, \dots, X_n)$  is absolutely continuous with respect to  $\nu \times \dots \times \nu$  ( $n$  times) under  $\mathbb{P}_x$  and determine the density.
- (2) Explain how this is related to Corollary 4.22.

**Exercise 61.**

Let  $S$  be a countable set, and  $P = \{p_{ij}\}_{i,j \in S}$  be a transition matrix. Assume that  $S$  can be decomposed as  $S = T \cup R$ , where

- $R$  is a non-empty set,
- all states in  $T$  are transient, and
- the submatrix  $\{p_{ij}\}_{i,j \in R}$  is a positive recurrent transition matrix.

- (1) Show that there exists a unique stationary distribution  $\pi = (\pi_i)_{i \in S}$ .
- (2) Argue that  $\pi_i = (\mathbb{E}_i[T_i])^{-1}$  for all  $i \in S$ .

**Exercise 62.**

Let  $S = \{0, 1, 2\}$ . Consider the transition matrix  $P$  given by

$$P = \begin{pmatrix} p & 0 & 1-p \\ \frac{1-p}{2} & p & \frac{1-p}{2} \\ 0 & 1 & 0 \end{pmatrix}.$$

Here,  $p$  is a parameter satisfying  $0 < p < 1$ .

- (1) Argue that there is a unique stationary distribution with point probabilities given by  $(\pi_0, \pi_1, \pi_2)$ , where

$$\pi_0 = \frac{1}{5-2p}, \quad \pi_1 = \frac{2}{5-2p}, \quad \pi_2 = \frac{2(1-p)}{5-2p}.$$

(2) Show that

$$\frac{1}{n} \sum_{m=1}^n X_m^2 \rightarrow (\pi_1 + 4\pi_2) \quad \text{for } n \rightarrow \infty,$$

$\mathbb{P}_\mu$ -a.s. no matter the initial distribution  $\mu$ .

**Exercise 63.**

Let  $S$  be countable and  $P = \{p_{ij}\}_{i,j \in S}$  be a transition matrix. Show that if  $\pi = (\pi_i)_{i \in S}$  is a stationary distribution then  $\pi_i = 0$  whenever  $i$  is a transient state.

**Exercise 64.**

- (1) Prove the results in Remark 6.15.
- (2) Let  $\pi$  be a probability measure which satisfies the detailed balance equation cf. Definition 6.25. Show that  $\pi$  is a stationary distribution.
- (3) Does  $\pi$  in Exercise 62 satisfy the detailed balance equation?

**Exercise 65.**

Let  $Q(x, A)$ ,  $x \in S$ ,  $A \in \mathcal{B}(S)$ , denote a transition kernel. Let  $g : S^{n+1} \rightarrow \mathbb{R}$  be bounded and measurable. Show that the map

$$\begin{aligned} \phi : S^n &\rightarrow \mathbb{R} \\ (x_0, \dots, x_{n-1}) &\mapsto \int_S g(x_0, \dots, x_{n-1}, x_n) Q(x_{n-1}, dx_n) \end{aligned}$$

is measurable.

*Hint:* Start by looking at  $g(x_0, \dots, x_n) = \mathbb{1}_{A_0 \times \dots \times A_n}(x_0, \dots, x_n)$  where  $A_0, \dots, A_n \in \mathcal{B}(S)$

**Exercise 66.**

Let  $Q$  be a transition kernel,  $\mu$  be a probability measure on  $(S, \mathcal{B}(S))$  and  $g : S^2 \rightarrow \mathbb{R}$  be measurable and bounded. Show that for  $n \in \mathbb{N}_0$ ,  $\mathbb{E}_\mu[g(X_n, X_{n+1}) | \mathcal{F}_n^X] = Qg(X_n)$ , where  $Qg(x) = \int_S g(x, y) Q(x, dy)$  for  $x \in S$ .

**Exercise 67.**

Let  $Y_{n+1} = \theta Y_n + \epsilon_{n+1}$ , where the  $\epsilon_n$ 's are i.i.d. and  $N(0, 1)$ -distributed,  $\theta \in \mathbb{R}$  is a constant, and  $Y_0 = 0$ .

- (1) Show that  $(Y_n)_{n \geq 0}$  is a Markov chain. Determine the state space and the transition kernel.
- (2) Use the same arguments as in Exercise 60 to show that for every  $n \in \mathbb{N}$  the vector  $(Y_1, \dots, Y_n)$  is absolutely continuous with respect to the Lebesgue measure on  $\mathbb{R}^n$  and determine the density.

- (3) Assume we have observed  $(Y_1, \dots, Y_n)$  for some  $n > 2$ . Compare this situation to a simple linear regression with no intercept, and use this to argue that

$$\hat{\theta}_n = \frac{\sum_{j=1}^n Y_{j-1} Y_j}{\sum_{j=1}^n Y_{j-1}^2}. \quad (\text{H.3})$$

is a reasonable estimate of  $\theta$ .

- (4) Derive the likelihood function and verify that (H.3) is the MLE.  
 (5) Let  $\theta = 0.8$ . Simulate and plot the process. Plot  $\hat{\theta}_n$  as a function of  $n$ .

**Exercise 68.**

Let  $S$  be a *finite* set and let  $P = \{p_{ij}\}_{i,j \in S}$  be a transition matrix. Let  $T \subseteq S$  be the set of transient states and let

$$\tau = \inf\{n \geq 0 : X_n \in S \setminus T\}.$$

Thus,  $\tau$  denotes the time when the Markov chain leaves  $T$ . Since  $S$  is finite,  $\mathbb{P}_\mu(\tau < \infty) = 1$  no matter the initial distribution  $\mu$ . In fact, it can be shown that  $\mathbb{E}_\mu[\tau] < \infty$ . (See Pedersen (2014, Chapter 7)). Put  $g(i) = \mathbb{E}_i[\tau]$  for  $i \in S$ .

- (1) Show that

$$\begin{cases} g(i) = 0, & \text{for } i \in S \setminus T, \\ g(i) = 1 + \sum_{j \in T} p_{ij} g(j), & \text{for } i \in T. \end{cases} \quad (\text{H.4})$$

*Hint:*  $\tau = 1 + \tau \circ \theta^1$  on  $\{X_0 \in T\}$ .

- (2) Let  $f : S \rightarrow \mathbb{R}$  satisfy (H.4) with  $g$  replaced by  $f$ . Show that  $(f(X_{\tau \wedge n}) + [\tau \wedge n])_{n \geq 0}$  is an  $(\mathcal{F}_n^X)_{n \geq 0}$ -martingale under  $\mathbb{P}_i$  for all  $i \in S$ .  
 (3) Let  $f$  be as in (2). Show that  $f(i) = g(i)$  for  $i \in S$ , i.e.,  $g$  is the unique solution to (H.4).

*Hint:* Calculate the expectation of the martingale in (2) and let  $n \rightarrow \infty$ .

**Exercise 69.**

Let  $\widetilde{\mathcal{G}}$  be a collection of events which is stable under intersection. Put  $\mathcal{G} = \sigma(\widetilde{\mathcal{G}})$ . Let  $X$  be an  $\mathbb{R}^d$ -valued random vector. Assume

$$\mathbb{E}[e^{i\langle \theta, X \rangle} \mathbb{1}_A] = \mathbb{P}(A) \mathbb{E}[e^{i\langle \theta, X \rangle}] \quad \text{for } \theta \in \mathbb{R}^d \text{ and } A \in \widetilde{\mathcal{G}}.$$

- (1) Show that  $A$  and  $X$  are independent for  $A \in \widetilde{\mathcal{G}}$ .  
 (2) Show that  $\mathcal{G}$  and  $X$  are independent.

**Exercise 70.**

Let  $d \in \mathbb{N}$ ,  $\nu$  be a probability measure on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  and  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space. Let  $\mathcal{G}$  be a sub- $\sigma$ -field in  $\mathcal{F}$  and  $X$  denote an  $\mathbb{R}^d$ -valued random vector.

Assume that

$$\mathbb{E}[e^{i\langle \theta, X \rangle} | \mathcal{G}] = \int_{\mathbb{R}^d} e^{i\langle \theta, x \rangle} \nu(dx), \quad \theta \in \mathbb{R}^d.$$

Show that  $X$  is independent of  $\mathcal{G}$  and that  $X \sim \nu$ .

**Exercise 71.**

Prove Proposition G.1.

*Hint:* Since  $[0, T]$  is compact for  $T \in (0, \infty)$ , every sequence in  $[0, T]$  has a converging subsequence.

**Exercise 72.**

Let  $\nu$  denote a probability measure on  $(S, \mathcal{B}(S))$  and define  $(P_t)_{t \geq 0}$  as

$$P_t(x, B) = \begin{cases} \delta_x(B) & \text{for } t = 0 \\ \nu(B) & \text{for } t > 0 \end{cases}$$

for  $x \in S$  and  $B \in \mathcal{B}(S)$ .

- (1) Show that  $(P_t)_{t \geq 0}$  is a transition semigroup.
- (2) Let  $(Y_t)_{t \geq 0}$  be i.i.d. random variables with common distribution  $\nu$ . Thus in particular  $\mathbb{P}(Y_t \in B) = \nu(B)$  for all  $t \geq 0$  and  $B \in \mathcal{B}(S)$ . Show that  $(Y_t)_{t \geq 0}$  is a Markov process with transition semigroup  $(P_t)_{t \geq 0}$ .

**Exercise 73.**

Let  $(Y_t)_{t \geq 0}$  be a Markov process with transition semigroup  $(P_t)_{t \geq 0}$  with respect to a filtration  $(\mathcal{F}_t)_{t \geq 0}$  and let  $\Delta > 0$ . Show that the discrete time process  $(Y_{n\Delta})_{n \geq 0}$  is a Markov chain, and determine its transition kernel.

**Exercise 74.**

Verify that  $(p_t)_{t \geq 0}$  in Example 7.6(3) is a transition semigroup. Describe the corresponding Markov process.

**Exercise 75 (The Gaussian Ornstein-Uhlenbeck transition semigroup).**

Let the state space be  $S = \mathbb{R}$  and let  $\lambda > 0$  be a fixed parameter. Define a transition kernel  $P_t$  on  $\mathbb{R}$  for  $t \geq 0$  as

$$P_t(x, \cdot) = N\left(xe^{-\lambda t}, \frac{1}{2\lambda}[1 - e^{-2\lambda t}]\right) \quad \text{for } x \in \mathbb{R} \text{ and } t \geq 0. \quad (\text{H.5})$$

That is,  $P_t(x, \cdot)$  is the normal distribution with parameters as indicated.

- (1) Show that  $(P_t)_{t \geq 0}$  is a transition semigroup.

*Hint:* Calculate the characteristic function of  $P_t P_s(x, \cdot)$  for  $x \in \mathbb{R}$  and  $t, s \geq 0$ .

- (2) Determine a probability measure  $\nu$  on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$  so that  $X_t \xrightarrow{d} \nu$  as  $t \rightarrow \infty$  under  $\mathbb{P}_\mu$  for every probability measure  $\mu$  on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ . Here, we use the notation in Theorem 7.4.

**Exercise 76.**

Let the state space be  $S = \mathbb{R}$  and let  $\lambda > 0$  be a fixed parameter. Let  $(Y_t)_{t \geq 0}$  be a Markov process with transition semigroup  $(P_t)_{t \geq 0}$  defined in (H.5). Let  $\Delta > 0$  be a small number.

- (1) Show that for  $0 \leq s \leq t$ ,  $Y_t - e^{-\lambda(t-s)}Y_s$  is independent of  $\mathcal{F}_s^Y$  and determine the distribution of  $Y_t - e^{-\lambda(t-s)}Y_s$ .
- (2) Let  $Y_0 = 0$ . Describe how you can use item (1) to simulate  $Y_\Delta, Y_{2\Delta}, \dots$
- (3) Simulate  $Y_\Delta, Y_{2\Delta}, \dots$ . Describe what you see, in particular in relation to the paths and the influence of  $\lambda$ .

**Exercise 77.**

Consider a filtration  $(\mathcal{F}_t)_{t \geq 0}$ , and a real and adapted stochastic process  $(B_t)_{t \geq 0}$ . Assume that  $B_t - B_s$  is independent of  $\mathcal{F}_s$  and  $B_t - B_s \sim N(0, t - s)$  for  $0 \leq s < t$ , and that  $B_0 = 0$  a.s. (This means essentially that  $(B_t)_{t \geq 0}$  is a standard Brownian motion). Let  $\mu \in \mathbb{R}$  and  $\sigma > 0$ .

- (1) Determine

$$\mathbb{E}\left[e^{\sigma B_t + \mu t} \mid \mathcal{F}_s\right]$$

for  $s < t$  and for  $s \geq t$ .

- (2) Determine the set of  $(\mu, \sigma^2)$  for which  $(e^{\sigma B_t + \mu t})_{t \geq 0}$  is a martingale, that is satisfies

$$\mathbb{E}\left[e^{\sigma B_t + \mu t} \mid \mathcal{F}_s\right] = e^{\sigma B_s + \mu s}$$

for  $s < t$ .

**Exercise 78.**

Consider a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  and let  $U : \Omega \rightarrow \{1, 2\}$  be a random variable satisfying  $\mathbb{P}(U = 1) = \mathbb{P}(U = 2) = 1/2$ . Define the real stochastic process  $(Y_t)_{t \geq 0}$  as  $Y_t = 0$  for  $t \in [0, 1]$  and for  $t > 1$  let

$$Y_t = \begin{cases} (t-1) & \text{if } U = 1 \\ -(t-1) & \text{if } U = 2. \end{cases}$$

Let  $(\mathcal{F}_t^Y)_{t \geq 0}$  denote the corresponding natural filtration, that is  $\mathcal{F}_t^Y = \sigma(Y_s \mid s \leq t)$  for  $t \geq 0$ .

(1) Argue that

$$\mathcal{F}_t^Y = \begin{cases} \{\emptyset, \Omega\} & \text{for } t \leq 1 \\ \{\emptyset, \Omega, \{U = 1\}, \{U = 2\}\} & \text{for } t > 1 \end{cases}$$

and determine  $(\mathcal{F}_{t+}^Y)_{t \geq 0}$ .

(2) Let

$$\tau = \inf\{t > 0 \mid Y_t > 0\}.$$

Argue that  $\tau$  is an extended stopping time with respect to  $(\mathcal{F}_t^Y)_{t \geq 0}$ . Is  $\tau$  a stopping time with respect to  $(\mathcal{F}_t^Y)_{t \geq 0}$ ?

**Exercise 79.**

Let  $(P_t)_{t \geq 0}$  be a transition semigroup on  $S$  and  $(Y_t)_{t \geq 0}$  be a stochastic process with state space  $S$ . Show that  $(Y_t)_{t \geq 0}$  is a Markov process with transition semigroup  $(P_t)_{t \geq 0}$  if and only if for every  $n \in \mathbb{N}$ ,  $0 \leq t_0 \leq t_1 \leq \dots \leq t_n$  and  $f : S \rightarrow \mathbb{R}$  bounded and measurable,  $\mathbb{E}[f(Y_{t_n}) \mid Y_{t_{n-1}}, \dots, Y_{t_0}] = P_{t_n - t_{n-1}} f(Y_{t_{n-1}})$ .

**Exercise 80.**

Prove Proposition 8.4(4).

**Exercise 81.**

Verify (7.6) in the case  $j = n$ .

**Exercise 82.**

Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration.

- (1) Show that if  $\sigma$  and  $\tau$  are stopping times, then so are  $\sigma \wedge \tau$  and  $\sigma \vee \tau$ .
- (2) Show that if  $(\tau_n)_{n \geq 1}$  is a sequence of stopping times then  $\sum_{n=1}^{\infty} \tau_n$  and  $\sup_{n \in \mathbb{N}} \tau_n$  are stopping times as well.
- (3) Show that if  $(\tau_n)_{n \geq 1}$  is a sequence of extended stopping times then  $\inf_{n \in \mathbb{N}} \tau_n$  is an extended stopping time as well.

**Exercise 83.**

Consider a filtration  $(\mathcal{F}_t)_{t \geq 0}$ .

- (1) Let  $s \in [0, \infty)$  be a constant and set  $\tau = s$ . Is  $\tau$  a stopping time? An extended stopping time?
- (2) Let  $(Y_t)_{t \geq 0}$  denote a real-valued adapted process with right-continuous sample paths. Assume  $Y_0(\omega) = 0$  for all  $\omega \in \Omega$ . Let

$$\tau = \inf\{t > 0 : Y_t > c\},$$

where  $c > 0$ .

- (1) Explain why  $\tau$  is an extended stopping time.
- (2) Let  $\omega \in \Omega$  and assume  $\tau(\omega) < \infty$ . Must we have  $Y_\tau(\omega) > c$ ,  $Y_\tau(\omega) = c$ , or something similar? (Hint: Sketch a sample path.)
- (3) Assume in addition that  $t \mapsto Y_t(\omega)$  is continuous. Must we have  $Y_\tau(\omega) > c$ ,  $Y_\tau(\omega) = c$ , or something similar? (Hint: Sketch a sample path.)

**Exercise 84.**

Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration.

- (1) Show that if  $\tau$  is a stopping time then  $\mathcal{F}_\tau$  is a  $\sigma$ -field and  $\tau$  is  $\mathcal{F}_\tau$ -measurable.
- (2) Let  $\sigma$  and  $\tau$  be stopping times. Show that  $\{\tau < \sigma\} \in \mathcal{F}_\sigma \cap \mathcal{F}_\tau$ . Deduce that  $\{\tau \leq \sigma\}$  and  $\{\tau = \sigma\}$  are in  $\mathcal{F}_\sigma \cap \mathcal{F}_\tau$ .
- (3) Let  $\sigma$  and  $\tau$  be stopping time satisfying  $\sigma(\omega) \leq \tau(\omega)$  for  $\omega \in \Omega$ . Show that  $\mathcal{F}_\sigma \subseteq \mathcal{F}_\tau$ .

**Exercise 85.**

- (1) Prove Remark 8.12(1).
- (2) Prove Remark 8.12(5).
- (3) Prove Remark 8.12(6).
- (4) Prove Remark 8.12(7).

**Exercise 86.**

Let  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$  denote a filtered probability space and let  $(M_t)_{t \geq 0}$  be a submartingale with respect to  $(\mathcal{F}_t)_{t \geq 0}$ .

- (1) Let  $\mathcal{B}$  denote a sub- $\sigma$ -field of  $\mathcal{F}$  and assume  $\mathcal{B}$  is independent of  $\mathcal{F}_t$  for  $t \geq 0$ . Show that  $(M_t)_{t \geq 0}$  is a submartingale with respect to  $(\mathcal{H}_t)_{t \geq 0}$ , where  $\mathcal{H}_t = \sigma(\mathcal{F}_t \cup \mathcal{B})$  for  $t \geq 0$ .
- (2) Show that  $(M_t)_{t \geq 0}$  is a submartingale with respect to  $(\overline{\mathcal{F}}_t)_{t \geq 0}$ .
- (3) Assume  $t \mapsto M_t(\omega)$  is right-continuous for  $\omega \in \Omega$ . Show that  $(M_t)_{t \geq 0}$  is a submartingale with respect to  $(\mathcal{F}_{t+})_{t \geq 0}$ .

*Hint:* Apply the following version of Lévy's theorem: If  $Z \in L^1(\mathbb{P})$  and  $(\mathcal{G}_n)_{n \geq 1}$  is a decreasing family of  $\sigma$ -fields in  $\mathcal{F}$  then  $\mathbb{E}[Z | \mathcal{G}_n] \rightarrow \mathbb{E}[Z | \cap_{k=1}^\infty \mathcal{G}_k]$  as  $n \rightarrow \infty$  a.s. and in  $L^1$ .

**Exercise 87.**

Consider a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  and let  $\mathcal{N} = \{B \in \mathcal{F} : \mathbb{P}(B) = 0\}$ . Let  $\mathcal{B}$  denote a sub- $\sigma$ -field of  $\mathcal{F}$  and put  $\overline{\mathcal{B}} = \sigma(\mathcal{B} \cup \mathcal{N})$ .

Show that

$$\overline{\mathcal{B}} = \{B \in \mathcal{F} : \text{there is a } B' \in \mathcal{B} \text{ s.t. } B \Delta B' \in \mathcal{N}\}.$$

*Hint:* Note that on the right-hand side  $B = [B \setminus B'] \cup [B' \setminus (B' \setminus B)]$ .

Another way of writing the result above is that

$$\overline{\mathcal{B}} = \{B \in \mathcal{F} : \text{there is a } B' \in \mathcal{B} \text{ s.t. } \mathbb{1}_B = \mathbb{1}_{B'} \text{ a.s.}\}.$$

**Exercise 88.**

Let  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$  denote a filtered probability space, and put  $\mathcal{G}_t = \mathcal{F}_{t+}$  and  $\mathcal{H}_t = \overline{\mathcal{F}}_t$  for  $t \geq 0$ . Show that  $\overline{\mathcal{G}}_t = \mathcal{H}_{t+}$  for  $t \geq 0$ .

*Hint:* If, for  $n \in \mathbb{N}$ ,  $A_n \in \mathcal{F}_{t+1/n}$  then  $\limsup A_n \in \mathcal{F}_{t+}$ .

**Exercise 89.**

Let  $(Y_t)_{t \geq 0}$  denote a  $d$ -dimensional Lévy process defined on  $(\Omega, \mathcal{F}, \mathbb{P})$ . Fix  $s \in [0, \infty)$ .

- (1) Let  $\mathcal{F}_\infty^Y = \sigma(Y_t | t \geq 0)$ . Show that for  $Z : \Omega \rightarrow \mathbb{R}$  bounded and  $\mathcal{F}_\infty^Y$ -measurable,  $\mathbb{E}[Z | \mathcal{F}_s^Y] = \mathbb{E}[Z | \mathcal{F}_{s+}^Y]$  a.s.

*Hint:* First consider  $Z = \mathbb{1}_A \mathbb{1}_B$  where  $A \in \mathcal{F}_s^Y$  and  $B \in \sigma(Y_t - Y_s | t \geq s)$ .

- (2) Show Corollary 9.19(1).

**Exercise 90.**

Let  $d \in \mathbb{N}$  and  $\mu$  and  $\nu$  denote probability measures on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ .

- (1) Let  $X, Y$  be independent  $\mathbb{R}^d$ -valued random vectors such that  $X \sim \mu$  and  $Y \sim \nu$ . Argue that  $X + Y \sim \mu * \nu$ .
- (2) Verify Equation (B.3).

**Exercise 91.**

Let  $(Y_t)_{t \geq 0}$  and  $(Z_t)_{t \geq 0}$  be stochastic processes with state space  $S = \mathbb{R}^d$ .

- (1) Assume  $(Y_t)_{t \geq 0}$  and  $(Z_t)_{t \geq 0}$  are versions. Show that  $(Y_t)_{t \geq 0}$  has stationary independent increments if and only if  $(Z_t)_{t \geq 0}$  has stationary independent increments.
- (2) Assume  $(Y_t)_{t \geq 0}$  and  $(Z_t)_{t \geq 0}$  are versions and that  $(P_t)_{t \geq 0}$  is a transition semigroup on  $\mathbb{R}^d$ . Show that  $(Y_t)_{t \geq 0}$  is a Markov process with transition semigroup  $(P_t)_{t \geq 0}$  if and only if the same holds for  $(Z_t)_{t \geq 0}$ .
- (3) Assume that  $(Y_t)_{t \geq 0}$  and  $(Z_t)_{t \geq 0}$  have stationary independent increments. Show that  $(Y_t)_{t \geq 0}$  and  $(Z_t)_{t \geq 0}$  are versions if and only if  $Y_t \stackrel{d}{=} Z_t$  for  $t \geq 0$ .

**Exercise 92.**

Let  $(Y_t)_{t \geq 0}$  denote a Lévy process. Do we have  $Y_2 \stackrel{d}{=} 2Y_1$ ? Do we have  $Y_1 \stackrel{d}{=} nY_{\frac{1}{n}}$  for  $n \in \mathbb{N}$ ?

**Exercise 93.**

Let  $(Y_t)_{t \geq 0}$  denote a Lévy process. Show that  $(Y_t)_{t \geq 0}$  is continuous in probability in the sense that if  $t_n \rightarrow t$  then  $Y_{t_n} \rightarrow Y_t$  in probability.

*Hint:* It suffices (why?) to consider the two cases " $t_n \geq t$  for all  $n$ " and " $t_n \leq t$  for all  $n$ ".

**Exercise 94.**

- (1) Show that  $(\nu_t)_{t \geq 0}$  defined in Equation (9.5) is a regular convolution semi-group.
- (2) Show that  $(\nu_t)_{t \geq 0}$  defined in Equation (9.6) is a regular convolution semi-group.

**Exercise 95.**

Let  $(X_t)_{t \geq 0}$  be a real stochastic process (i.e., the state space is  $S = \mathbb{R}$ ). This process is called *square-integrable* if  $X_t \in L^2$  for all  $t \geq 0$ . Recall that  $(X_t)_{t \geq 0}$  is called *Gaussian* if  $(X_{t_1}, \dots, X_{t_n})$  follows an  $n$ -variate normal distribution for all  $n \in \mathbb{N}$  and  $0 \leq t_1 \leq \dots \leq t_n$ .

- (1) Show that if  $(X_t)_{t \geq 0}$  is square-integrable and has independent increments then  $\text{Cov}(X_s, X_t) = \text{Var}(X_s)$  for  $0 \leq s \leq t$ .
- (2) Show that if  $(X_t)_{t \geq 0}$  is Gaussian then the process has independent increments if and only if  $\text{Cov}(X_s, X_t) = \text{Var}(X_s)$  for  $0 \leq s \leq t$ .

**Exercise 96.**

Let  $(Y_t)_{t \geq 0}$  denote a real-valued Lévy process.

- (1) Assume  $Y_t$  is integrable for every  $t \geq 0$ . Let  $\mu = \mathbb{E}[Y_1]$ .
  - (a) Determine  $\mathbb{E}[Y_n]$  for  $n \in \mathbb{N}$ .
  - (b) Determine  $\mathbb{E}[Y_{\frac{1}{2}}]$ .
 

*Hint:* Use the decomposition  $Y_1 = Y_{\frac{1}{2}} + (Y_1 - Y_{\frac{1}{2}})$ .
  - (c) Determine  $\mathbb{E}[Y_{\frac{1}{n}}]$  for every  $n \in \mathbb{N}$ , and show that  $\mathbb{E}[Y_{\frac{m}{n}}] = \frac{m}{n}\mu$  for all  $m, n \in \mathbb{N}$ .
- (2) Now assume that  $Y_t$  is square-integrable for all  $t \geq 0$ . Show that  $\text{Cov}(Y_s, Y_t) = \text{Var}(Y_s)$  for all  $0 \leq s \leq t$ .
- (3) Now assume that  $Y_t \sim N(0, t)$  for every  $t \geq 0$ . Determine  $\mathbb{E}[Y_4^2 | \mathcal{F}_3^Y]$ .

**Exercise 97.**

Does there exist a real Lévy process  $(Y_t)_{t \geq 0}$  such that  $Y_t \sim N(0, t)$  for  $t \in [0, 1]$  and  $Y_t \sim N(0, t^2)$  for  $t > 1$ ?

**Exercise 98.**

Let  $(B_t)_{t \geq 0}$  denote a Brownian motion and consider the filtration  $(\mathcal{F}_t^B)_{t \geq 0}$ . Which of the following processes are progressively measurable?

- (1)  $(B_t)_{t \geq 0}$  and  $(B_t \cos t)_{t \geq 0}$ .
- (2)  $(B_{t+1})_{t \geq 0}$ .
- (3)  $(B_{t \wedge \tau})_{t \geq 0}$ , where  $\tau$  is an extended stopping time.
- (4)  $(\mathbb{1}_{\{B_t > 7\}})_{t \geq 0}$ .
- (5) The process  $(Y_t)_{t \geq 0}$  defined as follows: Let  $\tau_1$  and  $\tau_2$  denote extended stopping times and put  $Y_t(\omega) = \mathbb{1}_{(\tau_1(\omega), \tau_2(\omega)]}(t)$ .

**Exercise 99.**

Let  $X$  and  $Y$  be independent random variables and let  $p > 0$ . Show that  $X$  and  $Y$  are in  $L^p$  if and only if  $X + Y$  is in  $L^p$ .

*Hint:* By the “useful rule”,  $\mathbb{E}[|X + Y|^p] = \mathbb{E}[H(Y)]$ , where  $H(y) = \mathbb{E}[|X + y|^p]$ .

**Exercise 100.**

Let  $(Y_t)_{t \geq 0}$  be a real Lévy process and let  $p > 0$ . Show that  $Y_t \in L^p$  for all  $t \geq 0$  if and only if  $Y_1 \in L^p$ .

We say that  $(Y_t)_{t \geq 0}$  is *integrable* (resp. *square-integrable*) if  $Y_1$  is integrable (resp. square-integrable).

**Exercise 101.**

Let  $(Y_t)_{t \geq 0}$  be a real square-integrable Lévy process and put  $\sigma^2 = \text{Var}(Y_1)$ . Show that  $\text{Var}(Y_t) = \sigma^2 t$  for  $t \geq 0$ .

*Hint:* Consider the three cases (a):  $t = 1/n$  for  $n \in \mathbb{N}$ , (b):  $t = m/n$  for  $m, n \in \mathbb{N}$ , and (c):  $t \geq 0$ .

As in the previous exercise it can be shown that if  $(Y_t)_{t \geq 0}$  is integrable then  $\mathbb{E}[Y_t] = \mu t$  for  $t \geq 0$  where  $\mu = \mathbb{E}[Y_1]$ .

**Exercise 102.**

Let  $(Y_t)_{t \geq 0}$  be a real Lévy process. Assume  $Y_1 \geq 0$  a.s.

- (1) Using the same steps as in Exercise 101 show that  $Y_t \geq 0$  a.s. for  $t \geq 0$ .
- (2) Show that  $t \mapsto Y_t$  is non-decreasing a.s.

**Exercise 103.**

Let  $(Y_t)_{t \geq 0}$  be a  $d$ -dimensional Lévy process. In the following, treat  $Y_t$  as a column vector.

- (1) Let  $A$  be an  $m \times d$  matrix with real entries. Show that  $(AY_t)_{t \geq 0}$  is an  $m$ -dimensional Lévy process.

- (2) Let  $(Z_t)_{t \geq 0}$  be a  $d$ -dimensional Lévy process independent of  $(Y_t)_{t \geq 0}$ . Show that  $(Y_t + Z_t)_{t \geq 0}$  is a Lévy process.

**Exercise 104.**

Let  $(Y_t)_{t \geq 0}$  denote a one-dimensional integrable  $(\mathcal{F}_t)_{t \geq 0}$ -Lévy process. Assume  $\mathbb{E}[Y_t] = 0$  for all  $t$ .

- (1) Show that  $(Y_t)_{t \geq 0}$  is an  $(\mathcal{F}_t)_{t \geq 0}$ -martingale.
- (2) Assume  $(Y_t)_{t \geq 0}$  is square-integrable and put  $\sigma^2 = \text{Var}(Y_1)$ . Using Exercise 101 show that  $(Y_t^2 - \sigma^2 t)_{t \geq 0}$  is an  $(\mathcal{F}_t)_{t \geq 0}$ -martingale.

**Exercise 105.**

Let  $(Y_t)_{t \geq 0}$  be a stochastic process with state space  $\mathbb{R}^d$  for some  $d \in \mathbb{N}$ .

- (1) Let  $(\nu_t)_{t \geq 0}$  be a regular convolution semigroup on  $\mathbb{R}^d$  and let  $(Y_t)_{t \geq 0}$  have stationary independent increments and satisfy  $Y_t \sim \nu_t$  for every  $t \geq 0$ . Show that  $(Y_t)_{t \geq 0}$  is continuous in probability.
- (2) Let  $(Y_t)_{t \geq 0}$  be a Lévy process. Argue that  $(Y_t)_{t \geq 0}$  is continuous in probability.

**Exercise 106.**

Let  $(B_t)_{t \geq 0}$  denote a standard Brownian motion and let  $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  denote a non-decreasing continuous function with  $\phi(0) = 0$ .

Use  $(B_t)_{t \geq 0}$  to construct a Gaussian process  $(Y_t)_{t \geq 0}$  with independent increments satisfying  $Y_t \sim N(0, \phi(t))$  for all  $t \geq 0$  and that  $t \mapsto Y_t(\omega)$  is continuous for every  $\omega$ . What is the distribution of  $Y_t - Y_s$  for  $0 \leq s \leq t$ ?

**Exercise 107.**

Prove Theorem 10.12. In item (2) first show that  $(B_t)_{t \geq 0}$  and  $(\tilde{B}_t)_{t \geq 0}$  are versions.

**Exercise 108 (The Law of Large Numbers for Brownian motion).**

Let  $(B_t)_{t \geq 0}$  denote a Brownian motion with parameter  $\sigma^2 > 0$ . Show that, almost surely,  $\lim_{t \rightarrow \infty} \frac{B_t}{t} = 0$ .

*Hint:* Apply Theorem 10.12.

**Exercise 109.**

Let  $(B_t^i)_{t \geq 0}$  ( $i = 1, 2$ ) denote independent standard Brownian motions. Define

$$X_t = B_t^1 + B_t^2 \quad \text{and} \quad Y_t = B_t^1 - B_t^2 \quad \text{for } t \geq 0.$$

- (1) Show that  $(X_t)_{t \geq 0}$  and  $(Y_t)_{t \geq 0}$  both are Brownian motions and determine in both cases the parameter  $\sigma^2$ .

(2) Are  $(X_t)_{t \geq 0}$  and  $(Y_t)_{t \geq 0}$  independent?

*Hint:* The processes are independent if and only if  $(X_{t_1}, \dots, X_{t_n})$  and  $(Y_{t_1}, \dots, Y_{t_n})$  are independent for all  $n \in \mathbb{N}$  and all  $0 \leq t_1 \leq \dots \leq t_n$ .

**Exercise 110.**

Let  $(B_t)_{t \geq 0}$  denote a standard Brownian motion. For  $n = 1, 2$  let

$$\alpha_n = \inf\{t \geq 0 : B_t \geq n\}.$$

Later we show that  $\alpha_n$  is a.s. finite for  $n = 1, 2$ . You may take this for granted in the following.

- (1) Show that  $\{\alpha_1 > 3\}$  and  $B_{\alpha_1+7}$  are independent.
- (2) Show that  $\alpha_2 - \alpha_1$  and  $\alpha_1$  are independent and identically distributed.

**Exercise 111.**

Let  $(B_t)_{t \geq 0}$  denote a Brownian motion with parameter  $\sigma^2$ . Let  $a > 0$  and  $\tau$  denote the hitting time to  $(a, \infty)$ , i.e.,

$$\tau = \inf\{t > 0 : B_t > a\}.$$

It follows e.g. from Durrett (2010, Theorem 4.1.2) that  $\tau$  is finite with probability

1. Let  $B_{\tau+t} = a$  on the null set  $\{\tau = \infty\}$ .

Show that  $(B_{\tau+t} - a)_{t \geq 0}$  is a Brownian motion with parameter  $\sigma^2$ .

**Exercise 112.**

Let  $(B_t)_{t \geq 0}$  be a standard Brownian motion defined on  $(\Omega, \mathcal{F}, \mathbb{P})$ . Define

$$\tau = \inf\{t \geq 0 : B_t \geq 2\}.$$

This stopping time is finite with probability one. (Why is it a stopping time—and with respect to which filtration?). More generally, any stopping time of the form

$$\inf\{t \geq 0 : B_t \geq c\}, \quad c > 0,$$

is a.s. finite. (This follows e.g. from Durrett (2010, Theorem 4.1.2).)

- (1) Show that for any  $t > 0$ ,  $B_{\tau+t} \sim N(2, t)$ .
- (2) Put  $\sigma = \inf\{t \geq 0 : B_t \leq -1\}$ . Show that the event  $\{\sigma < \tau\}$  is independent of the process  $(B_{\tau+t})_{t \geq 0}$ .
- (3) Show that the event  $\{B_3 < 0, \tau > 3\}$  is independent of  $(B_{\tau+t})_{t \geq 0}$ .
- (4) For each  $n \in \mathbb{N}$ , put  $\alpha_n = \inf\{t \geq 0 : B_t \geq n\}$ . Argue that  $\alpha_1, \alpha_2 - \alpha_1, \alpha_3 - \alpha_2, \dots$  is an i.i.d. sequence.

**Exercise 113.**

Let  $(N_t)_{t \geq 0}$  denote a Poisson process with parameter  $\lambda > 0$ . Suppose we look back in time at time  $T$ , where  $T > 0$ , and let  $L_T$  denote time since last jump. (Thus, if at time  $T$  we have seen precisely  $n$  jumps, where  $n \in \mathbb{N}$ , then  $L_T = T - T_n$ ). Put  $L_T = T$  if there are no jumps in the process on  $[0, T]$ .

- (1) Determine the distribution of  $L_T$ . (That is, you may e.g. determine the cdf.)
- (2) Show that  $L_T$  converges in distribution as  $T \rightarrow \infty$  and determine the limiting distribution.

**Exercise 114.**

Let  $(N_t)_{t \geq 0}$  denote a Poisson process with parameter  $\lambda > 0$ . Show that  $N_t \rightarrow \infty$  as  $t \rightarrow \infty$  a.s.

*Hint:* Use the Law of Large Numbers (for i.i.d. random variables) to show that  $N_n \rightarrow \infty$  a.s. as  $n \rightarrow \infty$  (here  $n$  denotes an integer). Then use the non-decreasing paths to infer the result.

**Exercise 115.**

Show formula (11.11) in the case  $n = 2$ .

*Hint:* Condition on  $(N_{t_1}, N_{t_2} - N_{t_1})$ .

**Exercise 116.**

Let  $(N_t)_{t \geq 0}$  denote a Poisson process with parameter  $\lambda$ . Determine

- (1)  $\text{Cov}(N_1, N_2 + (N_6 - N_5)^7)$ ,
- (2)  $\mathbb{P}(N_1 = N_2 = 0)$ ,
- (3)  $\mathbb{P}(T_n \leq t < T_{n+1})$  and  $\mathbb{P}(T_n < t < T_{n+1})$  for any  $n \in \mathbb{N}$  and  $t > 0$ ,
- (4)  $\mathbb{E}[N_2 | N_1 > 0]$ ,
- (5)  $\mathbb{P}(N_1 = 1, H_2 > 2)$ ,
- (6)  $\mathbb{P}(N_{T_n+t} = n)$  for any  $n \in \mathbb{N}$  and  $t > 0$ .

**Exercise 117.**

Let  $f : [0, \infty) \rightarrow \mathbb{R}$  be a non-increasing function that satisfies  $f(0) = 0$  and  $f(s+t) = f(s) + f(t)$  for all  $s, t \geq 0$ . Show that  $f(t) = tf(1)$  for all  $t \geq 0$ .

**Exercise 118.**

For  $j = 1, 2$  let  $(N_t^{(j)})_{t \geq 0}$  be a Poisson process with parameters  $\lambda_j$ . Assume these processes are independent.

What is the probability that the two processes jump at the same time, i.e., that there is a  $t > 0$  such that  $\Delta N_t^{(1)}$  and  $\Delta N_t^{(2)}$  are both positive?

**Exercise 119.**

Show (a)–(b) in Example 11.19.

*Hint:* In (b) note that you have a triangular matrix when calculating the Jacobian; to calculate the determinant it hence suffices to consider the diagonal entries.

**Exercise 120.**

Show the result in Remark 12.4.

*Hint:* Apply the following version of Lévy's theorem: If  $Z \in L^1(\mathbb{P})$  and  $(\mathcal{G}_n)_{n \geq 1}$  is a decreasing family of  $\sigma$ -fields in  $\mathcal{F}$  then  $\mathbb{E}[Z | \mathcal{G}_n] \rightarrow \mathbb{E}[Z | \cap_{k=1}^{\infty} \mathcal{G}_k]$  as  $n \rightarrow \infty$  a.s. and in  $L^1$ .

**Exercise 121.**

Verify (12.13).

**Exercise 122.**

Let  $(Y_n)_{n \geq 0}$  and  $(Z_n)_{n \geq 0}$  be real stochastic processes that are versions. Show that  $(Y_n)_{n \geq 0}$  is an  $(\mathcal{F}_n^Y)_{n \geq 0}$ -martingale if and only if  $(Z_n)_{n \geq 0}$  is an  $(\mathcal{F}_n^Z)_{n \geq 0}$ -martingale.

Does this result transfer to the case where the index set is  $T = \mathbb{R}_+$ ? (See Chapter 8 for a definition of versions in the continuous time case).

**Exercise 123.**

Let  $(Y_n)_{n \geq 0}$  be a Markov chain with transition kernel  $Q$  with respect to some filtration  $(\mathcal{F}_n)_{n \geq 0}$ . Show that for every  $n \in \mathbb{N}_0$ ,  $A \in \mathcal{F}_n$  and  $B \in \sigma(Y_m | m \geq n)$ ,

$$\mathbb{P}(A \cap B | Y_n) = \mathbb{P}(A | Y_n) \mathbb{P}(B | Y_n).$$

What is the interpretation of this result?

**Exercise 124.**

Let  $(\mathcal{F}_t)_{t \geq 0}$  be a filtration.

- (1) Let  $s \in [0, \infty]$  be a fixed number. Put  $\tau = s$  and show that  $\tau$  is a stopping time and an extended stopping time. Determine  $\mathcal{F}_\tau$  and  $\mathcal{F}_{\tau+}$ .
- (2) Let  $f : \mathbb{R}_+ \rightarrow \mathbb{R}$  be  $(\mathcal{B}(\mathbb{R}_+), \mathcal{B}(\mathbb{R}))$ -measurable. Put  $Y_t(\omega) = f(t)$  for  $t \geq 0$  and  $\omega \in \Omega$ . Show that  $(Y_t)_{t \geq 0}$  is progressively measurable.
- (3) Prove the following generalization of Proposition 8.10(3): Let  $(Y_t)_{t \geq 0}$  be progressively measurable and  $\tau$  be an extended stopping time. Then  $(Y_{\tau \wedge t})_{t \geq 0}$  is progressively measurable.

**Exercise 125.**

Let  $S$  be finite and  $P = \{p_{ij}\}_{i,j \in S}$  be an irreducible transition matrix. Let  $\sigma = \inf\{n \in \mathbb{N} : X_n \neq X_0\}$  be the first time our Markov chain shifts state.

(1) Argue that  $\mathbb{P}_i(\sigma > n) = (p_{ii})^n$  for  $n \in \mathbb{N}$  and  $i \in S$ .

(2) Show that

$$\mathbb{P}_i(X_\sigma = j) = \frac{p_{ij}}{1 - p_{ii}} \quad \text{for } i, j \in S \text{ with } i \neq j.$$

### Exercise 126.

In practice, Markov processes are often specified via their *infinitesimal behavior*. This approach is discussed in detail in Chapter 12. In the following we consider the case when the state space is  $S = \{0, 1, 2\}$ , and state 2 for simplicity is taken to be absorbing. This situation occurs e.g. in the so-called *illness-death model* in survival analysis where state 0 corresponds to *healthy*, state 1 to *ill* and state 2 to *dead*.

In the following,  $(X_t)_{t \geq 0}$  is assumed to be a Markov process with transition semigroup  $(p_t)_{t \geq 0}$  under  $\mathbb{P}_x$  for  $x \in S$  where subscript  $x$  denotes that we start in  $x$ . Assume  $p_t$  is of the form

$$p_t = \begin{pmatrix} 1 - (q_{01} + q_{02})t & q_{01}t & q_{02}t \\ q_{10}t & 1 - (q_{10} + q_{12})t & q_{12}t \\ 0 & 0 & 1 \end{pmatrix} + o(t). \quad (\text{H.6})$$

Here,  $o(t)$  signifies a matrix that is very small as  $t \rightarrow 0$ . Moreover, we assume that the  $q_{xy}$ s are positive. The interpretation of (H.6) is as follows:

- If current state is 0 the probability that we jump to state 1 in a small interval of length  $t$  is approximately  $q_{01}t$  and the probability that we jump to 2 is approximately  $q_{02}t$ ; the probability that we remain in 0 is approximately  $1 - (q_{01} + q_{02})t$ .
- If current state is 1 the probability that we jump to state 0 in a small interval of length  $t$  is approximately  $q_{10}t$  and the probability that we jump to 2 is approximately  $q_{12}t$ ; the probability that we remain in 1 is approximately  $1 - (q_{10} + q_{12})t$ .
- State 2 is absorbing.

Let  $\tau = \inf\{t > 0 : X_t \neq X_0\}$  be the first time we jump away from the initial state.

The idea is that you should give heuristic arguments in the first two questions below using Exercise 125 and the assumption that, instead of observing the process continuously, we observe the Markov chain  $X_0, X_\Delta, X_{2\Delta}, \dots$  where  $\Delta > 0$  is very small (and  $\Delta \rightarrow 0$ ).

- (1) Give an argument that  $\mathbb{P}_0(\tau > t) = e^{-(q_{01} + q_{02})t}$  for  $t > 0$ . That is, under  $\mathbb{P}_0$   $\tau$  is exponentially distributed with parameter  $q_{01} + q_{02}$ .

(2) Argue that  $\mathbb{P}_0(X_\tau = 1) = \frac{q_{01}}{q_{01}+q_{02}}$  and  $\mathbb{P}_0(X_\tau = 2) = \frac{q_{02}}{q_{01}+q_{02}}$ .

Points (1) and (2) describe the behavior of  $X_t$  up to time  $\tau$  under  $\mathbb{P}_0$ . Namely, we should stay in 0 for an exponentially distributed amount of time,  $\tau$ , where the parameter is  $q_{01} + q_{02}$ . At time  $\tau$  jump to 1 with probability  $\frac{q_{01}}{q_{01}+q_{02}}$  and to 2 with probability  $\frac{q_{02}}{q_{01}+q_{02}}$ .

(3) Describe how the process evolves after time  $\tau$ . What do you implicitly assume?

(4) Simulate the process in the case  $q_{01} = 10$ ,  $q_{02} = 1$ ,  $q_{10} = 5$ ,  $q_{12} = 2$ .

### Exercise 127.

Consider a filtered probability space  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$  and let  $(X_t)_{t \geq 0}$  be a real progressively measurable process. Assume that  $\sup_{s \leq t} |X_s(\omega)| < \infty$  for every  $\omega \in \Omega$  and  $t \geq 0$ . Let  $\mu$  denote a measure on  $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$  satisfying that  $\mu(\{0\}) = 0$  and  $\mu([0, t]) < \infty$  for every  $t \geq 0$ .

Put

$$\begin{aligned} Y_t(\omega) &= \int_0^t X_s(\omega) ds && \text{for } t \geq 0 \text{ and } \omega \in \Omega, \\ Z_t(\omega) &= \int_0^t X_s(\omega) \mu(ds) && \text{for } t \geq 0 \text{ and } \omega \in \Omega. \end{aligned}$$

Here we put  $\int_0^t = \int_{[0, t]} = \int_{(0, t]}$ , where the last equality is due to that  $\mu(\{0\}) = 0$ . Also recall that  $ds$  means integration with respect to the Lebesgue measure.

- (1) Show that  $(Z_t)_{t \geq 0}$  is  $(\mathcal{F}_t)_{t \geq 0}$ -adapted and that  $t \mapsto Z_t(\omega)$  is càdlàg with  $Z_0(\omega) = 0$  for every  $\omega \in \Omega$ . Find an expression for  $\Delta Z_t(\omega)$  for  $t > 0$  and  $\omega \in \Omega$  in terms of  $\mu$  and  $(X_t)_{t \geq 0}$ .
- (2) Argue that  $(Y_t)_{t \geq 0}$  is  $(\mathcal{F}_t)_{t \geq 0}$ -adapted and that  $t \mapsto Y_t(\omega)$  is continuous with  $Y_0(\omega) = 0$  for every  $\omega \in \Omega$ .

Questions like the above are important in *stochastic calculus*. In this field it is of interest to replace  $\mu$  by a random measure or an object that is not even a measure.

### Exercise 128.

Let  $(B_t)_{t \geq 0}$  denote a standard Brownian motion, and let  $z \in \mathbb{R}$  and  $\lambda > 0$ . We say that a real stochastic process  $(X_t)_{t \geq 0}$  solves the *stochastic differential equation* (SDE)

$$dX_t = -\lambda X_t dt + dB_t \tag{H.7}$$

with initial condition  $X_0 = z$  if

$$X_t = z - \lambda \int_0^t X_s ds + B_t \quad \text{for all } t \geq 0 \quad (\text{H.8})$$

almost surely.

- (1) Show that the unique (up to indistinguishability) solution to the SDE (H.7) with initial condition  $X_0 = z$  is

$$X_t = B_t + e^{-\lambda t} \left[ z - \lambda \int_0^t B_s e^{\lambda s} ds \right] \quad \text{for } t \geq 0. \quad (\text{H.9})$$

*Hint:* Calculate  $\frac{\partial}{\partial t}(X_t - B_t)$ . As for the uniqueness, subtract two possible solutions.

- (2) Let  $X_t$  be given by (H.8). Show that  $X_t$  has a normal distribution with mean  $e^{-\lambda t}z$  for every  $t$ .

With some effort it can even be shown that  $\text{Var}(X_t) = \frac{1}{2\lambda}(1 - e^{-2\lambda t})$ . That is,  $X_t \sim N(e^{-\lambda t}z, \frac{1}{2\lambda}(1 - e^{-2\lambda t}))$ . You can use this freely in the following.

- (3) Let  $X_t$  be given by (H.8) for all  $t \geq 0$ . Show that  $(X_t)_{t \geq 0}$  is a Markov process with transition semigroup  $(P_t)_{t \geq 0}$  given by (H.5) with respect to  $(\mathcal{F}_t^B)_{t \geq 0}$ .

*Hint:* Let  $s, t \geq 0$ . Then

$$X_{s+t} = (B_{s+t} - B_s) + e^{-\lambda t} \left[ X_s - \lambda \int_0^t (B_{s+u} - B_s) e^{\lambda u} du \right].$$

Now apply Theorem E.7 to calculate  $\mathbb{E}[f(X_{s+t}) | \mathcal{F}_s^B]$  for  $f: \mathbb{R} \rightarrow \mathbb{R}$  measurable and bounded.

### Exercise 129.

Let  $(N_t)_{t \geq 0}$  denote a Poisson process with parameter  $\lambda > 0$ . Since  $t \mapsto N_t$  is non-decreasing and càdlàg, this map induces a measure, to be denoted  $\Lambda$ , on  $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$ . Recall that  $\Lambda$  is uniquely determined by the two properties

$$\Lambda(\{0\}) = 0 \quad \text{and} \quad \Lambda((u, v]) = N_v - N_u \quad \text{for } 0 \leq u \leq v. \quad (\text{H.10})$$

In the following we will use the standard notation

$$\int_0^\infty f(t) dN_t = \int_0^\infty f(t) \Lambda(dt) \quad (\text{H.11})$$

for  $f: \mathbb{R}_+ \rightarrow \mathbb{R}$  measurable, bounded and with bounded support. In particular  $\Lambda(A) = \int_0^\infty \mathbb{1}_A(t) dN_t$  for any bounded Borel set  $A$  in  $\mathbb{R}_+$ .

As always, we suppress  $\omega$ . Note however that  $\Lambda$  as well as the integrals do depend on  $\omega$ , i.e., they are random variables.

- (1) Let  $f : \mathbb{R}_+ \rightarrow \mathbb{R}$  be measurable, bounded and have bounded support. Show that the characteristic function of the integral in (H.11) is

$$\mathbb{E}[e^{iz \cdot \int_0^\infty f(t) dN_t}] = \exp \left[ \lambda \int_0^\infty (e^{izf(t)} - 1) dt \right] \quad \text{for } z \in \mathbb{R}.$$

*Hint:* It suffices (explain why without proving it in detail) to consider the case  $f(t) = \mathbb{1}_{(u,v]}(t)$  for  $0 \leq u < v < \infty$ .

- (2) Let  $n \in \mathbb{N}$  and  $f_1, \dots, f_n : \mathbb{R} \rightarrow \mathbb{R}$  be bounded, measurable and have bounded supports. Assume in addition that  $f_i \cdot f_j = 0$  for  $i, j = 1, \dots, n$  with  $i \neq j$ . Show that  $\int_0^\infty f_1(t) dN_t, \dots, \int_0^\infty f_n(t) dN_t$  are independent.

*Hint:* Note that for  $z = (z_1, \dots, z_n) \in \mathbb{R}^n$ ,  $f = (f_1, \dots, f_n)$  and  $t \geq 0$ ,

$$e^{i\langle z, f(t) \rangle} - 1 = \sum_{j=1}^n (e^{iz_j f_j(t)} - 1).$$

- (3) Let  $n \in \mathbb{N}$  and  $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R}_+)$  be bounded and disjoint. Show that the random variables  $\Lambda(A_1), \dots, \Lambda(A_n)$  are independent and  $\Lambda(A_j) \sim \text{po}(\lambda \text{Leb}(A_j))$  for  $j = 1, \dots, n$ , where  $\text{Leb}$  denotes the Lebesgue measure.

### Exercise 130.

Let the situation be as in Proposition 12.7. Show that  $Y_1$  is independent of  $H_1 = T_1$  under  $\mathbb{P}_i$ .

### Exercise 131.

Let  $(X_t)_{t \geq 0}$  be a Markov jump process with  $Q$ -matrix  $Q$ . Denote the jump chain by  $(Y_n)_{n \geq 0}$  and the jump times by  $(T_n)_{n \geq 0}$ .

- (1) Let  $i, j \in S$  with  $i \neq j$ . Assume  $q_{ij} > 0$ . Show that for arbitrary  $t > 0$ ,

$$\mathbb{P}_i(Y_1 = j, T_1 < t, T_2 > t) > 0.$$

Deduce that  $p_t(i, j) > 0$ .

- (2) We call  $Q$  *irreducible* if for arbitrary  $i, j \in S$  there is an  $n \in \mathbb{N}$  and a sequence  $i_0, i_1, \dots, i_n \in S$  with  $i_0 = i$  and  $i_n = j$  such that  $q_{i_{k-1}i_k} > 0$  for  $k = 1, \dots, n$ .

Use Chapman–Kolmogorov (Proposition 12.5) to show that if  $Q$  is irreducible then  $p_t(i, j) > 0$  for all  $i, j \in S$  and  $t > 0$ .

(To make the argument more transparent you may assume  $n = 2$  in the definition of irreducibility).

### Exercise 132.

Let  $(B_t)_{t \geq 0}$  denote a standard  $(\mathcal{F}_t)_{t \geq 0}$ -Brownian motion.

- (1) Show that  $(B_t)_{t \geq 0}$  is a standard  $(\overline{\mathcal{F}_t})_{t \geq 0}$ -Brownian motion.
- (2) Show that  $(B_t)_{t \geq 0}$  is a standard  $(\mathcal{F}_{t+})_{t \geq 0}$ -Brownian motion.  
*Hint:* Use that for  $t > 0$  and  $s \geq 0$ ,  $B_t - B_s = \lim_{n \rightarrow \infty} (B_t - B_{s+\frac{1}{n}})$  pointwise.
- (3) Show that for any  $\alpha \in \mathbb{R}$  the process  $(\exp(\alpha B_t - \frac{1}{2}\alpha^2 t))_{t \geq 0}$  is an  $(\mathcal{F}_t)_{t \geq 0}$ -martingale.

**Exercise 133 (Doob's inequality in continuous time).**

Let  $(X_t)_{t \geq 0}$  denote a square-integrable  $(\mathcal{F}_t)_{t \geq 0}$ -martingale. Assume  $t \mapsto X_t(\omega)$  is right-continuous for every  $\omega \in \Omega$ .

Using the discrete-time version of Doob's inequality show that

$$\mathbb{E}[(\sup_{s \leq t} |X_s|)^2] \leq 4\mathbb{E}[X_t^2]$$

for every  $t \geq 0$ .

**Exercise 134.**

Let  $S = \{1, 2\}$  and  $q \in (0, 1)$ . Does there exist a Markov jump process where  $(p_t)_{t \geq 0}$  is given by (7.10)?

**Exercise 135.**

Consider a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  and let  $\tau : \Omega \rightarrow [0, \infty)$  denote a random variable with CDF  $F$ . Define  $(N_t)_{t \geq 0}$  as

$$N_t(\omega) = \mathbb{1}_{[\tau(\omega), \infty)}(t) \quad \text{for } t \geq 0 \text{ and } \omega \in \Omega.$$

Assume that  $F(0) = 0$  and  $F(t) < 1$  for every  $t \geq 0$ .

- (1) Sketch a path of  $(N_t)_{t \geq 0}$ . Show that  $\tau$  is an  $(\mathcal{F}_t^N)_{t \geq 0}$ -stopping time.
- (2) Show that the process  $(\int_0^{t \wedge \tau} \frac{1}{1-F(u-)} dF(u))_{t \geq 0}$  is  $(\mathcal{F}_t^N)_{t \geq 0}$ -adapted and has càdlàg paths. (Here we put  $\int_0^t = \int_{(0, t]}$ ).
- (3) Show that  $\mathcal{F}_s^N$  ( $s \geq 0$ ) consists precisely of the following events:
  - (a)  $\{\tau \in B\}$  where  $B \in \mathcal{B}([0, s])$ ;
  - (b)  $\{\tau > s\}$ ;
  - (c)  $\{\tau > s\} \cup \{\tau \in B\}$  where  $B \in \mathcal{B}([0, s])$ .
- (4) Show that for  $s \leq t$  and  $A$  as in (a) or (b),

$$\mathbb{E}[(N_t - N_s) \mathbb{1}_A] = \mathbb{E}\left[\int_{(s \wedge \tau, t \wedge \tau]} \frac{1}{1-F(u-)} dF(u) \mathbb{1}_A\right].$$

Show that the process

$$N_t - \int_0^{t \wedge \tau} \frac{1}{1 - F(s-)} dF(s) \quad (t \geq 0) \quad (\text{H.12})$$

is an  $(\mathcal{F}_t^N)_{t \geq 0}$ -martingale.

**Exercise 136.**

For  $n \in \mathbb{N}$  let  $p_n \in (0, 1)$  and assume there is a positive  $\lambda$  such that  $np_n \rightarrow \lambda$  for  $n \rightarrow \infty$ . Show that  $\text{Bin}(n, p_n) \xrightarrow{\sim} \text{po}(\lambda)$  for  $n \rightarrow \infty$ .





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